Feeding the City: How Rural Productivity Shapes Urban Structural Change*

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Abstract

Agricultural productivity in the countryside can promote structural transformation in the cities it supplies with food. I exploit as-if-random variation in the agro-climatic match between settlers' origins and destinations under Indonesia's Transmigration Program, which generated differences in village productivity. I show that more productive rural hinterlands trigger structural changes in nearby cities, where the expansion of manufacturing and service employment exceeds the net addition of workers. Urban households consume more food while devoting a smaller share of spending to it, consistent with a relaxation of subsistence constraints and greater demand for non-agricultural goods. Together with higher household income per worker and spending, these consumption changes suggest improvements in urban living standards. These employment gains are larger in cities with weaker access to international rice markets, in line with their greater reliance on local food supply. I use a calibrated spatial model to examine this mechanism, showing how food trade supports urban structural transformation with local populations held fixed. Thus, rural productivity gains can translate into expanded services and manufacturing in cities without substantial population growth, especially where food trade relaxes subsistence constraints.

JEL codes: O18, R12, R23. *Keywords:* structural transformation, agricultural productivity, rural-urban spillovers, migration, urbanization.

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1 Introduction

Development programs of governments and intergovernmental organizations often target rural areas: raising farm productivity, expanding cultivation, and improving rural livelihoods. In agriculture-based countries, governments directed about 4% of their public spending to agriculture (World Bank 2007). These policies are typically evaluated based on the rural outcomes they achieve. Yet, when goods and labor markets are linked across space, rural productivity gains can also reshape the towns and cities nearby. Higher productivity may attract and retain workers in agriculture earning a higher income, while also reducing liquidity constraints for farmers to finance migration and leave for cities. Higher agricultural output can lower the local cost of living (especially when staple foods are locally sourced), raise rural incomes and demand for non-agricultural goods and services, and alter migration incentives and urban labor supply. The more cities rely on local food supply, the more likely rural spending is also urban policy. Most existing research estimates the effects of agricultural productivity gains *within* the region that received them, so the effect on cities cannot be separated from the effect on the villages themselves. How trade in goods and in labor spreads across space, and whether these channels translate into urban growth and structural transformation, are open questions.

In this paper, I study rural-to-urban spillovers by using a policy that generated a large and spatially dispersed rural shock, produced plausibly exogenous differences in rural productivity across the newly settled villages, and can be linked to urban outcomes over several decades. Indonesia's Transmigration Program provides this setting, having resettled over 2 million people from the more central islands of Java and Bali to the Outer Islands. Although the program was designed as a rural resettlement and agricultural development policy, it created sharp, localized changes in rural population and agricultural production in the hinterlands of many towns. Transmigrant households did not choose their destination villages, and farming households who ended up in destination villages more similar to their places of origin are more productive. Higher agro-climatic similarity between origin and destination raises village rice productivity by roughly 20% per standard deviation, for reasons unrelated to local economic conditions, since farming skills transfer better to places that resemble home in geography and climate (Bazzi et al. 2016). These villages trade in food with the cities around them, increasing the available food supply in a setting where households spend half of their household budget on agricultural goods.

I estimate rural-to-urban spillovers by comparing cities exposed to a more or less productive rural hinterland, measured as the average agro-climatic similarity of the transmigration villages within a five-hour journey of the city. I use three decades of Indonesian census microdata and village and household surveys along with the satellite-derived urban footprints of the Global Human Settlement Layer to construct a consistent-boundary sample of urban centers on the Outer Islands. I compare city population and employment in 2010, controlling for baseline population, employment, and urban structure in 1980. Baseline balance tests detect an association with the 1980 urban population share, which I control for. Most other baseline differences are imprecisely estimated. Cities surrounded by more productive villages change their sectoral composition more than their size: employment moves out of

urban agriculture into manufacturing and services with a positive but imprecisely estimated population response. Per standard deviation of hinterland agro-climatic similarity, the agricultural employment share falls by about 5.1 percentage points, against a 2010 mean of 37.4% and about a quarter of the average city's shift out of agriculture over the three decades, and modern-sector employment rises by about 17%. Comparing the growth in the workforce to the growth in modern-sector employment, I show using back-of-the-envelope calculations that new arrivals cannot account for the structural shift with the incumbent workforce fixed and without offsetting departures or labor-force entry and exit. Much of the fall in the agricultural share is within birthplace groups. Repeated cross-sections do not identify individual moves or occupational transitions.

The food-consumption evidence is consistent with trade in food supporting urban structural transformation, and the response concentrates in the closed cities, those less connected to global markets, where food availability and food trade are more likely to bind. Urban residents consume more food, measured in quantities and calories, while the food expenditure share falls. Higher overall household spending is consistent with the higher demand for non-agricultural goods and services, pulling labor into the modern sector. The countryside moves the opposite way. Villages with higher agro-climatic similarity keep at least as large a share of their workers in agriculture and produce more food. The settler-retention response to village agro-climatic similarity is imprecisely estimated. Among urban residents born between 1955 and 1990, the share born elsewhere in the same province rises, while the share born on Java or Bali falls.

To provide further evidence on the role of food trade in urban employment composition, I compare the effects of productivity exposure in cities that are more vs. less connected to global trade networks. I conjecture that less connected ("closed") cities must rely more on local food supply to alleviate the subsistence constraint, and thus respond more strongly to rural productivity gains. I define city-level trade openness using proximity to international rice-shipping routes, measuring each city's travel time by land and sea to the Thai and Vietnamese hubs that ship most of Indonesia's rice imports. The coefficient estimates for population and sectoral employment reallocation are larger in the closed cities, by more than double on modern-sector employment. These estimates provide further suggestive evidence on trade openness shaping the urban response to rural productivity, complementing the predictions of Matsuyama (1992) and the cross-country evidence in Glaeser (2014).

I use a spatial model with two sectors, agriculture and combined manufacturing and services, to study how rural productivity affects urban employment through trade and migration. Both goods are tradable. Workers choose a location and a sector, and Stone-Geary preferences impose a subsistence food requirement. A rural productivity gain changes the cost of food and the relative returns to agriculture and non-agriculture in surrounding cities. The model separates sectoral adjustment within locations from population adjustment across locations. Numerical comparisons in Section 6 illustrate these mechanisms in a variant with partially tradable non-agricultural output; those calculations are distinguished from the two-good main specification.

I make two contributions. First, my finding that a village productivity gain transforms the sectoral composition of the cities it feeds adds a supply-side spillover to the literature on spillovers of local shocks across space, which has followed such shocks through demand and through capital, and adds the hinterland's productivity to the explanations of urbanization without growth. Local shocks have been followed through demand, from cash transfers that raise consumption in nearby non-recipient villages (Egger et al. 2022) to resource booms that raise income and employment in surrounding counties (Feyrer et al. 2017; Allcott & Keniston 2018) and new jobs in one urban sector that create jobs in others (Moretti 2010), and agro-processing plants that raise employment in the villages around them (Edwards 2024; Kraus et al. 2024), and through capital, as agricultural windfalls flow through the banking system into industrial growth elsewhere (Bustos et al. 2020). The shock here reaches cities through the supply of the food sold by the nearby villages, and in related work on India, my co-authors and I study the reverse direction of urban employment shocks that reduce rural population but increase non-agricultural employment in surrounding villages (Asher et al. 2026). Studies of local productivity shocks therefore need to account for spillovers on population and sectoral employment in the places that buy goods from affected villages. For a government deciding how much of its budget goes to villages, the return to a rural program includes the shift in the sectoral structure of the cities those villages feed, which is largest where the cities have little access to imported food. Research on urbanization and structural transformation documents cities that gain population without commensurate industrialization (Gollin et al. 2016; Jedwab & Vollrath 2015; Gollin 2023; Henderson & Turner 2020), while spatial accounts of structural change emphasize the importance of sectoral reallocation within regions (Eckert & Peters 2025). I identify a rural source of urban structural change: productivity gains in the hinterland shift city employment toward manufacturing and services, with a comparatively modest population response. More broadly, the finding speaks to the macro-development literature on agricultural productivity and structural transformation (Gollin et al. 2007; Restuccia et al. 2008; Gollin et al. 2014; Herrendorf et al. 2014; McArthur & McCord 2017; Gollin et al. 2021), which studies a country as one closed economy, in which the countryside that becomes more productive and the cities it supplies form a single aggregate. Closest to this paper, Bazzi et al. (2016) establish the productivity variation itself. Settlers assigned to agro-climatically familiar destinations farm more productively, raising village rice yields and nightlights within the settlements. I take that village-level variation as the shock and follow it out of the settlements, asking what the productivity of a city's surrounding villages does to the city itself. Complementary work in progress studies the villages' own outcomes under the same program's variation in market access, where better-connected villages shift out of agriculture in place yet their income per capita falls (Mitchell 2026), while I show that hinterland productivity moves the cities those villages feed.

Second, I show that the urban response concentrates in cities weakly connected to outside food markets. The idea that cities need food to grow is not novel, and the evidence for it is cross-country and historical (Glaeser 2014; Gollin et al. 2007; Nunn & Qian 2011); I establish it with causal evidence from an exogenous shock to productivity. It also adds to the literature on internal trade and migration a case in which goods trade transmits the shock while labor flows stay small. Matsuyama (1992) predicts that higher agricultural productivity moves labor into manufacturing in a closed economy and into agriculture in a small open economy, where comparative advantage sets the manufacturing share; in

both cases the positive productivity shock is within the same economy where outcomes are measured. Existing studies estimate what an agricultural shock does inside the locality with the source of the shock, where the sign of the non-farm response varies with which markets happen to be open (Bustos et al. 2016; Emerick 2018; Santangelo 2019; Foster & Rosenzweig 2004; Hornbeck & Keskin 2015); across Indian districts, Moscona (2019) finds that the Green Revolution's productivity gains raised agricultural employment and land use and reduced urbanization and manufacturing employment, most in the districts more involved in trade. The empirical tests of that prediction (Bustos et al. 2016) and the within-country tests that interact an agricultural shock with regional openness (Moscona 2019; Liu et al. 2023; Marden 2016) measure the response on the shocked side. The city-and-hinterland economy of Glaeser (2014) is closer to my setting: the farms around the city become more productive and the city responds, and urbanization depends on rural prosperity when that economy is closed and on rural desperation when it is open. Matsuyama's open and closed cases have been tested on cross-country data, and Glaeser extends the theory from the sectoral response, agricultural contraction and the growth of the modern sector, to the growth of urban areas. In his closed economy urbanization increases with farm productivity, which is mostly consistent with my result: the population estimate is larger in the closed cities, and the change in sectoral employment is much greater than the change in population. The literature on internal trade and migration (Tombe & Zhu 2019; Bryan & Morten 2019; Morten & Oliveira 2024; Lagakos et al. 2023; Budí-Ors 2026) quantifies how goods-market and labor-market integration shape aggregate income; here goods trade transmits the shock while labor flows stay small, and in a world where migration is low, trade still accounts for the urban structural change, whereas migration without trade does not enable that transformation. The importance of food trade as the channel has implications for studying trade and migration frictions in spatial models of structural transformation (Tombe 2015; Nath 2025).

The rest of the paper is organized as follows. Section 2 describes the program and the urban setting, Section 3 the data and the construction of the exposure measure, Section 4 the framework and the empirical strategy, Section 5 the reduced-form results, Section 6 the model and its quantification; Section 7 concludes.

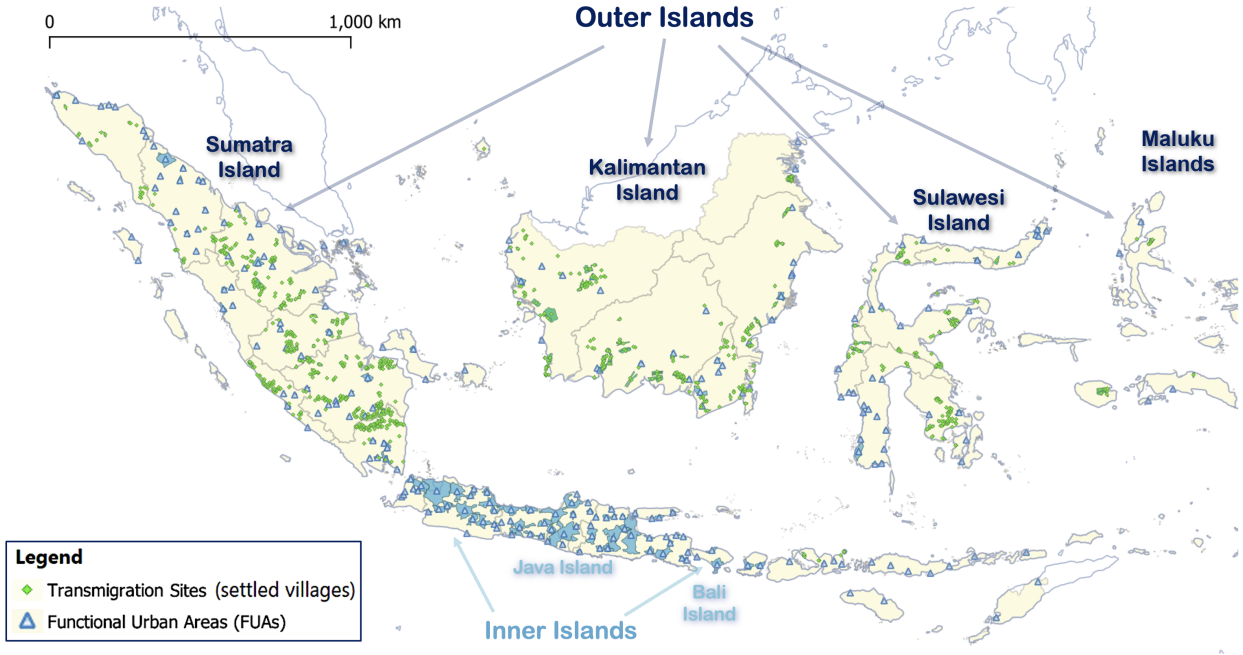
2 Setting

Indonesia's Transmigration Program resettled over two million people in new villages on the Outer Islands, and because households did not choose their destinations, the match between their farming origins and the destination's agro-climate, and with it village productivity, varied across villages for reasons unrelated to local economic conditions. The cities nearby are the units the paper follows: in 1980, 56.0% of their workers farmed on average, so a shift of city employment out of agriculture is structural transformation within the city.

2.1 The Transmigration Program and Rural Productivity

The Transmigration Program aimed to alleviate population pressure on Java and Bali by resettling households in less densely populated parts of Indonesia, including Sumatra, Kalimantan, Sulawesi, and Papua. During its peak period, from 1979 to 1988, it moved more than 2,000,000 volunteer migrants to newly established villages on the Outer Islands. Participating households received a land plot, a house, and free transportation to the destination. The program expanded agricultural settlement, largely on previously unused land, with the aim of increasing food production in the receiving regions. Its expansion in the early 1980s was followed by a sharp contraction as falling oil revenues reduced the funds available for resettlement (Bazzi et al. 2016). The settled villages were geographically dispersed across the Outer Islands (Figure 1).

Figure 1: Transmigration Villages and Urban Agglomerations



Notes: Diamonds mark the settled transmigration villages, from the Transmigration census (Bazzi et al. 2016). Triangles and shaded areas mark functional urban areas of the Global Human Settlement Layer (Moreno-Monroy et al. 2021). Papua is not shown: it is excluded from the analysis throughout (Section 2.2.1).

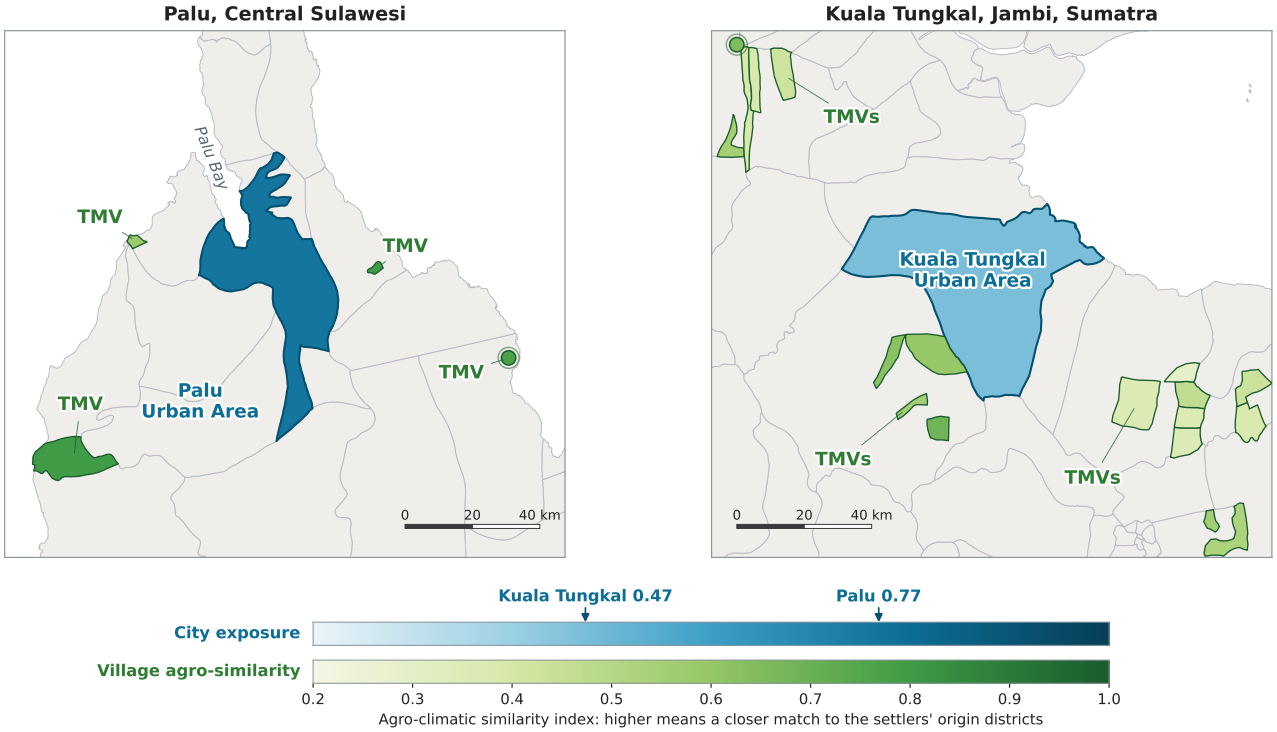
Transmigrant households did not choose their destination villages: which of the newly opened settlements a household was sent to depended on when it arrived at the resettlement center, and trains and ships departed for different destinations from one day to the next. Since migrants had no way to express any preference, and the government was not intentionally matching migrants from origins to destinations, there is variation in how similar or different the places of origin and destination are in their agro-climatic characteristics, soil and geography among them. Conditional on where the government chose to open a settlement site, the composition of settlers arriving there was therefore as-if random. Consistent with this assignment process, Bazzi et al. (2016) find that the agro-climatic

similarity between settlers' origins and their destinations is uncorrelated with predetermined village characteristics, including potential crop yields and development in the surrounding area.

Agro-climatic similarity measures how closely a destination's growing conditions resemble those in the migrants' regions of origin, providing a proxy for the transferability of their farming skills. Bazzi et al. (2016) estimate that a one-standard-deviation increase in similarity raises village rice productivity by roughly 20%. The replication data used here cover 814 settled villages where roughly 1,500,000 transmigrants were initially placed. Section 3.3 defines the similarity index.

I construct a city-level measure of exposure to rural productivity by estimating travel times between cities and transmigration villages along land and sea routes. For each city, I then average the agro-climatic similarity of the villages within 300 travel minutes.[1] In Figure 2, the villages around Palu in Central Sulawesi have higher average similarity than those around Kuala Tungkal in Sumatra.

Figure 2: Sample Areas of Urban Agglomerations and Nearby Transmigration Villages



Notes: Transmigration villages are shaded by agro-climatic similarity between settlers' origins and destinations. Urban subdistricts are shaded by their city's mean similarity across villages within 300 travel minutes, used as a proxy for rural productivity exposure. Palu, Central Sulawesi (left), and Kuala Tungkal, Jambi, Sumatra (right), are drawn at a common scale. Mean similarity, on the 0 to 1 index, is 0.77 across 9 villages for Palu and 0.47 across 14 for Kuala Tungkal. The map frames differ from the travel-time catchments: the left panel shows 4 settler villages, of which 3 enter Palu's exposure; the right shows 18, of which 8 enter Kuala Tungkal's. A village whose boundary is absent from the village polygon layer is drawn as a circle at its recorded location.

To compare resettlement with the size of the receiving countryside, I define a city's rural hinterland as rural subdistricts outside every urban center and within 135 km of the city. This straight-line

[1] I augment an inherited land-travel friction surface with maritime connections and calculate the pairwise distances used here; Appendix B.1 describes the spatial construction and source limitations. Appendix Figure A1a shows the joint variation in agro-climatic similarity and settler exposure across cities.

radius gives the closest approximation, among the distances examined, to the set of city-village pairs within the 300-minute travel-time window.[2] Across the 134 cities, settlers placed in these hinterlands amount to about 7% of the hinterlands' 1980 population, with roughly a seventh of cities above 15%. The 1980 census precedes the settlement of 84% of the program's villages, so the denominator includes some early arrivals.

2.2 Urbanization and Structural Transformation on the Outer Islands

The Outer Islands entered the study period less urban and more agricultural than Java and Bali. In 1980, agriculture accounted for 67% of employment on the Outer Islands, compared with 51% on Java and Bali. The urban population shares were 18% and 25%, respectively.[3] By 2010, the Outer Islands' urban share had risen to 37%, and their agricultural employment share had fallen to 52%. The shift toward non-agricultural employment was concentrated in services, whose share rose from 23% to 35%, while manufacturing's fell from 6% to 5%. On Java and Bali, manufacturing's employment share rose from 11% to 15%.

2.2.1 The Urban Sample

The analysis follows urban centers on the Outer Islands. Java and Bali do not have settled transmigration villages in the program sample. I exclude Papua because the 1980 census measures needed for baseline controls are unavailable and early spatial coverage is sparse. Indonesia's administrative hierarchy consists of provinces, districts (*kabupaten* and *kota*), subdistricts (*kecamatan*), and villages (*desa* and *kelurahan*). I aggregate subdistricts into cities to match the geographic detail available consistently across the data sources.

I define the satellite-based cities using the urban-center polygons of the Global Human Settlement (GHS) Urban Center Database. The analysis uses the 2020 footprints, with city membership held fixed across census years. I overlay these polygons with subdistrict boundaries and aggregate the overlapping subdistricts into urban centers, including the rural parts of those subdistricts. The functional urban areas shown in Figure 1 are a separate GHS layer that includes commuting zones around urban centers (Moreno-Monroy et al. 2021). [4]

The satellite layers are based on partial interpolation of the built-up area and the gridded population map, and the historical record for Indonesia has a gap (measured in the Online Supplement), so the layer does not give a full picture of which areas were urban in 1980. I therefore go through an additional reclassification using the fine subdistrict-level data from 1980 and two meaningful distinctions of cities, density and the share of agricultural employment. A subdistrict is added to the urban

[2] The matching criterion penalizes both pairs omitted by the distance cutoff and pairs included by it that lie beyond the travel-time window.

[3] These aggregates use the harmonized census files described in Section 3 and exclude Papua in both years (Section 2.2.1). Badan Pusat Statistik (BPS), Indonesia's national statistical agency, classifies villages as urban or rural using population density, the share of farm households, and access to urban facilities.

[4] Appendix A.1 details the polygon-to-city mapping. Section 3.2 explains outcome aggregation, and Section 4.3 discusses the implications of using endline geography.

sample when its 1980 population density exceeds 500 persons per square kilometer and its agricultural employment share is below one half. Because both rules use only 1980 observables, the sample does not condition on the treatment. Density identifies the communities that are more urban, and the share of agriculture the ones that have likely started urbanizing but not fully; together they provide a simple but meaningful check on which other areas could be viewed as urban but might not be captured by an urban layer with partially missing historical satellite data. The sectoral results hold when the reclassified areas are excluded and when both criteria are tightened at the same time on a smaller sample (Appendix Table A2); Appendix Figure A2a has the diagnostics behind the rule, and Appendix A.1 names the added and re-attached centers.

The main city sample retains centers whose nearest transmigration village lies within 300 travel minutes and whose 1980 baseline is directly observed in the census. Section 3.3 describes the exposure and sample restrictions. Table 1 summarizes these cities.[5]

Table 1: Summary Statistics: The Main City Sample

Panel A: Levels (counts and the travel time to the nearest transmigration village)

	Mean	SD	Median	Min	Max	N
Total population 1980	118,652	220,180	67,979	5,785	2,078,951	134
Total population 2010	229,366	404,467	115,447	28,284	3,803,645	134
Total employed 1980	32,841	57,395	18,874	2,226	537,303	134
Total employed 2010	90,259	152,082	48,181	10,952	1,419,054	134
Agricultural employment 1980	17,376	21,328	10,622	261	148,091	134
Agricultural employment 2010	27,455	29,925	18,530	1,298	161,274	134
Modern-sector employment 2010	53,037	110,425	23,658	1,533	1,061,804	134
Minutes to nearest transmigration village	109	86	78	1	300	134

Panel B: Shares, the unemployment rate, and the exposure measures

	Mean	SD	Median	Min	Max	N
Urban population share 1980	0.260	0.228	0.239	0.000	0.807	134
Agricultural employment share 1980	0.560	0.219	0.579	0.045	0.981	134
Agricultural employment share 2010	0.374	0.183	0.362	0.050	0.899	134
Modern-sector employment share 2010	0.516	0.156	0.523	0.094	0.827	134
Unemployment rate 2010	0.122	0.040	0.116	0.038	0.223	134
Settler count (thousands per travel minute)	0.466	0.684	0.130	0.001	4.125	134
Agro-similarity	0.008	0.996	-0.094	-2.930	2.692	134

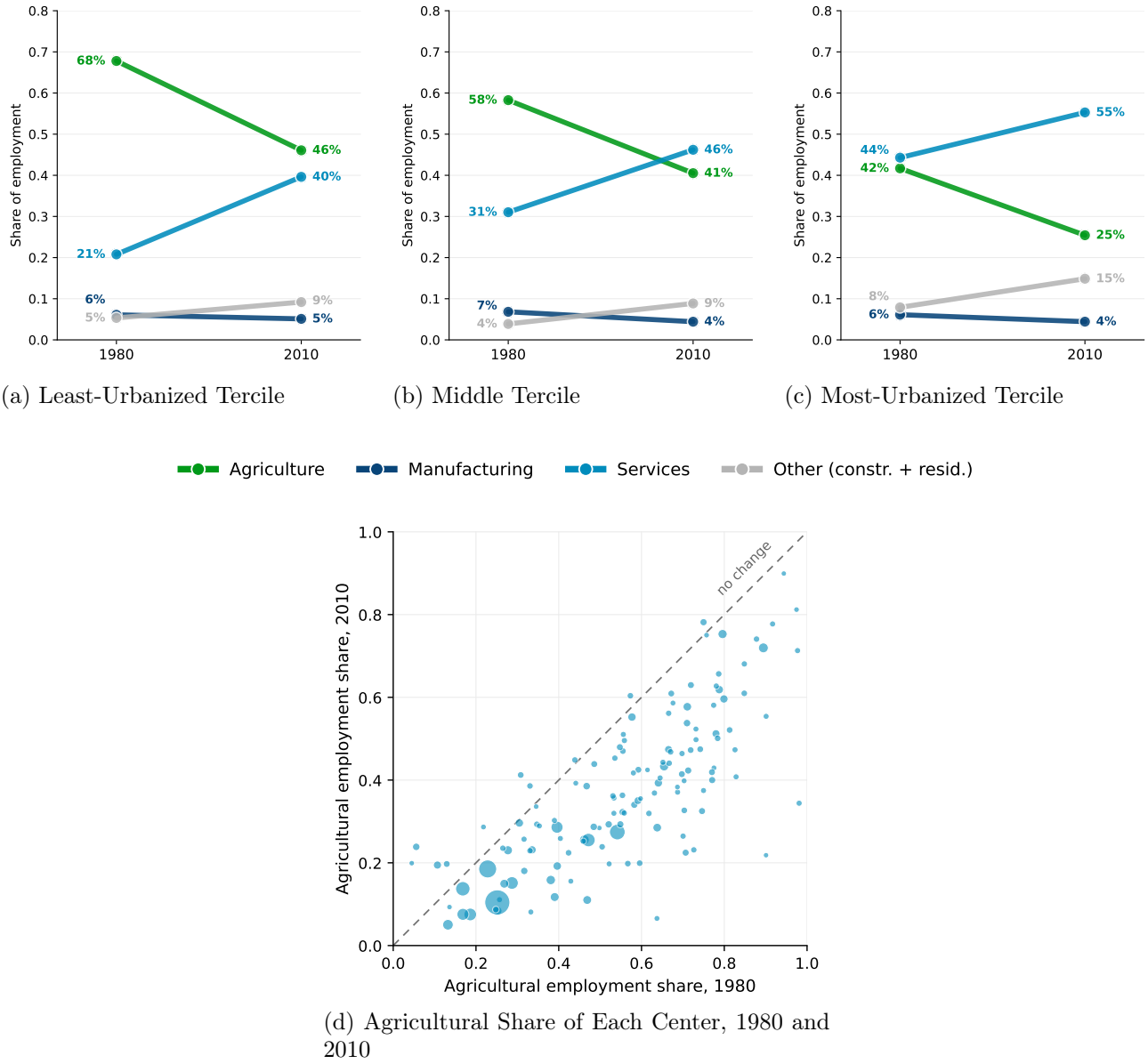
Notes: The main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline); every row is measured on the whole sample. The rows are the seven main 2010 outcomes, the exposure measures, the four 1980 baselines of the main specification, and the 1980 agricultural employment share. The settler count sums, over the transmigration villages within 300 travel minutes, the number of settlers placed in each divided by its travel minutes to the city, in thousands, so a village of 1,000 settlers 100 minutes away adds 0.010; agro-similarity is the standardized mean agro-climatic similarity of those villages. The modern sector is manufacturing plus services. Sources: 1980 and 2010 population censuses and the Transmigration census. Comparison with the other outer-island urban centers: Appendix Table A5.

Agricultural employment shares fell and service shares rose in each tercile of cities ranked by their 1980 administrative urban population share (Figure 3, panels (a)-(c)). Even in the most urbanized tercile, agriculture accounted for 42% of employment in 1980 and 25% in 2010. The decline extended

[5] Appendix Table A5 compares them with the other outer-island urban centers, Appendix Table A3 summarizes the program villages, and Appendix Table A4 compares cities, rural subdistricts hosting a transmigration village, and rural subdistricts without one.

across more than 90% of cities, with a median of 19 percentage points (panel (d)). The measured extent of urban agriculture also depends on how much rural territory these city boundaries include.

Figure 3: Sectoral Employment Change Across Urban Centers, 1980 to 2010



(a) Least-Urbanized Tercile

(b) Middle Tercile

(c) Most-Urbanized Tercile

(d) Agricultural Share of Each Center, 1980 and 2010

Notes: The sample is the main city sample. Panels (a)-(c) split the urban centers into terciles by 1980 urban population share (cutoffs 11% and 37%), held constant across both years; each line is the mean employment share of a sector across the centers of a tercile from 1980 to 2010, and "Other" combines construction with mining, utilities, and undefined industry codes. Panel (d): each center's 1980 (horizontal) and 2010 (vertical) agricultural employment share, marker area proportional to 2010 census population; the dashed line is the 45-degree line. Source: 1980 and 2010 population censuses.

2.2.2 Urban Agriculture

I use urban agriculture to denote agricultural employment among city residents. The census records where workers live, so these measures do not identify the location of their jobs or farm plots. When whole member subdistricts are counted, the mean agricultural employment share in the estimation

sample fell from 56.0% in 1980 to 37.4% in 2010. Restricting the calculation to villages that touch the satellite footprint gives a 2010 mean of 18%; requiring those villages also to be classified as urban by BPS gives 14% (Appendix Table D3). These alternatives count all workers in each selected village, and the set of cities observed varies across definitions.

To measure variation within these city boundaries, I classify mapped villages by whether their interior points fall inside the footprint. Each selected village is counted in full, including any territory extending beyond the footprint. For each city, I pool workers separately across the two sets of villages. Across the 120 cities with a measured share for the inside group, the median city-level agricultural employment share is 8% for that group and 49% for the outside group.[6]

Figure 4 shows the village-level variation in Palembang, Sumatra, and Gorontalo, Sulawesi. Among mapped villages whose interior points lie inside the footprints, agriculture accounts for 3.4% of employed residents in Palembang and 7.2% in Gorontalo; the corresponding shares for the outside villages are 50.4% and 29.0%. The maps show higher shares toward the edges of both footprints, with considerable variation among the inside villages: 6 of 100 in Palembang and 33 of 79 in Gorontalo have agricultural employment shares above 10%.

When city outcomes are recalculated from less agricultural member subdistricts, the estimated agricultural-share response to exposure remains negative and statistically significant (Appendix Table D4). These regressions use smaller city samples, and the modern-employment estimate becomes imprecise under the stricter restriction.

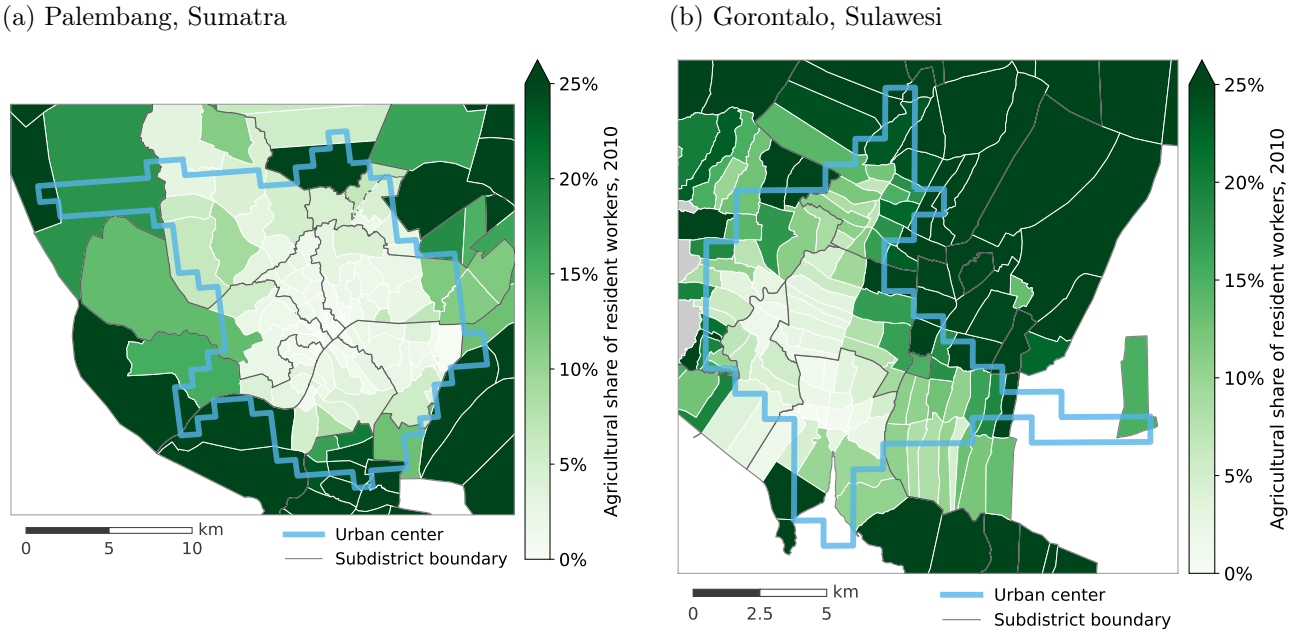
Urban agriculture is also documented outside Indonesia. In Nairobi, 20% of households grew crops within city limits in the 1984-1985 survey (Foeken & Mwangi 1998). More recently, Gallaher et al. (2015) document vegetable cultivation in sacks in densely populated Kibera, where access to land is limited.[7]

Limited access to affordable food is one possible reason that urban residents continue farming. Indonesia's household food expenditure share fell as its urban share rose, a pattern also observed in the large countries shown in Figure 5. This aggregate association does not establish that food scarcity explains agricultural employment in the sample cities.

[6] McGee (1991) describes Southeast Asian areas combining rice cultivation and non-agricultural activity at urban densities as *desakota*, from the Indonesian words for village and city. Across countries, agricultural employment within cities declines with development and city size (Gollin et al. 2016; Jedwab et al. 2025).

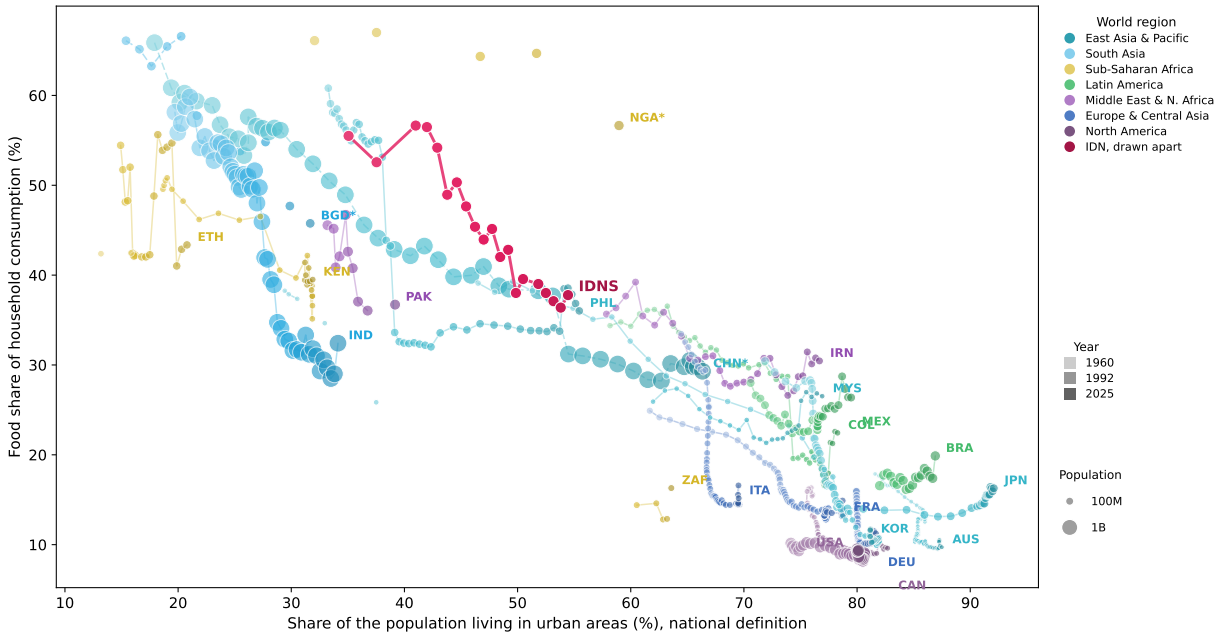
[7] The Nairobi percentage measures household participation in crop cultivation; the Indonesian shares measure agricultural employment among residents.

Figure 4: Agricultural Employment Among Village Residents in Palembang and Gorontalo



Notes: Agricultural workers as a share of employed residents in each mapped 2010 village of the cities' member subdistricts, from the 2010 population census. The wide outline is the Global Human Settlement urban-center footprint; thin dark lines separate subdistricts. Both maps use a common color scale from 0 to 25%; higher shares receive the darkest shade. The inside group consists of whole villages whose interior points lie inside the footprint, including any parts extending beyond it. Of these villages, 1 in Palembang and 3 in Gorontalo exceed the color limit. Gray denotes villages without measured employment; white areas fall outside the mapped member villages. The maps identify workers' residential villages; workplace and farm-plot locations are unobserved.

Figure 5: Food Share of Household Consumption vs. Urban Population Share, Largest Countries



Notes: Country paths plot the food share of household consumption against the urban population share under national definitions. Marker size indicates population; darker shades indicate later years. Indonesia (IDNS) uses the adjusted Susenas series. Dashed paths and asterisks identify broader food definitions whose levels are not directly comparable. Appendix A.3 documents country coverage, data sources, definition adjustments, and interpolation.

3 Data and Outcomes

I combine population censuses, transmigration records, household and village surveys, and spatial data to measure city outcomes and exposure to nearby settlements. The urban boundaries and subdistrict membership are defined in Section 2.2.1.

3.1 Data Sources

The population censuses provide individual records for a 5% sample in 1980 and the full population in 2000 and 2010. I use the 1980 records, expanded by census weights, for baseline controls and the 2010 records for city population, employment, and demographic composition. The 2000 census supplies village agricultural-employment and settler-retention measures. I link all waves to year-2000 subdistricts using the historical BPS code links in Olken (2009), supplemented by census codes and village administrative histories.[8]

The Transmigration Census records settlement locations, establishment years, and the number of individuals initially placed in each village. I obtain these records and the village agro-climatic similarity index from the replication data of Bazzi et al. (2016) and Bazzi et al. (2019). The National Socioeconomic Survey (Susenas) supplies household consumption and earnings; the Village Potential Statistics (PODES) supply rice production. Table 2 summarizes their years, geographic detail, and coverage.

Spatial data serve two purposes. The GHS layer provides urban boundaries and the larger functional urban areas used in descriptive maps. I augment a general land-travel friction surface with shipping connections and calculate travel times between the paper's geographic locations (Appendix B.1).

3.2 Urban Outcomes

I aggregate the constituent subdistricts into a city area that approximates an urban labor market. The subdistrict is the finest unit the 1980 baseline is available for, and since the exposure is based on travel time around the city, a reasonably large distance, there is little meaningful variation across the subdistricts within one city; dissolving them captures the one urban labor market. Population and employment are sums across residents, survey outcomes pool the individuals or households within these areas with their survey weights, and the cities available vary by outcome and wave (Table 2 and Appendix A.1).

Population and employment. The headline employment count and the sector counts use the different definitions in Table 2. Sector totals therefore need not add to headline employment. Modern employment comprises manufacturing and services; agriculture is reported separately. Construction, mining, utilities, and other industry codes remain in the sector-share denominator. The sectoral anal-

[8] The Online Supplement documents coverage and comparability.

Table 2: Outcome Definitions and Source Coverage

Outcome and source	Location detail	Definition	Coverage
Population, 2010 census	Village or urban neighborhood	Resident count	Main city sample
Total employment, 2010 census	Village or urban neighborhood	Worked or ran a business in the preceding week	Main city sample
Sector employment, 2010 census	Village or urban neighborhood	Counts by reported industry; shares divide by everyone with a valid industry code	Main city sample
Food outcomes, Susenas 1996, 2002, and 2011	Household, located by subdistrict	Weighted means of spending and quantities; price indices from item unit values	Surveyed cities, varying by wave and outcome
Wages and rents, Susenas 2011	Individuals and households, located by subdistrict	Weighted median monthly earnings among positive earners; median rent relative to the national median	Cities with the required survey measure
Rice production, PODES 2003	Village	Kilograms of paddy per person, using population in villages with observed output	Cities with observed rice production
Rice net imports, Susenas 2002 and PODES 2003	Subdistrict	Rice consumption minus production, in tons of milled rice	Shared subdistrict coverage, aggregated to cities

Notes: The main specifications apply the city sample's exposure and baseline-control requirements. Survey and production outcomes additionally require the stated source coverage; their observation counts are reported with the estimates. Source geography is mapped to year-2000 subdistricts and then to cities. Household statistics use survey weights. BPS denotes Indonesia's statistical office; Susenas is the National Socioeconomic Survey and PODES the Village Potential Statistics. Appendix Table A6 reports survey coverage by wave. Sources: the 2010 population census, Susenas, and PODES.

ysis also separates service activities, including trade and transportation. I report the unemployment rate and use $\ln(1+x)$ for log outcomes.

Birthplace and recent migration. The 2010 census records birthplace and residence five years earlier at the district level. Birthplace shares describe residents' origins; five-year residence identifies recent moves across district or province boundaries. Both distinguish Java and Bali from other origins. These measures do not locate moves within districts or identify former transmigrants among city residents. The censuses do not follow individuals between waves. Worker-specific decompositions restrict the denominator to workers and are labeled accordingly. The 1980 birthplace measures identify origins at the province level.

Household spending and consumption. Susenas provides consumption per person, household spending, and the food share, household food spending divided by total spending. The 1996 wave supplies the main food-spending, calorie, staple-quantity, and agricultural-price measures before the 2004 rice import ban. It covers more of the main city sample than the 2002 wave. The latter provides the consumption quantities paired with 2003 production. The 2011 wave supplies later spending measures, wages, and rents. These outcomes lack comparable 1980 measures, so the regressions compare their levels across cities.

Prices and wages. Agricultural price indices use survey-weighted medians of household item unit values. In 1996, these divide consumption value by quantity, including own production valued by BPS. Each index compares an item's survey-weighted median unit value over the city's households with its median over all urban-center households and aggregates the items under the expenditure weights of all urban centers, the city's own weights, or equal weights; the rice price is the value of rice consumed over the kilograms consumed across the city's households. Later waves distinguish purchases, permitting purchase unit values. The indices can reflect both prices and the quality and composition of consumption. I deflate 2011 monthly earnings by local price indices to measure real wages. The housing-rent index divides the survey-weighted city median monthly rent, including actual and imputed rents, by the national weighted median; it serves as a proxy for non-tradable prices.

Rice production and net imports. The production measure and the net-import calculation use different coverage rules. Production per person uses all villages with observed rice output and their recorded population. The net-import calculation covers the subdistricts observed in both sources, applying the 2002 survey's rice consumption per person to the 2003 population of their villages with a rice output figure and subtracting their 2003 production, both in tons of milled rice. Rice surplus is production minus consumption; the net-import proxy in Figure 9 reverses that sign, so a positive value indicates a local shortfall. This accounting measure does not observe shipments between a city and its hinterland.[9]

3.3 Exposure: Agro-Climatic Similarity, Settler Counts, and Cost Distance

I measure exposure to rural agricultural productivity using the agro-climatic similarity index of Bazzi et al. (2016). The index measures the match between settlers' origin environments and their destination village, which predicts agricultural productivity through the transferability of farming skills.

Village agro-climatic similarity. Let x_o and x_v denote agro-climatic characteristics of origin district o and destination village v, and let ω_{ov} be the share of settlers in v born in o. The underlying similarity index is

$$AS_v^{\text{raw}} = -\sum_o \omega_{ov}\, D(x_o, x_v), \tag{1}$$

where $D(\cdot)$ sums absolute differences across standardized characteristics. The negative sign makes a closer match a higher value. I use the min-max rescaling stored in the replication data, denoted AS_v, which ranges from 0 to 1 across the 814 program villages. It is a proxy for agricultural productivity.

Travel time and city exposure. I start with an inherited land-travel friction surface, add shipping connections, and calculate least-cost travel times for the paper's subdistricts, urban areas, and transmigration villages. The result is a full travel-time matrix between villages and cities that is representative of the actual time it takes to travel from one place to the next: two cities on opposite islands that sit on a shipping route next to ports are more connected than two that are close to the water but far from a port.

[9] The Online Supplement gives the detailed outcome definitions.

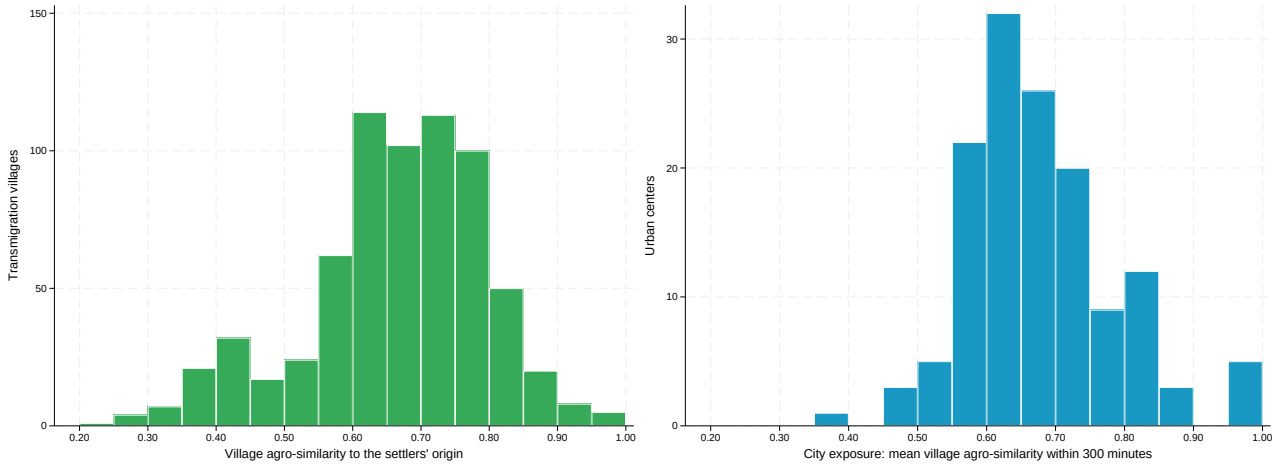
The matrix selects the villages inside each city's catchment, supplies the inverse-travel-time weights of the settler count, and sets which cities have a village within 300 minutes. Maritime connections matter because straight-line distance does not reflect the routes linking Indonesian islands.[10]

I average village agro-similarity over the settlements within 300 travel minutes of city c, giving each village equal weight:

$$AgSim_c = \frac{1}{|V_c|} \sum_{v \in V_c} AS_v, \qquad V_c = \{v : d_{cv} \leq 300 \text{ minutes}\}, \tag{2}$$

where d_{cv} is the travel time used to construct exposure and $|V_c|$ is the number of villages in the catchment.[11] The city mean remains between 0 and 1. For estimation, I subtract its mean and divide by its standard deviation across all urban centers with measured exposure, before restricting the sample to cities with complete baseline controls. One standard deviation of city agro-similarity is 0.103 on the 0 to 1 scale of the index, 0.76 of the standard deviation across the program villages (0.136), because the catchment mean averages over several villages. The estimates are reported per standard deviation of city agro-similarity. Applying the 20% higher rice productivity per village standard deviation that Bazzi et al. (2016) estimate, one city standard deviation corresponds to about 15% higher rice productivity in the surrounding villages. The two sample regions in Figure 2 illustrate the comparison, and Figure 6 shows the village and city distributions.

Figure 6: Transmigration-Village Agro-Similarity and Aggregated Urban Exposure



Notes: Left panel: agro-similarity to the region of origin of the settlers, an index from 0 to 1, for the 680 of the 814 transmigration villages that lie within 300 travel minutes of an outer-island urban center; the other 134 villages lie beyond 300 minutes of every urban center. Right panel: the exposure of each of the 138 urban centers with a village in that window, the mean agro-similarity of those villages. Averaging narrows the city distribution, which starts at 0.38 where the village index starts at 0.22. Bins are 0.05 wide on the same horizontal scale in both panels.

[10] Appendix B.1 documents the construction, distance adjustments, and source-date limitations.

[11] For exposure construction, travel time is capped at 1 minute for villages with more than 0.1% of their polygon area inside a city polygon. Times implying speeds below 2 kilometers per hour are capped at the time implied by great-circle distance from the city centroid to the village at that speed. Sample eligibility uses travel times before these adjustments.

Settler counts and weighting. I also measure the scale of settlement around each city using the number of individuals initially placed in nearby villages. Let N_v denote that initial count in thousands. The control sums these counts with inverse-travel-time weights over the same catchment:

$$SettlerCount_c^{IDW} = \sum_{v \in V_c} \frac{N_v}{\max\{1, d_{cv}\}}, \tag{3}$$

where the superscript IDW stands for the inverse-travel-time weights and travel time is floored at 1 minute. Its unit is thousands of settlers per travel minute. It increases with the number and size of nearby settlements and gives greater weight to closer villages.

Equal village weights measure the average match quality of nearby settlements. Conditional on the settler-count and destination controls, the comparison is between cities whose catchments offer different average conditions for transferring farming skills. The contribution of each village to urban food supply can also depend on its size, marketed output, and transport costs. A productive village that is farther away is not the same as a low-productivity village: the higher trade costs to a more remote village are not equivalent to that village being less productive. The two are different channels, and the distance dimension is in some ways accounted for by the weighted total number of migrants placed, which is a scale effect. Distance weights mute the estimates, which keep their sign and a comparable magnitude. Their standard errors stay close to those of the main specification, so where a weighted estimate is less significant, its coefficient is smaller and its precision unchanged. Catchments of 400 and 500 minutes keep both signs with the agricultural share less precise (Table 4).

Measured exposure and the main city sample. Applying the 300-minute cutoff to unadjusted nearest-village travel time includes 138 of the 187 urban centers, of which 134 have a 1980 baseline recorded directly in the census. The main city sample consists of the 134 cities with a nearest-village travel time within the window, measured exposure, the required baseline controls, and measured 2010 outcomes. When V_c is empty, the catchment mean is undefined and exposure is left unmeasured. The regressions compare degrees of exposure among cities with measured catchments.[12] The estimates describe outcomes in the sampled urban centers and do not identify the effect on the formation of new cities.

3.4 Baseline Controls

The main specification includes the settler-count control and 16 destination agro-climatic characteristics averaged over the same villages as exposure. For each village characteristic X_v, I construct $X_c = |V_c|^{-1} \sum_{v \in V_c} X_v$. They comprise three slope measures, ruggedness, log elevation, soil organic carbon, exchangeable sodium percentage, soil pH, two soil-texture and two drainage measures, distances to coasts and rivers, and mean rainfall and temperature over 1948-1978. They condition on destination

[12] The distribution of nearest-village travel times and the full sample selection are reported in Appendix Figure A1b and Appendix Table A1.

characteristics that also enter the origin-destination match. The tables refer to them as agro-climatic transmigration-village controls.

The city controls are four 1980 baselines: log population, log employment, the urban population share, and log agricultural employment. Some villages were settled in 1979 and 1980, before the census that records these baselines; Section 4.3 discusses the timing. A sensitivity check adds the 1980 birthplace shares of the population (Appendix Table C3).

The urban population share distinguishes the census administrative classification from city membership. For city c it is

$$UrbanShare_c = \frac{\sum_{k \in c} Pop_k^{\text{urban}}}{\sum_{k \in c} Pop_k},$$

where k indexes its constituent subdistricts and Pop_k^{urban} is their 1980 population in villages classified as urban by BPS. The satellite boundary determines membership in the city; the census village classification determines the urban share within that geography.

3.5 Trade and Openness Measures

Access to imported rice can reduce a city's dependence on its hinterland. I measure this access using international rice routes and realized rice imports, and describe general port access using domestic travel times.[13]

Route openness. The preferred measure adds least-cost travel time from the city to its nearest customs port of entry to the onward sea leg to the nearer of Bangkok and Ho Chi Minh City, following the rice-route approach in Bazzi (2017). Both legs are expressed in hours. The domestic leg is bounded below at half an hour, and the international leg approximates sea travel by great-circle distance divided by a shipping speed of 40 kilometers per hour. Shorter total travel time means greater openness. The high-openness group lies below the median travel time among cities in the exposure window; continuous route proximity reverses the sign and is standardized so that higher values mean greater openness.

The international leg distinguishes cities near different ports by their distance from the rice exporters. The port set includes the full customs list with coordinates, without selecting on realized rice imports. Its opening dates are unrecorded. Historical maps identify ports with cargo activity in 1973 or a harbor inventory in 1986 (Table B1). The domestic leg uses the land and shipping inputs described in Section 3.3, so the measure combines international geography with domestic links observed after settlement.

Realized-import access and general port access. Gravity market access sums port rice imports recorded in 2000-2010, dividing each port's volume in kilograms by travel time from the city in hours, bounded below at half an hour. I take $\ln(1+x)$ of this sum and standardize it across the full set of cities in the source data, with higher values indicating greater access. It supplies the alternative openness

[13] Appendix B.2 reports the construction and comparison of these measures.

split in Table 5, with high openness defined above the median within the exposure window. Realized imports respond to demand and supply, so this measure is partly an equilibrium outcome. I use it to show which way that bias runs, and the openness result relies on the route measure (Appendix B.2). The balance comparisons in Section 4.3 also use travel hours to the nearest customs port and the log of one plus summed inverse travel hours to all customs ports (with a half-hour lower bound in that sum). Both describe general port access.

Rice from other Indonesian regions provides another potential source of urban food. These import and port-access indices do not measure the availability of that supply. The appendix comparisons with domestic port access describe how much international-route geography adds to general port proximity. The instrumental-variable interpretation in Section 5.2 requires that, conditional on the main controls, route proximity affects urban outcomes and their response to exposure only through measured rice-import access. Access to other markets could violate that restriction.

4 Conceptual Framework and Empirical Strategy

This section sets out how a productivity gain in a city's hinterland can reach the city (Section 4.1), the equation that estimates its effect (Section 4.2), and the identification argument behind it (Section 4.3).

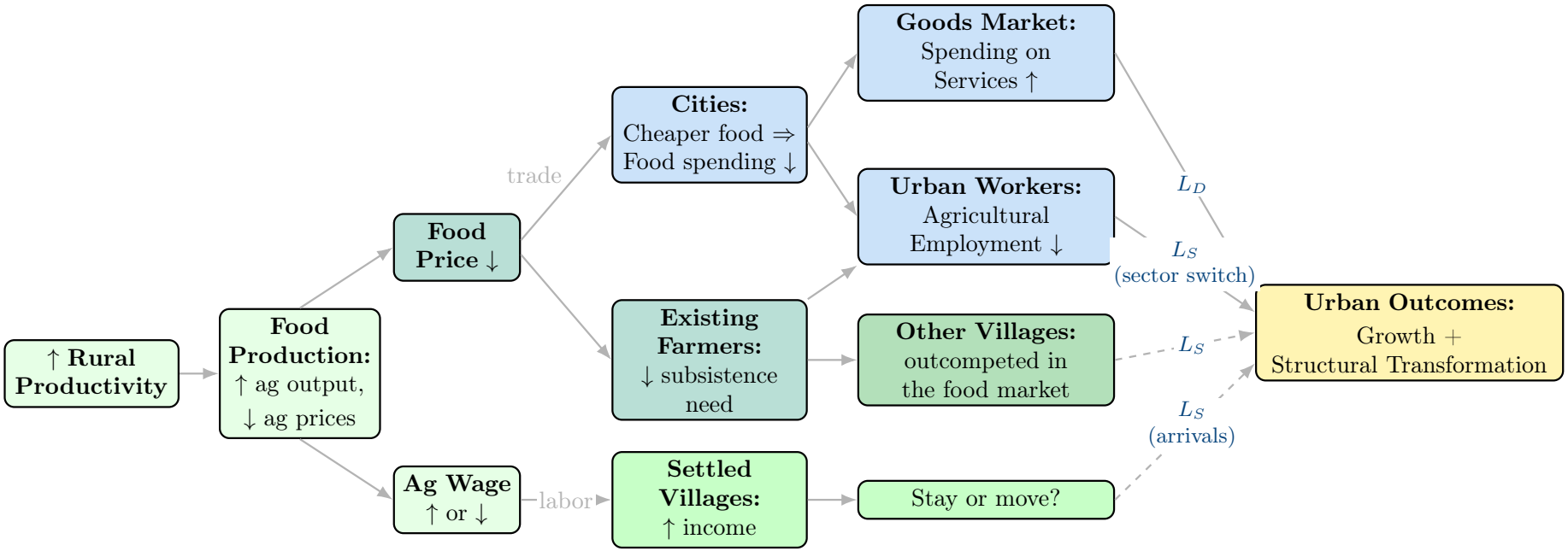
4.1 Framework: Mechanisms Linking Rural Productivity to Urban Outcomes

The productivity shock directly affects food production in the villages that receive it. The region is the city and its rural hinterland, which together make up one economy, as in Glaeser (2014). The farms around the city feed it, and the city responds when they become more productive. Food trade can then change the returns to urban agriculture and the demand for non-agricultural goods, while changes in rural earnings affect workers' incentives to move. The relative strength of these responses is ambiguous.[14] Figure 7 summarizes the mechanisms within a city and its hinterland; the spatial equilibrium model in Section 6 also allows trade and migration between urban regions.

Cheaper food reaches the city through trade and moves urban employment through two price effects, a labor supply effect in urban agriculture and a labor demand effect in the modern sector. As rural productivity rises, food production goes up, meaning that there is more agricultural output available at lower prices. This affects cities first through trade, as cities have access to cheaper food. That means that cities themselves do not need to produce as much food locally, and the remaining peripheral urban agriculture can decline as workers exit urban agriculture, a labor supply effect in the cities. Households can also reallocate spending to services and other non-agricultural goods when cheaper food reduces the cost of meeting their food needs. The resulting increase in demand can raise

[14] The food-problem literature emphasizes the role of agricultural productivity in releasing labor (Schultz 1953; Gollin et al. 2007), and the dual-economy model sustains non-agricultural employment through cheap food (Lewis 1954). Trade opportunities and the factor bias of technical change can alter this relationship (Matsuyama 1992; Bustos et al. 2016). The Online Supplement discusses these accounts.

Figure 7: Mechanism Map



Notes: The figure summarizes adjustments within a city and its rural hinterland. Signs on the food-trade branch describe the case in which additional rural supply lowers urban food prices and the cost of meeting food needs. Agricultural wages depend on the food price and the marginal product of labor; wages and rural incomes can rise or fall as prices, land use, and employment adjust. L_S denotes labor available to urban non-agricultural sectors through sector switching or migration; L_D denotes labor demand through spending on services. Shading distinguishes village agriculture, urban markets, and competition faced by other farmers. Dashed arrows identify migration responses whose direction is ambiguous.

employment in those sectors. These supply and demand responses allow structural transformation among existing urban residents.

Rural wage and income effects also change workers' location choices. Places that received the shock have higher incomes coming from agriculture, net of the lower food price (Section 6 derives the equilibrium response). When incomes rise, farmers have a stronger incentive to stay in those villages, but may also be better able to finance migration (Bryan et al. 2014). At the same time, other rural areas that do not experience the shock are now at a disadvantage, so they are facing the increased competition from places that have the shock, while also being negatively affected by lower prices.

Each outer-island city with its rural hinterland resembles the city-and-hinterland economy of Glaeser (2014), and how closed it is varies with the city's access to outside food markets. Cheaper food from nearby villages can have a larger effect when alternative sources are costly.[15] Comparing more open and more closed urban regions therefore provides the variation to test the theoretical predictions within one country.

The price channel moves a city's employment composition without moving anyone, while the labor channel moves composition and population together and empties the villages that became better at farming. Each employment response considered separately is consistent with several mechanisms, so the data have to show three patterns together, in the order Section 5 tests them. First, cities exposed to more productive villages move employment out of agriculture and into manufacturing and services with little population growth (Section 5.1), and their food prices fall (Section 5.2); with food-supply conditions held fixed, an inflow of workers raises food demand and does not imply this price decline. Second, the response is larger in cities with costly access to outside food (Section 5.2). Third, arrivals cannot account for the increase in modern-sector employment once their number and employment composition are counted, and the settled villages do not show a distinct migration response by productivity and keep farming (Section 5.2 and Section 5.3). Birthplace data help assess the third requirement and whether arrivals came from the regions the hypothesis predicts. Repeated cross-sections do not identify individual moves or occupational transitions, so the model in Section 6 compares the responses under different trade and migration costs.

4.2 Estimating Equation

I compare urban outcomes across cities surrounded by villages with different agro-climatic match quality, conditional on settlement scale, destination characteristics, and the city's 1980 baseline. The main population and employment outcomes are measured in the 2010 census. For urban center c, I estimate

[15] District price differences provide descriptive evidence of food-market frictions. Using 2011 Susenas unit values for pairs of outer-island districts, the coefficient on log travel time in a 95th-quantile regression of the absolute log rice-price gap is 0.087; the corresponding coefficient for the food basket is 0.065. Travel time explains at most 12% of the variation in pairwise price gaps. These gradients do not identify trade costs or establish that transport links are predetermined, and the quantitative exercise in Section 6 uses the trade-cost range from the literature.

$$y_c = \beta\, AgSim_c + \phi\, SettlerCount_c^{IDW} + C_{c,1980}\,\xi + X_c\,\zeta + \varepsilon_c \tag{4}$$

where y_c is an urban outcome aggregated from the city's subdistricts and $AgSim_c$ is standardized city agro-climatic similarity, constructed as in Section 3.3. The estimated β captures how urban outcomes change in response to one standard deviation higher agro-similarity of the villages within the 300-minute travel-time window of Section 3.3, about 15% higher rice productivity in those villages. There are different crops (Appendix Table E5 reports staples, rice, and legumes separately), and different ways to measure output: harvest or yield, per person or per hectare. I treat the estimates as a reduced form, without taking a stance on which of these measures is the relevant one, and Section 6 puts their magnitudes into economic units.

The settler count, $SettlerCount_c^{IDW}$, accounts for the scale and proximity of settlement, which the program chose, so β is identified off the composition of nearby settlement, whether the villages a city happened to draw were well or poorly matched; X_c contains the catchment's agro-climatic and geographic controls, and both are constructed as in Section 3.3. The baseline vector $C_{c,1980}$ contains log population, log total employment, the urban population share, and log agricultural employment. I add these 1980 baseline controls primarily for precision and for the long-run difference interpretation. The same four baselines enter every column, so the 1980 value of the outcome is among them for population, total employment and agricultural employment, and the one characteristic that is unbalanced at baseline, the 1980 urban population share, enters as well (Section 4.3). Together, baseline agricultural and total employment also determine the agricultural employment share, even though that share does not enter separately. For the unemployment rate, modern employment and the modern share, the own-outcome row of Table 4 adds the 1980 value of the outcome to these four. The results are robust to each of these decisions: to controlling for the 1980 shares beside or in place of the levels, to three or four baselines, and to the other control sets of the specification curve. I use Conley standard errors allowing spatial correlation within 200 kilometers.

The focus of the paper is rural-to-urban spillovers, and I do not expect them to be discontinuous at district or province borders: the catchments that define exposure follow travel time and run across those borders. The specification therefore compares cities across the whole sample, with no district or province fixed effects, since those borders are not meaningful boundaries for the spillover and a fixed effect would impose a discontinuity at them. The meaningful boundary is the island, which no catchment crosses, and the estimates are robust to island fixed effects (Table 4).

Prices, wages, and expenditure have no 1980 measurement to condition on, so those regressions compare levels across cities at the end of the period, and I read them as suggestive evidence on the mechanisms. For the food share, the one survey outcome estimated precisely, the estimate is unchanged when a 1980 proxy for the missing baseline is added, population density, the agricultural employment share, or schooling, one at a time or together (Appendix Table E8). Alternative specifications add island fixed effects and vary the treatment construction, controls, and inference (Appendix Figure D3).

4.3 Identification

The key identifying assumption is that, conditional on the number of settlers and the other covariates, the match quality of the villages around a city is plausibly exogenous. In the language of an experiment, the cities are the units; the productivity of the farming communities placed within five hours of a city is the treatment; the province and settlement wave of each site are the strata, within which a cohort could have gone to any site that opened at the time; which cohort reached which site is the randomization; and the 1980 census is the baseline. While the assignment to destinations was as-if random, the total number of transmigration villages and their locations around cities were not random. I therefore control for the number of settlers, because I am not claiming that all village placements were as-if random, only the agro-similarity of the settlers each village received, averaged across the settlers in the village and across the villages around the city, so the settler count holds the people part of the program fixed and leaves the variation in what the settlers produce. That measure is the match quality between the places settlers came from and the places they were sent. I also control for the characteristics of the destination villages. Among cities with equally large nearby settlements on similar land, some drew cohorts whose origin conditions resembled their new home and some did not, and that difference is the treatment.

In the aggregate, urbanization, falling food prices and rising agricultural productivity move together as a country goes through the structural transformation, so a country's own history cannot separate the effect of agricultural productivity from the rest; here the variation is only in the agricultural productivity piece. The random component of the treatment is established at the village level. The program did not match migrants to destinations on agro-climatic similarity, so where each cohort ended up was as-if random (Bazzi et al. 2016), and within a cohort it is plausible that people could have been assigned to another village among those that opened up around the same time (Section 2.1 describes the assignment process). Families from a rice-producing region such as Bali who were sent to a destination also suited to rice are a high-productivity match; their neighbors, who signed up for the program a few weeks later and were sent to an area more suited to cassava, are a lower-quality match; and families from Java who used to grow cassava and were sent to that cassava-producing region are again a high-quality match. The variation I leverage is the match quality between origin and destination; the productivity of the destination itself is held fixed by the agro-climatic controls.

Agro-climatic similarity predicts agricultural productivity. Bazzi et al. (2016) find that a one-standard-deviation increase in village similarity is associated with 20% higher rice productivity. The village-level validation for this paper also shows a positive relationship with village crop yields relative to each crop's national average yield (Appendix Figure C2 and Table C4).

Once I average the shock from each village to the city, a non-random component appears inside that averaging. The exposure incorporates the centrality of the city and how many villages happened to be selected around it. In my case, a city surrounded by transmigration sites with land more similar to the Java and Bali origin mix might be bound to have higher exposure regardless of how settlers are reshuffled across the as-if randomly assigned destinations, because of its pre-existing characteristics. I condition

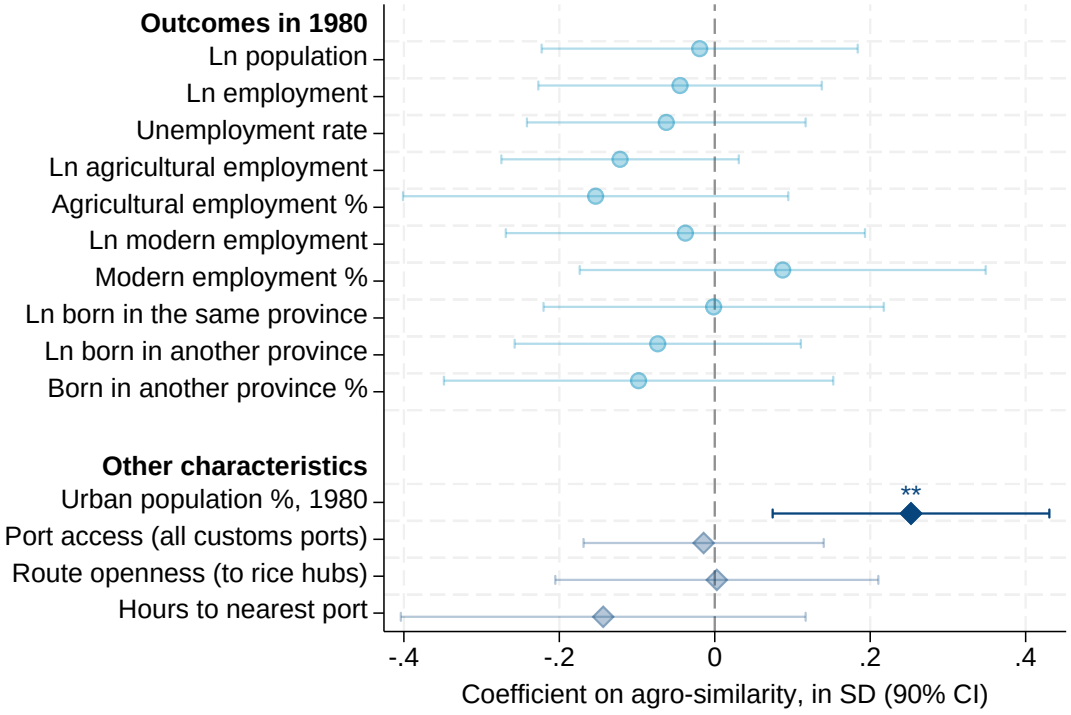
on those characteristics separately, and that helps with identification; to do it more systematically, I run the test of Borusyak & Hull (2023). A village's match has two parts: how close its land is to the origins the program drew on, which the program chose when it placed the site, and which settlers arrived, which it did not. The estimates on each part are reported beside the main results (Section 5.1 and Appendix C.1), and the modern-sector result holds on the origins that arrived.

Two features of the geography could have responded to the program, and the design holds both fixed. The exposure is built on the cost-distance travel time between each village and each city, so it controls for where the villages are (Appendix B.1). Exposure does not predict a city's access to outside food: the three trade-access rows of Figure 8 are not distinguishable from zero, and the road network around more exposed cities did not improve more than its earlier level predicts. The sample is the fixed set of urban-center footprints described in Section 2.2.1, held constant across the census years. The response to exposure is in employment composition, with city size nearly unchanged (Section 5.1), so the shock is changing cities that already exist, and the sectoral results hold when each city is rebuilt from its less agricultural subdistricts. The estimates describe outcomes in the sampled urban centers and do not identify the effect on the formation of new cities.

The placebo and balance checks use the 1980 baseline data to check whether urban outcomes at baseline were already correlated with the agro-similarity exposure, that is, whether there was selection at baseline; if there were, it would pose more serious concerns about the exogeneity of the later productivity exposure. Figure 8 plots each 1980 baseline characteristic's standardized coefficient on exposure, conditional on the settler count and the agro-climatic controls; thirteen of the fourteen rows are not distinguishable from zero, and the exception, the 1980 urban population share, is a baseline the estimating equation conditions on, so the estimates below already hold it fixed. Settlement began in 1979, so the 1980 census is a baseline that already includes the first settlements: 127 of the 814 villages, accounting for 19.5% of the settlers, had been placed by the time of the count. A separate check excludes the villages already settled by the time of the count from the construction of the exposure, and the balance is unchanged (Appendix Table C2); in any case, it is plausible that village effects need some time to reach cities. The compositional results are unchanged with the 1980 birthplace shares added to the controls.[16] Overall, baseline levels and shares are uncorrelated with the agro-similarity exposure, which supports the empirical design and reduces concerns about confounders.

[16] The 1980 census records province of birth, which limits the baseline comparison to a broader geographic classification than the district-level decomposition available in 2010.

Figure 8: Baseline Balance: 1980 City Characteristics and Exposure



Notes: Each row is one regression of a standardized city characteristic on standardized agro-similarity, with the settler count and 16 agro-climatic covariates within 300 minutes as controls, as in the levels test of Appendix Table C2. Bars are 90% confidence intervals from Conley SE within 200km, a solid marker is a row whose interval excludes zero, and stars above a marker give * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The upper block (circles), the outcomes in 1980, holds the ten 1980 baseline outcomes of the placebo table, in its order (the birthplace rows split the 1980 population by province of birth, so the share born in the same province is one minus the plotted share born in another province). The lower block (diamonds), the other characteristics, holds the 1980 urban population share, a main specification control, and three trade-access measures: port access (the log of summed inverse travel hours to every customs port of entry), route openness to the Thai and Vietnamese rice hubs, and travel hours to the nearest customs port of entry. Every row uses the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline).

5 Reduced-Form Results

Cities surrounded by more productive villages, compared with cities surrounded by less productive ones, differ substantially in employment structure, with more employment in manufacturing and services and imprecisely estimated differences in population and total employment (Section 5.1). Rural productivity can affect cities through food trade and migration. I examine consumption, food prices and employment by trade openness, then migration and the contribution of additional workers to modern-sector employment (Section 5.2). I then compare rural and urban adjustment and assess public provision, capital and rural demand as additional links from the villages to the cities (Section 5.3). The spatial model examines trade and migration as mechanisms of these responses (Section 6).

5.1 Urban Structural Transformation

Across the cities in the main sample, exposure to more productive villages shifts employment from urban agriculture to manufacturing and services. Each column of Table 3 compares 2010 outcomes across cities whose exposure differs by one standard deviation, conditional on their 1980 characteristics and the other controls. The agricultural employment share is lower by 5.1 percentage points, against a 2010 mean of 37.4%, and agricultural employment is lower by about 15%, a marginally significant estimate. Modern-sector employment, combining manufacturing and services, is higher by about 17%, and its employment share by 6.6 percentage points, against a mean of 51.6%. These are conditional endline differences; the regressions do not follow individual workers as they change jobs.

Population and total employment are higher by about 5% and 2% per standard deviation of exposure, with both estimates statistically indistinguishable from zero. The unemployment response is also imprecisely estimated. The increase in the modern employment share establishes that modern-sector employment expands relative to total employment, while the estimates of city size allow uncertainty about how much employment and population expand overall.

The sectoral differences are large relative to the structural transformation these cities experienced. The average city's agricultural employment share fell 18.6 percentage points between 1980 and 2010, from 56.0% to 37.4%. Cities at the 90th percentile of exposure sit 2.3 standard deviations above those at the 10th, implying an agricultural share 11.6 percentage points lower, about 63% of the average decline. This comparison describes the magnitude of the exposure gradient within the sample; zero exposure lies far outside its observed range. Services account for most of the increase in the modern employment share, with manufacturing accounting for about one percentage point (Appendix Table E4).

Most of the estimated reallocation remains after I control for where the government placed the settlements. Settlers were assigned quasi-randomly across sites, but the sites themselves were chosen for development. A city near many suitable sites could therefore have high exposure under many assignments. Controlling for its expected exposure under reassignment within province and settlement period gives an agricultural-share coefficient of -0.040 and a modern-share coefficient of $+0.052$, compared with -0.051 and $+0.066$ in the main specification. Both remain statistically significant, at about four fifths of the main estimates (Appendix Table C1). A randomization test that reshuffles village similarity among these sites rejects a zero effect on the modern share and does not reject it on the agricultural share.[17]

The main sensitivity checks distinguish the sectoral results from the less stable response of city size (Table 4). Agricultural employment shares fall and modern-sector employment rises across the alternative control sets, distance weights, and catchments in the table. The agricultural share is statistically or marginally significant in 10 of the 12 alternatives, and modern employment in 10. Wider

[17] The randomization p-values based on the controlled coefficient's t-statistic are 0.050 and 0.196, respectively (Online Supplement). Subtracting expected exposure leaves little variation and produces smaller standardized coefficients (Appendix Figure C1). As discussed in Appendix C.1, this changes the scale of the exposure variation as well as the comparison.

Table 3: Urban Growth and Structural Transformation: From Agriculture to the Modern Sector

	Population + Employment			Agric Empl		Modern Empl	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	ln pop	ln emp	unemp rate	**ln L agric**	**agric %**	**ln L modern**	**modern %**
Agro-similarity	0.051	0.024	0.008	-0.166*	-0.051***	0.158**	0.066***
	(0.059)	(0.057)	(0.007)	(0.093)	(0.018)	(0.062)	(0.015)
Settler count	✓	✓	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agro-climatic controls	✓	✓	✓	✓	✓	✓	✓
Outcome mean	11.842	10.933	0.122	9.822	0.374	10.232	0.516
Outcome SD	0.854	0.840	0.040	0.896	0.183	0.975	0.156
Observations	134	134	134	134	134	134	134

Notes: Each column is one regression of a 2010 population census outcome on standardized agro-similarity on the main city sample, with the main specification controls marked by the check marks: the settler count and 16 agro-climatic covariates within 300 minutes, and four 1980 baselines. Outcomes are log population, log total employment and the unemployment rate (columns 1 to 3), log agricultural employment and the agricultural employment share (columns 4 and 5), and log modern-sector (manufacturing and services) employment and the modern-sector employment share (columns 6 and 7); shares and the unemployment rate are fractions, and the four sector columns are shaded. In the labels: ln (the natural log), L (a count of workers), emp and Empl (employment), unemp rate (the unemployment rate), and agric % and modern % (the two sector shares); Baseline Pop 1980, Total Empl 1980 and Agri Empl 1980 are the logs of 1980 population, total employment and agricultural employment, and Urban share 1980 is the urban share of the 1980 population; SD is the standard deviation. Conley SE within 200km in parentheses, with stars from the same standard errors: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Appendix C.1 reports randomization inference for the design-based estimates, the Online Supplement every randomization-inference p-value, and Appendix Table D6 other spatial-inference radii; variants with polynomial terms in the settler count are in the Online Supplement (Section S3.7).

catchments strengthen modern-employment and population responses, with population statistically significant at both 400 and 500 minutes. Province fixed effects reduce the agricultural-share coefficient to -0.017 and the coefficient on log modern employment to 0.084, leaving both statistically insignificant with standard errors of similar size. Island fixed effects preserve both results.

The sectoral pattern also survives alternative city samples (Appendix Table A2), and most proximity windows of 300 minutes or wider (Appendix Figure D5). Baseline birthplace controls, alternative forms of the settler-count control, and the settlers' share of the pre-existing rural population give similar compositional results (Appendix Table C3; Appendix Table D1; Online Supplement). Modern employment responds more in cities with larger baseline populations, while its response does not differ detectably by initial agricultural intensity (Appendix Table D5). Alternative spatial standard errors generally preserve the sectoral results; inference is weaker under the wild bootstrap over the small number of catchment components.[18]

The reallocation is also present within more urban parts of the cities. Keeping member subdistricts that were at least one quarter non-agricultural in 1980 gives an agricultural-share decline of 6.2 percentage points, compared with 5.1 for all member subdistricts, and modern employment still rises. Raising the threshold to one half gives a decline of 7.1 percentage points and a less precise modern-

[18] Appendix Table D6 reports each inference method. Under the component bootstrap, log modern employment remains statistically significant and the modern employment share marginally significant. Appendix Table D2 defines the specifications, and Appendix Figure D3 extends them to every outcome. The wider robustness material is in Appendix D.2.

Table 4: Estimates' Robustness Under Main Specification Variations

	(1) Log population	(2) Agricultural employment share	(3) Log modern-sector employment	Observations
Main specification	0.051	-0.051***	0.158**	134
	(0.059)	(0.018)	(0.062)	
Controls				
Unweighted settler count	0.047	-0.053***	0.150**	134
	(0.071)	(0.018)	(0.074)	
Weighted covariates	0.101*	-0.045***	0.187***	134
	(0.060)	(0.014)	(0.060)	
Own-outcome baseline	0.051	-0.051***	0.177***	134
	(0.059)	(0.011)	(0.063)	
Exposure weights				
Population-weighted	0.057	-0.044***	0.155**	134
	(0.057)	(0.015)	(0.061)	
1/d	0.002	-0.052***	0.110*	134
	(0.045)	(0.018)	(0.062)	
$1/d^2$	-0.023	-0.047***	0.075	134
	(0.036)	(0.017)	(0.057)	
Catchment radius				
Within 400 min	0.118**	-0.023	0.188**	134
	(0.059)	(0.032)	(0.094)	
Within 500 min	0.137***	-0.044*	0.268***	134
	(0.048)	(0.026)	(0.052)	
FE and inference				
Province FE	0.060	-0.017	0.084	134
	(0.074)	(0.017)	(0.056)	
Island FE	0.043	-0.049**	0.151**	134
	(0.059)	(0.020)	(0.060)	
Cluster: province	0.051	-0.051**	0.158*	134
	(0.077)	(0.022)	(0.084)	
Island FE, cluster province	0.043	-0.049*	0.151*	134
	(0.078)	(0.025)	(0.084)	
Outcome mean	11.842	0.374	10.232	
Outcome SD	0.854	0.183	0.975	

Notes: Each cell is the coefficient on standardized agro-similarity from one city-level regression of the 2010 census outcome in the column heading, where the modern sector is manufacturing plus services, with its standard error in parentheses. The first row is the main specification of Table 3, on the main city sample, with the full set of controls and Conley SE within 200km. Each row below it changes the design choice its label names, as Appendix Table D2 defines; the exposure-weight rows replace the equal weights of the villages within 300 minutes, with d the travel time from a city to a transmigration village, and the fixed-effect rows add an indicator for each province or for each island. Standard errors are clustered by province (17 provinces) in the two rows labeled as clustered, whose stars use the t distribution. Shaded cells clear the 10% level, and the shade deepens at the 5% and 1% levels. The Observations column gives the number of cities in each row. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

employment response (Appendix Table D4). The sectoral result therefore extends to established urban areas, weakening a purely peripheral land-conversion explanation.

The evidence is clearest for a change in urban employment composition, with services accounting for most of the modern-sector expansion. The response of city size is less precisely estimated. The results therefore establish sectoral adjustment more clearly than overall urban growth.

5.2 Food Trade and Migration

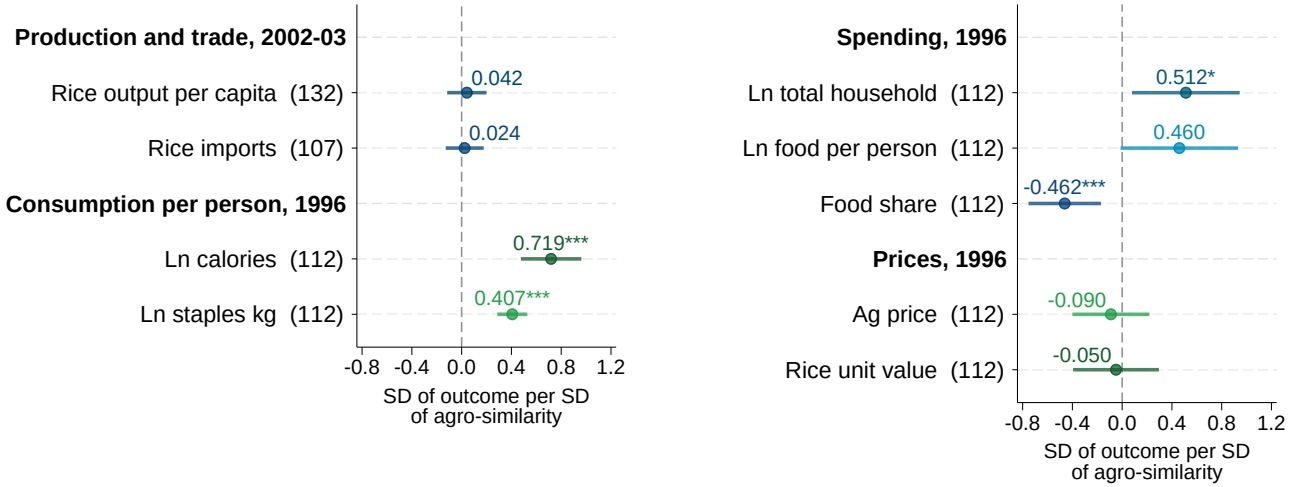
A city that can import food cheaply has less to gain from a more productive hinterland. Its food supply depends less on nearby harvests, so the food-market mechanism predicts a smaller employment response.

In more exposed cities the food expenditure share falls, and households consume more food. In the 1996 household survey, before the 2004 rice import restrictions, the food share moves by −0.462 of a standard deviation per standard deviation of exposure, calories per person by 0.719, and staple kilograms per person by 0.407, all statistically significant (Figure 9). Total household spending is higher, with a marginally significant estimate. Importantly, the food expenditure share is going down while households consume more calories, which is consistent with food consumption rising less than proportionately with household resources. In the food-market mechanism, cheaper food lets residents meet their basic needs and spend more on non-agricultural goods, raising the demand for services and manufactured products and, with it, the demand for labor in those sectors.

Figure 9: The Goods-Market Mechanism: Responses to Exposure, in SD

(a) Production, Trade, and Consumption

(b) Spending and Prices



Notes: Each row is one regression of the standardized outcome on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Bars are 90% confidence intervals from the same standard errors; the number above each marker is the coefficient, and parentheses give the number of urban centers. Rice output per capita is 2003 village-census (PODES) rice production in kilograms of unhusked paddy, divided by the population the same census records for the villages with a rice output figure, and rice imports are 2002 household-survey (Susenas) rice consumption per person times the population of those villages, minus that production in milled-rice weight, at the official rate of 0.6402 tons of milled rice per ton of paddy, each over the subdistricts with surveyed households and a rice output figure. Every other row is from the 1996 Susenas wave, before the 2004 rice import ban: calories over the whole diet and staple kilograms per person, total household spending, food spending per person, the food share of spending, the city's raw-food price index, and the rice unit value (consumption value over consumption kilograms). The 1996 wave records each food's consumption whatever its source, so spending, the food share and the unit values include what households produced themselves, valued by BPS. Logs are ln(x+1). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

I do not observe trade flows between a city and its hinterland, so the evidence on rice imports from the production and consumption measures is suggestive. Rice output per person moves by 0.042 of a standard deviation and rice consumption less local production, in milled-rice weight, by 0.024, both statistically insignificant; the latter combines 2002 consumption with 2003 production into an

implied net requirement and does not identify where the rice comes from. The log of aggregate city rice production is higher by +0.735 (Appendix Table E3).[19]

The expenditure pattern persists in the later survey, while wage and rent responses remain imprecise. In 2011 the non-food expenditure share is higher by 1.6 percentage points, statistically significant, and staples per person do not differ detectably with exposure. The non-agricultural wage, overall wage, real wage, and housing-rent estimates are statistically insignificant; rents are also imprecise in 1996 (Appendix Table E2; Appendix Table E6; Appendix Table E7).

The food-price estimates have the predicted negative signs and are imprecise. The raw-food price index moves by −0.090 of a standard deviation and the rice unit value by −0.050, with both statistically indistinguishable from zero. The other 1996 price measures also have small, negative, insignificant estimates (Appendix Figure D4). Unit values combine price and quality choices, and the rice-grade shares are consistent with households choosing higher-quality rice (Table S24 of the Online Supplement).

In the more closed cities, which rely more on their rural hinterlands for food, agricultural prices decline in 1996 (Figure 10), while prices are not lower in open cities. The gap between the more closed and the more open cities is marginally significant. These price declines in closed cities and mostly lower price levels in rural subdistricts with transmigration villages are consistent with the price mechanism through food trade. Across the rural subdistricts, I am underpowered to detect stronger price effects since I can only construct price data at the subdistrict level which includes newly settled and preexisting villages, capturing the overall effect.

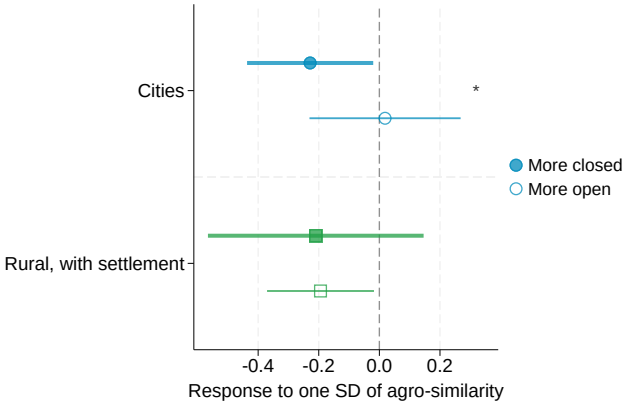
At the median split, the 1996 gap between the more closed and the more open cities is marginally significant; across thirds of route distance the 1996 price responses do not differ (Appendix Table B3).

The 1996 and 2002 price waves cover fewer cities than the census, and the pooled employment responses are weaker on those observed-price samples. The larger modern-employment response in the more closed cities remains significant on both samples (Appendix Table E9 and Table E10). The price contrast can also reflect retail and transport costs, household income responses, and changes in purchased quality.

I test the prediction of a smaller employment response in more open cities using route proximity to the Thai and Vietnamese hubs that supply most of Indonesia's rice imports, following Bazzi (2017) (Table 5, Panel A). This comparison supports the mechanism if route proximity does not also capture other determinants of the response to rural productivity, including urban labor demand. The transport-source and timing qualifications in Section 4.3 apply to this measure.

[19]Home-grown food, produced by the household itself or received as a gift, is a small share of the raw food of city households in the 2002 survey, 10.7% of its value on average across cities for all households of the member subdistricts and 4.3% for the households the survey classifies as urban, and exposure does not lower it. The share moves by +0.017 per standard deviation of exposure for all households and by +0.003 for urban households, both statistically insignificant.

Figure 10: The Agricultural Price by Trade Openness in Cities and Subdistricts With a Settlement, 1996



Notes: Each marker is the coefficient on agro-similarity for the more closed or the more open group of cities or of rural subdistricts with a settlement, split at the median route distance to the Thai and Vietnamese rice hubs within each location type. The coefficients come from one regression per location type of the 1996 agricultural price index (raw-food basket) on agro-similarity interacted with indicators for the two groups. Filled markers denote the more closed group and hollow markers the more open group. Bars are 90% confidence intervals based on Conley SE within 200km. Stars indicate that the coefficients of the two groups differ: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The price is standardized across all cities and rural subdistricts of the outer islands, and agro-similarity separately across the cities and across the rural subdistricts. The eligible samples are the 134 cities of the main city sample and the 191 rural subdistricts with a settlement; the regressions use the locations with measured prices. The city median is computed over all urban centers within 300 travel minutes of a settlement before restricting to the main sample and measured prices. Prices are from the 1996 wave of the household survey (Susenas), before the 2004 rice import ban. Regressions are in the main specification with the full set of controls, except that the settler count of a subdistrict with a settlement is the number of settlers placed in the transmigration villages that overlap it.

The median split estimates a separate exposure response in the more closed and more open cities:

$$\begin{aligned} y_c = \beta_1 \, AgSim_c + \beta_2 \, AgSim_c \times HighOpen_c + \chi \, HighOpen_c \\ + \phi \, SettlerCount_c^{IDW} + C_{c,1980} \, \xi + X_c \, \zeta + \varepsilon_c \end{aligned} \tag{5}$$

Here $HighOpen_c$ indicates cities with above-median openness, and the controls are those of Equation 4. The exposure response is β_1 in the more closed cities and $\beta_1 + \beta_2$ in the more open cities. The interaction β_2 tests whether the responses differ.

Modern-sector employment responds more in the cities less connected to international rice markets. In the more closed cities it rises by 0.269 log points per standard deviation of exposure, and the agricultural employment share falls by 6.5 percentage points, both statistically significant. In the more open cities modern employment rises by 0.099 log points, imprecisely, while the agricultural share falls by 4.4 points and the modern share rises. The modern-employment difference is -0.169 log points (standard error 0.095), marginally significant; the modern-share difference is also marginally significant, and the agricultural-share difference is imprecise. Within-group significance therefore provides stronger evidence of reallocation than the between-group tests provide of heterogeneity. Both groups retain the same signs of sectoral response. In the more closed cities the population and employment estimates are positive and larger than in the more open ones, and neither difference is precisely estimated; the

pattern is consistent with urbanization rising with farm productivity in Glaeser's closed economy. In his model the urban sector produces traded manufacturing goods; here the added modern-sector workers are mostly in services, so cities can also grow in services (Gollin et al. 2016; Jedwab & Vollrath 2015) as the cost of meeting food needs falls.

Table 5: Urban Responses by Access to Rice-Import Routes and Markets

	(1) ln pop	(2) ln emp	(3) unemp rate	(4) **ln L agric**	(5) **agric %**	(6) **ln L modern**	(7) **modern %**
Panel A: High openness = below-median route distance to the Thai and Vietnamese rice hubs							
Agro-similarity, more closed cities: β_1	**0.118**	**0.101**	**0.004**	**-0.164**	**-0.065*** **	**0.269*** **	**0.093*** **
	(0.086)	(0.082)	(0.009)	(0.132)	(0.024)	(0.089)	(0.022)
Agro-similarity, more open cities: $\beta_1+\beta_2$	0.014	-0.018	0.011	-0.170*	-0.044**	0.099	0.052***
	(0.059)	(0.060)	(0.007)	(0.092)	(0.021)	(0.070)	(0.016)
Difference, open − closed: β_2	-0.104	-0.119	0.007	-0.007	0.021	-0.169*	-0.040*
	(0.075)	(0.080)	(0.009)	(0.110)	(0.025)	(0.095)	(0.021)
High openness (=1)	0.028	0.009	-0.004	0.053	0.004	-0.007	-0.013
	(0.067)	(0.083)	(0.010)	(0.151)	(0.028)	(0.083)	(0.019)
Panel B: High openness = above-median market access over realized rice-import volumes							
Agro-similarity, more closed cities: β_1	**0.109**	**0.073**	**0.014****	**-0.223****	**-0.078*** **	**0.231****	**0.083*** **
	(0.085)	(0.085)	(0.007)	(0.105)	(0.018)	(0.092)	(0.016)
Agro-similarity, more open cities: $\beta_1+\beta_2$	-0.002	-0.022	0.002	-0.106	-0.025	0.085	0.049***
	(0.068)	(0.057)	(0.007)	(0.084)	(0.020)	(0.077)	(0.018)
Difference, open − closed: β_2	-0.111	-0.096	-0.012*	0.117**	0.053**	-0.146	-0.034*
	(0.091)	(0.082)	(0.007)	(0.054)	(0.021)	(0.121)	(0.020)
High openness (=1)	0.025	0.010	-0.005	0.084	0.008	-0.036	-0.008
	(0.074)	(0.067)	(0.011)	(0.082)	(0.026)	(0.096)	(0.023)
Panel C: Two-stage least squares on market access (z), at the mean closed and open city							
Agro-similarity, mean closed city	**0.141**	**0.125**	**0.004**	**-0.138**	**-0.061*** **	**0.307*** **	**0.103*** **
	(0.099)	(0.101)	(0.010)	(0.161)	(0.021)	(0.096)	(0.021)
Agro-similarity, mean open city	-0.032	-0.071	0.014	-0.233**	-0.050**	0.046	0.047**
	(0.067)	(0.075)	(0.009)	(0.118)	(0.023)	(0.075)	(0.021)
Difference, open − closed	-0.173	-0.197	0.010	-0.095	0.012	-0.261*	-0.056*
	(0.118)	(0.132)	(0.014)	(0.187)	(0.026)	(0.137)	(0.028)
Market access (z)	0.122	0.147	-0.015	0.322*	0.045	0.019	-0.057**
	(0.120)	(0.125)	(0.015)	(0.171)	(0.028)	(0.133)	(0.025)
First-stage F (Kleibergen-Paap)	8.13	8.13	8.13	8.13	8.13	8.13	8.13
Outcome mean	11.842	10.933	0.122	9.822	0.374	10.232	0.516
Outcome SD	0.854	0.840	0.040	0.896	0.183	0.975	0.156
Observations	134	134	134	134	134	134	134

Notes: Each column of each panel is one regression of a 2010 outcome of Table 3 on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Panels A and B interact agro-similarity with an indicator for high openness, as in Equation 5, split at the median over all 138 urban centers within the 300-minute window: route distance (Section 3.5) is the travel time from a city through its nearest customs port to the nearer of Bangkok and Ho Chi Minh City, and market access (Appendix B.2) sums the rice imports of every customs port, discounted by the travel time from the city. Realized imports can respond to local food substitution, so Panel B is a sensitivity comparison. Panel A is the primary route-based comparison, subject to the transport-source and timing qualifications in Section 4.3. Panel C is two-stage least squares, with standardized market access and its interaction with agro-similarity instrumented by standardized route proximity and its interaction; its two slopes are the effect of agro-similarity at the mean market access of the cities below and above the median of market access (−0.84 and 0.86 SD). The difference row is the continuous interaction times the gap between these evaluation points. The Kleibergen-Paap first-stage F under the Conley variance is 8.13. SE in parentheses; the four sector columns are shaded. (z) means standardized to mean zero and SD one, and the beta terms are the coefficients of Equation 5. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Realized rice imports give a stronger agricultural-share contrast, but can respond to the mechanism being tested. Panel B splits cities by gravity market access, constructed from realized import volumes. A city whose productive hinterland replaces imported rice can appear more closed because of

that substitution. The larger agricultural-share difference in this panel is consistent with this source of endogeneity; it does not measure the bias in the route-based estimate. Modern-sector results have similar signs across the two measures (Appendix B.2).

In Panel C, market access and its interaction with exposure are instrumented by standardized route proximity and its interaction with exposure. This requires route proximity to affect urban outcomes and their response to exposure only through market access, conditional on the main controls. The estimating equation and first stages are in Appendix B.2.

The table evaluates the exposure response at the mean market access of the more closed and more open groups. The modern-employment interaction is −0.153 log points per standard deviation of market access (standard error 0.080), marginally significant, as is the modern-share interaction. The first-stage Kleibergen-Paap statistic is 8.13; the median-split instrumental-variable specification has a weaker first stage (Appendix Table B2).

The geographic overlap between trade access and other regional characteristics limits this comparison. With island indicators interacted with exposure, the modern-employment difference remains −0.203 log points (standard error 0.126) and loses precision. Baseline food-price and availability data are unavailable, so I cannot directly test whether food supply constrained the more closed cities more strongly. The openness pattern supports the food-market mechanism, conditional on other mechanisms having comparable responses across the two groups after controls. Migration, amenities, and rural demand could also vary with trade access. Migration is examined next; rural demand and public provision are considered in Section 5.3.

Rural productivity can also affect migration through labor supply in the settlements and labor demand in the cities. A labor-saving productivity gain would release settlers to the cities, as in Bustos et al. (2016). A pull from the cities, as in Harris & Todaro (1970), would show up as migration into the more exposed cities. Food trade can raise urban labor demand and induce migration from origins outside the settlements. The variation here is in the settlers' farming skills at their destination, conditional on the number of settlers placed, so it is a productivity shock to the people the program moved and holds their number fixed. For the settlers a better match raises what farming pays where they are, so the labor-saving release applies to them in reverse; the standard response is predicted instead for the surrounding countryside whose output competes with theirs. If anything, these people are more mobile than most; that they do not move from their places of settlement is evidence that productivity retains them.

Recent migration is more common among residents of more exposed cities. On the main city sample, a one-standard-deviation increase in exposure raises the share of 2010 residents who lived in another province five years earlier by +0.52 percentage points (standard error 0.26), and the share who lived in another district by +1.92 points (0.49). Both estimates are statistically significant with the main spatial standard errors; with Bartlett distance weights, the province estimate is marginally significant and the district estimate remains significant (Appendix Table E11). District moves include

moves within a province as well as across provinces. These measures describe recent arrivals among current residents and do not identify departures, earlier moves, or the settlement villages from which migrants came.

Recent migration rises more in the more closed cities, but the differences between the openness groups are imprecise for both province and district moves (Appendix Table E12). Recent district moves are also more common among workers in more exposed cities. This comparison uses workers with an observed district of residence five years earlier and includes return migrants; its openness difference is imprecise under both spatial inference methods. These estimates leave migration as a possible contributor to the larger employment response in the more closed cities.

The recent interprovincial share also rises significantly in the groups with larger initial populations or higher initial agricultural employment shares (Appendix Table D5, column 4). The estimates for smaller and less agricultural cities are positive and imprecise, and the differences between groups are imprecisely estimated in both splits.

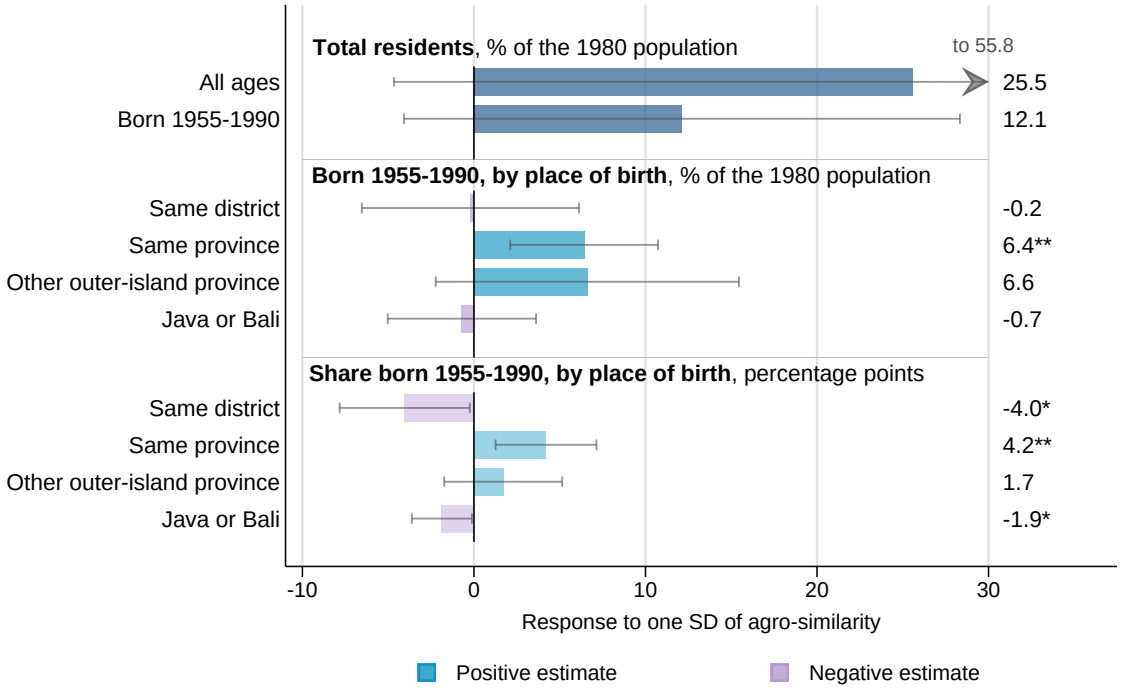
Birthplace composition provides a longer-run view of who lives in the more exposed cities. Among residents born from 1955 to 1990, the share born on Java or Bali is lower and the share born elsewhere in the same province is higher (Figure 11). Java or Bali birthplace is a proxy for the original settlers, excluding their children born on the Outer Islands and including other migrants from those islands. The pattern does not support disproportionate settlement of Java- or Bali-born residents in the more exposed cities. It cannot locate migrants' previous villages or distinguish transmigrants' children from other residents born in the same province.

The measured settler stay-share does not fall detectably with agro-similarity. Its linear gradient is statistically insignificant (Online Supplement); Figure 12 plots the local-linear fit of a broader definition. The definitions in the figure and Online Supplement count Java- or Bali-born residents relative to the original settlers, excluding locally born children. They do not follow the original settlers individually or locate the origins of urban in-migrants. The measure therefore provides limited evidence on the labor release from settlements that a labor-saving productivity gain would predict.

The two migration patterns fit together once the origin of the shock is kept in view. In the settlements, people are not leaving when they are more productive, so the migration that does happen is not coming from the productive areas. The arrivals the cities record come from elsewhere in the same province (Figure 11), and the rural subdistricts without a settlement shift employment out of agriculture while their total employment does not rise (Figure 15): these are the areas that are now out-competed by the more productive settlements, and theirs is the standard response. The censuses do not locate the migrants' origin villages, so this is a pattern across places; the flows themselves are not traced.

A calculation based on net employment growth assesses how much modern employment additional workers could supply under a fixed incumbent workforce. At the mean city, the main coeffi-

Figure 11: 2010 Urban Residents by Birthplace



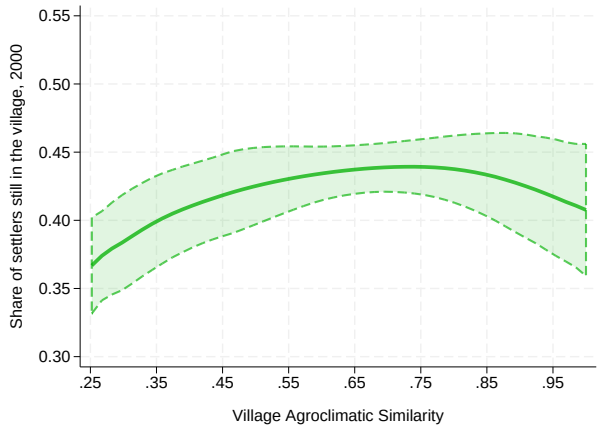
Notes: Each bar is one regression of the row's outcome on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Whiskers are 90% confidence intervals from the same standard errors; the whole-population interval runs past the axis and ends in an arrowhead, with its upper bound written beside it. The first block is the city's whole population and its residents born from 1955 to 1990 in the 2010 census, in % of its 1980 census population; the whole-population response is 10% of that outcome's mean of 244%, against a log population response of 5% in Table 3, since the two outcomes weight the cities differently and a regression in levels gives more weight to the cities that grew most. The second block splits that cohort, in the same units, into four birthplace groups that add up to it: born in the same district as the city, elsewhere in the same province, in another outer-island province, and on Java or Bali; the third block gives each group's share of the cohort in percentage points, with standard errors in Appendix Table C3. The hue of a bar marks the sign of its estimate, and its shade marks the block. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

cients imply about 9,100 additional modern-sector jobs and 4,200 fewer agricultural jobs per standard deviation of exposure, alongside 2,200 additional workers (Figure 13). The gap between the modern-employment and total-employment responses is 6,900 jobs, with a 90% interval of 1,900 to 11,900. If the incumbent workforce is fixed, departures and labor-force entry and exit do not offset arrivals, and every additional worker takes a modern-sector job, the workers added could fill 24% of the modern-job gain, with a one-sided 95% upper bound of 82%. This rejects a restricted explanation in which additional workers fill the entire gain.

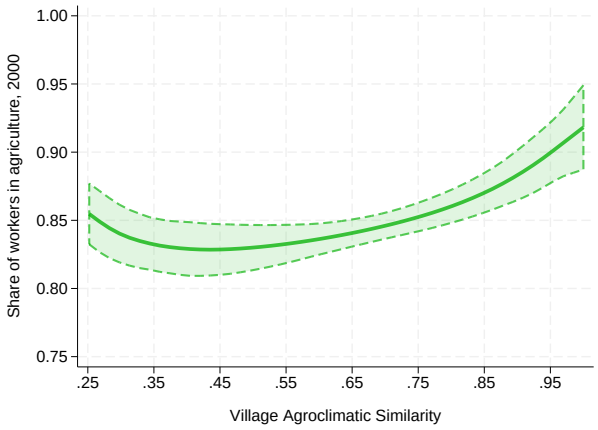
Net employment growth does not measure gross migration when arrivals replace departing workers or coincide with changes in labor-force participation. At the measured modern-employment rate of in-migrants, the workers added could fill 16% of the gain, with an upper bound of 54%. Counts of residents aged 15 and over are less informative because they include nonworkers, and they do not reject that the residents added could fill the entire modern-job gain if every additional resident worked in that sec-

Figure 12: Settler Stay-Share and the Agricultural Employment Share Against Agro-Climatic Similarity

(a) Settler Stay-Share



(b) Agricultural Employment Share, 2000



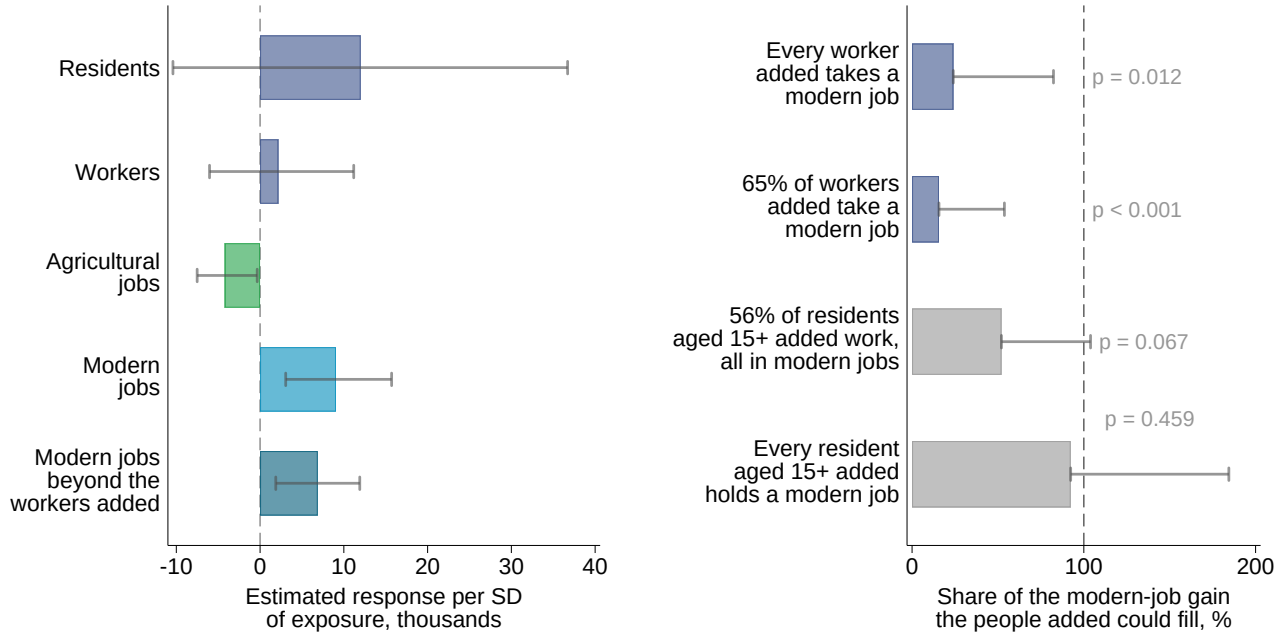
Notes: Semiparametric local linear fit (Epanechnikov kernel, bandwidth 0.25) of the village settler stay-share in panel (a), and of the share of a village's workers in agriculture in the 2000 population census in panel (b), on the village's agro-similarity (0 to 1 scale), with their pointwise 90% confidence bands, partialing out the village's ruggedness, slope classes, log elevation, distance to the coast and 1948 to 1978 mean temperature, estimated on the 808 transmigration villages with a measured 2000-census agricultural employment share and a measured stay-share below 2 and plotted over the 0.25 to 1 similarity range. The bands treat the villages as independent; panel (a) of Appendix Figure E1 reports the linear gradient of the agricultural share with its island-clustered, wild-bootstrap and Conley inference. Appendix Figure D1 draws both panels at a bandwidth of 0.30 and with each village weighted by its 2000 population. The plotted stay-share is a village's 2000-census residents born on Java or Bali over the settlers the program placed there; the stay-share named in the text also removes residents who lived in Java or Bali five years before the census and residents born after the village was settled. The linear gradient under each definition, with island-clustered, wild-bootstrap, and Conley inference, is in the Online Supplement.

tor. With the expected-exposure control used above, the worker-based upper bound includes the whole gain when every additional worker takes a modern-sector job (Appendix E.2; Appendix Table E14). Employment among workers born in the city's district and children and youths per native-born adult do not rise detectably with exposure. These proxies cannot exclude labor-force or fertility responses (Appendix Table E15).

Most of the agricultural-share response occurs within birthplace groups. Decomposing the full 2010 workforce by birthplace and industry attributes 84% of the fall to differences within groups, 16% to workforce composition, and 1% to their interaction (Appendix Figure E4). A separate partition by arrival recency attributes 88% of the fall to differences within groups: workers who never left their birth district, migrants present five years before the census, and migrants arriving within five years (Online Supplement). This partition distinguishes lifetime migrants by arrival recency, while the recent district-migration measure also includes return migrants. Both decompositions use the full workforce, including birth cohorts outside the 1955 to 1990 range of Figure 11. Repeated cross sections combine individual sectoral switching, cohort replacement, and selection within groups. They cannot identify

Figure 13: Workers Added and Modern Jobs Gained at the Mean City

(a) Estimated Responses of People and Jobs per SD (b) Share of New Modern Jobs People Added Could Fill



Notes: The main city sample, in the main specification with the full set of controls and Conley SE within 200km. Panel (a) turns columns 1, 2, 4 and 6 of Table 3 into the estimated response of each outcome's level at the mean city per SD of exposure (the 2010 mean times $e^b - 1$), with 90% intervals; the last bar is the modern jobs gained less the workers added, with its interval from the joint covariance of the two coefficients. Panel (b) gives the share of the modern-job gain the people added could fill under a fixed incumbent workforce, no offsetting departures or labor-force entry and exit, and the sector mix stated by each row, its one-sided 95% upper bound (whisker) and the one-sided p-value of a share of 100% or more (dashed line); the first two bars count workers and the last two residents aged 15 and over. Appendix E.2 gives the construction. Source: 1980 and 2010 population censuses.

the share due to each. This distinction matters in Indonesia, where Budí-Ors (2026) attributes much of aggregate agricultural-share decline to generational replacement.[20]

Food consumption, the price contrast and the employment response by openness support a food-trade mechanism. Recent migration can contribute to the urban response, including through labor demand generated by food trade. The conditional workforce calculation and the within-group decompositions suggest additional adjustment in employment composition, while incomplete migration histories prevent a separation of individual sector changes from arrivals, departures and cohort replacement.

5.3 Rural Adjustment and Alternative Channels

The village agricultural employment share has a positive, statistically insignificant gradient with agro-similarity, while the city agricultural share declines (Figure 14; Appendix Figure E1 gives the linear gradients). The village data provide suggestive evidence of continued agricultural specialization. In Matsuyama's world, the villages are the open economy that receives the shock: they trade their food

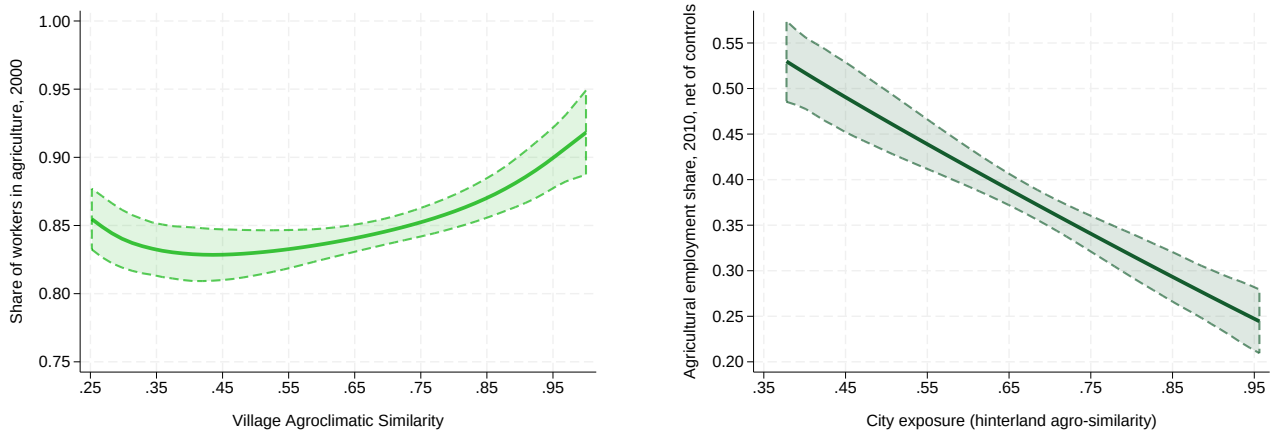
[20] Hamory et al. (2021) use individual histories in the Indonesian Family Life Survey to study sectoral switching. Its coverage of the outer-island cities studied here is insufficient for the city-level exposure design.

with the cities and, if anything, specialize further in agriculture, their comparative advantage. The cities they feed respond as a city does in the closed economy of Glaeser (2014), moving employment out of agriculture and into the modern sector, and a test confined to the shocked villages misses this pairing.

Figure 14: The Agricultural Employment Share Against Agro-Similarity in the Transmigration Villages and in the Nearby Cities

(a) Transmigration Villages, 2000

(b) Nearby Cities, 2010



Notes: Semiparametric local linear fits (Epanechnikov kernel) of the agricultural employment share on agro-similarity, with pointwise 90% confidence bands, after partialing out each panel's controls. **(a)** The share of a village's workers in agriculture in the 2000 population census against the village's own agro-similarity (0 to 1 scale), on the 808 transmigration villages with a measured agricultural share and a measured settler stay-share below 2, partialing out the village's ruggedness, slope classes, log elevation, distance to the coast and 1948 to 1978 mean temperature, at a bandwidth of 0.25, as in panel (b) of Figure 12. **(b)** The agricultural employment share in the 2010 population census against city exposure, the mean agro-similarity of the transmigration villages within 300 travel minutes, on the main city sample, partialing out the controls of the main specification, at a bandwidth of 0.30. The bands treat the villages and the cities as independent, so they are narrower than intervals from Conley SE. At a bandwidth of 0.30 the kernel weights every city at each point of panel (b), so its curve is close to linear by construction. Appendix Figure E1 gives each linear gradient with its clustered, wild-bootstrap and Conley inference.

The fitted curve of the village agricultural share is flat over the lower half of the similarity range and rises over the upper half, most steeply among the best-matched villages (panel (b) of Figure 12). Its statistically insignificant linear gradient limits the evidence for stronger agricultural specialization; the city estimates provide clearer evidence of structural transformation.

Better-matched settlement villages are more likely to grow rice, the staple consumed in the cities, under both the village controls used here and those of Bazzi et al. (2016); that study finds less similar villages allocating more crop revenue to cash crops.[21]

A national average of agricultural contraction and modern-sector expansion conceals within-country variation in how the gains to agricultural productivity accrue, differing between the regions that experience the growth and the regions that benefit from it indirectly through trade. The broader

[21] The rice-growing probability rises by 0.079 (standard error 0.017) with the village geography controls, by 0.058 (0.026) with province-by-placement-year cells, and by 0.127 (0.022) under the Table 7 controls of Bazzi et al. (2016). The village-control estate-crop estimates are imprecise.

comparison includes entire rural subdistricts with and without a settlement, alongside the cities (Figure 15). Both rural groups lie outside every city and within 300 travel minutes of a settlement village. Outcomes and exposure are standardized separately among cities and among the pooled rural subdistricts. The coefficients describe changes relative to each population's variation, so their magnitudes are not effects in common physical units. Regressions use the same controls, with the settler count measured within 300 travel minutes; for rural subdistricts with a settlement, it is the number of settlers placed in the transmigration villages that overlap the subdistrict.

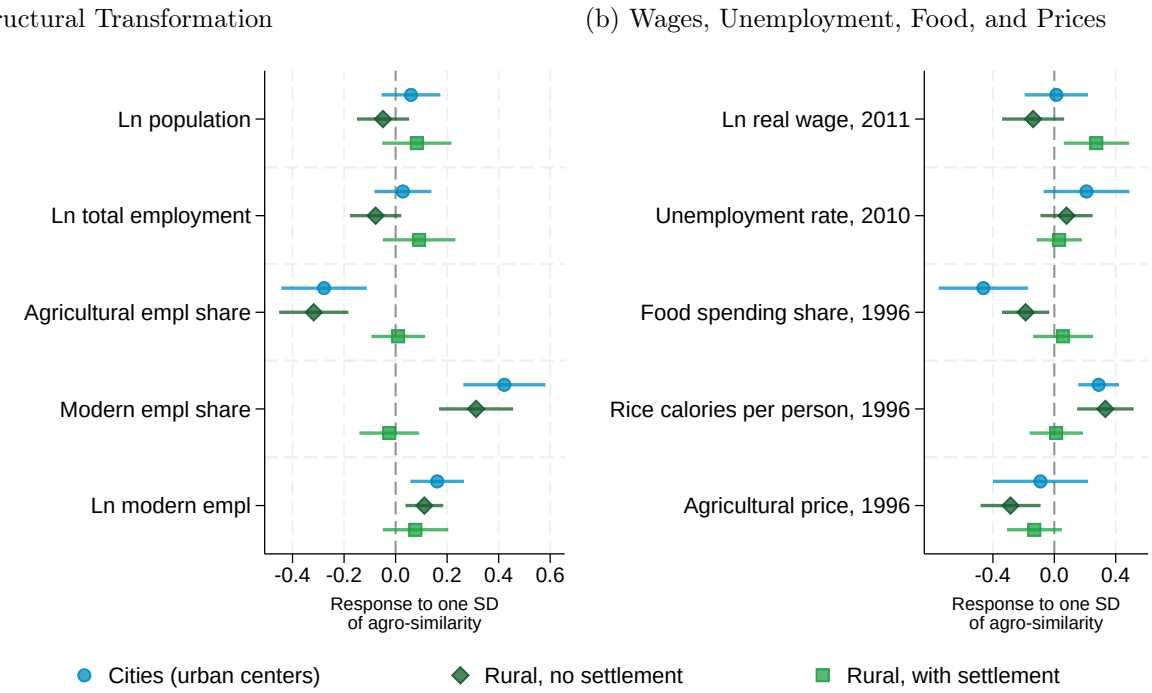
In cities the log modern-employment coefficient exceeds the log population coefficient, whose estimate is imprecise. Rural subdistricts without a settlement also reallocate employment: the agricultural share falls by 0.32 of a standard deviation and the modern share rises by 0.31, both precisely estimated. The standardized modern-employment response is positive, while the total-employment estimate is negative. In subdistricts with a settlement, population and employment estimates are positive and imprecise, and the composition estimates are close to zero with smaller standard errors than in the rural group without settlements. The contrast is consistent with productive villages continuing to farm while nearby places expand their modern sector.

Real wages rise detectably in rural subdistricts with a settlement. Among observed wage earners there, the 2011 real wage rises by 0.27 of a standard deviation, while the nominal-wage estimate is positive and imprecise. The raw-food price estimates come from 1996, when the index is lower by 0.13 of a standard deviation in subdistricts with settlements, imprecisely, and by 0.29 without settlements, significantly (Appendix Table E1). Nominal-wage imprecision does not establish a zero nominal gain, and the earlier food prices cannot decompose the later real-wage response. Real-wage responses in cities and in rural subdistricts without settlements are statistically indistinguishable from zero.

The food expenditure share falls where employment moves out of agriculture. In 1996 it falls by 0.46 of a standard deviation in cities and by 0.19 in rural subdistricts without settlements, the latter significant; the estimate in subdistricts with settlements is imprecise. Calories per person rise in cities and do not move detectably in either rural group; rice calories (Figure 15, panel (b)) rise in cities and rural subdistricts without settlements (Appendix Table E1). Unemployment responses are imprecise throughout. These outcomes locate the employment, consumption, and earnings responses, while providing an incomplete basis for ranking welfare across places.

Rural responses also differ with access to outside food markets. Among rural subdistricts with a settlement, the more closed subdistricts have larger total-employment and agricultural-employment responses, with both differences statistically significant (Figure 16). The population difference is marginally significant. Spending per person rises in the more closed subdistricts and falls in the more open ones, and the food-expenditure-share response is lower in the more closed subdistricts. The expenditure differences remain statistically significant with alternative spatial standard errors and province clustering, while the real-wage responses do not differ detectably. Each location type has its own median route-distance split, so these tests compare responses within location types at potentially different levels of access. Employment is measured in 2010, and wages and spending in 2011. The larger

Figure 15: The Response by Location Type: Cities and Rural Subdistricts With and Without a Settlement



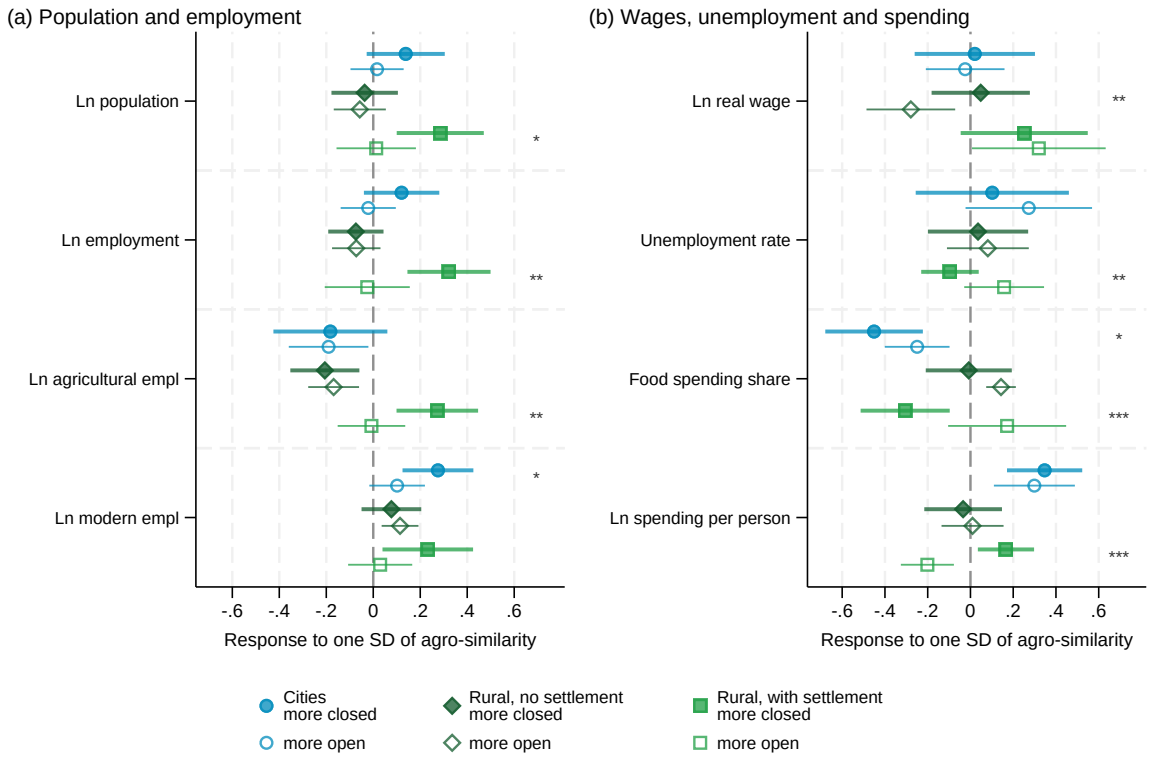
Notes: Each marker is the response to a one-SD increase in agro-similarity, the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, and its bar is the 90% confidence interval from Conley SE within 200km. Cities are the main city sample; the rural subdistricts with and without a settlement lie outside every city and within 300 travel minutes of a transmigration village. Outcomes and agro-similarity are standardized once among the cities and once among the pooled rural subdistricts, at exposure SDs of 0.103 (138 cities with a measured exposure) and 0.100 (787 subdistricts), so city and rural coefficients use different standardization scales; the two rural groups share a scale. The city regressions are in the main specification with the full set of controls; the subdistrict regressions have the same controls, except that the settler count of a subdistrict with a settlement is the number of settlers placed in the transmigration villages that overlap it. **(a)** Every row is a 2010 census outcome on the main city sample, 191 rural subdistricts with a settlement and 596 without. **(b)** The unemployment rate is from the 2010 census; the real wage is the median earner's wage over the local price index, from the 2011 household-survey (Susenas) wave used here, among observed positive earners in retained subdistricts; the food share of spending, rice calories per person (rice kilograms at a fixed calorie content) and the agricultural price (the raw-food index) are from the 1996 wave, before the 2004 rice import ban. Each row uses the units its wave covers: 112 to 134 cities, 82 to 191 subdistricts with a settlement and 236 to 596 without. The rural wage source retains survey subdistricts with at least five observed workers and a positive share of workers in urban survey blocks. Every outcome is in Appendix Table E1 with its SE.

agricultural-employment response in more closed settlement areas is consistent with continued farming where outside food access is weaker.

The spatial comparison is consistent with productive settlements continuing to supply food while cities and rural places without settlements reallocate employment. The village specialization estimates remain imprecise, and the earnings measures do not establish a welfare ranking across places. Public investment, capital through banks, and rural demand could contribute to this adjustment. I assess each by its predicted response in the settlements or the rural subdistricts that overlap them.

If the state invests more in productive settlements, their roads, electricity, schools and health centers would improve more. The settlement-village estimates give imprecise responses for electricity, schools and health; the share with a main road passable by car rises, with a marginally significant estimate (Figure 17). The 90% intervals bound those responses: per standard deviation of agro-similarity,

Figure 16: Trade Openness and the Response by Location Type: Population, Employment, Wages, and Spending



Notes: Each marker is the response of its row's outcome to a one-SD increase in agro-similarity, in the specification and samples of Figure 15: the main city sample and the rural subdistricts with and without a settlement, which lie outside every city, with each outcome standardized over the cities or over the pooled rural subdistricts. Each location type is split at the median route distance to the Thai and Vietnamese rice hubs among its units within 300 travel minutes of a transmigration village. The coefficients come from one regression per outcome and location type of the outcome on agro-similarity interacted with indicators for the more closed and the more open group. Filled markers on thick intervals denote the more closed group and hollow markers on thin intervals the more open group; intervals are 90%, from Conley SE within 200km. Stars indicate that the coefficients of the two groups differ: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Panel (a): the 2010 census outcomes. Panel (b): the unemployment rate (2010 census) and three household-survey outcomes; the real wage and the two spending rows are from the 2011 household-survey (Susenas) wave, which covers fewer subdistricts than the census. Wages summarize observed positive earners. The rural wage source retains survey subdistricts with at least five observed workers and a positive share in urban survey blocks. The Online Supplement reports each row's coverage and the test of each difference under two further variance estimators. Regressions are in the main specification with the full set of controls, except that the settler count of a subdistrict with a settlement is the number of settlers placed in the transmigration villages that overlap it.

the share of villages on the electricity grid is at most 5.8 percentage points higher, against a mean of 42%; the share with a health center at most 6.2 points, against 40%; the share with a junior secondary school at most 5.3 points, against 31%; and the road share between 0.1 and 5.4 points, against 90%. That road response becomes imprecise with Bartlett distance weights or province clustering (Appendix Table E21). The village comparisons do not have comparable pre-settlement provision baselines. The 2011 estimates for rural subdistricts with a settlement are statistically insignificant when each measure's 1980 level is held fixed (Appendix Table E19). The city road-density and road-condition estimates are imprecise, and the pre-1980 INPRES school allocation does not favor more exposed cities (Appendix E.4).

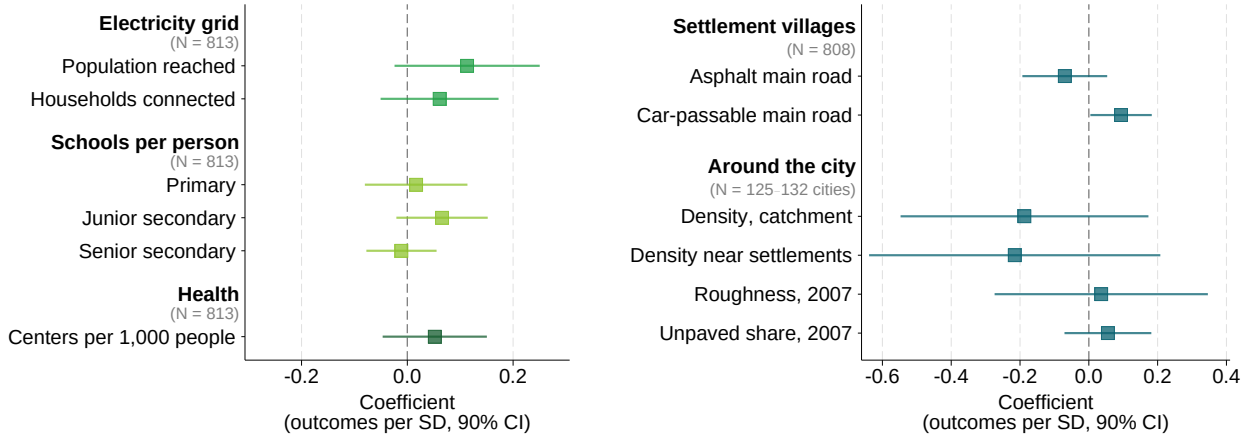
The countryside around more productive settlements did gain grid access and car-passable roads after the baseline (Appendix Table E22). These responses can reflect both public supply and household

demand for connections. Public investment remains a possible contributor to rural food production; the tests do not quantify its contribution to city employment.

Figure 17: Public Provision in Settlement Villages and Roads Around the City

(a) Electricity, Schools, and Health

(b) Roads

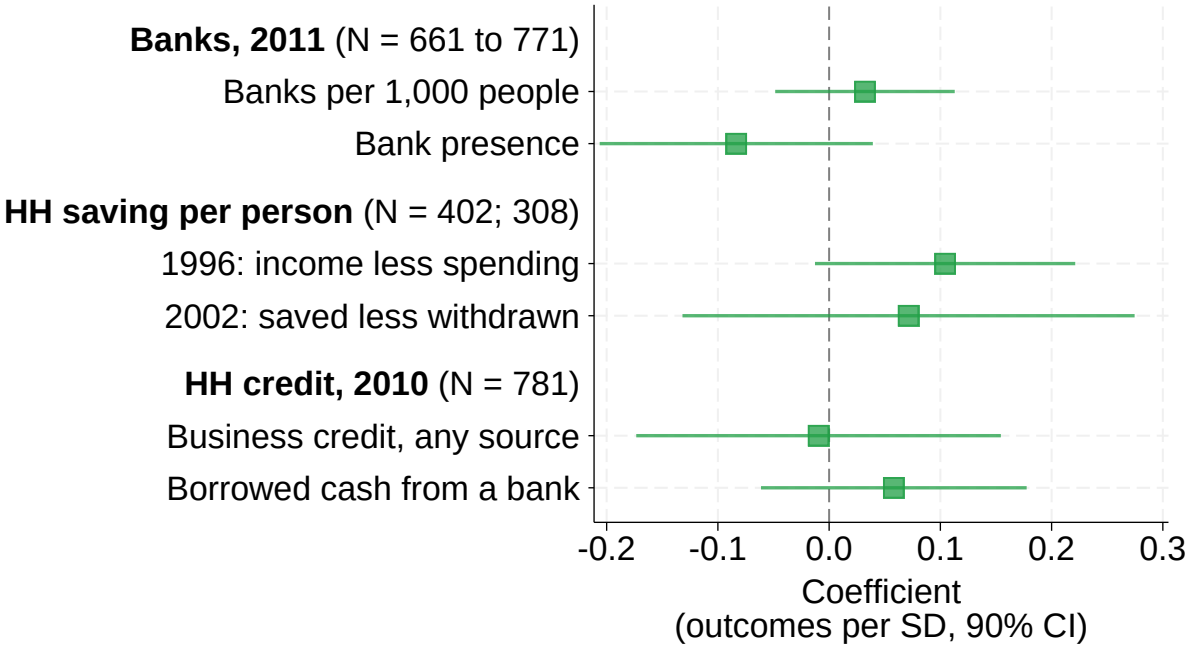


Notes: Markers give responses per standard deviation of agro-similarity, with outcomes standardized over each regression sample; bars are 90% confidence intervals from Conley standard errors within 200km. The settlement-village rows relate 2011 provision to each village's own agro-similarity, standardized over the settlement villages, with village geography controls. These comparisons do not include pre-settlement provision baselines. The household grid-connection share divides connected households by all measured households in the village. Their sample labels count villages. City-road rows use the main city sample, its exposure and full main controls: road density holds the 1950 density fixed, and road condition is the 2007 survey, controlling for its 1990 level (the regression's outcome is the 1990 to 2007 change). Their labels count cities. The car-passable-road response becomes imprecise with Bartlett weights or province clustering (Appendix E.5); the baseline-controlled subdistrict evidence remains in Appendix Table E19, where grid connections are conditional on villages reached by the grid. These outcomes do not measure all public investment, earlier provision changes, or service quality. Sources: village census (PODES), 2011; historical and current road maps; Indonesian Road Management System surveys, 1990 and 2007.

A capital-supply channel begins with greater saving in productive rural areas, which banks lend to firms elsewhere (Bustos et al. 2020). In rural subdistricts with a settlement, household saving and the changes in bank density and presence are imprecisely estimated (Figure 18). The bank-presence interval runs from -3.5 to $+0.7$ percentage points of the population per standard deviation, against a mean change of 2.0 points since 1980, so it excludes an expansion of bank presence larger than a third of the average change. Saving is measured as income less spending in 1996 and as money saved less withdrawn in 2002. In 2010, business credit from any source and cash borrowing from a bank are imprecisely estimated. The saving intervals allow increases, and these household measures do not cover all deposits available to banks. The village censuses record bank presence but do not trace deposits or lending from rural savers to city firms. The evidence therefore leaves capital transmission possible.

A rural-demand mechanism predicts higher purchases of city goods by households in productive settlements. Total spending and its non-food share provide indirect measures of this demand. Their 2011 exposure responses are statistically insignificant in both the rural subdistricts with a settlement and the whole hinterland (Appendix E.5). These late spending measurements leave earlier rural-demand responses unresolved.

Figure 18: The Capital Channel in Rural Subdistricts With a Settlement



Notes: Each marker is the coefficient on standardized agro-similarity for rural subdistricts with a settlement, from one regression per row over the rural subdistricts of the main sample: the outcome, standardized to mean 0 and SD 1 over the row's subdistricts, on agro-similarity times an indicator for a settlement in the subdistrict and times its complement, the indicator, the settler count (the settlers placed in the subdistrict's own settlement villages where it has one and, where it has none, the settlers of the villages within 300 travel minutes weighted by inverse distance), the four 1980 census baselines of the subdistrict and the sixteen agro-climatic covariates; 90% intervals from Conley SE within 200km. Banks: the 2011 village census, controlling for the outcome's 1980 level (the regression's outcome is the 1980 to 2011 change, standardized over the row's subdistricts); bank presence is the population share living in a village with a bank, and it leaves out West Kalimantan and the West Sumatra subdistricts recorded as nagari in 2011. HH saving: monthly household (HH) income less expenditure in 1996, and net saving (money saved less savings withdrawn) in 2002, each in Rupiah per person per month as its inverse hyperbolic sine (Susenas 1996 and 2002). HH credit: the shares of survey households with business credit from any source and that borrowed cash from a bank (Susenas 2010). The sample sizes in the headings count all rural subdistricts in each regression, including those without a settlement; the plotted coefficients describe the subdistricts with a settlement. Sources: village censuses (PODES) of 1980 and 2011; Susenas 1996, 2002 and 2010.

The geographic checks leave the own-city sectoral results broadly similar after controlling for neighboring exposure, while some specifications show negative city-size responses to neighboring exposure. The slopes do not differ detectably in the conflict provinces; excluding them reduces the sectoral coefficients and their precision. Becoming a new district seat after 1999 is unrelated to exposure, and the modern-employment result survives excluding education, health, social and government services and cities with new district seats (Appendix Table E25; Appendix E.5).[22] These checks do not cover processing of farm output, commodity booms, or every government program.

The productive villages and their neighbors adjust differently. Food consumption, lower agricultural prices in more closed cities and their larger modern-employment response support a food-market explanation for the urban adjustment. Recent migration and rural infrastructure responses can also

[22]The conflict comparison covers Aceh, West and Central Kalimantan, and Central Sulawesi (Barter & Côté 2015; van Klinken 2007).

contribute, while the conditional accounting and incomplete migration histories leave the quantitative contribution of each mechanism unresolved. Section 6 examines trade and migration within a spatial model.

6 A Spatial Equilibrium Model with Trade and Migration

The empirical results show a substantial shift from agriculture to modern employment in cities exposed to more productive rural hinterlands, alongside a modest and imprecisely estimated population response. I use a spatial model to examine how agricultural productivity gains in rural areas can change employment within nearby cities. Households choose both location and sector, allowing the model to distinguish changes in a city's employment composition from changes in its population.

Cities need to feed their residents. They can obtain food from their rural hinterlands, buy from more distant suppliers, or farm within the urban boundary. Access to these sources changes how agricultural productivity affects structural transformation. In Matsuyama (1992), openness changes the relationship between agricultural productivity and industrialization; Glaeser (2014) shows why rural prosperity supports urbanization when outside food is unavailable. Here, the productivity gain occurs in rural locations supplying the city. Cheaper food can reduce the return to farming within the city and allow residents to work in non-agricultural activities. Affordable supplies from outside the hinterland reduce the city's dependence on its rural neighbors.

I hold urban productivity fundamentals fixed to study this transmission through food markets. Food and one non-agricultural good are traded across locations at a cost, and different elasticities govern sector and location choices. The central comparison holds each location's population fixed while wages, prices, employment composition, and land use adjust.[23]

I use differences in rural agro-climatic match quality to study how agricultural productivity affects employment in nearby cities. That focus connects the city-hinterland relationship in Nagy (2023), who studies city formation and growth, to the spatial structural change analyzed by Eckert & Peters (2025), where technology diffusion shapes regional adjustment. The distinction between goods-market and labor-market adjustment also follows Tombe & Zhu (2019), who quantify the aggregate productivity effects of trade and migration frictions. Here, the model explains how cities' exposure to rural productivity gains through food markets can affect their employment composition separately from their population response.

Section 6.1 describes the economic environment and equilibrium. Section 6.2 explains how the data and empirical targets inform the model's parameters. Section 6.3 compares sectoral and population responses under different adjustment restrictions and presents a historical magnitude benchmark.

[23] The trade and choice structure draws on Allen & Arkolakis (2014); Redding (2016); Khanna et al. (2025), and food demand is non-homothetic (Kongsamut et al. 2001; Gollin et al. 2007). The numerical comparison below uses a partially tradable modern-output variant, with the restrictions stated for that variant.

6.1 Setup and Equilibrium

The economy consists of J locations and two sectors: agriculture, A, and a non-agricultural sector, S, combining manufacturing and services. Each sector produces one good, and both goods are tradable across locations. Trade in non-agricultural output provides a source of earnings for cities purchasing food from their hinterlands.

A fixed mass $\bar{L}$ of workers chooses where to live and in which sector to work. Total employment at location j is $N_j = L_{jA} + L_{jS}$, with $\sum_j N_j = \bar{L}$. The model abstracts from unemployment and natural population growth, so changes in N_j represent migration. Each location has a fixed land endowment $\bar{T}_j$, divided between agriculture, T_{jA}, and non-agricultural uses, T_{jH}. Both production technologies use land and labor:

$$Y_{jA} = (A_j^A T_{jA})^{1-\alpha} L_{jA}^{\alpha}, \qquad Y_{jS} = \bar{A}_j^S N_j^{\eta} (T_{jH})^{1-\gamma_S} L_{jS}^{\gamma_S}, \tag{6}$$

where $0 < \alpha, \gamma_S < 1$. Agricultural productivity is A_j^A, and $\bar{A}_j^S$ governs non-agricultural productivity. The baseline sets $\eta = 0$, so population changes do not raise productivity through agglomeration. The productivity experiment changes A_j^A in rural locations hosting transmigration settlements while holding urban productivity fixed.

A common land market allocates $T_{jA} + T_{jH} = \bar{T}_j$ across active uses. Land rental revenue accrues to local residents and is distributed equally as a per-person rebate R_j. Land can adjust between sectors even when population is held fixed. Appendix F.1 describes technology and endowments, and Section F.7 gives the land-market conditions.

Preferences and expenditure. Households consume food, c_A, and the non-agricultural good, c_S. Stone-Geary preferences introduce a subsistence food requirement, the form Matsuyama (1992) made standard in models of structural transformation (see also Kongsamut et al. 2001):

$$U(c_A, c_S) = (c_A - \bar{a})^{\psi_A} c_S^{\psi_S}, \qquad \psi_A + \psi_S = 1, \qquad \bar{a} > 0 \tag{7}$$

The parameter $\bar{a} > 0$ is the minimum food requirement. Let y denote the resources available for expenditure.[24] For $y > p_j^A \bar{a}$, expenditure above the subsistence bill is divided between the two goods in proportions ψ_A and ψ_S. At given prices, the food expenditure share falls with income:

$$\frac{p_j^A c_A}{y} = \psi_A + (1 - \psi_A) \frac{p_j^A \bar{a}}{y} \tag{8}$$

At a given income, cheaper food lowers the subsistence bill and raises expenditure on modern output. Indirect utility is proportional to $(y - p_j^A \bar{a})/P_j$, where $P_j = (p_j^A)^{\psi_A} (p_j^S)^{\psi_S}$. Setting $\bar{a} = 0$ gives Cobb-Douglas preferences with expenditure shares independent of income. Appendix Section F.2 gives the household demand derivation, and Section F.9 gives the trade-price equations for both goods.

[24] Spending is $W_j + R_j$, where $W_j = (w_{jA}^{\theta_S} + w_{jS}^{\theta_S})^{1/\theta_S}$ is the sector-inclusive value of living in j, and the supply-price indices of both goods are inverse marginal labor productivity; Appendix Section F.9 states the income concept and the accounting that follows from it.

Sector and location choice. Worker heterogeneity coming from Fréchet shocks allows workers facing the same wages and prices to make different choices, while resulting in the closed-form choice probabilities that determine the aggregate allocation of workers across locations and sectors. The probability $\pi_{A|j}$ gives the agricultural employment share conditional on living in location j, with $1 - \pi_{A|j}$ giving the corresponding modern-sector share. The sectoral allocation rule is

$$\pi_{A|j} = \frac{w_{jA}^{\theta_S}}{w_{jA}^{\theta_S} + w_{jS}^{\theta_S}}, \qquad L_{jA} = N_j \pi_{A|j}, \qquad L_{jS} = N_j(1 - \pi_{A|j}), \tag{9}$$

where w_{js} is the sectoral wage. The parameter θ_S governs the response of employment composition to relative sectoral wages, the sector-choice elasticity. A larger θ_S implies a larger response of the agricultural employment share to a given change in w_{jA}/w_{jS}.

The probability π_j gives the share of all workers in the economy choosing location j, so that $N_j = \bar{L}\pi_j$. Let v_j denote the deterministic location value, incorporating sectoral earning opportunities, the subsistence food bill, local consumption prices, land income, and amenities Φ_L^j. The location allocation rule is

$$\pi_j = \frac{v_j^{\theta_L}}{\sum_k v_k^{\theta_L}}, \qquad N_j = \bar{L}\pi_j \tag{10}$$

The parameter θ_L governs the population response to differences in location values. A lower value implies less population adjustment for a given change in relative location values. Appendix F.13 considers explicit origin-destination moving costs informed by IFLS migration flows; Appendix Section F.8 gives the choice rules.

Food trade and prices. Food varieties are differentiated by origin and combined with elasticity of substitution $\sigma > 1$. Shipping from k to j incurs an iceberg cost $\tau_{kj} \geq 1$. I parameterize these costs using the travel-time measure introduced in Section 3.3: $\tau_{kj} = \max\{d_{kj}, 1\}^{\delta}$ for $k \neq j$, with $\tau_{jj} = 1$. Let q_k^A denote the origin food-supply index, constructed from inverse agricultural marginal productivity. The food price index at location j is

$$p_j^A = \left[\sum_k (\tau_{kj} q_k^A)^{1-\sigma}\right]^{1/(1-\sigma)}, \qquad \sigma = 4 \tag{11}$$

The sensitivity of p_j^A to an origin's supply price equals that origin's share of food expenditure at j. Productivity gains therefore have greater scope to affect a city when the affected villages account for a larger share of its food purchases. This sourcing relationship provides the link between the geography of the shock and the geography of urban exposure. Appendix Section F.4 describes the trade-cost specification and the origin supply index. The model's endogenous food-expenditure weights differ from the geographic exposure weights used in the empirical regression.

Equilibrium and transmission. The model jointly determines wages, prices, land allocation, rents, and employment across sectors and locations. Non-agricultural varieties are traded with cost $\tau_{kj}^S =$

$\max\{d_{kj}, 1\}^{\delta_S}$, with unit cost within each location. Their supply-price index is $q_k^S = 1/(\partial Y_{kS}/\partial L_{kS})$, following the same convention as food. The non-agricultural price is

$$p_j^S = \left[\sum_k (\tau_{kj}^S q_k^S)^{1-\sigma}\right]^{1/(1-\sigma)}. \tag{12}$$

The main specification sets $\delta_S = \delta$. Both price levels follow from the quantity-based supply indices, so a separate reference-wage normalization is not imposed. These price equations determine consumption prices without imposing goods-market clearing. The income convention and its accounting implications are stated in Section F.9.

A fall in the urban food price lowers the subsistence bill and raises demand for modern output. Competition from rural origins also lowers the return to urban agriculture. Cities need to feed their residents: where the rural hinterland is not productive and internationally traded food is not readily available, urban agriculture persists to feed people in the city, and rising rural productivity frees those workers for other sectors. Workers respond to relative sectoral wages, while differences in location values induce migration. Household income, land rents, and modern-good prices adjust together, so the direction and size of employment changes come from the full calculation.

Comparative statics. A rural productivity gain affects cities directly through their food purchases. Consider an increase in agricultural productivity A_v^A in village v and its effect on a city $j \neq v$, holding the city's productivity fundamentals fixed. At given agricultural land and employment in the village, higher A_v^A lowers its food-supply index q_v^A. Holding other origins' indices fixed, the CES food-price index implies

$$\frac{\partial \ln p_A^j}{\partial \ln A_v^A} = s_{vj} \frac{\partial \ln q_v^A}{\partial \ln A_v^A} < 0, \qquad s_{vj} = \frac{(\tau_{vj} q_v^A)^{1-\sigma}}{\sum_k (\tau_{kj} q_k^A)^{1-\sigma}}, \tag{13}$$

where s_{vj} is the share of city j's food expenditure allocated to village v, and $\sigma > 1$ is the elasticity of substitution across food origins. A given reduction in the village's supply index therefore lowers food prices more in cities that buy a larger share of their food from that village. Access to alternative suppliers can weaken this transmission by reducing the city's reliance on the affected villages.

At a given nominal household income, cheaper food reduces the subsistence bill and increases expenditure on modern output in the city. Urban employment responds to the change in relative wages induced by the rural productivity gain. The sector-choice equation gives:

$$\frac{d\pi_{A|j}}{d \ln A_v^A} = \theta_S \pi_{A|j}(1 - \pi_{A|j}) \frac{d \ln(w_{jA}/w_{jS})}{d \ln A_v^A}, \tag{14}$$

where $\pi_{A|j}$ is the city's agricultural employment share, w_{jA}/w_{jS} is its relative agricultural wage, and $\theta_S > 0$ is the sector-choice elasticity.

Proposition 1 (Urban Sectoral Reallocation with Fixed Population)**.** *For a city with positive employment in both sectors and fixed population, holding θ_S fixed, agricultural employment falls if and only if*

its agricultural wage falls relative to its non-agricultural wage. Let $\widetilde{\pi}_{A|j}$ denote the sector-choice probability evaluated before production inputs adjust. For a productivity gain at a rural origin $v \neq j$, its direct response at the initial production inputs in every location is

$$\frac{\partial \widetilde{\pi}_{A|j}}{\partial \ln A_v^A} = -\theta_S \pi_{A|j}(1 - \pi_{A|j})(1 - \alpha) s_{vj} < 0 \quad \textit{when } s_{vj} > 0.$$

The quantities on the right are evaluated at the initial allocation.

The relative-wage condition follows from Equation 9. For the direct effect, higher A_v^A lowers the origin's inverse marginal-productivity index with elasticity $-(1 - \alpha)$. At the initial production inputs, the non-agricultural supply indices remain fixed, and Equation 13 gives the food-price response. This direct response precedes adjustment of production inputs. Once inputs and prices adjust across locations, the relative-wage condition continues to determine the sign of the city's sectoral change. Urban purchases of the traded non-agricultural good can be supplied from other locations, so local employment is determined by relative returns after those adjustments.

6.2 Taking the Model to the Data

The spatial economy combines urban centers and rural district remainders, using census population and employment, household-survey consumption and wages, village-survey crop area, and geographic information. The empirical exposure gradients inform the response of population and sectoral employment. The numerical figures in this section report a partially tradable variant with separate local and traded non-agricultural activities. Its parameter estimates and counterfactuals are results for that variant; they do not constitute an estimated two-good main specification. Appendix F.11 records the numerical comparison and its sample.

Table 6 distinguishes observed inputs, constructed location characteristics, fixed restrictions, and jointly selected choice parameters.

Preferences, technology, and local fundamentals. Susenas expenditure shares discipline the food weight ψ_A and the non-agricultural weight $\psi_S = 1 - \psi_A$. The subsistence requirement uses the BPS food-poverty threshold for 2000 normalized by the reference wage. Labor shares govern production. The substitution elasticity is $\sigma = 4$, within the range estimated by Broda & Weinstein (2006). The two-good specification applies the food trade-cost convention to the whole non-agricultural good. The measured price-gap gradients in Section 4.3 inform the trade-cost assumptions; they do not identify those costs directly.

Location characteristics determine productivity and amenities. Agricultural productivity uses agricultural employment per unit of crop land, non-agricultural productivity uses non-agricultural employment and its land scale, and location amenities combine geographic and infrastructure measures. The two-good specification has one non-agricultural productivity parameter per location. The addi-

Table 6: Mapping the Two-Good Model to the Data

Object	Role in the Model	Source or Restriction
Population and employment	Location size and agricultural and non-agricultural employment	Population census, 2010; urban centers and rural district remainders
Land and location amenities	Constructed land endowments and location attractiveness	PODES 2003 crop area and satellite built-up area, with the model's crop-area minimum; satellite measures around 1980 and PODES 1983 infrastructure for amenities
Travel time d_{kj}	Geography of sourcing and exposure	Cost-distance construction in Section 3.3; distances in minutes
Consumption and wages	Food spending and the subsistence normalization	Susenas 2011 expenditures and wages; BPS 2000 food-poverty threshold
Productivity and amenities	Constructed location characteristics	Agricultural employment per unit of crop land; non-agricultural employment and land scale; geographic and infrastructure indices
$\psi_A, \psi_S, \bar{a}$	Preference parameters	Food spending disciplines ψ_A; $\psi_S = 1 - \psi_A$; the food-poverty threshold disciplines $\bar{a}$
$\alpha, \gamma_S, \eta, \sigma, \delta, \delta_S$	Fixed technology and trade restrictions	Labor shares and substitution elasticity are fixed; $\eta = 0$ and $\delta_S = \delta$ in the main specification
θ_L, θ_S	Jointly selected choice elasticities	Exposure slopes for log population and the agricultural employment share; both outcomes respond to both parameters
Targets and units	Regression coefficients for comparison	Log population and employment shares have different natural units; the displayed numerical comparisons use standardized exposure slopes
Productivity experiment	Rural agricultural-productivity change	A common gain at settlement-hosting rural locations in the displayed comparison; agro-similarity describes city exposure, without identifying the historical productivity gain

Notes: The main specification has agriculture and one tradable non-agricultural good. Constructed location characteristics use observed inputs and model restrictions; they are not direct observations of productivity or amenities. The displayed numerical figures use a separate partial-tradability specification and its own calibration sample. Their estimates are not estimates of this two-good model. Sources: the population census, Susenas household survey, PODES village census, satellite measures, and BPS (Statistics Indonesia); Appendix F.11 gives the numerical comparison's sample and parameter selection.

tional productivity calibration for a traded-modern subset belongs to the partially tradable comparison in Appendix F.12.

Estimating the choice elasticities. I choose the sector-choice and location-choice elasticities jointly using urban exposure gradients. Population is informative about adjustment across locations, while the agricultural employment share is informative about adjustment within locations. Both parameters affect both outcomes through equilibrium wages and prices.

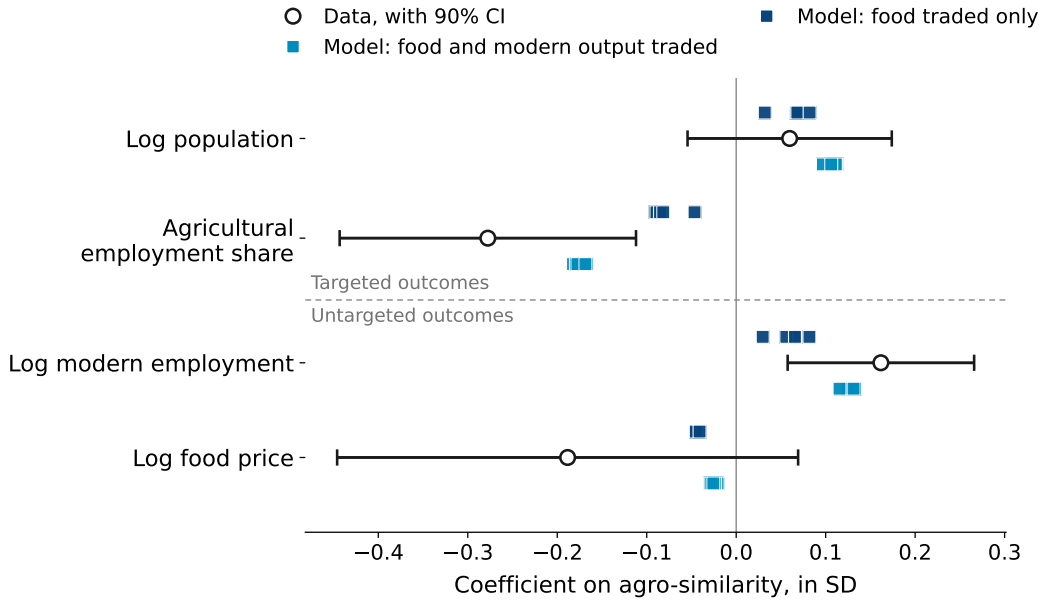
For each candidate pair $\theta = (\theta_L, \theta_S)$, the model is solved before and after a rural agricultural productivity change. The resulting urban outcomes are related to exposure, and the parameters are chosen to minimize the distance between the model and empirical coefficients. This is indirect inference: the coefficients of an auxiliary regression are the estimation targets (Hansen 1982; Gourieroux et al.

1993; Smith 1993). Let $\widehat{b}$ collect the targeted empirical coefficients and $b(\theta)$ the corresponding model coefficients. The estimates minimize

$$\widehat{\theta} = \arg\min_{\theta} \left[b(\theta) - \widehat{b}\right]' \Omega \left[b(\theta) - \widehat{b}\right], \tag{15}$$

where Ω weights the coefficients by their empirical precision. Figure 19 presents the existing comparison between a food-only-trade economy and the partially tradable non-agricultural variant. The reported choice elasticities and fit refer to those numerical specifications. In the two-good main specification both goods are traded, and the same estimation criterion defines the population and composition targets without adopting the comparison's parameter estimates.

Figure 19: Numerical Comparisons Under Partial Non-Agricultural Tradability



Notes: The model calculations use a separate partially tradable non-agricultural specification. Circles are the main-sample estimates of Appendix Table B4, with 90% confidence intervals; squares are the model's responses in the full economy, with traded modern output (light), and in the restricted economy with only food traded (dark), each re-estimated at one of several food trade costs and land-income conventions. Population and the agricultural employment share are the targets; modern employment and the food price are untargeted. The quantity rows relate standardized outcome changes to standardized exposure; the food-price row relates post-shock log food prices to exposure with controls on the price sample.

Additional outcome comparisons. Modern employment and the food price are untargeted in these re-estimates. In the full economy both responses have the data's sign and sit inside its 90% intervals; the food-price estimate is imprecise, so that row tests the sign of the response. The food price also falls with the cropland share around a city (Appendix Figure F1).

The partially tradable numerical variant compares sectoral adjustment with and without population movement. This comparison illustrates an implication of the trade mechanism that the reduced form cannot observe directly.

6.3 Counterfactuals

The numerical comparisons use the partially tradable variant of Appendix F.12. The agricultural-productivity parameter rises by 0.05 log points in rural locations hosting transmigration settlements, holding urban productivity and the remaining exogenous inputs fixed. Within that variant, fixing population preserves nearly the full standardized exposure gradient in urban agricultural employment shares. This is a result of the comparison specification; the two-good main model is defined in Section 6.1.

Table 7 states the restrictions behind the displayed numerical comparison. The same productivity shock and parameterization apply to each row.

Table 7: Restrictions in the Reported Channel Comparison

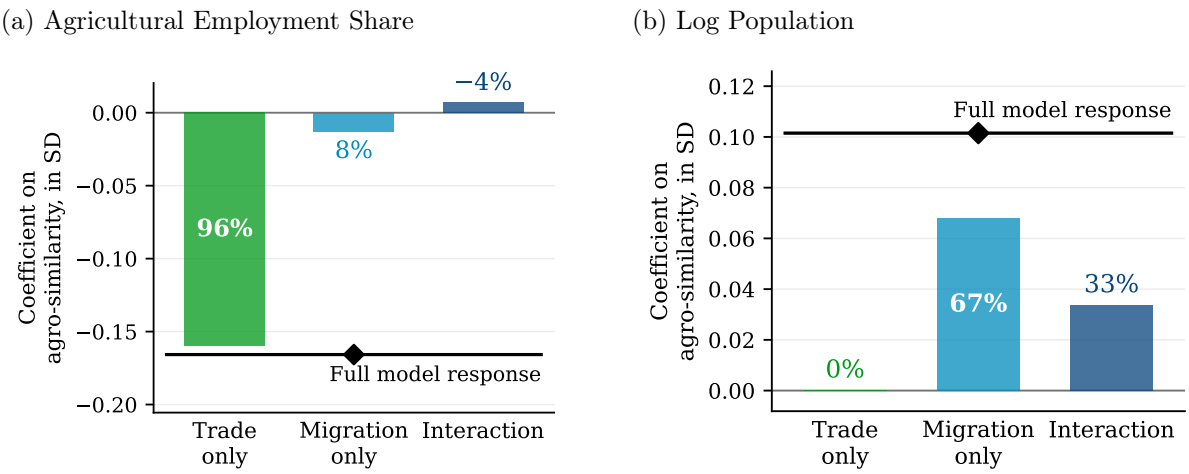
Figure Label	Fixed Quantities	Quantities Allowed to Adjust
Full	Exogenous inputs apart from the productivity gain	Population, goods prices, wages, land use, and sectoral employment
Trade only	Population in every location at its unshocked value	Goods prices, wages, land use, and sectoral employment shares
Migration only	Food prices and the price and employment of the variant's traded non-agricultural component, all at unshocked values	Population, local non-agricultural prices, wages, land use, and employment in agriculture and the local component

Notes: These rows describe the partial-tradability comparison in Figure 20. They do not add a separate non-agricultural component to the two-good main specification. The productivity shock and parameters are common across rows. The fixed-price restriction does not impose autarky. The trade-only population response is zero by construction. Each outcome change is standardized separately; the reported interaction is the full coefficient minus the two restricted coefficients. The figure does not allocate an employment change measured in common units between the channels. Source: the calculation definitions in Appendix F.12.

The figure reports the standardized exposure gradient of the urban response under each calculation, so the bars are read as relative magnitudes: how much of the full gradient each single-channel calculation reproduces. The interaction is the full gradient less the two single-channel gradients.

The displayed numerical comparison uses the services basket and distributes land income locally.

Figure 20: Channel Comparison Under Partial Non-Agricultural Tradability



Notes: These calculations use the partially tradable non-agricultural variant. Each panel decomposes one solved counterfactual, a rise in agricultural productivity in rural locations hosting transmigration settlements, in the model with services only in the traded-modern basket: location and sectoral choice elasticities $(\theta_L, \theta_S) = (0.25, 0.82)$, trade-cost elasticities of 0.5 for food and 1.0 for traded-modern goods, 287 locations, and land income rebated locally. Bars are slopes of the standardized change in the outcome on standardized agro-similarity, each change standardized within its own calculation, and are labeled as shares of the model's full response, marked by the diamond. Trade only holds population fixed, so its population bar is zero by construction; migration only holds food and traded-modern prices and traded-modern employment fixed; the interaction is the full response less the other two. These model calculations do not report sampling uncertainty.

Trade with population fixed. In the partially tradable comparison, Figure 20a shows that the fixed-population calculation preserves nearly the full standardized exposure gradient in the agricultural employment share. The model compares a geographic pattern of responses. Since each outcome change is standardized within its own calculation, the displayed ratios do not measure shares of an employment change in common units. The comparison does not establish the quantitative response of the two-good main specification.

Table 8 reports the saved aggregate responses in percentage points and employment equivalents under full adjustment and fixed population. The comparison uses the same reporting set, shock, and parameterization in both columns; its units differ from the standardized exposure gradients above.

This comparison connects the model to the empirical combination of urban sectoral change and limited net population adjustment: lower food costs raise urban demand for modern output and lower the return to urban farming, so employment moves within existing cities. For city size the pattern reverses. The trade-only calculation holds population fixed by construction, and the migration-only calculation reproduces two thirds of the population gradient, with the interaction the remaining third (Figure 20b).

Table 8: Full and Fixed-Population Responses in Common Units

	Full Adjustment	Fixed Population
Agricultural employment share (percentage points)	-0.017	-0.014
Agricultural employment equivalents	-6,512	-1,480
Non-agricultural employment equivalents	-4,359	+1,480
Population (persons)	-30,400	+0

Notes: Services-basket partial-tradability calculations with the same shock and parameters in both columns. The shock raises the land-augmenting productivity parameter by 5.127%, equivalent to 2.532% more agricultural output at fixed inputs. The reporting set is 110 cities with complete archived counts; all 118 cities remain in the model. These are pooled changes, distinct from the exposure slopes in Figure 20. The share denominator includes agriculture, manufacturing, and services. Employment equivalents use the fixed observed ratio of employment in those sectors to population, 0.3576. The historical benchmark uses a different reporting sample and all industry-reported employment in its denominator. These conditional calculations do not estimate sampling uncertainty or participation responses. Source: the saved common-unit responses of the retained model.

Historical magnitudes. Between 1980 and 2010, the aggregate agricultural employment share fell from 48.27% to 29.92% in the 134 main-sample cities, a change of -18.35 percentage points. Agricultural employment rose by 1,350,646 workers over the same period.[25]

Regressing the change in the agricultural share on exposure with the main specification's controls gives -5.13 percentage points per standard deviation of exposure, with a 95% interval of [-7.23, -3.04]. A lower-match comparison illustrates its scale. Suppose each city's exposure above the full main sample's tenth percentile is reduced to that reference, with lower exposures held fixed. Applying the estimated slope and holding observed 2010 employment weights fixed gives a difference of -5.54 percentage points between the observed and lower-match profiles, equal to 30.20% of the observed aggregate decline. The coefficient-based 95% interval is [-7.81, -3.28] percentage points. This reference is an observed exposure level; it has not been established as an absence of match-related productivity gains. Shared village catchments also mean this city-level comparison need not correspond to a feasible reassignment of settlers.

Historical attribution requires the match-related productivity component at both endpoints and its effect after prices, wages, and employment weights adjust. A persistent matching advantage can have different urban effects as other economic conditions change. Cross-city exposure differences do not identify an effect common to all cities. The magnitude comparison therefore does not identify the share of the historical decline caused by agricultural similarity. A model calculation of that share requires productivity mappings and the corresponding counterfactual at both endpoints, with settlement placements and initial settler counts held fixed.

Table 9 compares the observed historical decline with the lower-match exposure benchmark.

[25]The denominator includes all workers with a reported industry, and the aggregate share divides total agricultural employment by total employment. Counts follow the main analysis's fixed city membership, historical crosswalk, and allocation weights. Some later-created member codes lack a separate historical observation because their predecessor population is assigned to another member. A sensitivity requiring every member code to be observed retains 121 cities and gives a change of -18.68 percentage points; label completeness does not establish more accurate historical boundaries.

Table 9: Observed Structural Change and the Lower-Match Benchmark

	Observed Change	Lower-Match Benchmark
Comparison	2010 minus 1980	Observed minus lower-match exposure
Share change or difference in changes (percentage points)	−18.35	−5.54
95% coefficient-based interval	—	[−7.81, −3.28]
Ratio to the observed decline	Reference denominator	30.20%
Employment weights	Observed at each date	Observed 2010 weights held fixed

Notes: Both columns describe the 134 main-sample cities. The observed share divides total agricultural employment by total industry-reported employment. The benchmark lowers exposure above the full main sample's tenth percentile to that reference and preserves exposure below it. It is the predicted difference in share changes, observed exposure minus the lower-match profile. The ratio divides this difference by the signed observed decline. The displayed interval for the difference in share changes propagates the change regression's Conley uncertainty conditional on the reference, exposure profile, employment weights, and observed denominator. The dash indicates a descriptive historical change without a reported coefficient-based interval. This magnitude comparison does not identify the historical contribution of agricultural similarity or a feasible reassignment of settlers. Source: the saved census and exposure calculations underlying this subsection.

7 Conclusion

Governments and development organizations spend large amounts of money on programs that raise agricultural productivity and reduce rural poverty. Most existing research tells us how agricultural productivity within a region affects outcomes in that same region; we know much less about how it affects cities outside those areas. Investing in rural areas could create incentives for people to stay there, or it could help cities by producing enough food to supply a growing urban population. I use a population resettlement program in Indonesia that generated as-if random variation in productivity across villages. Cities surrounded by more productive villages barely grow in population, but they change the types of jobs people have, shifting from agricultural to non-agricultural jobs in manufacturing and services. The policy only targeted villages; the cities around them restructured their employment anyway.

Rural productivity transforms cities through trade, not migration. Urban residents, when surrounded by more productive villages, spend relatively less on food, without lower consumption, and more on non-agricultural items. I do not observe trade flows directly, so I compare cities more and less likely to rely on local trade. The modern-sector response concentrates in cities weakly connected to outside food markets and is much smaller in cities better connected to the Thai and Vietnamese rice hubs. Migration exists, but it is small: there is no distinct migration response among the settlers by village productivity, and the new arrivals cannot account for the shift. In the data I cannot randomly assign some cities to trade while preventing people from migrating. In the partially tradable numerical comparison, holding population fixed preserves nearly the full standardized exposure gradient of the agricultural employment share. This comparison illustrates sectoral adjustment without population movement; its numerical ratios refer to that variant.

The classical theories of structural transformation treat the country as one economy, in which the food the countryside supplies is never a local constraint; here the constraint is local, and the gain reaches the city through goods. The evidence that food availability constrains the shift out of agriculture has mostly been cross-country; this natural experiment shows it causally within one country. When governments invest in villages, they rarely consider the consequences for cities; I show these are interconnected. As governments think about reducing barriers to growth and structural transformation, trade barriers and migration barriers may affect urban growth very differently. When cities lack sufficient access to food, investments in rural areas have more direct impacts on the structure of urban economies.

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Appendix

Contents

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A Data and Sample

This part documents the sources behind every number in the paper and the rule that defines the estimation sample.

A.1 Urban Sample Construction

Cited in the main text. This section: Section 2.2, p. 7; Section 3.2, p. 12. Figure A1a: Section 2.1, p. 6. Figure A1b: Section 3.3, p. 16. Figure A2a: Section 2.2, p. 8. Table A1: Section 3.3, p. 16. Table A2: Section 2.2, p. 8; Section 5.1, p. 26.

Single inclusion rule. The urban sample is defined only by Urban Center overlap. A subdistrict enters the urban sample if it overlaps a GHS Urban Center polygon; overlapping subdistricts are aggregated into one analysis unit per urban center; and the objective reclassification described in Section 2.2.1 re-attaches dense administratively urban edge subdistricts and adds the new nodes, giving the 187-center sample, every center of which has a travel time to a program village. The Urban Center layer itself follows one global rule: on a one-kilometer population and built-up grid, contiguous high-density cells (at least 1,500 inhabitants per square kilometer) forming clusters of at least 50,000 people are classified as urban centers, and the polygon layer collects these clusters (Dijkstra et al. 2021), so a single satellite-based rule identifies cities consistently across the world.

Aggregation to the urban center. To identify what an urban center is, I combine data at three levels. The outcome data are available at the village and urban-neighborhood level (administrative level 4), the finest spatial units available; these comprise the subdistricts (administrative level 3) that the 1980 baseline data cover. Since an important part of the identification is to control for baseline levels, I need to combine the baseline and endline units in a comparable way, so the finest level the data afford is the level-3 subdistrict. The household surveys are at most available at the subdistrict level as well. Multiple contiguous subdistricts overlap with the same urban area, and since the analysis looks at the market overall, with exposure based on distance around the city, there is little meaningful variation across subdistricts within a city. I therefore dissolve contiguous subdistricts into one urban-center area, the unit that corresponds to the meaningful urban labor market.

The 2010 totals each city is built from. A city's 2010 population and employment are the sums of its member year-2000 subdistricts, and those subdistrict totals rely on the village-level link described in the Online Supplement. Under the boundary overlay, 203 outer-island subdistricts recorded a 2010 population under half or over two and a half times their year-2000 population, against a median move of 1.15; under the administrative link that falls to 54, of which 38 are in Aceh, in northern Sumatra, where the year-2000 census itself is short.[26]

Table A1: The Urban Sample, From the Satellite Layer to the Main City Sample

	Urban centers
Urban Center polygons on the Outer Islands (GHS layer)	182
Less a census twin, two polygons on one subdistrict	−1
Satellite-defined urban centers	181
Plus centers the reclassification adds as standalone units	+6
Analysis universe	187
Less centers more than 300 travel minutes from a transmigration village	−49
Within the 300-minute window	138
Less centers without a 1980 baseline the census recorded directly	−4
Main city sample	134

Notes: Counts of outer-island urban centers at each step of the sample construction. The Urban Center polygons are the GHS Urban Center Database layer; the census twin is two polygons that fall on a single subdistrict in the census geography and are treated as one unit. The reclassification adds two kinds of standalone center: dense administrative centers that fall below the satellite population threshold, and rural remainders of administrative cities (kota) whose 1980 urban population share exceeds one half. It also re-attaches 42 dense edge subdistricts to 22 existing centers, which leaves the count unchanged. The window keeps the centers at most 300 travel minutes from a transmigration village. The main city sample keeps the in-window centers whose three 1980 baselines, log population, log employment and the urban-population share, the census records directly; the text of this subsection gives the reason each of the other four is left out.

From the satellite layer to the analysis sample. The GHS layer has 182 Urban Center polygons on the Outer Islands, and one of them leaves before the analysis, a census twin: two polygons fall on a single subdistrict in the census geography, so they are treated as one unit throughout. That leaves 181 unique satellite-defined centers. A year-2000-population size floor had separately removed one Aceh city on Sumatra, Bireuen: its year-2000 census count of 357 is a known SP2000 enumeration shortfall, against a measured 78,876 people in 1980 and 78,997 in 2010. No size, inclusion, or outlier cutoff reads a year-2000 quantity, so Bireuen is in the satellite-defined count above. The reclassification rule (1980 population density above 500 persons per square kilometer with a 1980 agricultural employment share below one half) adds six dense administrative centers as standalone units, for 187 urban centers, and merges 42 dense urban edge subdistricts, drawn from 25 districts, into 22 parent rows. Figure A2a shows the rule in the data. One center, in the Sangihe island group off North Sulawesi, has no transmigration village within reach of its island group. Its travel time to the nearest program village, 1,120 minutes, is recovered from the all-subdistrict travel-time matrix rather than the urban-center matrix, and places it far outside any window used here. The analysis universe is therefore all 187 centers. Two restrictions then define the estimation sample, and they are separate. The 300-minute travel-time window leaves 138 centers, of which 134 have a 1980 baseline the census recorded directly, which is the estimation sample (Table A1). A center enters only if all three of the 1980 baselines that define the rule are observed, log population, log employment, and

[26] The 2000 census enumerated only 40.3% of Aceh's villages, so a 2000-to-2010 comparison for Aceh cannot be repaired by any subdistrict link. The paper uses the year-2000 census as Bazzi et al. (2016) do, for the settled villages: the village agricultural employment share of Figure 14 and the settler stay-share of Figure 12. Its subdistrict map and its village list are the geography every city total is built on, and its counts serve here only as a check on that link. Every city-level quantity taken from the population census comes from its 1980 and 2010 rounds.

the urban-population share, and a partial baseline does not qualify it. The main specification conditions on a fourth 1980 baseline, log agricultural employment, which is a control and not part of the inclusion rule; it is missing in the stored file for two in-sample centers whose 1980 agricultural employment level is measured, and I rebuild it from that level. This is an inclusion rule rather than an incidental data gap: the specification conditions each 2010 outcome on the pre-program level of the same quantity, which a center without a 1980 measurement cannot support, so the requirement is not relaxed on any other ground. Four in-window centers are excluded, and for two distinct reasons. Three, in Bengkulu on Sumatra, West Kalimantan and Central Kalimantan, have no 1980 census record at all, and their baselines cannot be recovered from the earlier village censuses, which count households by activity and do not count workers. The fourth, a reclassification node in West Sumatra, has a 1980 baseline supplied by the apportioned-parent fill, and a filled baseline does not qualify a center either. No Aceh center is excluded on this ground: the extension to the 1980 crosswalk described in Section 3.1 recovered them. Of the six added units, three are dense administrative centers that fall below the satellite population threshold, and a second, equally pre-treatment screen applies the same logic to the rural remainders of administrative cities (*kota*) whose 1980 urban population share exceeds one half: three enter as their own units (Palangka Raya on Kalimantan, Sawahlunto on Sumatra, and Batam off Sumatra), two re-attach to their city's existing urban center (Palu on Sulawesi and Samarinda on Kalimantan), and one is dropped because its kota is fully urban, so the node would duplicate the city itself (Pangkalpinang).

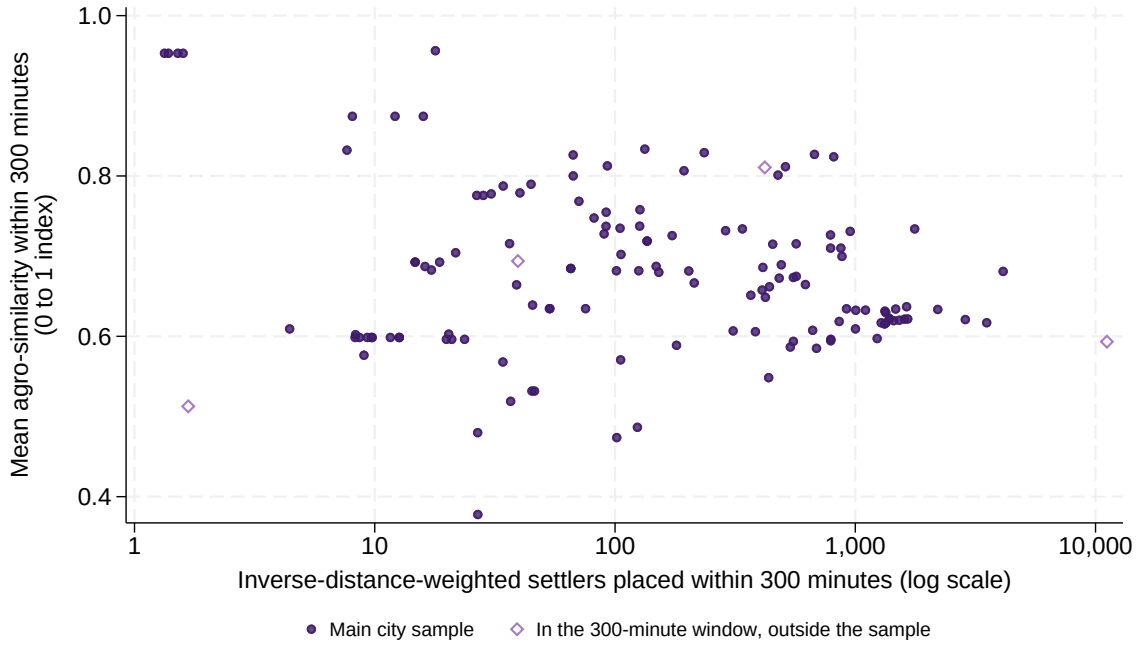
Distance to the nearest transmigration village, and the 300-minute cutoff. The 300-minute cutoff falls at the 73.8th percentile of travel time to the nearest transmigration village across the 187 centers, keeping the closer three-quarters of the sample. The cutoff is not a natural break in this distribution; it is chosen as a meaningful, achievable travel distance. The median center is 145 minutes from its nearest program village (Figure A1b).

Robustness to the sample construction. A reader may reasonably ask whether the reclassification accounts for the result, or whether the estimates rest on centers that were barely urban to begin with. Figure A2b re-estimates the main effects on subsamples that answer both questions: first dropping every center the reclassification touched, then imposing a 1980 population-density floor, a 1980 agricultural-share ceiling, and finally both. The two composition responses, the agricultural and modern employment shares, hold their sign, magnitude and significance in every one of them, and each estimate falls inside the interval of the headline it is being compared against. The agricultural-share screen is the demanding one, since it conditions on the 1980 level of the agricultural share itself and so works against finding a response in that share, and the response survives it.

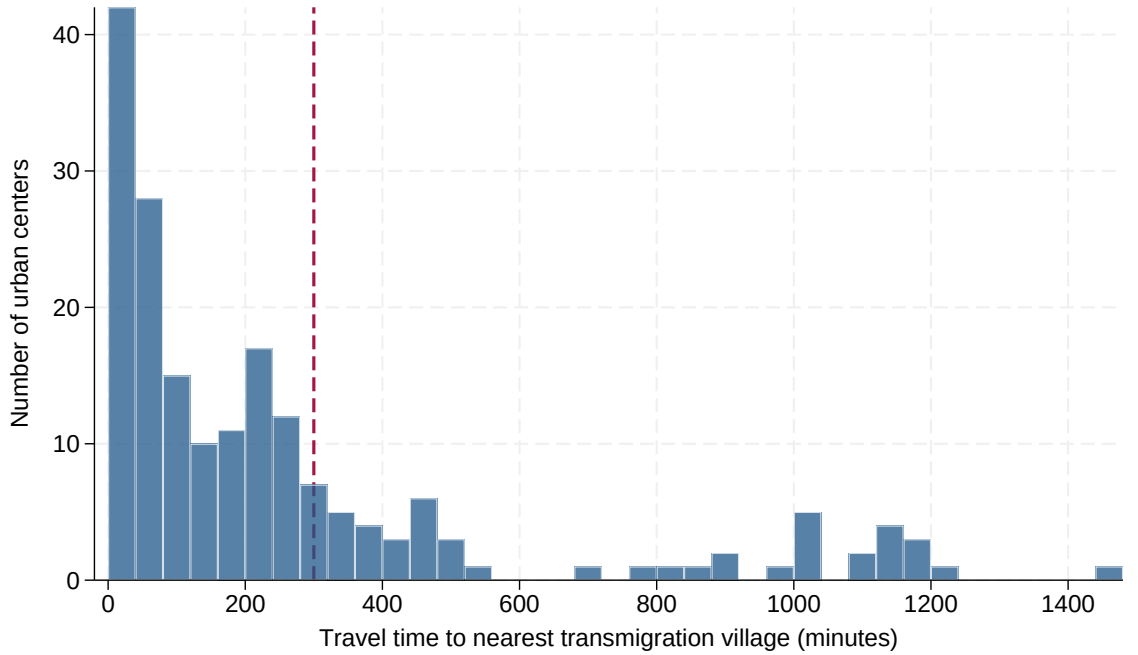
The same comparison, with the coefficient and its standard error for every outcome, is also given as a table for each of the three samples: the headline, the sample with every reclassified center removed, and the denser, less agricultural centers within that sample (Table A2).

Figure A1: Settler Placement and the Travel Time From Each City to the Nearest Program Village

(a) Agro-Similarity Against the Settler Count, Cities Within 300 Minutes



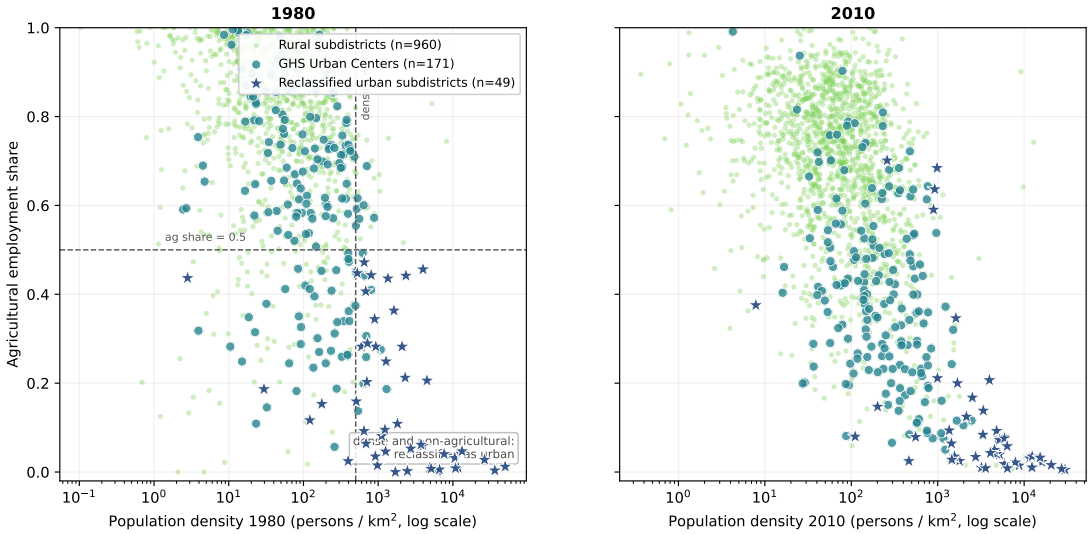
(b) Urban Centers by Travel Time to the Nearest Transmigration Village



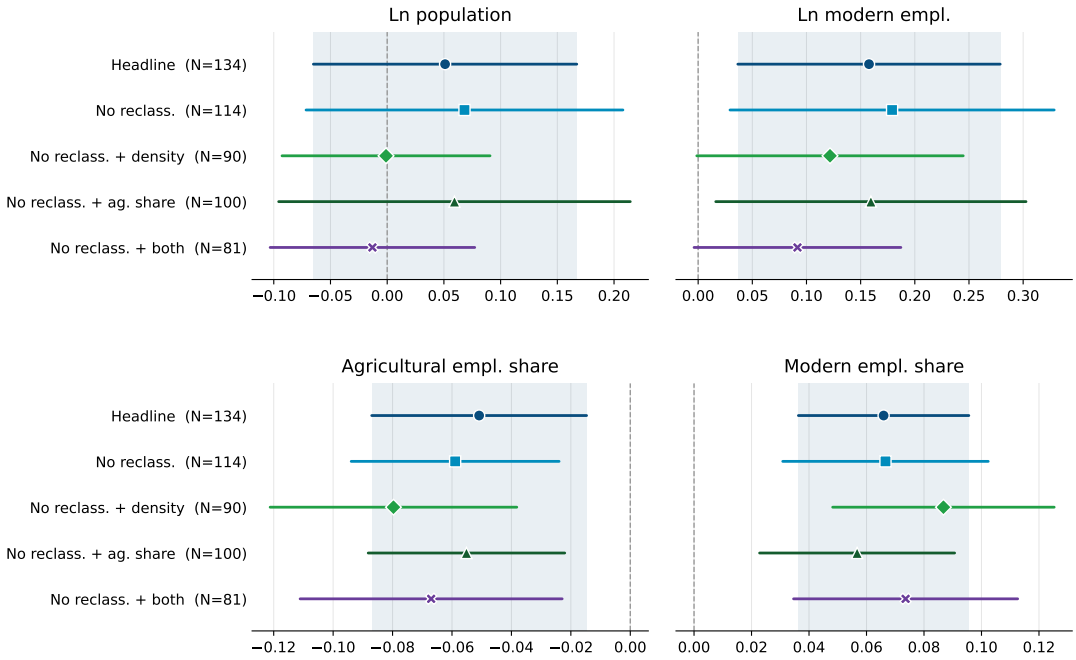
Notes: Panel (a): the 138 urban centers within 300 travel minutes of a transmigration village, with the main city sample (those with a 1980 census baseline) drawn as solid circles and the four in-window centers it excludes as hollow diamonds. The vertical axis is agro-similarity, the mean agro-climatic similarity of the transmigration villages within 300 minutes on its 0 to 1 index scale, and the horizontal axis is the number of settlers placed in those villages, weighted by inverse travel time, on a log scale. Panel (b): the 187-center analysis universe by travel time in minutes to the nearest transmigration village, in 40-minute bins. The dashed line marks the 300-minute cutoff, which keeps 138 of the 187 centers, the 73.8th percentile of this distribution; the median center is 145 minutes away.

Figure A2: The Reclassification Rule in the Data and the Results Under Tighter Urban Samples

(a) Reclassification Diagnostics: Density and Agricultural Employment Share



(b) Main Effects Under Tighter Urban-Sample Restrictions



Notes: Panel (a): each point is a place, positioned by population density (horizontal axis, log scale) and agricultural employment share (vertical axis) in 1980 (left) and 2010 (right), the two observables the reclassification rule uses; small circles are rural subdistricts, larger circles the satellite-defined urban centers, and stars the reclassified urban subdistricts. The dashed lines on the left mark the rule's two criteria in 1980, density above 500 persons per square kilometer and an agricultural employment share below 0.5, and the stars cluster in the dense, non-agricultural quadrant. Panel (b): each marker is the coefficient on agro-similarity, standardized once so that every row shares one scale, from one regression of the 2010 outcome named above it, in the main specification with the full set of controls and Conley SE within 200km; bars are 95% confidence intervals from the same standard errors. The top row, repeated as the shaded band, is the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline); the four restricted samples drop the centers the reclassification created or altered, and three of them add a 1980 population-density floor of 25 residents per square kilometer, a 1980 agricultural employment-share ceiling of 80%, or both.

Table A2: Main Estimates Under Tighter Urban-Sample Definitions

	Population + Employment			Agric Empl		Modern Empl	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	ln pop	ln emp	unemp rate	ln L agric	agric %	ln L modern	modern %
Panel A: the main city sample							
Agro-similarity	0.051	0.024	0.008	-0.166*	-0.051***	0.158**	0.066***
	(0.059)	(0.057)	(0.007)	(0.093)	(0.018)	(0.062)	(0.015)
Observations	134	134	134	134	134	134	134
Panel B: excluding every area the reclassification created or enlarged							
Agro-similarity	0.068	0.036	0.012	-0.152	-0.059***	0.179**	0.067***
	(0.071)	(0.067)	(0.008)	(0.093)	(0.018)	(0.076)	(0.018)
Observations	114	114	114	114	114	114	114
Panel C: Panel B further restricted to at least 25 residents per km^2 and at most 80% agricultural employment, both in 1980							
Agro-similarity	-0.013	-0.052	0.012	-0.304***	-0.067***	0.092*	0.074***
	(0.046)	(0.048)	(0.008)	(0.099)	(0.023)	(0.049)	(0.020)
Observations	81	81	81	81	81	81	81
Settler count	✓	✓	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agro-climatic controls	✓	✓	✓	✓	✓	✓	✓

Notes: Each cell is the coefficient on standardized agro-similarity from one regression of the 2010 census outcome in its column, in the main specification with the full set of controls and Conley SE within 200km. Standard errors are in parentheses. Agro-similarity is standardized once over every urban center with a measured exposure, so the panels share one scale. Panel A is the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline); Panel B drops every center the reclassification created or enlarged; Panel C keeps the Panel B centers with at least 25 residents per square kilometer and at most 80% of employment in agriculture in 1980. Columns 1 to 3 are population, employment and unemployment, columns 4 and 5 agriculture, and columns 6 and 7 the modern sector (manufacturing plus services); the four sector columns are shaded. In the labels: ln (the natural log), L (a count of workers), emp and Empl (employment), unemp rate (the unemployment rate), and agric % and modern % (the two sector shares); Baseline Pop 1980, Total Empl 1980 and Agri Empl 1980 are the logs of 1980 population, total employment and agricultural employment, and Urban share 1980 is the urban share of the 1980 population. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2 Transmigration-Village and Sample-Group Descriptives

Cited in the main text. Table A3: Section 2.2, p. 8. Table A4: Section 2.2, p. 8. Table A5: Section 2.2, p. 8. Table A6: Section 3.2, p. 13.

The transmigration villages whose agro-climatic similarity to the settlers' origin regions generates the city-level agro-similarity exposure are small agricultural settlements (mean 1,885 settlers placed at founding, 69% of employment in agriculture) established between 1979 and 1988. The program-wide total of more than two million resettled people cited in Section 2 is the official aggregate, which includes placements outside this village census. The village sample underlying the exposure measure records approximately 1.5 million placed individuals, matching the annual placements plotted in the Online Supplement. The summary statistics cover all 814 program villages, with the village-population rows available for the 716 villages with digitized polygon footprints (Table A3).

Table A3: Summary Statistics: The Transmigration Villages

Panel A: Levels (settlers placed, village population)

	Mean	SD	Median	Min	Max	N
Settlers placed (count)	1,885	1,011	1,792	136	8,852	814
Village population 1980	1,279	6,057	282	0	130,914	716
Village population 2010	2,235	5,228	817	0	85,296	716

Panel B: Economic structure, similarity, and agro-climatic covariates

	Mean	SD	Median	Min	Max	N
Agro-climatic similarity to migrant origins	0.667	0.136	0.684	0.000	1.000	814
Agricultural employment share	0.689	0.241	0.758	0.009	0.993	814
Urban village (share)	0.029	0.169	0.000	0.000	1.000	814
Years of schooling	3.998	0.899	3.978	0.713	7.653	814
Agricultural productivity (normalized)	0.996	2.647	0.435	0.001	27.316	770
Ethnolinguistic fractionalization	0.428	0.202	0.440	0.021	0.889	814
Ruggedness	0.311	0.129	0.310	0.006	0.691	814
Distance to coast (index)	10.553	1.138	10.844	5.795	11.925	814
Soil organic carbon	4.765	6.214	1.685	0.584	25.065	814
Soil pH	5.069	0.435	4.950	4.256	7.790	814
Rainfall, 1948–78	224.998	35.188	227.616	90.999	331.816	814
Temperature, 1948–78	26.253	1.709	26.748	17.225	28.174	814

Notes: Unit of observation is the transmigration village: the 814 villages from which the agro-similarity exposure of the urban centers is built, settled between 1979 and 1988 (mean 1983). Columns are the mean, standard deviation, median, minimum, maximum and number of villages with a value. Village population in 1980 and 2010 is estimated from the Global Human Settlement population grid for the 716 villages with digitized settlement footprints, and normalized agricultural productivity is available for 770 villages. Agricultural productivity is the unitless yield index of Bazzi et al. (2016), equal to one for a village that farms every crop at the benchmark yield of that crop. Similarity to the origin regions of the settlers is an index from 0 to 1, soil organic carbon is a percentage and temperature is in degrees Celsius. Settler counts, placement years, agro-climatic similarity and the covariates come from the transmigration program records and the construction of Bazzi et al. (2016). N is the number of villages with a value; the rainfall and temperature rows are averages over 1948 to 1978, and the urban village row is the share of villages classified as urban.

The three sample groups entering the structural-transformation regressions differ on observable characteristics: cities are larger and far less agricultural than rural subdistricts, while the agro-similarity exposure is balanced across all three groups, consistent with the quasi-random settler allocation (Table A4).

The main city sample and the other outer-island urban centers. Table A5 sets the main city sample beside the urban centers outside it, on the 1980 baselines and the 2010 outcomes of Table 1.

Table A4: Descriptive Statistics by Location Type: Cities and Rural Subdistricts With and Without a Settlement

	Cities		Rural, no TMV		Rural, with TMV		All	
	mean	sd	mean	sd	mean	sd	mean	sd
Population 1980	118,652	(220,180)	25,601	(18,604)	17,949	(13,326)	37,552	(91,647)
Population 2010	229,366	(404,467)	36,812	(24,079)	52,180	(35,899)	68,015	(169,605)
Employed 2010	90,259	(152,082)	16,101	(10,894)	22,864	(15,625)	28,293	(64,284)
Agric employment share 2010	0.374	(0.183)	0.656	(0.168)	0.694	(0.130)	0.623	(0.193)
Manuf employment share 2010	0.047	(0.042)	0.032	(0.038)	0.025	(0.025)	0.032	(0.037)
Services employment share 2010	0.470	(0.145)	0.253	(0.120)	0.218	(0.085)	0.277	(0.143)
Urban population share 1980	0.260	(0.228)	0.027	(0.102)	0.011	(0.052)	0.058	(0.148)
Agro-similarity (raw)	0.680	(0.103)	0.677	(0.098)	0.675	(0.107)	0.677	(0.101)
Travel time to nearest TMV (min)	109	(86)	162	(79)	1	(0)	121	(96)
Observations	134		596		191		921	

Notes: Columns are the samples of the structural-transformation regressions by location type: cities (the main city sample: outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline), rural subdistricts without a settlement, rural subdistricts with a settlement (one of the 814 transmigration villages), and all three pooled. The rural subdistricts are in 2000 boundaries, lie outside every city and within 300 travel minutes of a transmigration village, and have a measured 1980 baseline and 2010 outcomes. Cells give the group mean with the standard deviation in parentheses; the final row gives the number of observations, and every row is measured on all of them. Employment counts those who worked or ran a business in the previous week, and employment shares divide each sector count by everyone with an industry recorded. Agro-similarity is the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, before standardization. Travel time is in minutes to the nearest transmigration village, set to one in a subdistrict with a settlement. The population and employment rows come from the 1980 and 2010 population censuses. In the labels, TMV is a transmigration village.

Coverage of the household survey by wave. The 2011 wave of the household survey (Susenas) includes the whole main city sample but follows Indonesia's 2004 rice import ban, which moved food prices nationwide for reasons unrelated to local agro-climatic conditions. The agricultural price in the main results therefore comes from a wave before the ban, and the 2002 wave also matches the timing of the 2003 village-census (PODES) production data. Of the two waves before the ban, the 1996 wave covers more cities and more rural subdistricts, with and without a settlement, than the 2002 wave (Table A6).

Price measures from household unit values. The price measures of Figure D4 are built from the 1996 household survey (Susenas). A household's unit value for a food item is the value of its consumption of the item, own production included, divided by the quantity; within each item the unit values are winsorized at the 5th and 99th percentiles within province and urban status, and the survey-weighted median over the city's households gives the city's price of the item. An index divides each item's city price by the item's median over the households of all the urban centers the survey covers, the national price, and aggregates the price relatives over the raw agricultural food items under those urban centers' pooled expenditure weights (Laspeyres), under the city's own expenditure weights (Paasche) and under equal weights (a geometric mean); the crops index takes the geometric mean over the rice, corn and cassava items and their processed forms. The rice unit value is the value of the rice consumed, the local and the higher-quality varieties with own production valued by BPS, over the kilograms consumed, each summed over the city's households with survey weights. As unit values, every measure reflects price and quality together.

Table A5: The Main City Sample Against the Other Outer-Island Urban Centers

	Main city sample	Other centers	Difference	SE	N sample	N other
1980 baselines						
Log population, 1980	11.093	11.139	-0.046	(0.205)	134	50
Log employment, 1980	9.846	9.861	-0.015	(0.200)	134	50
Log agricultural employment, 1980	9.249	9.280	-0.031	(0.204)	134	50
Urban population share, 1980	0.260	0.237	0.023	(0.032)	134	50
Agricultural employment share, 1980	0.560	0.548	0.012	(0.036)	134	50
2010 outcomes						
Log population, 2010	11.842	11.717	0.125	(0.168)	134	52
Log employment, 2010	10.933	10.779	0.154	(0.177)	134	52
Log agricultural employment, 2010	9.822	9.753	0.069	(0.223)	134	52
Log modern-sector employment, 2010	10.232	10.069	0.163	(0.171)	134	52
Agricultural employment share, 2010	0.374	0.406	-0.032	(0.035)	134	52
Modern-sector employment share, 2010	0.516	0.510	0.007	(0.030)	134	52
Unemployment rate, 2010	0.122	0.133	-0.012	(0.013)	134	52

Notes: Each row compares the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline) with the 53 other outer-island urban centers: 49 lie farther than 300 travel minutes from every transmigration village, and 4 lie inside that window and lack either a 1980 census baseline or a 2010 endline. The first two columns are group means; the third is the coefficient from a regression of the row variable on an indicator for the main city sample, with its Conley SE within 200km (uniform kernel) in parentheses; the last two count the centers in each group with a value for the row. Population and employment counts are in natural logs, and shares and the unemployment rate are fractions; the modern sector is manufacturing plus services. A Wald test that the five 1980 rows jointly predict membership in the main city sample has $p = 0.735$ under the uniform kernel, which weights every pair of centers inside the cutoff equally, and $p = 0.727$ under the Bartlett kernel, which weights a pair by one minus its distance over the cutoff. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Household Survey Coverage by Wave and Sample

		Rural subdistricts	
	Main city sample	Without a settlement	With a settlement
Universe	134	596	191
Sample, Susenas 1996	112	285	117
Sample, Susenas 2002	109	204	104
Sample, Susenas 2011	134	576	188

Notes: Counts of units. Universe: the units of each sample that hold agro-similarity and every control of the main specification. The main city sample is the outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline; the rural subdistricts with and without a settlement lie outside every city and within 300 travel minutes of a transmigration village. Sample: the units of the universe that a wave's surveyed households live in, pooled to the city from its own households and to the year-2000 subdistrict, on the wave's fullest outcome; a few outcomes cover fewer. The 2011 wave follows the 2004 ban on rice imports.

A.3 Cross-Country Food Consumption and Urbanization

Cited in the main text. This section: Section 2.2, p. 11.

Figure 5 plots national food expenditure shares against urban population shares over the years available for each country, between 1960 and 2025. The main food-share source is the United Nations National Accounts household final consumption expenditure by purpose (Table 3.2). I divide expenditure on food and non-alcoholic beverages by total

household consumption expenditure. The World Bank's World Development Indicators supply population and the urban population share, using each country's administrative definition of urban. Marker size represents population and darker shades indicate later years. Differences in national urban definitions and food-expenditure measurement limit comparisons of levels across countries.

The country set starts with the 20 most populous countries in the United Nations panel, ranked by the latest available population observation. Poland and Iraq are omitted because their urban shares change little over the period. China, the United States, Pakistan, Bangladesh, and South Africa are added from national statistical sources, giving 23 countries. Russia, which the United Nations source also lacks, is omitted because of price-measurement problems in its household data. Egypt, the Democratic Republic of the Congo, Viet Nam, Türkiye, and Thailand are absent because the United Nations source lacks their household-consumption breakdown.

Published household-survey rounds supply the United States (1984 to 2024), Pakistan (2001 to 2024), South Africa (2006 to 2023), Nigeria (1986 to 2019), and Bangladesh (1981 to 2022). The United States series uses the Bureau of Labor Statistics Consumer Expenditure Surveys: food at home is divided by the consumption subtotal, which excludes personal insurance, pensions, and cash contributions. Pakistan's Household Integrated Economic Survey series removes tobacco and ready-made food from earlier rounds using the expenditure components in the published reports; later rounds report the narrower food category directly. South Africa's published Income and Expenditure Survey and Living Conditions Survey rounds report food and non-alcoholic beverages relative to household consumption expenditure. Nigeria and Bangladesh retain their broader published food shares, which include meals away from home and tobacco. Nigeria's change from diary collection to a shorter recall period further limits comparisons over time.

China's series is the Engel coefficient published by its National Bureau of Statistics, covering 1978 to 2025. Before the national series begins in 2013, I combine the published urban and rural coefficients using urban and rural population weights. The urban and rural expenditure definitions differ, and the 2013 survey change introduces a break. China's food category includes tobacco, liquor, and meals away from home. Dashed paths and asterisks identify China, Nigeria, and Bangladesh because these broader definitions prevent direct comparison of their levels with the narrower food shares.

Indonesia (IDNS) uses the national household-survey series from Susenas for 1994 to 2017. I remove catering, alcohol, and tobacco using item-level consumption data available in 7 of the 20 observed years. The adjustment is the share of reported food spending retained after these exclusions: it is measured in 1996, 2002, 2011, and 2013 to 2016, interpolated linearly between those years, and held at the nearest measured value outside that interval. I multiply each year's broad food share by this retained share. I then add the mean difference between the United Nations and adjusted Susenas shares over their overlapping years, +6.3 percentage points, to the entire Susenas series. This constant adjustment aligns its level with the national-accounts source while preserving its changes in percentage points.

B Measures

This part sets out how each measure the main text uses is constructed.

B.1 Land-and-Sea Travel Times and Pairwise Distances

Cited in the main text. This section: Section 2.1, p. 6; Section 3.1, p. 12; Section 3.3, p. 15.

Cost distance measures the minimum accumulated travel time between two locations, allowing the fastest route to differ from the shortest geographic path. I construct these measures for the paper's mapped urban areas, subdistricts, and program villages before applying the estimation-sample restrictions.

Inherited land surface. I construct the pairwise travel measures used in this paper from the Malaria Atlas Project (MAP) 2015 land-travel friction surface, which assigns the time required to cross each location. The inherited surface

combines information on transport access, land cover, and terrain from global accessibility mapping (Weiss et al. 2018). The shipping speeds follow the construction described by Nelson (2008). My spatial construction starts with the land surface: I add the missing maritime connections, align the geographic layers, locate the paper's origins and destinations, and compute their pairwise cost distances. The published distance to the nearest city is a separate output of the global accessibility studies; the distances used here are calculated for this paper's locations.[27]

Maritime connections and geographic alignment. The land surface does not supply the shipping connections needed to travel between Indonesian islands. I add the commercial-shipping layer from Halpern et al. (2008), based on vessel observations during the 12 months beginning October 2004. I project the land and shipping layers into the same coordinate system for Southeast Asia, clip them to a common geographic extent, and align their grid cells. I express the land friction values in minutes per kilometer. For shipping, I set travel costs at 3 minutes per kilometer in cells with positive recorded intensity and 1.5 minutes per kilometer in major shipping corridors, the cells where the layer records a value of 3 or more, corresponding to assumed speeds of 20 and 40 kilometers per hour. I add the land and water friction values where both are present, including coastal cells, and set costs at 30 minutes per kilometer in cells without an observed component. This last value is an assumed cost of travel outside the mapped connections. The shipping observations describe vessel activity after settlement, and their coverage is incomplete.

Origins, destinations, and least-cost paths. I link the census subdistrict polygons, GHS urban-center boundaries, and transmigration-village locations to their administrative identifiers and project their coordinates onto the combined surface. I calculate distances to program villages separately for subdistricts and urban locations. For the urban calculations, I intersect each constituent subdistrict with the union of mapped urban footprints and use the centroid of that intersection when it lies inside the footprint, otherwise an interior point. Where the intersection is empty, the calculation retains the subdistrict centroid. For the surrounding-subdistrict calculations, origins are the subdistrict centroids; destinations are the transmigration-village locations matched to the replication data of Bazzi et al. (2016).

For each origin, the least-cost calculation accumulates travel costs across adjacent cells and selects the path with the smallest sum to each destination. The calculations use an approximately 1-kilometer grid and average the friction values of adjacent cells, with diagonal steps weighted by $\sqrt{2}$. They treat a horizontal or vertical step as one nominal kilometer.[28] I match the resulting pair records by subdistrict and village identifiers. For the revised urban endpoints, I calculate the change in least-cost travel time between the original and revised locations on the same surface and grid, calibrate it to the scale of the original calculation, and add that change to the pairwise times. This preserves the original travel-cost convention while changing the location represented by the urban endpoint. Pairs newly included by the endpoint change use the calibrated revised calculation, and I round the resulting values to whole minutes with a minimum of one minute. Travel time from a city to a village is the minimum across the city's constituent subdistrict locations, with membership updated to the city definition used in the analysis. For members without urban-origin pair records, I use the corresponding subdistrict-origin calculations.

Pairwise times and exposure. The pairwise calculations cover a broader set of locations than the estimation sample. The urban and subdistrict pair records cover journeys up to 2,000 travel minutes, a limit separate from the 300-minute catchment used to construct the main exposure measure. I separately calculate great-circle distances from geographic coordinates for the straight-line comparisons. I retain the original travel measure alongside the adjusted measure used in constructing exposure: the adjustments set travel time to one minute when more than 0.1% of a village's

[27] The land input agrees with the MAP 2015 surface; the VMap-based input of the earlier construction is not available. The earlier construction described by Nelson (2008) uses VMap0 roads, released in 1997 as an update to the Digital Chart of the World developed in 1992. The documentation of the land input identifies the MAP 2015 surface associated with Weiss et al. (2018), which uses later road sources. The available records therefore do not establish that the road component of the travel times is restricted to VMap0.

[28] The projected cells are approximately 1.09 kilometers wide. The calculation does not separately multiply its costs by that cell width. The reported travel-time measure retains this discretization convention.

polygon area overlaps the urban footprint and bound excessive travel times using the city-centroid-to-village great-circle distance at 2 kilometers per hour. The city's nearest-village eligibility test uses the original travel measure, while village catchment membership and inverse-time weights use the adjusted measure. The main exposure is the unweighted mean of village agro-similarity within the catchment; weighted variants are described in the main text.

Other geographic pairs. I also construct great-circle distances between urban-member subdistrict centroids and other subdistrict centroids in the same island group. For the spatial model, I construct pairwise land-and-sea travel costs among its urban and rural locations. The rural model locations aggregate subdistricts into the rural remainders of districts. These calculations have distinct geographic scopes from the urban-area-to-village and subdistrict-to-village travel measures described above.

Road timing and identification. Roads mapped closer to the settlement period limit the scope for subsequent investment responding to settlement or productivity gains to determine measured exposure. This advantage requires the earlier roads to be the ones represented in the inherited surface. It does not establish that those roads were unaffected by the program: roads mapped in the 1990s may already reflect investment during settlement. The later road sources and shipping observations prevent interpreting the combined surface as a transport network observed before the program. If transport responses affect catchment membership or nearest-village eligibility, holding those catchments fixed in the expected-exposure exercises does not remove the resulting selection.

Travel time and straight-line distance. Geographic connections matter for access around Tana Paser. Two transmigration villages inside the same 50-kilometer radius of the functional urban area have sharply different travel times (Figure B1). The northern village is labeled at about an hour, while the southwestern village is labeled at over five hours. This illustration uses travel time to the nearest functional urban area; the empirical distances between cities and villages use the urban-center definitions and pairwise construction described above. Figure B2 traces the least-cost route from three cities to their nearest transmigration village.

B.2 Trade-Openness Measure: Rice-Import Market Access

Cited in the main text. This section: Section 3.5, p. 17; Section 5.2, p. 31. Table B2: Section 5.2, p. 32. Table B1: Section 3.5, p. 17. Table B3: Section 5.2, p. 29.

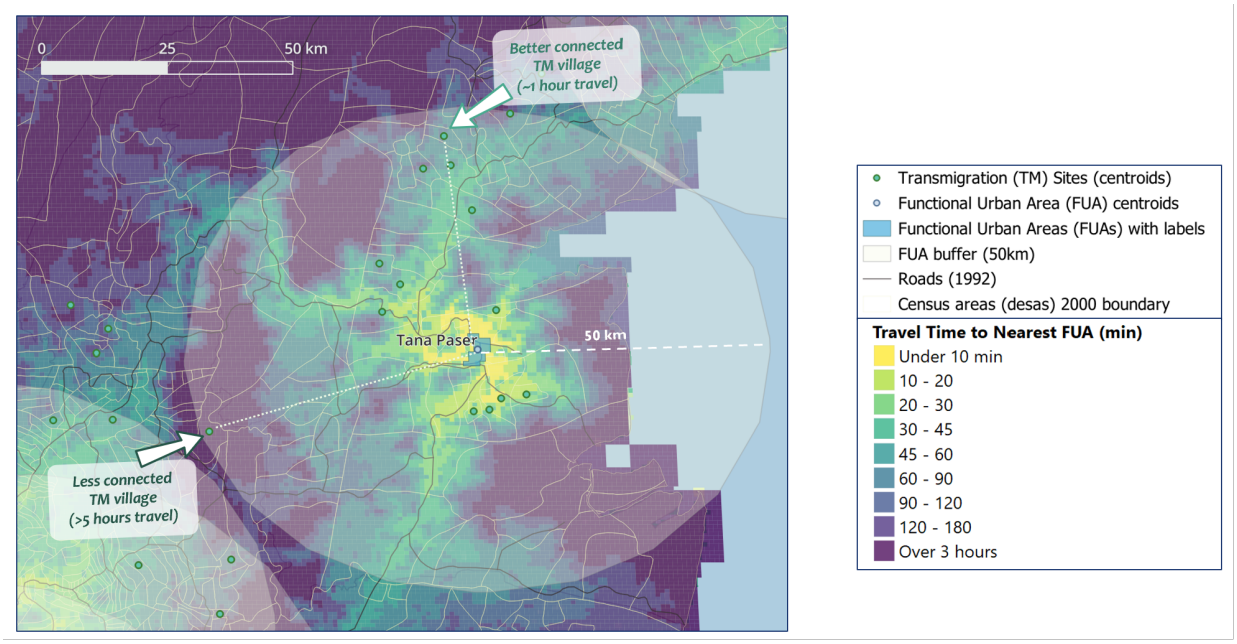
Source data. To measure how exposed each urban center is to imported food, I use administrative records of *rice imports by port of entry*: volumes (kg) and values disaggregated by importing port and origin for 2000-2010, with port coordinates (the "Ports & Prices" compilation). Because the traded commodity is rice, the resulting index is specifically a measure of *food*-trade openness, the form of openness most directly relevant to a local agricultural-productivity shock. Summing import throughput to the port level gives, for each port of entry p, a total volume kg_p and a location. The compilation stores port coordinates in degrees and minutes; taken as decimal degrees, each port falls 20 to 40 km from its berth, so every measure below uses the converted coordinates, which place the ports on the coast (median 2.8 km from the coastline against 13.0 km) and put the named terminals on their real berths.

Construction. Not every urban center is itself a port, so I separate *proximity* (being or abutting a port) from *trade intensity* (the volume of trade a city can reach), and compute three candidate metrics: (i) **proximity**, the great-circle distance $\texttt{dist_nport}_i$ from city i to its nearest customs port of entry (plus an indicator for a port within 50 km); (ii) **volume**, the import throughput of the reachable port(s); and (iii) a **gravity market-access** composite that combines the two,

$$\text{open}_i \;=\; \sum_p \frac{\text{kg}_p}{d_{ip}},$$

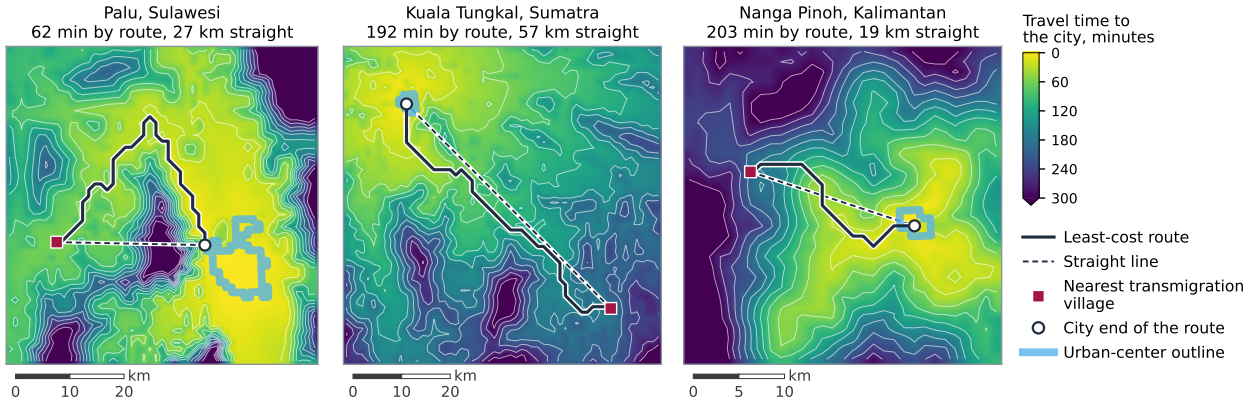
a distance-discounted sum of every port's import volume (decay parameter $\delta_{MA}= 1$; the nearest-port distance is floored at 10 km so that port-cities do not diverge). The gravity variable is $\ln \text{open}_i$, standardized over the urban centers inside

Figure B1: Straight-Line Distance and Travel-Time Connectivity Around Tana Paser



Notes: The background shows calculated travel time to the nearest functional urban area (FUA), Tana Paser in this region, in bins from under 10 minutes to over 3 hours. Filled dots identify transmigration-village locations, and the small ring identifies the urban-area centroid. The circle shows a 50-kilometer radius; dotted lines connect the two highlighted villages to the urban area. The 1992 roads are displayed for orientation and do not establish the road source underlying the travel-time surface. Village boundaries follow the 2000 census, and urban-area boundaries come from the Global Human Settlement Layer. Source: my geographic construction and travel-time calculations; transmigration-village locations from Bazzi et al. (2016). This is a descriptive map, with assumed travel costs and no statistical inference.

Figure B2: Least-Cost Routes From Three Cities to Their Nearest Transmigration Village



Notes: Each panel shows one city of the main city sample (Palu on Sulawesi, Kuala Tungkal on Sumatra, Nanga Pinoh on Kalimantan) and the transmigration village with the shortest travel time to it. The background is the travel time to the city in minutes, on the grid of the travel-time calculation, with white lines every 30 minutes. The solid line is the least-cost path between the village and the city's origin point, traced on the travel-speed friction surface; the dashed line is the straight line between the same two points. Panel titles give the travel time of the pair used in the paper and the great-circle length of the straight line. Source: my travel-time calculations; transmigration-village locations from Bazzi et al. (2016); urban-center boundaries from the Global Human Settlement Layer.

the headline cost-distance window ($N = 138$). It uses information from *all* reachable ports weighted by their size, where bare proximity collapses access to a single nearest-neighbor distance. Because it loads realized import volumes it is an

equilibrium outcome in part, so it enters in Panel B of Table 5 as a check on the sign of the bias that realized trade introduces; the preferred measure is route openness, constructed below.

The gravity and port-distance measures, in brief. The gravity measure loads realized trade. Administrative records of rice imports by port of entry over 2000-2010 give each port a total import volume, and the market-access index sums those volumes over every port, discounting by distance. Realized import volumes respond to the substitution this paper studies: a city whose hinterland becomes more productive buys more of its food there, its ports handle less imported rice, and the gravity measure classifies it as more closed. On the agricultural share the bias this builds into the split has a predicted sign. Closed cities are where the effect is large, so the gravity split is expected to overstate how much openness dampens the response, and it does so exactly where the route split finds no difference: on the agricultural employment share the openness offset is 0.053 (s.e. 0.021) on the gravity split against 0.021 (s.e. 0.025) on the route split. On the modern-sector outcomes the two splits agree in sign, and the gravity offsets are, if anything, slightly smaller: -0.034 against -0.040 on the modern employment share, both marginally significant, and -0.146 against -0.169 on log modern-sector employment, where only the route estimate is marginally significant. The gravity split therefore serves as a check on the sign of this bias, and the openness result relies on the route split. It is the alternative openness split in Panel B of Table 5, and it correlates at 0.60 with route openness. Rutz (1987) records a 1973 cargo turnover at the ports through which 93.5% of the 2000 to 2010 rice tonnage moved, six years before the main placement wave of 1979 to 1984. The historical maps date part of the port list; the domestic travel-time input uses later roads and shipping observations. The simplest measure is the distance to the nearest customs port of entry. Bare distance is nearly uninformative about reachable trade, because ports of entry are common (the median city sits 45 km from one) and the measure ignores what moves through them. It enters the balance tests of Section 4.3 as a trade-access covariate. The nearest port of entry is a 1973 main port for 81 of the 134 cities, and a harbor the Directorate General of Sea Communications already listed in 1986 for 118 of them (Table B1).

Table B1: Vintage of the Ports of Entry Behind the Two Openness Measures

	Route measure (route hours) cities	Balance covariate (great circle) cities	Rice imports (2000 to 2010) share of tons
1973 main port (Map 2)	88 (66%)	81 (60%)	93.5%
Other 1986 harbor (Map 1)	37 (28%)	37 (28%)	5.6%
On neither map	9 (7%)	16 (12%)	0.9%
In the 1986 inventory	125 (93%)	118 (88%)	99.1%
Cities	134	134	
Route measure rebuilt on a restricted anchor set			
1973 main ports only	same side of the median: 132 of 134		correlation 0.989
1986 inventory only	same side of the median: 134 of 134		correlation 0.998

Note: The sample is the main city sample. Each city's anchor is the nearest of the 183 customs ports of entry, by travel time over land and sea for the route measure and by great-circle distance for the port-distance balance covariate; the first two columns count cities by the category of that port, with the share of cities in parentheses, and the third gives the share of 2000 to 2010 rice-import tonnage through the ports of each category. A 1973 main port is the main port of one of 49 coastal regions at which Map 2 of Rutz (1987) draws a 1973 cargo circle; Map 1 prints the 237 harbors in their 1986 state. The main placement wave of the program ran 1979 to 1984. The lower panel rebuilds the route measure on the restricted anchor set and compares its below-median high-openness split with the full-list split. Each correlation is between the route measure rebuilt on the restricted ports and the full-list route measure across the cities.

Proximity versus the gravity composite. The gravity market-access index is negatively related to distance to the nearest customs port of entry, in the expected direction, but only weakly ($\rho = -0.15$): at a given distance there is wide variation in gravity access, because the composite also loads the *size* of the reachable ports and the contribution of

more distant ones, while bare proximity is nearly uninformative once every port of entry counts as a candidate port (the median city sits 45 km from one). Restricting the horizontal axis to the ports that recorded rice volume in 2000-2010 tightens it to $\rho = -0.54$, a tighter but conditioned comparison. Restricting to ports with realized rice volume conditions on the trade outcomes themselves, so the full-list figure is the one reported. Either way the conclusion is the same, and the full-list version shows it more clearly. The gravity measure is not a relabeling of raw proximity, and it contains independent trade-volume information. The alternative openness split in Panel B of Table 5 uses it for that reason.

Route proximity to the Thailand/Vietnam rice supply. The gravity composite loads realized import volumes, which are an equilibrium outcome. The preferred openness measure in Section 5.2 uses route proximity, following the "distance to Ho Chi Minh City/Bangkok (port)" construction in Bazzi (2017) (his original uses great-circle distance to the nearest port and averages the two hubs; I use least-cost travel time and the nearer hub): for each urban center, the route-aware least-cost travel time to its nearest customs port of entry (over the same land-and-sea friction surface used for every cost distance in the paper) plus the sea distance from that port onward to the nearer of Bangkok and Ho Chi Minh City, the hubs from which the bulk of Indonesian rice imports ship, converted to hours at the friction surface's shipping-lane speed so the two legs share one unit. Smaller totals mean closer alignment with the rice-import routes, hence higher measured openness; the high-openness indicator in Table 5 is a below-median split inside the estimation window. The international geography is fixed. The domestic leg uses the land-and-sea surface described in Section B.1, whose road and shipping inputs postdate settlement. The customs list has no vintage field, and the Rutz maps date the ports of 88 of the 134 cities to a main port with a recorded 1973 cargo turnover and 125 of them to the 1986 harbor inventory or earlier, leaving 9 that may post-date the program (Table B1). The international leg dominates the measure. Three properties matter in practice. First, the international leg dominates the composite (median 42 hours against 1.2 domestic), which adds the west-to-east proximity-to-the-routes gradient that nearest-port distance discards: route proximity correlates 0.60 with the gravity composite but only 0.18 with domestic-only nearest-port proximity, and the measure is insensitive to selecting the cheapest gateway rather than the nearest port (correlation 0.99). Second, the international leg is a great-circle approximation, which understates distances from western ports whose actual sea route rounds the Strait of Malacca. This compresses the open end of the gradient but preserves the ranking. Third, the customs-port set includes ports with and without recorded rice imports. Whether a given port appears in the 2000-2010 rice-import records is itself an equilibrium outcome, and one the mechanism of this paper could move, since a city that substitutes hinterland food for outside supply leaves its port less rice-import business. The port set is therefore the full customs port-of-entry list, every port with coordinates whether or not it ever recorded rice volume, which removes any dependence on realized import quantities. Restricting it to the 58 ports that did record rice volume changes little. The two sets place 134 of the 138 in-window cities on the same side of the openness median, and the four they disagree on split two in each direction. Restricting the port set instead to the 1973 main ports leaves 132 of the 134 on the same side of that median, at a correlation of 0.989 with the full-list measure. The high-openness offset in Table 5 goes from $+0.021$ to -0.006 on the agricultural share and from -0.040 to -0.021 on the modern share (standard errors 0.029 and 0.024 on the restricted set): the modern-share offset keeps its sign at about half its size and loses its significance at the 10% level, and the agricultural-share offset is near zero on both sets. The reason the two agree at all is the first property above, since the international leg accounts for 97% of the composite's cross-city variance and no port's rice-import status can move it. The historical-port comparison assesses sensitivity to port selection; it does not establish the timing of the domestic transport network.

The gravity measure as the split. The gravity market-access measure, in place of route proximity, gives a second openness split, above the median inside the estimation window. Built from realized rice-import volumes, the measure is in part an equilibrium outcome, so its estimates check the sign of the bias described above. They move in the same direction as the route split on both outcomes. On the agricultural share the offset is 0.053 ($p = 0.013$) against 0.021 ($p = 0.399$) on the route split, so the realized-trade measure finds a differential that the predetermined one does not, in the direction the substitution predicts. On modern employment the two agree, at -0.146 ($p = 0.229$) against -0.169 ($p = 0.074$) (Panel B of Table 5, whose Panel A is the route split).

Table B2: Agro-Similarity by Trade Openness: The Gravity Split Instrumented by the Route Split

	(1) ln pop	(2) ln emp	(3) unemp rate	(4) **ln L agric**	(5) **agric %**	(6) **ln L modern**	(7) **modern %**
Agro-similarity, more closed cities: β_1	**0.161**	**0.152**	**0.001**	**-0.163**	**-0.074****	**0.342****	**0.111*****
	(0.122)	(0.127)	(0.012)	(0.173)	(0.032)	(0.150)	(0.032)
Agro-similarity, more open cities: $\beta_1+\beta_2$	-0.051	-0.098	0.014	-0.161	-0.028	-0.020	0.021
	(0.090)	(0.108)	(0.013)	(0.139)	(0.029)	(0.112)	(0.022)
High openness (=1)	0.045	0.008	-0.008	0.100	0.010	-0.028	-0.029
	(0.116)	(0.153)	(0.018)	(0.271)	(0.052)	(0.161)	(0.039)
First-stage F (Kleibergen-Paap)	3.37	3.37	3.37	3.37	3.37	3.37	3.37
Outcome mean	11.842	10.933	0.122	9.822	0.374	10.232	0.516
Outcome SD	0.854	0.840	0.040	0.896	0.183	0.975	0.156
Observations	134	134	134	134	134	134	134

Notes: Two-stage least squares on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. The high-openness indicator of the gravity measure (rice-import market access discounted by travel time, above its median among the centers with a transmigration village within 300 travel minutes) and its interaction with standardized agro-similarity are instrumented by the high-openness indicator of the route measure (travel time to the Thai and Vietnamese rice hubs, below its median among the same centers) and its interaction with agro-similarity. Rows give the agro-similarity effect in the more closed and in the more open cities and the high-openness indicator; the outcomes are measured in 2010, with the four sector columns shaded. Standard errors are in parentheses. The Kleibergen-Paap F statistic of the first stage, under the same Conley variance, is below the Stock-Yogo 10% maximal-size critical value of 7.03 for two endogenous regressors and two instruments. Agro-similarity is standardized, and the beta terms are the coefficients of (5). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B3: 1996 Food Prices by Third of Route Distance to the Rice Hubs

	Cities			Rural, with settlement			Rural, without settlement		
	Agric. price	Staple crops	Rice u.v.	Agric. price	Staple crops	Rice u.v.	Agric. price	Staple crops	Rice u.v.
Agro-similarity, most open third	-0.098	0.003	-0.187	-0.105	-0.402**	-0.268	-0.393***	-0.121	-0.052
	(0.214)	(0.213)	(0.234)	(0.133)	(0.173)	(0.164)	(0.108)	(0.136)	(0.111)
Agro-similarity, middle third	-0.333	-0.005	-0.182	-0.113	-0.130	-0.173	-0.138	-0.072	-0.172
	(0.253)	(0.270)	(0.230)	(0.249)	(0.308)	(0.299)	(0.166)	(0.142)	(0.154)
Agro-similarity, most closed third	0.014	0.124	0.147	0.040	0.164	-0.057	-0.305*	-0.208	-0.100
	(0.199)	(0.274)	(0.296)	(0.247)	(0.265)	(0.247)	(0.162)	(0.236)	(0.204)
p-value, closed minus open	0.648	0.653	0.223	0.619	0.020	0.321	0.646	0.700	0.807
Observations	112	112	112	117	117	116	285	285	283

Notes: Each column is one regression of the 1996 price, standardized over the cities or over the pooled rural subdistricts, on agro-similarity interacted with indicators for the thirds of route distance to the Thai and Vietnamese rice hubs within the location type, the most open third being the nearest, in the main specification with the full set of controls and Conley SE within 200km in parentheses; for rural subdistricts with a settlement, the settler count is the number of settlers placed in the transmigration villages that overlap the subdistrict. Coefficients are in SD of the price per SD of agro-similarity. The p-value tests that the most closed and the most open third respond alike. The agricultural price is the raw-food index, the staple-crops price covers rice, corn and cassava, and the rice unit value is consumption value over consumption kilograms, all from the 1996 household survey (Susenas), before the 2004 rice import ban. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Continuous market access. Panel C of the main openness table replaces the high-openness indicator in Equation 5 with standardized rice-import market access:

$$y_c = \beta_1\, AgSim_c + \beta_2\, AgSim_c \times MA_c + \chi\, MA_c + \phi\, SettlerCount_c^{IDW} + C_{c,1980}\,\xi + X_c\,\zeta + \varepsilon_c \tag{16}$$

Here MA_c is market access in standard deviations, so β_1 is the exposure effect at mean market access and β_2 is its change per standard deviation of market access. Market access and its interaction with exposure are instrumented by standardized route proximity and its interaction with exposure:

$$\begin{aligned} MA_c &= \pi_1\, Route_c + \pi_2\, AgSim_c \times Route_c + \pi_3\, AgSim_c + W_c\,\kappa + u_c \\ AgSim_c \times MA_c &= \rho_1\, Route_c + \rho_2\, AgSim_c \times Route_c + \rho_3\, AgSim_c + W_c\,\lambda + v_c \end{aligned} \tag{17}$$

$Route_c$ rises with proximity to the rice hubs, and W_c includes the settler count, the 1980 baselines, and the agro-climatic controls of Equation 4. The exclusion restriction requires route proximity to affect urban outcomes and their response to exposure only through market access, conditional on these controls. The main table evaluates $\beta_1 + \beta_2 MA_c$ at the mean market access of the more closed and more open groups.

The instrumented median splits. The instrumented estimates of the main text use continuous market access and continuous route proximity. With the two median splits in their place, the gravity split of Panel B instrumented by the route split of Panel A, the rows are those of Panels A and B and can be compared one for one. The exposure effect on the modern employment share is 0.111 in the more closed cities and 0.021 in the more open ones (Table B2). The first stage of this specification is weak. Its Kleibergen-Paap F statistic is 3.37, below the critical value of 7.03, so the estimates lean toward the ordinary least squares estimates of Panel B and their standard errors understate the uncertainty. The Anderson-Rubin test stays valid with weak instruments, and it does not reject the hypothesis that both instrumented coefficients are zero at the 10% level in any of the seven columns.

Validation: the 2004 rice-import ban. The route measure has independent validation from the event it was designed around, Indonesia's 2004 rice-import ban. Post-ban (2002-2008) rice-price growth was faster in more route-open cities (+0.037 per standard deviation of proximity, marginally significant), the geography-driven price response documented by Bazzi (2017). The point estimate is the same on the rice-active port set, where the standard error is tighter and the response is statistically significant; the wider interval here comes from using the full customs list, and it is the interval I report. The same paper documents the pre-ban gradient with the sign reversed, an additional falsification the route measure inherits. The exposure treatment itself is orthogonal to that price shock: agro-similarity does not predict the ban-era price response (+0.006, statistically insignificant), and the route × agro-similarity interaction is small and insignificant (its negative sign is consistent with productive hinterlands buffering the forced-autarky price jump, but the cross-sectional variation cannot establish it). The route result is also stable across codings: with the continuous proximity in place of the high-openness dummy, the interaction is statistically significant on all four headline outcomes.

B.3 Standardized (z-on-z) Reduced-Form Estimates: Model Calibration Targets

Cited in the main text. Table B4: Section 6.2, p. 49.

The model's response slopes are read in standardized (z-on-z) units, with both the treatment and each outcome standardized, so that the slopes, which are unit-free, are compared with the data's in sign and relative size. Table B4 reports the reduced-form estimates in those units on the main city sample, with the Susenas- and PODES-based rows on the sample each source covers, reported per row. The elasticities are chosen to fit three of these slopes, for population, the agricultural employment share and modern employment (m_1 to m_3); at the chosen values the model matches all three in sign and within their 90% intervals on the main city sample (Section F.11). The remaining rows are untargeted.

Table B4: Reduced-Form Responses to Agro-Similarity in Standardized Units

	Main specification		*z*-on-*z*	
Outcome	Coef.	(SE)	response	N
Outcomes the calibration targets				
m_1 Log population	0.051	(0.059)	+0.060	134
m_2 Agricultural employment share	-0.051^{***}	(0.018)	−0.278	134
m_3 Log modern employment	0.158^{**}	(0.062)	+0.162	134
Untargeted outcomes				
Log agricultural employment	-0.166^{*}	(0.093)	−0.185	134
Modern employment share	0.066^{***}	(0.015)	+0.420	134
Housing rent	−0.011	(0.107)	−0.012	134
Agricultural (raw-food) price index	0.016	(0.022)	+0.133	134
Log wage	0.019	(0.032)	+0.062	134
Log real wage	0.003	(0.029)	+0.013	134
Household food expenditure share	-0.016^{***}	(0.006)	−0.313	134
Log rice production	0.735^{***}	(0.252)	+0.314	132

Notes: Each coefficient is from the main specification of Table 3, with the full set of controls and Conley SE within 200km, on the main city sample; the treatment is standardized agro-similarity. The *z*-on-*z* response divides each coefficient by the standard deviation of the outcome over its estimation rows, so it is the response in standard deviations of the outcome per standard deviation of agro-similarity; the model's response slopes are computed in the same form and are compared with these in sign and relative size. The structural estimate in Section 6.2 is fitted to the population, agricultural-employment-share and modern-employment slopes (m_1 to m_3) in this table, which it matches in sign and within their 90% intervals (Section F.11); the remaining rows are untargeted. The housing rent, price, wage and food-share rows are from Susenas 2011 and the rice-production row from the 2003 village census, each on the urban centers the source covers; N is the number of urban centers in each regression.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C Identification Support

This part collects the evidence that exposure is as good as randomly assigned once the controls are in place, together with the placebo and recentering tests that support it.

C.1 Design-Based Recentering and the Placebo Tests

Cited in the main text. This section: Section 4.3, p. 23; Section 5.1, p. 25. Table C1: Section 5.1, p. 25. Figure C1: Section 5.1, p. 25. Table C2: Section 4.3, p. 23. Table C3: Section 3.4, p. 17; Section 5.1, p. 26; Section 5.2, p. 34.

Design-based recentering. The exposure measure combines a quasi-random shock, the match between arriving settler cohorts and the sites that were ready to receive them, with components the program chose: where the sites sit, how many settlers each received, and their agro-climate. A city surrounded by transmigration sites with land more similar to the Java and Bali origin mix is bound to have higher exposure regardless of how settlers are reshuffled (Section 4.3). Borusyak & Hull (2023) show that the bias in such a measure runs through a single confounder, the exposure a city would expect under counterfactual assignments of cohorts to sites, and that recentering the treatment on that expected exposure, or controlling for it, restores identification.

I construct the expected exposure directly. The sites, their locations, when each was settled, how many settlers it received, and which sites lie within each city's travel-time catchment all stay where the program put them. Only the match moves: I reshuffle the village-level agro-climatic similarity across the sites of the same province and Repelita settlement wave, rebuild each city's unweighted 300-minute catchment mean under each of $S = 2{,}000$ reshuffles, and average them:

$$z_c = \frac{1}{|V_c|} \sum_{v \in V_c} AS_v, \qquad \mu_c = \frac{1}{S} \sum_{s=1}^{S} z_c^{(s)}, \tag{18}$$

where V_c is the set of program villages within 300 minutes of city c, AS_v is the agro-climatic similarity of village v, and $z_c^{(s)}$ is the city's exposure under draw s. I then either control for the expected exposure or subtract it from the treatment:

$$\begin{aligned} y_c &= \beta\, z_c + \gamma_P\, \mu_c^P + \gamma_{PW}\, \mu_c^{PW} + X_c'\zeta + \varepsilon_c, \\ y_c &= \tilde{\beta}\, \left(z_c - \mu_c^{PW}\right) + X_c'\zeta + \varepsilon_c, \end{aligned} \tag{19}$$

where X_c holds the main specification controls, μ_c^{PW} and μ_c^P are the expected exposures under reshuffles within province-by-wave cells and within provinces, and every treatment is standardized. Controlling for both expectations is the double-robust form of the correction, consistent if either the controls or the reshuffle model is correct. A second reshuffle holds each site's destination agro-climate fixed and re-draws only the settler-origin composition, using the origin-destination similarity grid of Bazzi et al. (2016). The control form estimates the coefficients on realized and expected exposure separately, whereas subtraction restricts them to be equal in magnitude and opposite in sign before standardization, and controlling for an expectation also removes its linear association with realized exposure, so the two sets of coefficient magnitudes cannot be compared as effects of the same one-standard-deviation change. The design-based interpretation requires the assignment assumptions stated in Section 4.3, which these regressions do not test.

The structural-transformation results hold under the double-robust form (Table C1). The agricultural employment share falls by 0.040 and the modern employment share rises by 0.052, both statistically significant, and log modern-sector employment rises by 0.112, marginally significant; each has the headline's sign at about three quarters of its size. Subtracting the expected exposure instead removes most of the variation in the treatment, because a city's realized and expected exposure move closely together: they correlate at 0.75 under the whole-match reshuffle and 0.96 under the origin re-draw. The recentered estimates are closer to zero, and only the modern employment share stays statistically significant under both (Figure C1).

The same 2,000 reshuffles also give an exact test of the double-robust estimate: I re-estimate the regression under each reshuffle and ask how often a placebo t-statistic lies at least as far from the placebo average as the observed one. The test rejects the hypothesis of no effect on the modern employment share ($p = 0.050$ with reshuffles within province and wave, and 0.052 within province) and does not reject it for the agricultural share ($p = 0.196$) or for log modern-sector employment ($p = 0.181$). The test compares t-statistics because, once its expectation and the controls are taken out, a reshuffled exposure has about half the residual standard deviation of the realized one, so a coefficient estimated on it is spread more widely than the observed coefficient; dividing each by its own standard error puts the observed and the placebo estimates on one scale. The Online Supplement reports every randomization-inference p-value, the coefficient form included.

Baseline placebo test, in levels. Agro-similarity across villages does not predict the baseline outcomes of the sample cities. I regress ten 1980 baseline outcomes, urban center-level aggregates of individual 1980 census records, on agro-similarity across villages. Two constructions of the exposure inside the 300-minute identification radius are reported, the headline unweighted mean and the population×inverse-distance weighting of the exposure equation, each over all 814 transmigration villages and again over the 687 settled after the count. The right-hand block of Table C2 also rebuilds every village-derived control over those 687 villages, so nothing entering the regression could have acted on a 1980 outcome. One of the forty cells rejects zero at the 10% level, where about four would under the null.

Table C1: Headline Estimates Controlling for the Expected Exposure, Double-Robust Form

	(1) ln pop	(2) ln emp	(3) unemp	(4) ln ag	(5) ag share	(6) ln modern	(7) modern share
Agro-similarity (double-robust)	0.035	0.024	-0.006	-0.112	-0.040**	0.112*	0.052***
	(0.077)	(0.074)	(0.006)	(0.121)	(0.020)	(0.063)	(0.017)
Settler count	✓	✓	✓	✓	✓	✓	✓
Base ln pop 1980	✓	✓	✓	✓	✓	✓	✓
Base ln empl 1980	✓	✓	✓	✓	✓	✓	✓
Base ln agri empl 1980	✓	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓	✓
Agro-climatic controls	✓	✓	✓	✓	✓	✓	✓
Observations	134	134	134	134	134	134	134

Notes: Each column is one regression of a 2010 population census outcome on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Check marks show the controls. Every column also controls for the expected exposure under two random reassignments of village agro-climatic similarity, one within province and one within province and five-year plan (1979 to 1983 or 1984 to 1988), 2,000 draws each and both standardized; this is the double-robust specification of Borusyak & Hull (2023), consistent if either the controls or the reassignment model is correct. Modern-sector employment is manufacturing and services; shares and the unemployment rate are fractions. Standard errors are in parentheses. Agro-similarity (double-robust) is standardized agro-similarity with both expected exposures controlled. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

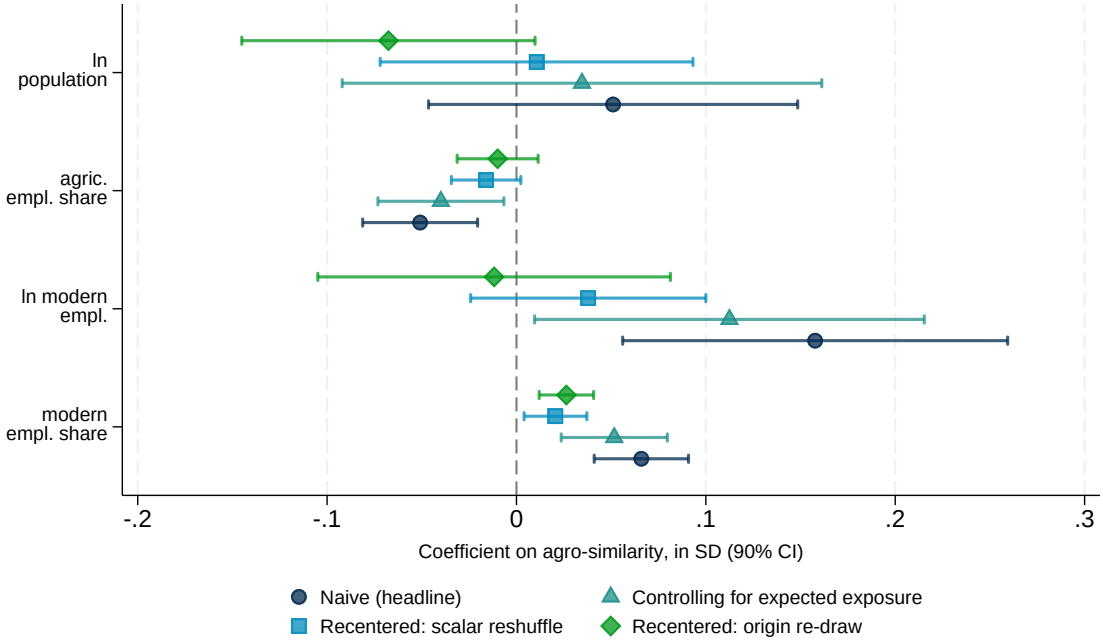
Controlling for the 1980 birthplace composition directly. The compositional headline does not depend on the 1980 birthplace composition of the population. With the 1980 birthplace shares added to the main specification, the coefficients on the agricultural and modern employment shares are unchanged at three decimals and remain statistically significant, and the 2010 birthplace pattern holds, with the locally born share still falling and the same-province share still rising (Table C3).

C.2 Village-Level Analysis

Cited in the main text. Table C4: Section 4.3, p. 22. Figure C2: Section 4.3, p. 22.

Across the transmigration villages, agricultural productivity rises with the agro-climatic similarity index, winsorized or raw, over the plotted range and over the full range. The productivity measure is the agricultural productivity index of Bazzi et al. (2016): the yield per hectare of each crop the village grows is divided by that crop's average yield across all Indonesian villages that grow it, and the index is the geometric mean of these ratios across crops, weighted by each crop's share of the village's crop revenue at national prices. A village whose crops all yield the national average scores one, and the median transmigration village scores 0.43. The index comes from the 2003 village census (PODES), which records the 2001-2002 growing season, and it is observed for 770 of the 814 program villages, while agro-similarity is defined for all 814. Table C4 reports this village-level relevance of agro-climatic similarity for agricultural productivity, which Section 4.3 cites.

Figure C1: Headline Estimates Under Design-Based Recentering



Notes: Rows are 2010 census outcomes; the modern sector is manufacturing plus services. Each marker is the coefficient on standardized agro-similarity, or on its recentered version, from one specification on the main city sample, in the main specification with the full set of controls and Conley SE within 200km; bars are 90% confidence intervals from the same standard errors. "Naive" is the headline specification. "Controlling for expected exposure" adds the expected exposure under the province and the province-by-settlement-wave reshuffles as controls (Table C1). "Scalar reshuffle" recenters the exposure against a permutation of the village match within province-by-settlement-wave cells. "Origin re-draw" recenters against a permutation that holds each site's destination agro-climate fixed and re-draws only the settler-origin composition, using the origin-destination similarity grid of Bazzi et al. (2016). Recentering subtracts from an exposure its average over 2,000 permutations and standardizes the recentered value, following Borusyak & Hull (2023). The Online Supplement reports the full recentered estimates.

Table C2: Placebo Baseline Test: 1980 Outcomes Under Two Exposure Constructions and Two Village Sets

	Outcome, main city sample			**All transmigration villages**		**Villages settled after the 1980 census**	
1980 baseline outcome	Mean	SD	*N*	Simple mean	Settler weighted	Simple mean	Settler weighted
ln population	11.093	0.99	134	-0.019	-0.022	+0.017	-0.109
				(0.122)	(0.166)	(0.154)	(0.165)
ln employment	9.846	0.96	134	-0.043	-0.042	+0.002	-0.118
				(0.107)	(0.151)	(0.140)	(0.154)
Unemployment rate	0.017	0.02	134	-0.001	-0.002	-0.002	-0.003*
				(0.002)	(0.002)	(0.002)	(0.002)
ln agricultural empl.	9.249	1.06	134	-0.129	-0.093	-0.109	-0.171
				(0.099)	(0.107)	(0.101)	(0.109)
Agricultural empl. share	0.560	0.22	134	-0.034	-0.021	-0.041	-0.020
				(0.033)	(0.032)	(0.026)	(0.028)
ln modern empl.	8.801	1.31	134	-0.050	-0.052	+0.002	-0.149
				(0.185)	(0.237)	(0.218)	(0.240)
Modern empl. share	0.383	0.20	134	+0.017	+0.007	+0.024	+0.005
				(0.031)	(0.031)	(0.027)	(0.029)
ln born same province	10.934	1.00	134	-0.001	-0.008	+0.034	-0.084
				(0.132)	(0.175)	(0.164)	(0.172)
ln born other province	8.342	1.88	134	-0.138	-0.125	-0.102	-0.266
				(0.211)	(0.235)	(0.225)	(0.264)
Born other province, share	0.134	0.14	134	-0.014	-0.011	-0.014	-0.020
				(0.021)	(0.015)	(0.019)	(0.017)
# significant / 10				*0*	*0*	*0*	*1*

Notes: Each cell is the coefficient from a separate placebo regression of a 1980 census outcome on one standardized construction of agro-similarity, with the settler count and 16 agro-climatic covariates within 300 minutes as controls, all built over the village set of the block; Conley SE within 200km in parentheses. Each column estimates on the main city sample where that construction and that outcome are measured, 134 urban centers. The ten outcomes are the seven headline outcomes at baseline (modern employment is manufacturing plus services) and three rows on province of birth, the finest level the 1980 census records. Mean, standard deviation and number of urban centers describe each outcome on the sample of the first coefficient column. In each village set, the first column is the exposure of the main specification, the simple mean of village agro-similarity within 300 travel minutes, and the second weights each village by its settler count times its inverse travel time. The left block builds every construction and control over all 814 transmigration villages; the right block rebuilds them over the 687 villages settled after the 1980 census (the other 127, placed by 1980, held 19.5% of all settlers). The last row counts the coefficients with at least one star in each column. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C3: Headline Estimates Controlling for the 1980 Birthplace Composition

	Main specification, as reported (1)	Main specification, same cities as (3) (2)	Adding the 1980 birthplace shares (3)
Panel A. Urban growth and structural transformation			
ln population	0.051	0.051	0.053
	(0.059)	(0.059)	(0.049)
p-value	0.389	0.389	0.275
ln total employment	0.024	0.024	0.027
	(0.057)	(0.057)	(0.049)
p-value	0.674	0.674	0.583
Unemployment rate	0.008	0.008	0.009
	(0.007)	(0.007)	(0.006)
p-value	0.216	0.216	0.159
ln agricultural employment	-0.166*	-0.166*	-0.161**
	(0.093)	(0.093)	(0.078)
p-value	0.075	0.075	0.039
Agricultural employment share	-0.051***	-0.051***	-0.051***
	(0.018)	(0.018)	(0.018)
p-value	0.006	0.006	0.005
ln modern employment	0.158**	0.158**	0.161***
	(0.062)	(0.062)	(0.060)
p-value	0.011	0.011	0.007
Modern employment share	0.066***	0.066***	0.066***
	(0.015)	(0.015)	(0.014)
p-value	0.000	0.000	0.000
Observations	134	134	134
Panel B. 2010 residents by birthplace			
Born in the same district	-0.040*	-0.040*	-0.044***
	(0.023)	(0.023)	(0.015)
p-value	0.080	0.080	0.003
Born elsewhere in the same province	0.042**	0.042**	0.043**
	(0.018)	(0.018)	(0.018)
p-value	0.019	0.019	0.016
Born in another outer-island province	0.017	0.017	0.015
	(0.021)	(0.021)	(0.012)
p-value	0.415	0.415	0.228
Born on Java or Bali	-0.019*	-0.019*	-0.014
	(0.011)	(0.011)	(0.009)
p-value	0.079	0.079	0.119
Observations	134	134	134

Notes: Each cell is one regression on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Beneath each coefficient are its standard error in parentheses and the two-sided Conley p-value, shaded by significance level. Column (3) adds two of the three 1980 census birthplace shares of the city's population (born in another outer-island province, born on Java or Bali; born in this province is the reference), and column (2) repeats the main specification on column (3)'s cities, those with the 1980 shares measured. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C3: Headline Estimates Controlling for the 1980 Birthplace Composition, Continued

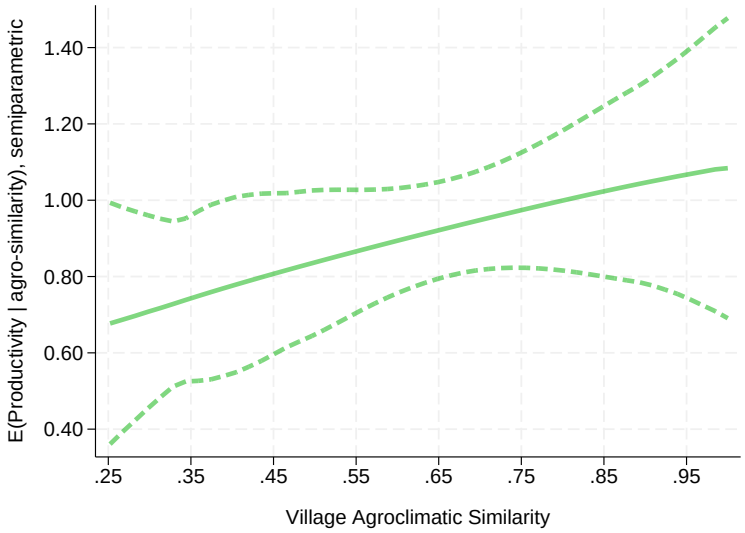
Notes, continued: Panel A gives the seven 2010 census outcomes of Table 3 and Panel B the four birthplace shares of Figure 11, whose 2010 shares sum to one over each city's cohort born 1955 to 1990; shares and the unemployment rate are fractions.

Table C4: Village Agricultural Productivity on Agro-Climatic Similarity: Linear Estimates

	Similarity 0.25–1 (plotted range)		Full similarity range	
	(1)	(2)	(3)	(4)
	winsorized	raw	winsorized	raw
Agro-climatic similarity	0.525***	0.767***	0.535***	0.772***
	(0.201)	(0.136)	(0.179)	(0.108)
Winsorized	Yes	No	Yes	No
Similarity range	0.25–1	0.25–1	All	All
Villages	768	768	770	770

Each column regresses the village agricultural productivity index on the village agro-climatic similarity index (0 to 1 scale) across the transmigration villages with a productivity measure in the similarity range of the column; the Villages row gives their number. Conley SE within 200km in parentheses. The productivity index of Bazzi et al. (2016), from the 2003 village census, is the revenue-weighted geometric mean across crops of the village's yield per hectare relative to the crop's national average yield, so a value of one means every crop yields the national average. The similarity index is built from the geographic, soil, and climate characteristics of the village, so the regression has the index as its only regressor. Winsorized columns cap productivity at its 1st and 99th percentiles. Column 1 is the specification plotted in Figure C2. Raw columns use the uncapped productivity index. *** 1%, ** 5%, * 10%.

Figure C2: Village-Level Relevance: Agricultural Productivity Against Agro-Climatic Similarity



Notes: Local linear fit (Epanechnikov kernel, bandwidth 0.30) of village agricultural productivity on agro-climatic similarity, an index from 0 to 1, across the 768 transmigration villages with a productivity measure and a similarity between 0.25 and 1; the solid line is the fit and the dashed lines its 90% confidence band. Productivity is the agricultural productivity index of Bazzi et al. (2016) from the 2003 village census (PODES), the revenue-weighted geometric mean across crops of the village's yield per hectare relative to the crop's national average yield, so a value of one means every crop yields the national average; it is winsorized at the 1st and 99th percentiles. Table C4 reports the same gradient in linear form, with Conley SE within 200km.

D Robustness

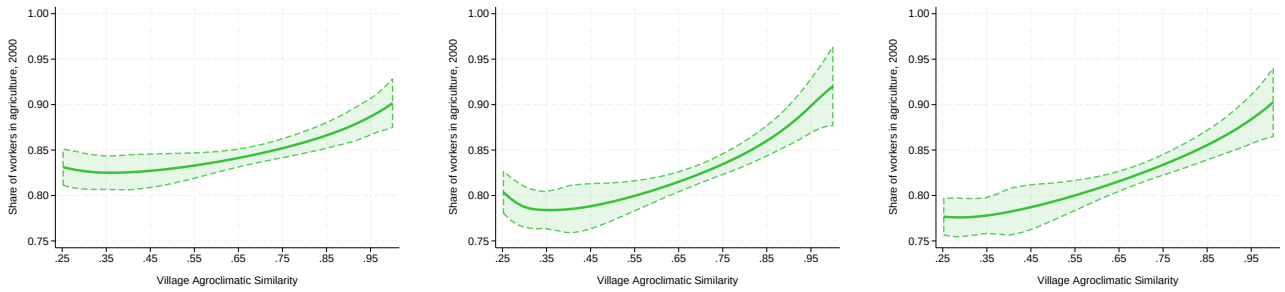
Cited in the main text. Figure D1: Section 5.2, p. 35.

This part re-estimates the headline results under alternative constructions of the exposure measure and alternative specifications.

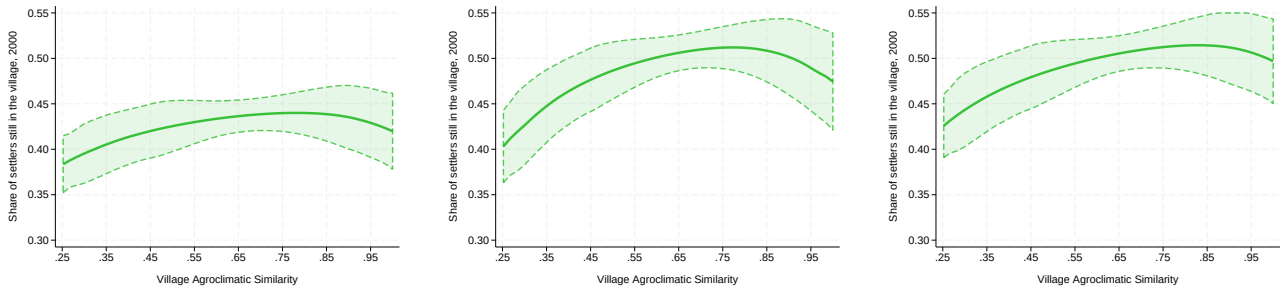
The two village curves of Figure 12 are drawn again unweighted at a bandwidth of 0.30, and with each village weighted by its 2000 population at bandwidths of 0.25 and 0.30 (Figure D1). In a regression of the village agricultural share on agro-similarity and its square, with the same controls on the same 808 villages, the squared term is +0.767 (Conley SE 0.189 within 200km), significant under those standard errors and insignificant under a wild cluster bootstrap over the four islands ($p = 0.142$).

Figure D1: The Village Agricultural Share and Settler Stay-Share at Bandwidth 0.30 and With Population Weights

(a) Unweighted, 0.30 (b) Population weights, 0.25 (c) Population weights, 0.30



(d) Unweighted, 0.30 (e) Population weights, 0.25 (f) Population weights, 0.30



Notes: Semiparametric local linear fits (Epanechnikov kernel) of the share of a village's workers in agriculture in the 2000 Population Census (top row) and of the settler stay-share (bottom row) on the village's agro-similarity, partialing out the village controls of Figure 12 on the same 808 transmigration villages: unweighted at a bandwidth of 0.30 (left), and with each village weighted by its 2000 population at bandwidths of 0.25 (middle) and 0.30 (right), where the weighted fits also remove the controls with the Epanechnikov kernel. Each band is a pointwise 90% confidence band that treats the villages as independent. The axes are those of Figure 12; a band that leaves its axis is cut at the axis limit, here in the population-weighted stay-share at 0.30 (band up to 0.552).

D.1 The Settlers' Size Relative to the Pre-Existing Rural Population

Cited in the main text. Table D1: Section 5.1, p. 26.

The settlers placed in a city's rural hinterland, meaning the rural subdistricts within 135 km, were a non-trivial share of the population that hinterland already held at the 1980 census: about 7% of it on average and roughly 5% at the median, with about a seventh of cities above 15%. That radius gives the closest approximation, among the distances

examined, to the 300-minute window the exposure measure uses. Cities therefore differ in how large the settlement was relative to the rural landscape it entered, and I test whether the main results change with that difference (Table D1).

They do not. With the share added to the main specification, the two employment shares and the unemployment rate are unchanged; log population, log employment and log modern employment are larger by 0.02 to 0.03; and log agricultural employment becomes marginally insignificant. Part of the reason is that the main specification already controls for the number of settlers placed near each city (Online Supplement), so the share's numerator is largely held fixed and the variable moves primarily through its denominator, the size of the countryside those settlers entered.

Table D1: Headline Estimates With a Control for the Settlers' Share of the Pre-Existing Rural Population

	Population + Employment			**Agric Empl**		**Modern Empl**	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	ln pop	ln emp	unemp rate	**ln L agric**	**agric %**	**ln L modern**	**modern %**
Main specification (main text)	0.051	0.024	0.008	-0.166*	-0.051***	0.158**	0.066***
	(0.059)	(0.057)	(0.007)	(0.093)	(0.018)	(0.062)	(0.015)
+ settler share of the hinterland	0.081	0.054	0.009	-0.149	-0.053***	0.177***	0.062***
	(0.056)	(0.055)	(0.007)	(0.093)	(0.019)	(0.063)	(0.015)
Observations	134	134	134	134	134	134	134

Notes: Each cell is the coefficient on standardized agro-similarity from one regression on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Standard errors are in parentheses. The first row is that specification. The second row adds the settlers placed in the rural hinterland of the city as a share of the 1980 census population of that hinterland, over the rural subdistricts within 135 kilometers, the radius that reproduces the 300-minute window; with the settler count already a control, this share varies primarily through its denominator. Outcomes are the seven 2010 census outcomes, with an employment level and share for agriculture and for the modern sector (manufacturing and services), each sector pair shaded; shares and the unemployment rate are fractions. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.2 The Specification Curve and Price-Measure Robustness

Cited in the main text. This section: Section 5.1, p. 26. Figure D3: Section 4.2, p. 21; Section 5.1, p. 26. Table D2: Section 5.1, p. 26. Figure D4: Section 5.2, p. 29.

Specification variants on the main city sample. The specification curve re-estimates the standardized agro-similarity coefficient across the cost-distance specification space, crossing the weight family, the travel-time cutoff, the control set, and island fixed effects on or off. For the three headline outcomes that space holds 61 specifications, every one of them at the 300-minute or 600-minute cutoff and on the main city sample. Straight-line distances, province fixed effects, unrestricted catchments, and the 180-minute cutoff are separate sensitivity exercises. The 180-minute family uses a smaller sample and includes wrong-signed estimates with 90% intervals excluding zero. The displayed comparison varies weights, catchments, controls, and island fixed effects while retaining the main city sample. Within this space no specification is both wrong-signed and significant for any of the three outcomes. The weight family crosses the headline's unweighted within-300-minute mean with an inverse-distance-weighted mean, a settler-weighted mean, and the inverse-distance and settler-weighted average, and every construction inside the 300-minute window keeps the agricultural-share estimate of the headline specification, while distance weights make the modern-employment estimate smaller and less significant.

The agro-similarity coefficient keeps its sign for the sectoral shares across essentially the whole specification space and is positive on average, with wider intervals, for population and prices. I extend the exercise to every outcome in one grid: each cell is the standardized agro-similarity coefficient for one outcome (columns) under one specification family (rows), built by crossing the exposure weighting, the control set, the travel-time cutoff, and island fixed effects on or off

Figure D2: Specification Curve by the Dimension Varied

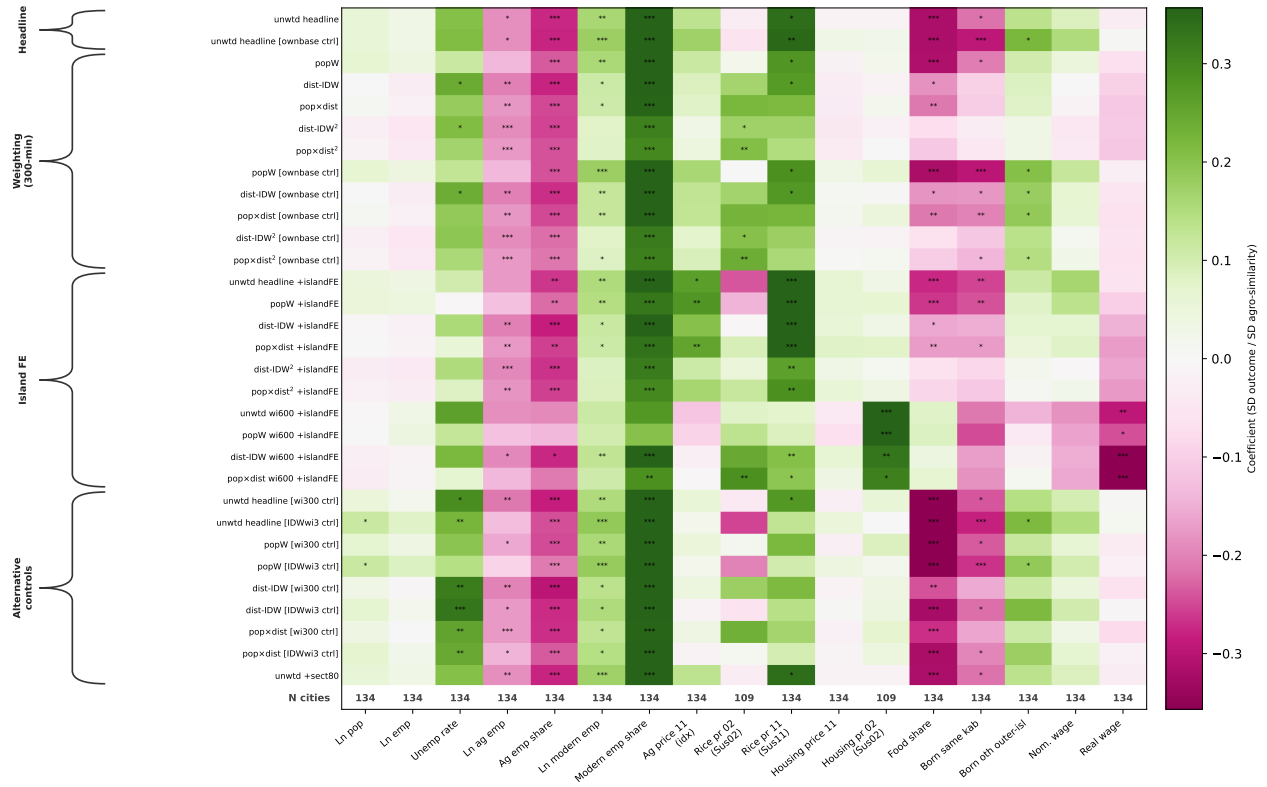


Notes: Each panel is one 2010 population census outcome. Each marker is the coefficient on standardized agro-similarity under one of 13 specifications on the main city sample, grouped by controls, exposure weights, catchment radius, and fixed effects and inference (Table D2). The diamond marks the main specification with the full set of controls and Conley SE within 200km; the 400- and 500-minute catchments widen the exposure and keep the controls at 300 minutes. Within a group, markers are ordered by the estimate, with the main specification first in the controls group; shaded bars are 90% confidence intervals from the same standard errors, or from standard errors clustered by province in two variants of the last group, and the dotted horizontal line marks the main estimate. For log population the own-outcome baseline coincides with the main specification, whose controls already include the 1980 population.

(Figure D3). Two patterns organize the grid. The sectoral columns are stable in sign: the agricultural employment share is negative and the modern-sector share positive across essentially the whole space. The population and price columns are positive on average with wider intervals; across the 61 specifications, counting each distinct estimate once, log population clears significance on 4 of 48 against 43 of 58 for the agricultural employment share. The displayed grid is 31 specifications across 17 outcomes. Province fixed effects and their clustered-standard-error variants are excluded from the grid, because province effects absorb the between-province variation identifying the treatment. Island fixed effects appear as their own row group here.

Robustness of the urban price result across measures. The urban food-price response is small, negative and imprecise under every measure of the local price. Figure D4 re-estimates the headline main specification reduced form with the dependent variable swapped across 5 1996 price measures: the broad agricultural-goods index under national (Laspeyres), city (Paasche), and equal-weight aggregation, the equal-weight crops index, and a rice unit value. The agro-similarity coefficient on the log price runs from -0.028 to -0.005 across the 5 measures, and none of them is statistically significant, in logs or in index levels. The four indices are price relatives with a median near one and enter as the log of one plus the index, so their coefficients read as about half of a proportional change; the rice unit value is a price in rupiah per kilogram, and its coefficient reads as a proportional change. The measures are also internally consistent (pairwise log correlations of 0.76 to 0.94 among the national and city agricultural indices and the rice unit value, and as low as 0.42 for a pair that includes an equal-weight index) with economically sensible spatial variation. Traded food prices are spatially compressed (coefficient of variation ≈ 0.11) relative to the non-traded housing benchmark (0.78). The 1996 survey also splits rice into a local and a high-quality variety. Per standard deviation of agro-similarity, the high-quality share of rice spending is 19.3 percentage points higher (s.e. 6.1), from a mean of 26.8%, and the share of rice buyers who buy any of it is 18.2 points higher (s.e. 6.3), from 26.2%, on 112 cities (Table S24 of the Online Supplement). Market rice prices (the World Bank's district-level market-price panel, matched to urban centers by market coordinates) do not move with agro-similarity in 2011 levels; the estimate is statistically insignificant. In the 2011 household survey, households near productive transmigration villages buy 4.2% fewer kilograms of rice per person, a marginally significant fall, while their spending on rice does not move, so they pay more per kilogram for less rice. Over 2002 to 2008 the rice price in the city's consumer price index does not grow faster than the index as a whole: both growth responses are statistically significant, and their difference is not.

Figure D3: Specification-Curve Heatmap: Constructions and Controls



Notes: Each cell is the coefficient, in SD of the outcome per SD of agro-similarity, from one regression of the column's outcome under the row's specification on the main city sample, held fixed across rows; the bottom row gives each column's number of cities. Row labels name the village weights of agro-similarity, unwtd for equal weights (the main specification), popW for settler weights, dist-IDW and dist-IDW^2 for inverse travel time and its square, and pop×dist and pop×dist^2 for settlers times those weights; wi600 builds exposure within 600 minutes and +islandFE adds island fixed effects; [ownbase ctrl] adds the outcome's own 1980 baseline, or for an outcome without one the 1980 agricultural share and log population density; [wi300 ctrl] uses the unweighted settler count within 300 minutes and [IDWwi3 ctrl] the 16 covariates weighted by inverse travel time; +sect80 adds the 1980 agricultural or modern-sector share to the matching sectoral outcomes. In the column labels, 02 and 11 are the 2002 and 2011 Susenas waves, idx an index, kab the district and outer-isl another outer-island province. Rows vary the village weights of agro-similarity, the control set and island fixed effects around the main specification with the full set of controls and Conley SE within 200km; four island fixed-effects rows measure agro-similarity within 600 minutes. Outcomes come from the 2010 population census and the 2002 and 2011 household-survey (Susenas) waves; the two 2002 columns run on 109 of those cities. Shading deepens with magnitude, one hue for positive and another for negative cells; stars use the same standard errors: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.3 City Boundary Definitions and the Matched Cutoff

Cited in the main text. Table D3: Section 2.2, p. 10. Table D5: Section 5.1, p. 26; Section 5.2, p. 33. Figure D5: Section 5.1, p. 26. Table D4: Section 2.2, p. 10; Section 5.1, p. 27.

Three checks bound what the city definition can do to the results. Table D3 measures the cities under four definitions, each on the cities it covers, from the whole member subdistrict to the built-up core intersected with the census's own urban classification: the level of urban farming falls as the city is drawn tighter, and its 2010 mean is still about 14% under the tightest cut. Most urban centers are a combination of a few subdistricts, so I also rebuild every city from its member subdistricts whose own 1980 employment was at least one quarter, or at least one half, outside agriculture, summing the 2010 outcomes and the four 1980 baselines over those subdistricts (Table D4). The agricultural employment share then falls by 6.2 and 7.1 percentage points per standard deviation of agro-similarity, against 5.1 for the whole city, and farm employment falls by more. Modern-sector employment rises at the quarter threshold, and its

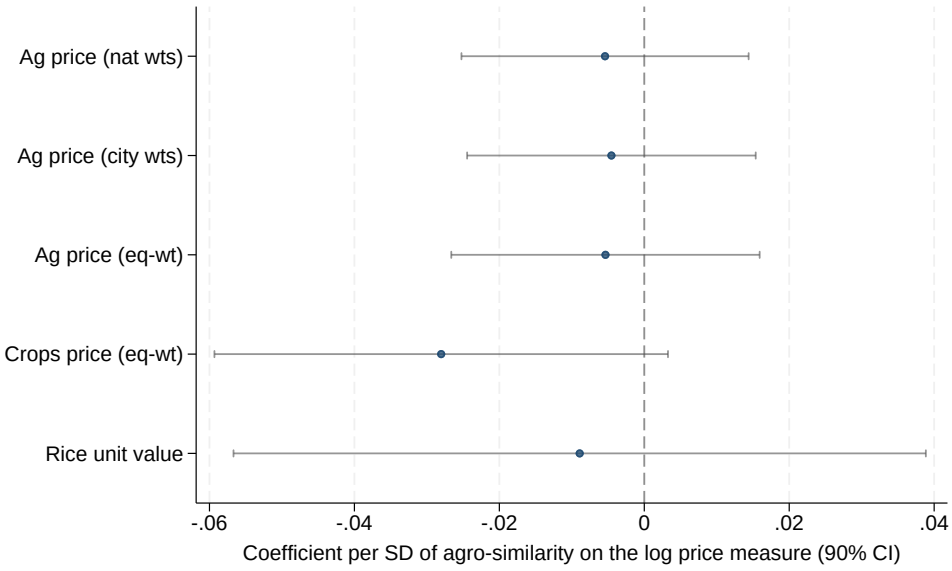
Table D2: Specification Variants in the Grouped Specification Curve

Variant	What it changes relative to the main specification	N
Controls		
Main specification	The settler count as an inverse-travel-time-weighted sum over the transmigrant villages inside the 300-minute catchment, the four 1980 baselines (log population, log employment, the urban-population share and log agricultural employment), and the 16 agro-climatic covariates as unweighted means over the same villages. Every other row changes one thing about this.	134
Unweighted settler count	The settler count entered as an unweighted sum over the same villages.	134
Weighted covariates	The 16 agro-climatic covariates entered as means over the same villages weighted by inverse travel time.	134
Own-outcome baseline	Adds the outcome's own 1980 level, or the 1980 agricultural employment share and log population density where the outcome has no 1980 level. For log population this coincides with the main specification, whose controls already include the 1980 population.	134
Exposure weights		
Population-weighted	Villages enter weighted by their settler count.	134
1/d	Villages enter weighted by the inverse of travel time, so nearer villages count for more.	134
$1/d^2$	The same, with the inverse of squared travel time, a steeper decay.	134
Catchment radius		
Within 400 min	Within 400 travel-time minutes, which every city in the estimation sample reaches.	134
Within 500 min	Within 500 travel-time minutes.	134
FE and inference		
Province FE	Adds province fixed effects. Province effects absorb the between-province variation that identifies the treatment, so this is never the headline.	134
Island FE	Adds island fixed effects.	134
Cluster: province	The same point estimate, with standard errors clustered at the province level instead of Conley.	134
Island FE, cluster province	Island fixed effects together with province-clustered standard errors.	134

Notes: Each row defines one bar in each panel of Figure D2, the coefficient on standardized agro-similarity under that specification for 2010 log population, the agricultural employment share and log modern-sector (manufacturing and services) employment, from the population census. The first row is the main specification with the full set of controls and Conley SE within 200km; every other row changes exactly one thing about it, which makes a group a controlled comparison. The last row combines two of these changes, island fixed effects and province-clustered standard errors. N is the number of urban centers, and every row estimates on the main city sample. Standard errors are those of the main specification, except in the two rows clustered by province. In the Variant column, d is the travel time from a city to a transmigration village and FE marks fixed effects.

estimate loses precision at the half. The half threshold also removes old town centers, since many town subdistricts were still mostly agricultural in 1980, and 40 of the 102 cities with a 1980 urban population lose the subdistrict that held most of it. The result holds when the city is drawn wider as well. Panel D adds to each city the rural subdistricts within 50 km, and in that region the agricultural employment share falls by 5.4 percentage points per standard deviation of agro-similarity, against 5.1 for the whole city, while modern-sector employment and the modern share rise by less and population and employment stay imprecisely estimated. Table D5 splits the sample at the median 1980 agricultural share, 57.9% of 1980 employment, and runs the split as an interaction, so both slopes come from one regression on all 134 centers and the difference between them has its own standard error. The less and the more agricultural cities respond alike. The agricultural-share coefficient is -0.046 in the less agricultural cities and -0.056 in the more agricultural ones, the modern-employment coefficient is 0.154 and 0.158, and no difference between them is distinguishable from zero, the smallest of the four p-values being 0.422. The same design on 1980 population does separate the sample. Log modern

Figure D4: The Effect of Agro-Similarity on Food Prices Across Price Measures, 1996



Notes: Each row is one regression of the log of one plus a 1996 price measure on standardized agro-similarity on the main city sample, 112 centers as the 1996 household survey (Susenas) covers each measure, in the main specification with the full set of controls and Conley SE within 200km. Bars are 90% confidence intervals from the same standard errors. Four measures are price indices built from the city's households: the survey-weighted median unit value of each raw agricultural food item over those households, relative to its median over the households of all urban centers, aggregated under national, city and equal weights, and over the rice, corn and cassava items under equal weights. These indices have a median near one, so a coefficient on their log is about half of a proportional change. The fifth measure is the city's rice unit value, the value of the rice consumed over the kilograms consumed, each summed over its households with survey weights, and its coefficient reads as a proportional change.

employment responds 0.152 more in the larger cities than in the smaller ones (s.e. 0.055) and log population 0.150 more (s.e. 0.041), both statistically significant, and the agricultural-share response is the same on both sides of that cut.

Table D3: Urban Agricultural Employment Share Under Four Definitions of the City

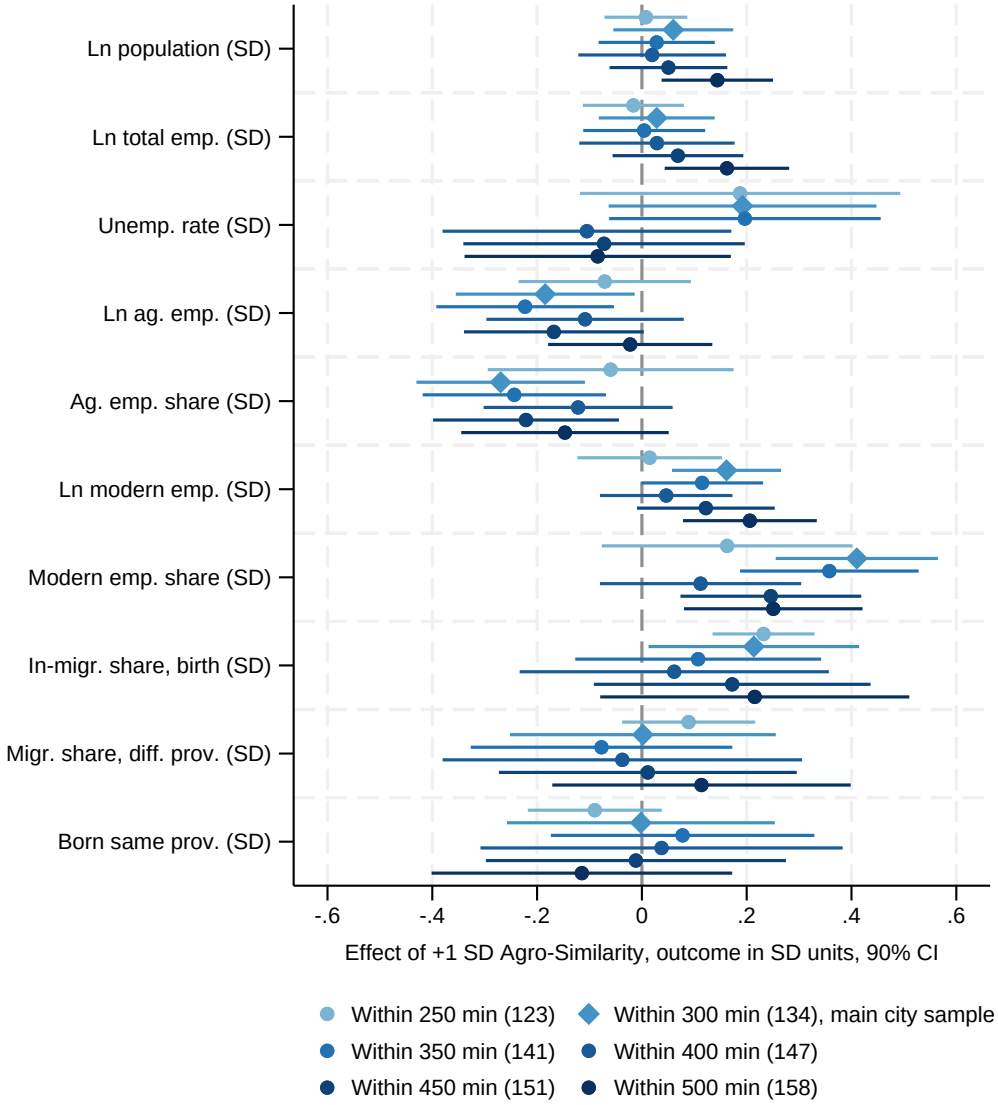
Year	Definition		City mean	Pooled	Cities
1980	K	whole subdistrict (as reported)	0.560	0.483	134
1980	U	BPS-urban villages	0.142	0.091	105
2010	K	whole subdistrict	0.374	0.299	134
2010	C	built-up core	0.181	0.113	130
2010	U	BPS-urban villages	0.159	0.100	132
2010	CU	core and BPS-urban	0.136	0.085	128

Notes: Agricultural employment share of the cities of the main city sample under four definitions of the city, from loosest to tightest: K sums every member subdistrict; C keeps the villages whose polygon meets the satellite footprint of the city; U keeps the villages the Indonesian census classifies as urban, a classification made independently of the satellite footprint; CU requires both. City mean averages the city shares, Pooled divides total agricultural employment by total employment, and Cities counts the cities measured; the data are the population censuses of 1980 (5% sample) and 2010.

D.4 Inference: The Conley Cutoff and Catchment Overlap

Cited in the main text. Table D6: Section 5.1, p. 26.

Figure D5: The Agro-Similarity Effect Across Nested Travel-Time Windows, 250 to 500 Minutes



Notes: Each row is one 2010 population census outcome, standardized over all urban centers so its SD stays fixed as the sample widens: the main results table's seven outcomes (modern-sector employment is manufacturing plus services) and the shares of residents born outside the current district, born in a different province, and born in the same province. The six points in a row are coefficients from separate regressions on standardized agro-similarity over nested catchments: the urban centers whose nearest transmigration village lies within 250 to 500 travel minutes, in 50-minute steps, with a 1980 census baseline and measured 2010 outcomes (123 to 158 urban centers, as the legend reports). At each radius the exposure, the settler count and the 16 agro-climatic covariates are rebuilt over the villages within it, and the four 1980 baselines enter as in the main specification. The 300-minute point, marked with a diamond, is the main specification on the main city sample. Bars are 90% confidence intervals from Conley SE within 200km.

Table D4: Main Results for Cities Rebuilt From Their Less Agricultural Subdistricts and From Their 50 km Regions

	Population + Employment			Agric Empl		Modern Empl	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	ln pop	ln emp	unemp rate	ln L agric	agric %	ln L modern	modern %
Panel A: Every member subdistrict							
Agro-similarity	0.051	0.024	0.008	-0.166*	-0.051***	0.158**	0.066***
	(0.059)	(0.057)	(0.007)	(0.093)	(0.018)	(0.062)	(0.015)
Outcome mean	11.842	10.933	0.122	9.822	0.374	10.232	0.516
Observations	134	134	134	134	134	134	134
Panel B: Subdistricts at least 25% non-agricultural in 1980							
Agro-similarity	0.042	0.015	0.008	-0.241**	-0.062***	0.138***	0.066***
	(0.060)	(0.059)	(0.007)	(0.110)	(0.018)	(0.048)	(0.015)
Outcome mean	11.762	10.828	0.128	9.474	0.298	10.272	0.582
Observations	115	115	115	115	115	115	115
Panel C: Subdistricts at least 50% non-agricultural in 1980							
Agro-similarity	-0.039	-0.040	0.009	-0.441***	-0.071***	0.080	0.077***
	(0.044)	(0.045)	(0.007)	(0.109)	(0.017)	(0.060)	(0.015)
Outcome mean	11.827	10.865	0.136	9.057	0.200	10.452	0.671
Observations	68	68	68	68	68	68	68
Panel D: The city and its rural subdistricts within 50 km							
Agro-similarity	0.041	0.010	0.006	-0.134*	-0.054***	0.129**	0.055***
	(0.056)	(0.048)	(0.006)	(0.069)	(0.020)	(0.056)	(0.014)
Outcome mean	12.724	11.851	0.111	11.152	0.519	10.881	0.390
Observations	134	134	134	134	134	134	134
Settler count	✓	✓	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agro-climatic controls	✓	✓	✓	✓	✓	✓	✓

Notes: Each column is one regression of the 2010 outcome in its heading on agro-similarity, on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Panel A repeats Table 3. Panels B and C rebuild each city from its member subdistricts whose own 1980 employment was at least one quarter (Panel B) or one half (Panel C) outside agriculture, summing the 2010 outcomes and the four 1980 baselines over them; agro-similarity, the settler count and the covariates remain those of the whole city, a subdistrict two cities share is split between them by its measured weights, and the 21 member subdistricts with no 1980 employment count stay in every panel. A city with no subdistrict above the threshold leaves the panel, 19 cities in Panel B and 64 in Panel C; Panel C also loses 2 cities whose kept subdistricts have no 1980 employment count, so a 1980 baseline cannot be formed. Panel D adds to each city the rural subdistricts, those outside every city, whose nearest city it is within 50 km in a straight line, measured to the nearest member subdistrict of each city on the same island, summing the 2010 outcomes and the four 1980 baselines over that region: 709 rural subdistricts join, 6 cities do not gain a rural subdistrict, and the 57 rural subdistricts within 50 km that the 1980 census does not record are left out. Agro-similarity is standardized once, so the four panels share one scale. SE in parentheses, with stars from the same SE: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: 1980 and 2010 population censuses.

The Conley cutoff. The headline uses a 200-kilometer Conley cutoff. Because the cutoff enters only the variance estimator, the point estimates do not depend on it, so the choice affects only inference. At a shorter 150-kilometer radius every standard error remains well defined and two of the seven are smaller than at 200 kilometers, so the wider cutoff is not uniformly the more conservative of the two: the two agricultural and the two modern-sector columns have slightly

Table D5: Exposure Effects by the City's 1980 Agricultural Share and by Its 1980 Population

	(1) Agric. share	(2) ln modern empl.	(3) ln population	(4) In-migrant share
Panel A: split at the median 1980 agricultural share, 57.9%				
Below median (less agricultural)	-0.046***	0.154***	0.055	0.004
	(0.017)	(0.060)	(0.063)	(0.004)
Above median (more agricultural)	-0.056***	0.158**	0.044	0.007***
	(0.018)	(0.080)	(0.060)	(0.002)
Above-median indicator (=1)	0.089***	-0.125**	-0.008	-0.002
	(0.022)	(0.061)	(0.056)	(0.003)
Difference (above minus below)	-0.010	0.003	-0.011	0.002
	(0.014)	(0.046)	(0.039)	(0.003)
p-value of the difference	0.463	0.940	0.781	0.422
p-value, Bartlett kernel	0.501	0.946	0.798	0.366
Observations	134	134	134	134
Panel B: split at the median 1980 population, 67,979 residents, the same design				
Below median (smaller)	-0.047**	0.089	-0.017	0.004
	(0.022)	(0.066)	(0.081)	(0.004)
Above median (larger)	-0.057***	0.241***	0.133**	0.007***
	(0.018)	(0.080)	(0.065)	(0.002)
Above-median indicator (=1)	-0.018	0.163*	0.150	0.001
	(0.018)	(0.098)	(0.098)	(0.003)
Difference (above minus below)	-0.010	0.152***	0.150***	0.003
	(0.018)	(0.055)	(0.041)	(0.002)
p-value of the difference	0.580	0.006	0.000	0.152
p-value, Bartlett kernel	0.566	0.013	0.013	0.201
Observations	134	134	134	134

Notes: Each column of each panel is one regression of a 2010 census outcome on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Standardized agro-similarity enters interacted with an indicator for each side of the split, so the below-median and above-median rows are the two slopes and the indicator row is the shift in the outcome on the above-median side; the row labeled Difference is the above-median slope minus the below-median slope, from the same regression. Each split is cut at the sample median: 57.9% of 1980 employment in agriculture in Panel A, 67,979 residents in 1980 in Panel B. Standard errors (uniform kernel) are in parentheses; the Bartlett-kernel *p*-value tests the same difference, weighting a pair of cities by one minus its distance over the cutoff, and its cells are shaded by significance level. Modern employment is manufacturing plus services; the agricultural share is a fraction of 2010 employment, and the in-migrant share is the fraction of the 2010 population that lived in a different province five years earlier. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

larger standard errors at 150 kilometers (the pair for each share, from the table), and those shares are the columns the compositional result relies on. The wider direction matters more, because the exposure measure averages over village catchments that neighboring cities share. The 150 kilometer row of Table D6 reports these standard errors beside every wider radius.

Catchment overlap and inference at wider radii. The relevant direction to vary the cutoff is wider, and the reason is in the construction of the treatment. Exposure is an unweighted mean over the transmigration villages inside a city's 300-minute catchment, and neighboring catchments intersect, so two cities can be built partly from the same village draws and their exposure errors are correlated at whatever distance that sharing reaches. Measured directly on the travel-time matrix the exposure itself is built from, the average city in the estimation sample draws on 30 in-window villages, 80% of those villages enter more than one city's catchment, and one village enters as many as 19. Of the 8,911 city pairs in the sample, 765 share at least one village, at a median separation of 143 kilometers and a maximum of

459; 27% of those sharing pairs sit farther apart than the 200-kilometer cutoff, which assumes independence beyond it. Grouping cities into the connected components of that sharing graph, the finest partition across which no village is shared at all, gives 14 components over the estimation sample, and the largest holds 75 cities: roughly half the sample is linked into a single block by shared catchments.

I therefore re-estimate the headline columns across the whole ladder, from 150 out to 500 kilometers, and add a variant clustered on those components together with a wild cluster bootstrap over them (Appendix Table D6). The agricultural share, log modern employment and the modern share are statistically significant under every Conley, Bartlett and component-clustered variance, although log modern employment has no uniform-kernel standard error at 450 and 500 kilometers; under the wild bootstrap log modern employment stays significant, the modern share is marginally significant and the agricultural share is not. Log population and log employment are insignificant except under the uniform kernel at 500 kilometers. The compositional standard errors are the more stable: both employment-share standard errors stay between 0.011 and 0.020 across the ladder and the clustering, and the agricultural share rejects at the 5% level at every radius ($p = 0.010$ at 150 kilometers and 0.006 at 200 under the uniform kernel, 0.002 at 500 under the Bartlett). Under the component clustering, log population has $p = 0.245$ with a wild cluster bootstrap p of 0.176, the agricultural share 0.014 and 0.140, and the modern share 0.000 and 0.058. Under the uniform kernel, which the headline uses, four of the table's 28 Conley cells have no standard error, because the estimated variance of the exposure coefficient is negative there, and all six of its standard errors at 500 kilometers are smaller than at 200 kilometers. At 200 kilometers the uniform variance matrix already has five or six negative eigenvalues out of 23, depending on the column, so it is not positive semi-definite, and the table's Conley rows at wide radii are not a reliable guide to precision. Every cell has a Bartlett standard error, and the Bartlett variance at 200 kilometers has no negative eigenvalue; from 200 to 500 kilometers the Bartlett standard error falls by less than the uniform one in five of the six columns with both standard errors at 500 kilometers, and for the unemployment rate it does not fall.

Table D6: The Headline Estimates Under a Ladder of Spatial-Inference Radii

	Population + Employment			Agric Empl		Modern Empl	
	ln pop	ln emp	unemp rate	ln L agric	agric %	ln L modern	modern %
Agro-similarity	0.051	0.024	0.008	-0.166*	-0.051***	0.158**	0.066***
Conley 200 km (main), SE	(0.059)	(0.057)	(0.007)	(0.093)	(0.018)	(0.062)	(0.015)
Conley 200 km (main), p	0.389	0.674	0.216	0.075	0.006	0.011	0.000
Conley 150 km, SE	(0.059)	(0.054)	(0.006)	(0.098)	(0.020)	(0.065)	(0.016)
Conley 150 km, p	0.389	0.657	0.188	0.091	0.010	0.015	0.000
Conley 450 km, SE	(.)	(.)	(0.007)	(0.059)	(0.014)	(.)	(0.011)
Conley 450 km, p	.	.	0.209	0.005	0.000	.	0.000
Conley 500 km, SE	(0.027)	(0.013)	(0.006)	(0.054)	(0.014)	(.)	(0.011)
Conley 500 km, p	0.055	0.071	0.186	0.002	0.000	.	0.000
Bartlett 200 km, SE	(0.055)	(0.052)	(0.006)	(0.094)	(0.019)	(0.059)	(0.016)
Bartlett 200 km, p	0.352	0.649	0.153	0.078	0.006	0.008	0.000
Bartlett 150 km, SE	(0.054)	(0.052)	(0.006)	(0.093)	(0.018)	(0.058)	(0.015)
Bartlett 150 km, p	0.345	0.647	0.133	0.074	0.005	0.006	0.000
Bartlett 450 km, SE	(0.049)	(0.045)	(0.006)	(0.081)	(0.017)	(0.052)	(0.014)
Bartlett 450 km, p	0.293	0.599	0.194	0.041	0.003	0.003	0.000
Bartlett 500 km, SE	(0.046)	(0.043)	(0.006)	(0.079)	(0.017)	(0.048)	(0.013)
Bartlett 500 km, p	0.267	0.576	0.195	0.036	0.002	0.001	0.000
Components, SE	(0.042)	(0.040)	(0.007)	(0.060)	(0.018)	(0.058)	(0.013)
Components, p	0.245	0.558	0.256	0.017	0.014	0.017	0.000
Components, wild bootstrap p	0.176	0.520	0.293	0.150	0.140	0.043	0.058
Observations	134	134	134	134	134	134	134

Notes: Every column repeats the main specification of Table 3, a regression of a 2010 census outcome on standardized agro-similarity on the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline), with the full set of controls; only the variance estimator changes across rows, so the coefficients are identical, and their stars come from the Conley p-value at 200 kilometers. The Conley rows use a uniform kernel, which gives every pair of cities inside the cutoff the same weight, and the Bartlett rows use the Bartlett kernel, which weights a pair by one minus its distance over the cutoff. Each p-value cell takes one shade at 0.10 or above and deepening shades below 0.10, 0.05 and 0.01. The radii bracket the distance over which cities share transmigration villages: 80% of the villages within the window enter more than one city's catchment, and two cities sharing a village sit up to 459 kilometers apart. The last three rows cluster on the connected components of the village-sharing graph, the finest partition that keeps every shared village inside one group: 14 components over the 134 cities, the largest holding 75, so the clustered standard error uses few effective groups and the wild cluster bootstrap over them, with Webb weights, is the version to read; it has little power at this count. Under the uniform kernel four of the 28 cells have no standard error, because the estimated variance of the exposure coefficient is negative there. In each row SE is the standard error of the coefficient in the first row and p its two-sided p-value, and (main) labels the radius of the main specification. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Additional Results

This part reports the outcomes, decompositions, and alternative explanations that the main text states but does not tabulate.

E.1 Additional Reduced-Form Outcome Tables

Cited in the main text. Table E1: Section 5.3, p. 38. Table E2: Section 5.2, p. 29. Table E3: Section 5.2, p. 29. Table E4: Section 5.1, p. 25. Table E5: Section 4.2, p. 21. Table E6: Section 5.2, p. 29. Table E7: Section 5.2, p. 29. Table E8:

Section 4.2, p. 21. Table E9: Section 5.2, p. 29. Table E10: Section 5.2, p. 29. Figure E1: Section 5.2, p. 35; Section 5.3, p. 36.

Table E1: All Outcomes by Location Type, Main Specification

	Cities	Rural, with settlement	Rural, no settlement
A. Size and composition			
ln population	+0.060	+0.082	−0.049
	(0.069)	(0.082)	(0.062)
ln employment	+0.028	+0.091	−0.077
	(0.067)	(0.086)	(0.061)
ln agricultural employment	−0.185*	+0.089	−0.200**
	(0.104)	(0.077)	(0.080)
Agricultural employment share	−0.278***	+0.010	−0.318***
	(0.101)	(0.063)	(0.082)
ln modern employment	+0.162**	+0.077	+0.112**
	(0.063)	(0.077)	(0.045)
Modern employment share	+0.422***	−0.024	+0.313***
	(0.097)	(0.070)	(0.087)
Observations	134	191	596
B. Prices, wages, and consumption			
Unemployment rate	+0.210	+0.032	+0.080
	(0.170)	(0.090)	(0.103)
	[134]	[191]	[596]
Agricultural price, 2002	+0.126	−0.228*	−0.052
	(0.262)	(0.117)	(0.080)
	[109]	[104]	[204]
Housing price, 2002	−0.016	−0.004	+0.039
	(0.141)	(0.085)	(0.077)
	[109]	[98]	[195]
ln nominal wage, 2011	+0.062	+0.154	−0.133**
	(0.101)	(0.178)	(0.066)
	[134]	[82]	[236]
ln real wage, 2011	+0.013	+0.274**	−0.138
	(0.126)	(0.129)	(0.123)
	[134]	[82]	[236]
Food spending share, 2011, stored measure	−0.313***	−0.232**	+0.167
	(0.106)	(0.105)	(0.114)
	[134]	[82]	[236]
ln food spending, 2011	+0.161	−0.079	+0.044
	(0.115)	(0.065)	(0.093)
	[134]	[188]	[576]
Food spending share, 2011, pooled households	−0.313***	−0.062	+0.113*
	(0.106)	(0.132)	(0.063)
	[134]	[188]	[576]
Rice calories per person	+0.471***	+0.071	+0.010
	(0.172)	(0.143)	(0.107)
	[108]	[104]	[204]

The response by location type, full set of outcomes. Every outcome behind Figure 15a and Figure 15b is estimated for the three location types, each with its standard error. The size and composition outcomes, which all three areas measure on the same units, form panel A, and the prices, wages and consumption outcomes, which come from the household survey and cover a different number of units in each row, form panel B. The agricultural and housing prices are the 2002 Susenas wave, which precedes the 2004 rice import ban; the real wage is the 2011 wave, because the subdistrict level has no 2002 wage. Panel C holds the eight 1996 rows; Figure 15b plots three of them, the food share, rice calories per person and the raw-food price, beside the 2011 real wage and the 2010 unemployment rate.

Linear gradients of the agricultural employment share. Figure E1 draws the binned scatter plots and linear fits behind the semiparametric curves of Figure 14, with the slope of each and its inference.

Table E1: All Outcomes by Location Type, Main Specification, Continued

	Cities	Rural, with settlement	Rural, no settlement
C. Household survey, 1996			
ln total household spending, 1996	+0.512*	+0.115	−0.150
	(0.263)	(0.124)	(0.100)
	[112]	[117]	[285]
ln food spending per person, 1996	+0.460	+0.116	−0.078
	(0.287)	(0.115)	(0.108)
	[112]	[117]	[285]
Food spending share, 1996	−0.462***	+0.057	−0.187**
	(0.177)	(0.119)	(0.094)
	[112]	[117]	[285]
Agricultural price (raw food), 1996	−0.090	−0.130	−0.285**
	(0.188)	(0.109)	(0.119)
	[112]	[117]	[285]
Rice unit value, 1996	−0.050	−0.200	−0.111
	(0.210)	(0.180)	(0.124)
	[112]	[116]	[283]
ln calories per person, 1996	+0.719***	+0.087	+0.002
	(0.148)	(0.178)	(0.146)
	[112]	[117]	[285]
Rice calories per person, 1996	+0.289***	+0.013	+0.333***
	(0.081)	(0.106)	(0.112)
	[112]	[117]	[285]
ln staple kg per person, 1996	+0.407***	−0.225	+0.133
	(0.073)	(0.151)	(0.102)
	[112]	[117]	[285]

Notes: Each coefficient is the response, in SD of the outcome, to a one-SD increase in agro-similarity, with Conley SE within 200km in parentheses. Cities are the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline); the rural subdistricts with and without a settlement lie outside every city and meet the same travel-time and baseline rule. The city column is in the main specification with the full set of controls; the two rural columns have the same controls, except that the settler count of a subdistrict with a settlement is the number of settlers placed in the transmigration villages that overlap it. Outcomes and agro-similarity are standardized once among the cities and once among both rural groups together, at exposure SDs of 0.103 over the 138 cities with a measured exposure and 0.100 over the 787 sample subdistricts, so one SD is about the same amount of exposure on either side, with outcomes on separate scales. Panel A and the unemployment rate are from the 2010 population census, the prices and rice calories from the 2002 household survey (Susenas), and the wage and food spending rows from its 2011 wave, with food spending and wages in logs. Panel C is the 1996 wave, pooled over the surveyed households of each city and subdistrict, with the agricultural price on the raw-food index and logs taken as ln(x+1). Panel A's observation count applies to every row of that panel; each Panel B and C row gives its own count in brackets. The stored 2011 food share exists only for the subdistricts where the 2011 survey found a wage earner living in an urban block of the statistics office, and the pooled food share averages all surveyed households of each subdistrict, as food spending does; for the cities the two are the same measure. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

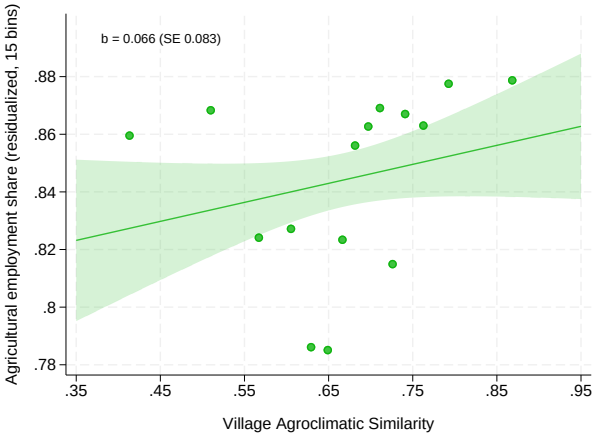
Village-census dynamics by location type. The village censuses show how population and the agricultural-household share respond over time (Figure E2).[29] In cities, both the population response and the decline in the share of agricultural households occur early and dissipate over time. The early population estimates are positive and the agricultural-household-share estimates negative, each statistically significant in one of the first two periods. In the later periods both effects are closer to zero and the population estimates are less precisely estimated. Rural subdistricts without settlements also have lower agricultural-household shares early in the period, without detectable population growth. These household classifications differ from the individual employment sectors measured in the population census, so their later attenuation does not establish that the worker-sector response also diminished.

This pattern is consistent with production in the transmigration villages replacing some of the farming by pre-existing households in these areas. The population responses in subdistricts with transmigration villages are imprecise,

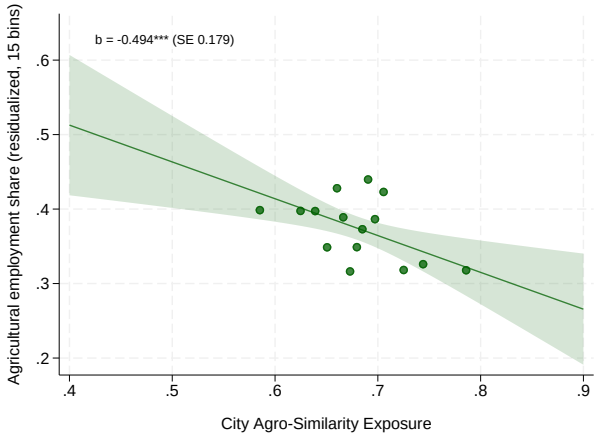
[29] These village PODES surveys record households reported by village heads, while the population census records individual workers by primary industry.

Figure E1: Linear Gradients of the Agricultural Employment Share in the Transmigration Villages and the Nearby Cities

(a) Transmigration Villages, 2000



(b) Nearby Cities, 2010



Notes: Binned scatter plots: dots are 15 quantile-bin means of the outcome and of agro-similarity on its 0 to 1 index scale, both residualized on the panel's controls with means added back. The line and its 90% band come from ordinary least squares on every residualized unit of the sample, and the in-plot coefficient is its slope with the panel's standard error and stars under that standard error (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). **(a)** The share of a village's workers in agriculture in the 2000 population census against the village's own agro-similarity, on the 808 transmigration villages with a measured agricultural share and a measured settler stay-share below 2, with village terrain, geographic and climate controls; the standard error is clustered by island (wild cluster bootstrap over the four island clusters, Webb weights, $p = 0.83$; Conley SE within 200km, $p = 0.32$). **(b)** The agricultural employment share in the 2010 population census against city exposure, the mean agro-similarity of the transmigration villages within 300 travel minutes, on the main city sample, in the main specification with the full set of controls and Conley SE within 200km; column (5) of Table 3 gives the same relationship in standardized units.

while their agricultural-household-share estimates are close to zero with tighter confidence intervals than in cities. The figure reports separate regressions and does not test differences between location groups.

The goods-market outcomes behind the mechanism discussion follow. Table E3 reports rice production, the production-minus-consumption surplus and food prices, and Table E6 reports consumption and the surplus by category. The non-food side (non-food spending, housing rents, and wages) is reported in Table E2, and the manufacturing/services composition of the modern sector in Table E4 below.

The main results treat the modern sector as a single object; its manufacturing and services components are reported separately here, in levels and shares, in the modern-sector layout (Appendix Table E4). Columns 5 to 7 report the shares of workers outside both sectors: construction, the remaining industry codes, and the two together. The agricultural share of Table 3, the two modern-sector shares and the share outside both sectors add up to one.

Finally, the agricultural production and food consumption/surplus responses are estimated across the crop-category hierarchy, in raw and standardized form, and the standardized versions define several of the model's calibration moments (Table B4). The raw estimates are in Table E5 and Table E6.

The earliest wave, household Susenas 1996 outcomes aggregated to the same urban centers, shows higher per-capita expenditure one to two decades after placement, while the 1996 rice price and the two food-price indices show little response (Table E7).

Table E2: Urban Prices, Spending, and Wages

	Prices & Spending		Wages		
	(1)	(2)	(3)	(4)	(5)
	non-food %	housing price	ln wage nonag	ln wage	ln real wage
Agro-similarity	0.016***	-0.011	0.027	0.019	0.003
	(0.006)	(0.107)	(0.030)	(0.032)	(0.029)
Settler count	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓	✓
Agro-climatic controls	✓	✓	✓	✓	✓
Outcome mean	0.420	1.586	14.062	13.968	13.749
Outcome SD	0.052	0.910	0.273	0.314	0.234
Observations	134	134	134	134	134

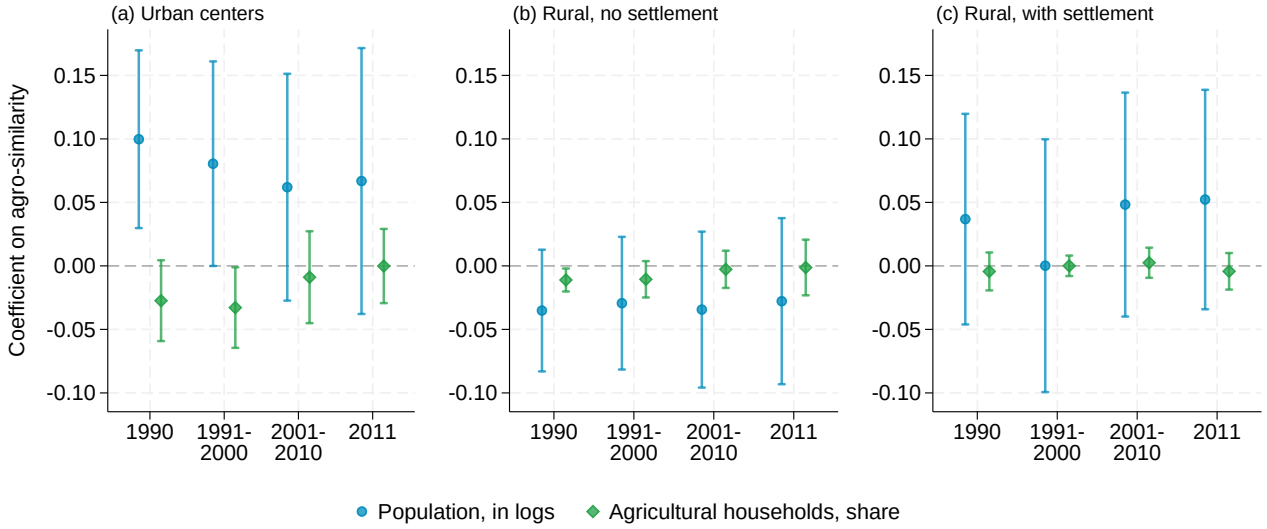
Each column is one regression of a 2011 household-survey (Susenas) outcome on standardized agro-similarity on the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline), in the main specification with the full set of controls and Conley SE within 200km. The Observations row gives the number of centers the survey covers. Standard errors are in parentheses. Non-food % is the non-food share of household expenditure, a fraction; housing price is the weighted median rent of the urban center relative to the national figure; the wage columns are the logs of the median monthly wage of non-agricultural workers, of all workers, and of that all-worker median deflated by a local food and housing price index. In the labels, ln is the natural log and nonag is non-agricultural workers; the settler count weights the settlers of each transmigration village within 300 travel minutes by the inverse of its travel time, in thousands; Baseline Pop 1980, Total Empl 1980 and Agri Empl 1980 are the logs of 1980 population, employment and agricultural employment; and SD is the standard deviation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E3: Urban Agricultural Goods: Rice Production, the Rice Surplus, and Food Prices

	Rice Output		Prices	
	(1)	(2)	(3)	(4)
	ln rice production	rice surplus, tons	rice price index	agric price index
Agro-similarity	0.735***	-335.323	0.041	0.016
	(0.252)	(1802.536)	(0.025)	(0.022)
Settler count	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓
Agro-climatic controls	✓	✓	✓	✓
Outcome mean	8.455	-5522.160	0.999	1.062
Outcome SD	2.343	21675.330	0.145	0.121
Observations	132	108	129	134

Notes: Each column is one regression of the outcome on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Standard errors are in parentheses. Column 1 is the log of one plus 2003 village-census (PODES) rice production, in tons of paddy, and column 2 is the rice surplus in tons, that production in milled weight (.6402 tons of milled rice per ton of paddy, from the 2018 conversion survey of Statistics Indonesia) minus 2002 household-survey (Susenas) rice consumption per person, over the city's surveyed subdistricts, times the 2003 population of its villages with a rice output figure. Columns 3 and 4 are the rice price index of each city and its agricultural price index, an index of raw-food prices, both from Susenas 2011. Survey outcomes are observed on the centers each source covers, reported in the Observations row. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure E2: Population and the Agricultural Household Share, 1990 to 2011, by Location Type



Notes: Each point is the coefficient on standardized agro-similarity from one regression, in the main specification with the full set of controls, on one cell of village-census (PODES) waves, with 90% confidence intervals from Conley SE within 200km. The cells are the 1990 wave, the 1993 and 1996 waves, the 2003, 2005 and 2008 waves, and the 2011 wave. Each panel keeps the units that report both outcomes in every cell: in panel (a) 126 urban centers of the main city sample, and in panels (b) and (c) 588 rural subdistricts without a settlement and 191 with one, lying outside every city and within 300 minutes of a transmigration village. The settler count of a subdistrict with a settlement is the number of settlers placed in the transmigration villages that overlap it. The other sectors are reported inconsistently across waves, so the agricultural share is the sectoral measure available over time. The series start in 1990, because earlier waves cover the agricultural share for less than two-thirds of the sample, and end in 2011, the last wave that releases population and household counts.

The cross-sectional outcomes have no 1980 baseline, so they are identified off balance in levels. I add a 1980 variable that proxies for the unobserved baseline of the outcome and re-estimate on the same urban centers (Table E8). The estimate is essentially unchanged under every proxy and remains statistically significant in every column.

Employment on the Observed-Price Samples. The earlier price surveys cover fewer main-sample cities. On those samples, the pooled agricultural-share and log modern-employment responses are smaller and imprecisely estimated, while the modern-employment differences by openness remain significant (Table E9 and Table E10). The openness groups and exposure scale are fixed before the coverage restriction. The 2011 raw-food price sample equals the main city sample. Bartlett distance-weighted results are in the Online Supplement.

Table E4: Modern Sector Split, and Employment Outside Agriculture and the Modern Sector

	Log employment		Modern-sector shares		Outside both sectors		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	ln L manuf	ln L serv	manuf empl %	serv empl %	constr %	other %	outside %
Agro-similarity	0.249**	0.144**	0.010**	0.056***	-0.002	-0.013*	-0.015*
	(0.120)	(0.067)	(0.005)	(0.018)	(0.004)	(0.008)	(0.009)
Settler count	✓	✓	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agro-climatic controls	✓	✓	✓	✓	✓	✓	✓
Outcome mean	7.556	10.136	0.047	0.470	0.060	0.049	0.109
1980 sector share	0.063	0.319	0.063	0.319	0.035	0.022	0.057
Outcome SD	1.356	0.966	0.042	0.145	0.027	0.065	0.078
Observations	134	134	134	134	134	134	134

Each column is one regression of a 2010 population census outcome on standardized agro-similarity on the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline), in the main specification with the full set of controls and Conley SE within 200km. Standard errors are in parentheses. Columns 1 and 2 are the logs of manufacturing and of services employment, the two parts of the modern sector, and columns 3 and 4 their shares of the employment of workers with an industry code (fractions). Columns 5 to 7 are the shares of construction (census industry code 10), of the other industry codes (mining and quarrying, electricity and gas, and the census code others: codes 7, 9 and 19), and of the two together, the employment outside agriculture and the modern sector; every share divides by the workers who report an industry, so the agricultural share of Table 3 and columns 3 to 6 sum to one, and column 7 is the sum of columns 5 and 6. The 1980 sector share row is the mean 1980 share of the same sector across the sample. In the column labels, ln is the natural log, L and empl are employment, manuf and serv are manufacturing and services, constr is construction, other the remaining codes and outside the two together, and % is a share of the employment of workers with an industry code, a fraction; the settler count weights the settlers of each transmigration village within 300 travel minutes by the inverse of its travel time, in thousands; Baseline Pop 1980, Total Empl 1980 and Agri Empl 1980 are the logs of 1980 population, employment and agricultural employment; and SD is the standard deviation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E5: Agricultural Production by Crop Category, 2003 Village Census

	Food (staples + legumes)			Non-food cash crops
	(1)	(2)	(3)	(4)
	Ln Staples	Ln Rice	Ln Legumes	Ln Coffee+Tea+Spices
Agro-similarity	0.608***	0.735***	0.424	0.103
	(0.197)	(0.252)	(0.360)	(0.501)
Settler count	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓
Agro-climatic controls	✓	✓	✓	✓
Outcome mean	8.953	8.455	4.250	4.341
Outcome SD	2.007	2.343	2.220	2.655
Observations	133	132	117	103

Notes: Each column is one regression of the outcome on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Standard errors are in parentheses. Outcomes are the log of one plus 2003 village-census (PODES) production in tons, by crop group: staples (rice, corn, cassava, sweet potato, taro, other tubers and sorghum), paddy rice, legumes (peanut, soybean and mungbean), and coffee, tea, pepper, nutmeg and clove. Coefficients are approximate semi-elasticities. Survey outcomes are observed on the centers each source covers, reported in the Observations row. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E6: Food Consumption per Person and the Production-Minus-Consumption Surplus, by Category

	Log consumption (kg per person)			$\sinh^{-1}$ surplus (output minus consumption, tons)	
	(1)	(2)	(3)	(4)	(5)
	Staples, 2002	Staples, 2011	Rice, 2002	Staples Surplus	Rice Surplus
Agro-similarity	0.129***	-0.028	0.112***	-0.147	-0.141
	(0.037)	(0.027)	(0.037)	(1.547)	(1.211)
Settler count	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓	✓
Agro-climatic controls	✓	✓	✓	✓	✓
Outcome mean	4.765	4.579	4.649	-3.403	-4.597
Outcome SD	0.203	0.162	0.208	9.319	8.282
Observations	108	132	108	108	108

Notes: Each column is one regression of the outcome on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Standard errors are in parentheses. Columns 1 to 3 are the log of one plus annual consumption per person in kilograms from the household survey (Susenas): staples (rice, corn, cassava and tubers) in 2002 and 2011, and rice in 2002. Columns 4 and 5 are the inverse hyperbolic sine of the staples and rice surplus in tons, 2003 village-census (PODES) production minus 2002 Susenas consumption per person, over the city's surveyed subdistricts, times its 2003 population (the whole city for staples, the villages with a rice output figure for rice), with rice output in milled weight (.6402 tons of milled rice per ton of paddy, 2018 conversion survey). Coefficients are approximate semi-elasticities. Survey outcomes are observed on the centers each source covers, reported in the Observations row. In the labels, $\sinh^{-1}$ is the inverse hyperbolic sine, used because a surplus can be negative. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E7: Household Survey Outcomes in 1996

	Ln rice	Ag price	All-food	House idx	Ln rent	Ln pcExp	Food share	Ag share	Mod share	Log wage
Agro-similarity	-0.009	-0.012	0.007	0.336	0.214	0.182**	-0.033***	0.048	-0.033	0.232***
	(0.029)	(0.025)	(0.025)	(0.238)	(0.140)	(0.087)	(0.013)	(0.040)	(0.040)	(0.072)
Outcome mean	6.84	1.06	1.07	1.56	9.84	11.14	0.65	0.44	0.47	12.43
N (cities)	112	112	112	112	112	112	112	119	119	105

Notes: Each column is a separate urban-center regression of a 1996 outcome from the national socioeconomic household survey (Susenas) on standardized agro-similarity, on the urban centers of the main city sample that the 1996 survey covers, counted in the bottom row, in the main specification with the full set of controls and Conley SE within 200km. Standard errors are in parentheses. Outcomes, in order: the log rice unit value (rupiah per kilogram); price indices of raw agricultural foods and of all foods, city prices relative to the median over the households of all urban centers; the median monthly housing rent relative to the national median, and the log of that median rent; log mean monthly spending per person; the mean household food spending share; the shares of workers in agriculture and in manufacturing and services; and log median household income per employed worker, which stands in for the wage. Ln is the natural log, and N (cities) is the number of urban centers in each regression. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E8: Baseline Proxy Controls and the Cross-Sectional Food-Share Estimate

	(1) Population density	(2) Agricultural employment share	(3) Schooling	(4) All three
Exposure, without the proxy	-0.0164	-0.0164	-0.0164	-0.0164
	(0.0055)	(0.0055)	(0.0055)	(0.0055)
Exposure, with the proxy	-0.0167	-0.0164	-0.0159	-0.0160
	(0.0057)	(0.0053)	(0.0056)	(0.0054)
Movement, in standard errors	-0.07	-0.01	0.09	0.06
Urban centers	134	134	134	134

Notes: The outcome is the 2011 household food share, the mean share of household spending on food in the national socioeconomic household survey (Susenas). Each column adds to the main specification one 1980 census proxy for the unobserved baseline of the outcome: population per square kilometer, the agricultural employment share, the number of residents who had completed secondary school, and all three together. Each regression is of the food share on standardized agro-similarity, in the main specification with the full set of controls and Conley SE within 200km. The two rows of each column use the same urban centers, those with a transmigration village within 300 travel minutes and a 1980 census baseline that have the food share and the proxy measured, counted in the bottom row. Movement is the with-proxy estimate minus the without-proxy estimate, over the without-proxy standard error. Standard errors are in parentheses.

Table E9: Employment Responses on the Observed-Price Samples

	Agricultural Share	Log Modern Jobs	Modern Share
Main City Sample and 2011 Prices			
Agro-similarity	-5.086***	0.158**	6.595***
	(1.845)	(0.062)	(1.513)
Cities	134	134	134
Cities With Observed Prices in 1996			
Agro-similarity	-3.854	0.071	5.726**
	(2.847)	(0.079)	(2.228)
Cities	112	112	112
Cities With Observed Prices in 2002			
Agro-similarity	-2.157	0.097	3.865
	(2.774)	(0.070)	(2.445)
Cities	109	109	109

Notes: The first panel uses the main city sample, which is also the observed-price sample in 2011. The other panels retain main-sample cities with the raw-food price index observed in the indicated wave. Exposure and outcome scales remain fixed. These are coverage sensitivities; differences across samples do not identify an effect of the import restrictions. Agro-similarity is standardized over the full city file before any sample restriction. All regressions include the settler count within 300 travel minutes, the 1980 population, employment, urban population share and agricultural employment, and sixteen agro-climatic controls. Agricultural and modern employment shares are in percentage points; modern employment is ln(count+1). Modern employment combines manufacturing and services. Sources: population censuses, 1980 and 2010; household survey (Susenas), 1996, 2002 and 2011. Conley standard errors within 200km are in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table E10: Employment Responses by Openness on the Observed-Price Samples

	Agricultural Share	Log Modern Jobs	Modern Share
Main City Sample and 2011 Prices			
More closed	-6.498***	0.269***	9.270***
	(2.401)	(0.089)	(2.157)
More open	-4.358**	0.099	5.244***
	(2.087)	(0.070)	(1.596)
More open minus more closed	2.139	-0.169*	-4.026*
	(2.539)	(0.095)	(2.149)
Cities	134	134	134
More-closed	67	67	67
More-open	67	67	67
Cities With Observed Prices in 1996			
More closed	-5.994**	0.190**	8.278***
	(2.738)	(0.079)	(2.481)
More open	-1.308	-0.098	2.840
	(3.405)	(0.093)	(2.621)
More open minus more closed	4.687	-0.288***	-5.438**
	(3.105)	(0.111)	(2.580)
Cities	112	112	112
More-closed	62	62	62
More-open	50	50	50
Cities With Observed Prices in 2002			
More closed	-4.228*	0.232**	7.131***
	(2.479)	(0.096)	(2.671)
More open	0.080	0.007	1.204
	(3.174)	(0.098)	(2.634)
More open minus more closed	4.308	-0.226**	-5.927**
	(2.754)	(0.103)	(2.433)
Cities	109	109	109
More-closed	62	62	62
More-open	47	47	47

Notes: The first panel uses the main city sample, which is also the observed-price sample in 2011. The other panels retain main-sample cities with the raw-food price index observed in the indicated wave. Exposure and outcome scales remain fixed. These are coverage sensitivities; differences across samples do not identify an effect of the import restrictions. The more open cities have route hours below the median among all cities within 300 travel minutes of a settlement; this division is fixed before restricting to measured prices. Both group slopes are estimated in one regression with a group indicator and common controls. The difference is more open minus more closed, using their covariance. Agro-similarity is standardized over the full city file before any sample restriction. All regressions include the settler count within 300 travel minutes, the 1980 population, employment, urban population share and agricultural employment, and sixteen agro-climatic controls. Agricultural and modern employment shares are in percentage points; modern employment is ln(count+1). Modern employment combines manufacturing and services. Sources: population censuses, 1980 and 2010; household survey (Susenas), 1996, 2002 and 2011. Conley standard errors within 200km are in parentheses. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

E.2 The In-Migration Required for the Agricultural-Share Fall

Cited in the main text. This section: Section 5.2, p. 35. Table E15: Section 5.2, p. 35. Table E14: Section 5.2, p. 35. Table E11: Section 5.2, p. 32. Table E12: Section 5.2, p. 33.

Direct Recent Migration. The five-year residence question measures recent arrivals among the residents observed in 2010. The province and district shares increase with exposure on the main sample (Table E11). They provide direct evidence of recent moves, while the employment-based calculations below ask how much adjustment additional workers could explain under restrictions on the incumbent workforce.

Table E11: Recent Migration Among City Residents

	Across provinces	Across districts
Agro-similarity (per SD)	+0.52**	+1.92***
	(0.26)	(0.49)
Main controls	Yes	Yes
Cities	134	134

Notes: Each column regresses the share of 2010 residents who lived in another province or district five years earlier on standardized agro-similarity, using the main city sample and the full main controls: settler count, 1980 population, employment, urban population share and agricultural employment, and the sixteen agro-climatic covariates. Coefficients and standard errors are in percentage points per standard deviation of exposure. Conley standard errors within 200km are in parentheses. With Bartlett distance weights, the province response is marginally significant and the district response remains significant. District moves include moves across provinces. These shares describe recent arrivals among current residents and do not measure departures, all gross moves, earlier migration, or individual sector switching. Source: 2010 population census. * p<0.10, ** p<0.05, *** p<0.01.

Recent Migration by Openness. Province and district moves have larger exposure responses in the more closed cities, with imprecise direct differences (Table E12). The worker comparison uses all workers with a recorded prior district and includes return migrants. It uses the same current fractional subdistrict allocation as the main employment outcomes, and its employment totals agree city by city. The lifetime-migrant recency partition used in the decomposition excludes return migrants from its recent-arrival category. The resident and worker shares therefore answer different questions. They do not identify individual sector changes.

The identity. A city has L workers in 2010, a share s of them in agriculture. Suppose it takes in M extra workers, a share a of whom also work in agriculture, while the incumbent workforce and its sector composition stay fixed. Its agricultural share becomes $(sL + aM)/(L + M)$. Writing the gross inflow rate as $g = M/L$, the fall in the share is $g(s - a)/(1 + g)$, so producing a fall of Δ requires

$$g(a) \;=\; \frac{\Delta}{s - a - \Delta}, \tag{20}$$

and a log employment response of $\ln(1+g(a))$. Equation 20 has two properties the test relies on. The requirement rises steeply in a, because arrivals who farm dilute the share less. And it is infinite once $a \geq s - \Delta$: an inflow with an agricultural share that high cannot pull the average down that far at any size.

The base s is the comparison city's *2010* share, 0.374 on the estimation sample. Both coefficients are cross-sectional 2010 differentials, so the share the extra arrivals dilute is the share the comparison city has when they arrive. Using the 1980 share instead would attribute the common 1980-2010 structural change to the inflow as well. The 1980 base is roughly 20 percentage points higher and lowers the requirement by more than a factor of three; I report it throughout as a conservative alternative.

Table E12: Recent Migration by Trade Openness

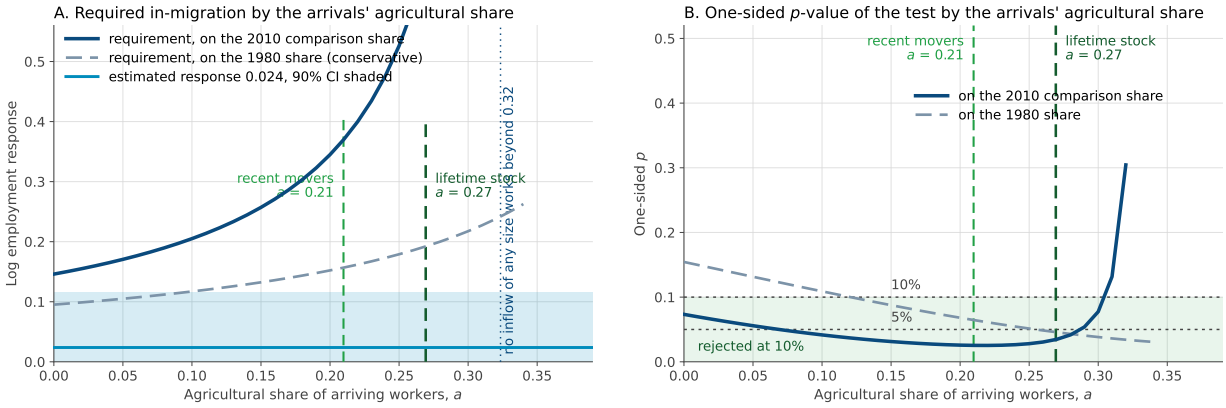
	Residents: Province	Residents: District	Workers: District
Main Spatial Inference			
More closed	0.86***	2.28***	2.35***
	(0.33)	(0.42)	(0.45)
More open	0.34	1.71***	1.64**
	(0.28)	(0.65)	(0.71)
More open minus more closed	-0.52	-0.58	-0.72
	(0.34)	(0.61)	(0.65)
Cities	134	134	134
More closed	67	67	67
More open	67	67	67
Bartlett Distance Weights			
More closed	0.86**	2.28***	2.35***
	(0.36)	(0.60)	(0.69)
More open	0.34	1.71**	1.64*
	(0.31)	(0.79)	(0.85)
More open minus more closed	-0.52	-0.58	-0.72
	(0.37)	(0.74)	(0.82)
Cities	134	134	134
More closed	67	67	67
More open	67	67	67

Notes: The first two columns regress the share of 2010 residents who lived in another province or district five years earlier on standardized agro-similarity interacted with the fixed openness groups, their group indicator and the full main controls. Shares are in percentage points. The more open cities have route hours below the median among all cities within 300 travel minutes of a settlement; this division is fixed before restricting to measured prices. Both group slopes are estimated in one regression with a group indicator and common controls. The difference is more open minus more closed, using their covariance. The sample is the main city sample in both panels. Standard errors in parentheses allow spatial correlation within 200km, with uniform weights in the first panel and Bartlett weights in the second. The third column uses workers with an observed district of residence five years earlier as its denominator; it includes return migrants. Prior district is observed for at least 94.90% of workers in each city; 1,901 workers have an unrecorded prior district. Current fractional subdistrict membership reproduces each city employment total. District moves include province moves. The measures describe arrivals among current residents and do not identify departures, moves within districts, earlier migration or individual sector changes. Source: 2010 population census. * p<0.10, ** p<0.05, *** p<0.01.

Measuring the arrivals' sector mix. No census aggregate crosses migrant status with industry. I therefore build that cross from the individual records of the 1980 and 2010 censuses, aggregating person to village, then to year-2000 subdistrict through the harmonized crosswalk, then to the reclassified urban centers. Recomputing each city's overall agricultural share from these person records matches the census share for the median city (correlation 0.999).

The test. Both Δ and the employment response are estimated, so the threshold is itself a random variable and the test needs their joint covariance under the same spatial kernel. I therefore stack the two equations: each city contributes two rows, one for log employment and one for the agricultural share, every regressor is interacted with an equation dummy, and one Conley regression is run on the stack. The two rows of a city sit at distance zero, so the kernel gives the cross-equation covariance directly. The stacked fit is saturated and reproduces the two separate regressions exactly. The delta method then propagates the covariance through Equation 20. The resulting standard error on the gap is 0.351, against 0.057 on the employment response, so the vertical distance between the two curves in Figure E3 understates the uncertainty.

Figure E3: Required In-Migration by the Arrivals' Agricultural Share, and the One-Sided Test



Notes: Log employment and the agricultural employment share in the 2010 census are each regressed on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km; inference uses the joint covariance of the two coefficients. The horizontal axis is a, the agricultural share of the arriving workers; solid curves take the 2010 mean agricultural share s as the base, and dashed curves the 1980 mean. Left panel: the curves are $\log(1+g(a))$, the log employment response Equation 20 requires; the flat line and shaded band are the estimated response to a one-standard-deviation rise in agro-similarity and its 90% confidence interval; the dotted line marks the asymptote $a = s - \Delta$, where Δ is the estimated fall in the agricultural share per standard deviation of agro-similarity, and beyond it an inflow of any size fails to produce that fall. Right panel: the one-sided p-value at each a, with reference lines at 10% and 5%. In both panels the dashed vertical lines mark the 2010 agricultural shares of recent movers (moved district within five years) and of the lifetime stock (born outside the district). The one-sided test rejects at 10% for every a up to 0.30 and at 5% from 0.07 to 0.28.

The calculation at the average city. A city with the sample's mean workforce, 91,751 workers with an industry code, and its mean agricultural share has 34,345 of them in agriculture. The high-exposure city has 2,210 more workers and 30,393 agricultural workers. If net employment growth equals arrivals, with the incumbent workforce fixed, and those arrivals farm at the measured 26.9%, they add 595 farmers, so agricultural employment would total about 34,940. It is at 30,393, a gap of roughly 4,547 workers relative to that calculation. Under those assumptions, 86,140 arrivals would be required, against net employment growth of 2,210, a shortfall of about 83,930 workers on a workforce of 91,751.

The modern-job accounting. Exposure adds modern-sector jobs to a city faster than it adds workers, and I test whether the workers a city gains could have filled those jobs with nobody already in the city changing sector. At the mean city, one standard deviation of exposure adds about 9,100 modern-sector jobs and 2,200 workers, the coefficients of Table 3 taken at the 2010 means. I count how many of the new jobs the workers added could fill, first if every one of them took a modern-sector job and then if they took such jobs at the rate the census measures for migrants, and I attach to each share a one-sided upper bound that uses the joint precision of the two coefficients, since both come from the same cities. The workers added fill about a quarter of the gain, 24%, and at most 82% at 95% confidence; at the migrants' rate they fill 16%, at most 54%. In the main specification, the test rejects that the net addition of workers fills the whole modern-job gain. This is an arrivals calculation with the incumbent workforce fixed and without offsetting departures or labor-force entry and exit. The checks below report how the conclusion varies with the variance estimator, job conversion, and design-based specification.

The share of the modern-job gain. Each log coefficient of Table 3 is approximately a proportional change, so at the mean city it is also a change in people: a coefficient b on an outcome with 2010 mean $\bar{X}$ moves that outcome by $\bar{X}(e^b - 1)$. On modern-sector employment (mean 53,037) the coefficient gives 9,100 jobs; on employment (mean 90,259) it gives 2,200 workers. If nobody already working in the city changes sector, only the workers added can fill the new modern-sector jobs, and the share they fill is the workers added, times the fraction of them in modern-sector work, divided by the jobs gained: $\phi = \mu\bar{L}(e^{b_L} - 1)/[\bar{M}(e^{b_M} - 1)]$, with $\bar{L}$ and $\bar{M}$ the two means, b_L and b_M the two coefficients and μ the fraction. With every worker added in a modern-sector job ($\mu = 1$) the share is 24%; at the rate at which workers born outside their district of residence hold modern-sector jobs in the 2010 census, 65%, it is 16%. The share is uncertain because the two coefficients are, and its numerator, the employment response, is the imprecise one, so a standard error of the ratio itself (50 percentage points by the delta method) is a poor guide. I therefore test each candidate share directly, asking whether the data reject that the workers added fill that much of the gain. The two coefficients move together across cities (correlation 0.82 in one regression that stacks the two outcomes, as in the test above), and the one-sided test uses their joint covariance. The largest share the test does not reject at 5% is the one-sided 95% upper bound: 82% with every worker added in a modern-sector job and 54% at the migrants' rate. A share of 100%, the workers added filling every new job, is rejected in both cases ($p = 0.012$ and $p < 0.001$). The net modern-job gain exceeds net employment growth by 6,900, with a 90% interval of 1,900 to 11,900. Repeated cross-sections do not identify individual moves or occupational transitions.

Table E13: The Share of the Modern-Job Gain by Variance Estimator and Job Conversion

		One-sided 95% upper bound				One-sided p, share of 100% or more			
Route	Share	Uniform	Bartlett	City	Province	Uniform	Bartlett	City	Province
Workers, every one in a modern job (headline)	24%	82%	79%	86%	93%	0.012	0.009	0.020	0.033
Workers, modern rate of migrants, employment-weighted	16%	54%	52%	56%	61%	<0.001	<0.001	0.001	0.004
Workers, modern rate of migrants, unweighted	15%	51%	49%	54%	58%	<0.001	<0.001	0.001	0.004
All workers with an industry code, every one modern	24%	84%	81%	87%	95%	0.014	0.010	0.022	0.037
Comparison city at the mean shares	27%	94%	90%	98%	106%	0.033	0.025	0.045	0.067
Share route, at the mean shares	30%	95%	92%	98%	106%	0.033	0.026	0.043	0.072
Residents aged 15 and over at the city rate of employment, every one modern	52%	104%	104%	110%	118%	0.067	0.066	0.088	0.123
Residents aged 15 and over, every one a worker in a modern job	92%	184%	185%	194%	210%	0.459	0.457	0.460	0.468

Notes: The main city sample, in the main specification with the full set of controls. Each row is the share of the modern-job gain that the people added could fill if nobody already working in the city changed sector, its one-sided 95% upper bound from inverting a one-sided test on the joint covariance of the two coefficients, and the one-sided p-value of a share of 100% or more. The headline row takes columns 2 and 6 of Table 3 at their mean levels with every worker added in a modern job; the next two use the rate at which migrants, workers born outside their district of residence, work in manufacturing or services; the fourth counts all workers with an industry code; the fifth and sixth put the comparison city at the mean shares, the fifth with the coefficient on log modern employment and the sixth with the coefficient on the modern share; the last two count residents. Variances: Conley SE within 200km with a uniform kernel (the paper's) and a Bartlett kernel, and errors clustered by city and by province, all on one stacked regression of the main specification. Source: 1980 and 2010 population censuses.

Variance estimators and job conversions. The strength of the rejection varies with the standard errors and the way jobs are counted. Under the paper's uniform Conley kernel, a Bartlett kernel and errors clustered by city, every row of Table E13 that counts workers rejects a share of 100% at 5%. Province clusters, the loosest of the four, still reject in the headline row ($p = 0.033$, bound 93%); only the two rows that put the comparison city at the mean of the

cities' shares do not reject at 5% ($p = 0.067$ and 0.072, bounds 106% and 106%), because larger cities are more modern, so the mean of the cities' modern shares is below the modern share of the mean city, and those two rows put fewer modern-sector jobs in the denominator. Counting residents aged 15 and over in place of workers weakens the comparison, because residents aged 15 and over grow by more than workers and only the working residents among them fill jobs. At the city's 0.563 workers per resident aged 15 and over, the residents added fill 52% of the gain, with a bound of 104%. The accounting counts workers because a job is filled by a worker.

Two moments. City size and the employment shares are two moments, and one inflow has to fit both. With net employment growth equated to arrivals and the incumbent workforce fixed, arrivals imply both the agricultural share and the modern share of the more exposed city. Each implied share is a restriction that holds when the estimate is consistent with it, and the one-sided Wald test of the two restrictions together, whose null distribution is a mixture of chi-squared distributions with weights set by the correlation of the two restrictions (Kodde & Palm 1986; Wolak 1989), rejects the hypothesis that arrivals with no switching produced both moments at 1% under each of the four variances, with p at most 0.002 (under province clusters). Testing the two shares jointly removes the concern that a one-moment test picks the share that happens to reject.

Table E14: The Bound in Workers Under the Design-Based Specifications

		One-sided 95% upper bound				One-sided p, share of 100% or more			
Specification	Share	Uniform	Bartlett	City	Province	Uniform	Bartlett	City	Province
Panel A. Every worker added takes a modern job									
Main specification	24%	82%	79%	86%	93%	0.012	0.009	0.020	0.033
Double-robust, whole-match expectation	35%	133%	137%	none	none	0.167	0.162	0.157	0.256
Double-robust, origin expectation	*log modern employment falls (−0.097), so there is no gain to fill*								
Recentered, whole-match	52%	none	none	none	none	0.359	0.359	0.353	0.410
Recentered, origin	*log modern employment falls (−0.012), so there is no gain to fill*								
Panel B. Workers added take modern jobs as often as migrants do									
Main specification	16%	54%	52%	56%	61%	<0.001	<0.001	0.001	0.004
Double-robust, whole-match expectation	23%	87%	89%	none	none	0.021	0.029	0.032	0.088
Double-robust, origin expectation	*log modern employment falls (−0.097), so there is no gain to fill*								
Recentered, whole-match	34%	none	none	none	none	0.167	0.209	0.207	0.291
Recentered, origin	*log modern employment falls (−0.012), so there is no gain to fill*								

Notes: The main city sample, with the main specification controls. Rows and columns as in Table E13, for the two routes in workers, under five specifications: the main specification; the two double-robust specifications of Appendix C.1, which add as a control the exposure a city would expect under counterfactual settler assignments, from the whole-match reshuffle or the origin re-draw; and the two recentered ones, which subtract that expected exposure from the treatment. The mean levels of employment and modern employment are the main specification's in every row. An upper bound of none means the bound is not finite. Source: 1980 and 2010 population censuses.

The design-based specifications. The bound needs precise coefficients, and the design-based specifications of Appendix C.1 give up precision by construction, since they add the exposure a city would expect under counterfactual settler assignments as a control, or subtract it from the treatment, and either way less of the exposure's variation is left to estimate the two responses. Table E14 repeats the two workers rows of Table E13 under the four design-based specifications. When every additional worker takes a modern-sector job, only the main specification rejects. With the whole-match expectation as a control the workers added could fill 35% of the gain and the bound passes 100% (133%, $p = 0.167$); at the migrants' rate the test still rejects at 5% under three of the four variances ($p = 0.021$ under the uniform kernel, bound 87%). With the recentered whole-match treatment the share is 52% with no finite upper bound ($p = 0.359$). In the two origin specifications log modern-sector employment falls, so there is no gain to fill, and the table shows the

coefficient in place of a share. These rows show changes in both estimates and precision; the point estimate of the share stays below 100% of the gain in every specification that has one.

Birthplace and age composition. The identity treats the extra workers as people who moved in. Transmigrant placement ran through the early 1980s, so a child born to an incumbent in that decade is a prime-age worker by 2010, and a fertility response would place locally born workers in the inflow. Table E15 reports statistically insignificant responses. Splitting 2010 employment by birthplace, native-born employment does not rise detectably, and the point estimate is concentrated among workers born elsewhere, whose employment rises by 20.8% while the migrant share of the workforce rises by 3.6 percentage points; neither is distinguishable from zero. The responses of children and youths per native-born adult are also statistically insignificant. These proxies do not rule out fertility responses or labor-force entry and exit. Both ratios are computed inside the native-born population, so adult in-migrants do not inflate the numerator or the denominator.

Table E15: In-Migration and Births as Sources of the Extra Workers

Outcome, 2010	Coefficient	SE	t
Panel A. Where the employment response comes from			
Log employment, all workers	0.0248	0.0582	0.43
Log employment, born in this district	-0.0369	0.0375	-0.99
Log employment, born elsewhere	0.1890	0.1307	1.45
Migrant share of employment	0.0359	0.0260	1.38
Panel B. Fertility, measured inside the native-born population			
Children aged 0 to 5 per native-born adult aged 16 to 50	0.0314	0.0253	1.24
Youth aged 5 to 16 per native-born adult aged 16 to 50	0.0299	0.0282	1.06

Notes: Each row is one regression of a 2010 population census outcome on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. The columns are the coefficient, its standard error and their ratio. Native-born means born in the district of residence and born elsewhere means born outside it; Panel A counts employed workers of the person-level census cross that measures the sector mix of arrivals, in logs (so its first row differs slightly from column 2 of Table 3), and the migrant share is the born-elsewhere fraction of employment. Panel B's ratios are computed within the native-born population. SE is the Conley standard error and t the coefficient divided by it.

Scope of the bound. The bound asks whether the *arrivals'* composition could produce the fall, holding the incumbent workforce fixed. The birthplace decomposition of Appendix E.3 measures a wider object, because its composition component also allows the locally born share of the workforce to fall. The two quantities are therefore not nested, and both are small: the bound puts the arrivals-only channel at 4.9% of the fall using the lifetime stock and 7.6% using five-year movers, against 16% measured directly.

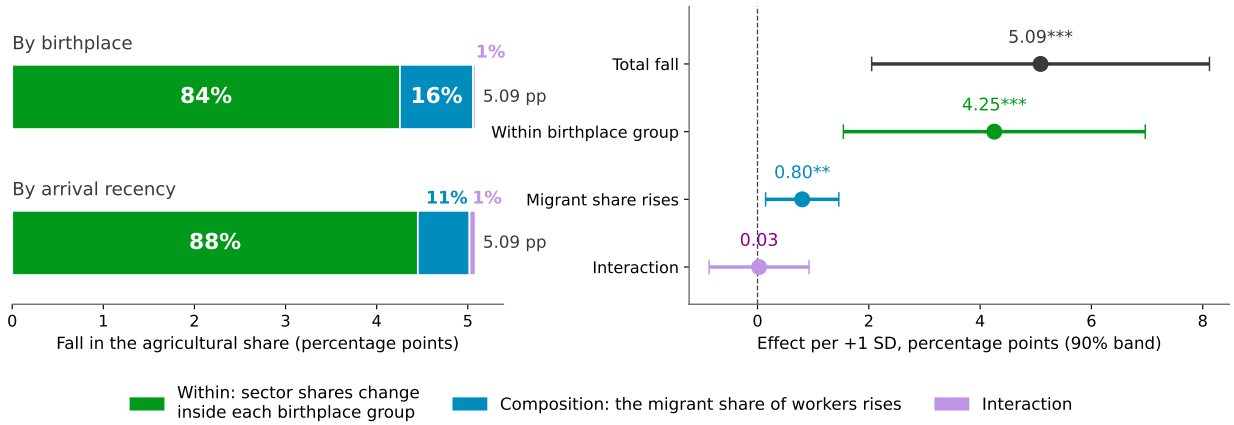
Two limitations remain. The requirement is computed on gross inflow with no offsetting outflow. A small net population response can conceal arrivals of modern-sector workers and departures of agricultural workers, producing sectoral reallocation without incumbent workers changing occupations. The native-employment coefficient in Table E15 does not identify those gross flows. And s and a enter as measured constants. They are means over the same 134 cities and are estimated far more precisely than the coefficients, but they contain sampling error that the reported standard errors omit.

E.3 Decomposing the Agricultural-Share Fall: Construction

Cited in the main text. Figure E4: Section 5.2, p. 35.

This appendix gives the construction behind Figure E4: the identity the decomposition relies on, the data it needs, the calculation it replaces, and the checks at each step. The exhibits label the two components by what they measure, the migrant share of the workforce and workers switching sector; in the text I use the literature's terms, composition and within, for the same two objects.

Figure E4: Decomposition of the Fall in the Urban Agricultural Employment Share, by Origin



Notes: The figure uses the main city sample in the 2010 population census; each effect is estimated by a regression across urban centers on standardized agro-similarity in the main specification with the full set of controls and Conley SE within 200km. Intervals are 90% confidence intervals from the same standard errors; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Each city's 2010 workforce is partitioned by birthplace (four groups) and, separately, by how recently a worker arrived (three groups). The stacked bars split the estimated fall in the agricultural employment share (percentage points per standard deviation, each piece labeled with its percentage of the fall) into a within component ("within birthplace group"), a composition component ("migrant share rises") and their interaction; the right half repeats the total and the three birthplace pieces with their intervals. Because the census is a repeated cross-section, the within component includes cohort replacement.

The identity

Index cities by c and partition each city's 2010 workforce into origin groups g. Let ω_{gc} be group g's share of city c's workers and a_{gc} that group's own agricultural employment share. The city agricultural share is then an exact accounting identity:

$$s_c \;=\; \sum_g \omega_{gc}\, a_{gc}. \tag{21}$$

Write each factor as its sample mean plus a deviation, $\omega_{gc} = \bar{\omega}_g + u_{gc}$ and $a_{gc} = \bar{a}_g + v_{gc}$, and Equation 21 becomes

$$s_c \;=\; \underbrace{\sum_g \bar{\omega}_g \bar{a}_g}_{\text{constant}} \;+\; \underbrace{\sum_g \bar{a}_g\, u_{gc}}_{\text{migrant share}} \;+\; \underbrace{\sum_g \bar{\omega}_g\, v_{gc}}_{\text{within origin}} \;+\; \underbrace{\sum_g u_{gc} v_{gc}}_{\text{interaction}}. \tag{22}$$

The second term is the *composition* component: it moves only because the local-versus-migrant makeup of the workforce changes, holding what each group does fixed. The third is the *within* component. It moves only because groups change what they do, holding that makeup fixed. I construct the three right-hand terms as variables and regress each on the *same* main specification used for the total. Because Equation 22 is an identity and the regressions share a design matrix, the three coefficients sum to the coefficient on s_c exactly, so no residual category absorbs the difference. Each component nonetheless has its own Conley standard error, so the split reports precision as well as magnitude.

Two partitions are used. The first is birthplace, in the four mutually exclusive groups of Figure 11 and Panel B of Table C3: born in the current district, born in a different district of the same province, born in a different outer-island province, and born on Java or Bali. The second is arrival recency, in three groups: never left the birth district, lifetime migrant already present five years before the census, and lifetime migrant who arrived within five years. The second is the sharper measure of the marginal inflow, since the birthplace groups pool arrivals from thirty years ago with last year's.

The interpretation comes with one caveat. The census is a repeated cross-section, so "born in this district" in 2010 is not the same list of people as in 1980. The within term measures sector composition among workers of a given origin, including whoever replaced whoever left. The within term therefore holds both sector changes inside a cohort and the replacement of one cohort by another; Budí-Ors (2026) puts cohort replacement at 68% of Indonesia's aggregate agricultural-share fall.

The data the identity needs

Evaluating Equation 21 requires each city's workers classified by origin *and* industry in the same data. The city-level census data hold the birthplace shares and, separately, the industry counts, but nothing crossing them, so the cross is built from the 2010 individual census records for the outer islands (93.5 million persons). Three definitional choices align it with the published outcome:

- The employment base is every person with a valid industry code (40.1 million workers), which is the denominator every sectoral share in this paper uses, rather than the stricter economically active count. Using the strict base would produce shares that do not sum to those of the main results
- Agriculture is industry codes 1 through 6, as everywhere else in the paper
- No birth-cohort restriction is imposed. The 2010 birthplace shares of Figure 11 and Table C3 restrict to the 1955-1990 cohort in order to match 1980 birth-district characteristics; imposing that here would place the cross on a different population than the published agricultural share and break the reconciliation below. Birthplace is recorded for 100% of the workers in the base, so no group has a thin denominator

The person-level counts are collapsed to the 2010 village, crosswalked to the year-2000 subdistrict geography the rest of the paper uses, and rolled up to urban centers over the reclassified membership. Both partitions recover the total workforce exactly in every one of the 1,999 subdistricts, and Equation 21 holds in every city.

The re-measured share is compared with the agricultural share of Table 3 before anything is split. Its mean across the estimation sample matches to three decimals (0.374 against 0.374), the two measures correlate at 1.000, and the main specification coefficient on the re-measured share, -0.0509 (standard error 0.0184), reproduces that table's -0.0509 (0.0184) on the same cities.

Subdistricts claimed by two urban centers

Urban centers are satellite-drawn built-up polygons while the census reports by subdistrict, and five outer-island subdistricts are large enough to contain two separate built-up areas. The sample records both claims, and city totals built by summing constituent subdistricts therefore give the whole subdistrict to each of the two claimants, so those workers enter two cities. Across the estimation sample this affects 95,728 workers, or 0.78% of the total, and both claimants lie inside the sample in two of the five cases.

The exhibits here divide such a subdistrict. Each of its villages is assigned to the claimant whose built-up polygon covers it, or, for a village inside neither, to the nearer claimant; the subdistrict's counts are then split in proportion to the employment in each city's villages. Total mass is preserved by construction and checked against the village-level source, so no worker is counted twice and none is dropped. With the subdistricts divided, the within share is 83.7%, and leaving out the 5 cities that draw part of their workforce from a shared subdistrict gives 83.7%.

E.4 Public Provision as an Alternative Explanation

Cited in the main text. This section: Section 5.3, p. 40.

The paper argues that agricultural productivity raised nearby city growth and reshaped urban employment through the goods market. The Indonesian state settled these villages, and the same state built the roads, schools, health posts and electricity lines that serve them. If public provision followed the settlements into the places where they turned out productive, public provision could contribute to the urban response. This subsection examines road quality, village facilities, and the allocation of the INPRES program.

A facility stock is the outcome of many decisions, and a connection can follow demand that is already there, so these tests cannot tell the state's supply apart from household demand. The INPRES test below examines a particular central-government program's allocation.

Roads. The trade channel runs on the road network, so the sharpest form of the objection is that exposed corridors received better roads. I test this with the Indonesian Road Management System panel assembled by Gertler et al. (2024), which gives road-quality-implied travel speeds on a fixed network in 1990 and 2007 for Sumatra and Sulawesi, the two outer islands the source covers. Each subdistrict takes the speed of its nearest 1990 district, and I average the speed over three areas of each city, weighting subdistricts by their 1990 population: the city itself, its countryside with settlements and its countryside without settlements (103, 103 and 97 cities). In 1990 road speed did not differ detectably with exposure in any of the three areas, at −0.46 km/h per standard deviation of exposure in the city (standard error 1.15), −0.94 km/h in the countryside with settlements (0.90) and −0.09 km/h in the countryside without settlements (0.96), against a mean of 60.5 km/h across rural subdistricts. The countryside with settlements sat farther out in the road network, by 39.4 km (standard error 22.5), marginally significant under both kernels and insignificant with province clusters. Holding the 1990 speed fixed, the 2007 speed differs by −0.92 km/h in the countryside with settlements (standard error 0.36) and by −0.73 km/h in the countryside without settlements (0.30), both statistically significant, and by −0.73 km/h in the city (0.50), statistically insignificant, against a mean of 62.3 km/h across rural subdistricts in 2007. Roads in the countryside of more exposed cities therefore improved less, and road quality on this network cannot account for the transformation of those cities. With controls for the 1983 baseline of road access in the village census, roads passable by a four-wheel vehicle, paved roads and distance to the subdistrict seat, which shows some imbalance in favor of the more exposed places, the two countryside estimates are −0.75 km/h (standard error 0.36) and −0.69 km/h (0.27). With the rural subdistrict as the unit, the same specification gives −0.47 km/h (standard error 0.43) over 576 subdistricts, statistically insignificant, also with standard errors clustered by 1990 district (85 districts), the level at which the speed is measured (Table E16).

The test has two limits. Network distance is identical to the last digit across 1990 and 2007 for every Sumatran and Sulawesi district in the source, so the measure re-weights a fixed graph by road quality and cannot detect a road that was newly built. It covers Sumatra and Sulawesi.

Facilities. I track what the state puts inside a village, city amenities from the 1980 village census to the 2003 and 2011 waves. Column (2) regresses the 1980 level itself on exposure, so the table shows whether public provision already differed before the settlements matured, and columns (3) and (4) give the change over each window with the amenity's own 1980 level held fixed (Appendix Table E17). Facility counts are per 1,000 people at the population of each year.

Two facts come out of it. The 1980 levels differ in three of the seven government-provided amenities. Senior-secondary schools and polyclinics show a 1980 gradient, at 0.0149 (standard error 0.0042) and 0.0151 (standard error 0.0051) per 1,000 people and per standard deviation of exposure, against means of 0.072 and 0.073, and the population share on a road passable by a four-wheel vehicle is higher by 5.2 percentage points (standard error 3.0), marginally significant, against a mean of 80%. The exposed cities entered the sample period with slightly more of two facilities the state provides. With each amenity's own 1980 level held fixed, the changes are statistically insignificant over both windows in schooling, in health centers and in both road-surface measures.

Table E16: Road Speed and Exposure in the City and Its Countryside

	City	Countryside with settlements	Countryside without settlements	Rural subdistricts
	(1)	(2)	(3)	(4)
Road speed 2007, 1990 speed held fixed	−0.73	−0.92***	−0.73**	−0.47
	(0.50)	(0.36)	(0.30)	(0.43)
p-value	0.143	0.009	0.015	0.273
N	103	103	97	576
with the 1983 road measures	−0.67	−0.75**†	−0.69**	−0.45
	(0.54)	(0.36)	(0.27)	(0.41)
p-value	0.218	0.038	0.011	0.279
N	103	103	97	568
Road speed 1990	−0.46	−0.94	−0.09	0.25
	(1.15)	(0.90)	(0.96)	(0.79)
p-value	0.688	0.301	0.926	0.755
N	103	103	97	576
Network distance 1990 (km)	47.1	39.4*†	27.1	41.3
	(33.5)	(22.5)	(27.9)	(28.8)
p-value	0.160	0.080	0.331	0.152
N	103	103	97	576
Change in road speed, 1990 to 2007	−0.35	−0.18	−0.66	−0.66
	(1.06)	(0.77)	(0.74)	(0.86)
p-value	0.738	0.820	0.372	0.442
N	103	103	97	576

Notes: Each cell is one regression of the outcome on standardized agro-similarity in the main specification with the full set of controls and Conley SE within 200km; the first two blocks hold the area's 1990 speed fixed, and the second adds the three 1983 village-census road measures (roads passable by a four-wheel vehicle, paved roads, distance to the subdistrict seat). Speeds are km/h on a fixed network from the Indonesian Road Management System panel of Gertler et al. (2024), Sumatra and Sulawesi; each subdistrict takes the speed of its nearest 1990 district, and an area averages its subdistricts by 1990 population. Columns (1) to (3) are the cities of the main city sample on the two islands; column (4) is the rural subdistricts of the paper's rural tables, with the settler control of those tables and exposure standardized over them. Conley SE (uniform kernel) in parentheses and the uniform-kernel p-value beneath, shaded by band (0.10 or more, below 0.10, below 0.05, below 0.01). The stars use the uniform-kernel SE: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. † significant at 10% with the uniform kernel and not significant at 10% with the Bartlett kernel, with province clusters or, in column (4), with 1990-district clusters.

Polyclinics are the one exception, positive and significant by 2011, at 0.0139 (standard error 0.0060) against a 1980 mean of 0.073, an estimate that stays significant with the Bartlett kernel and marginally significant with province clusters. A facility stock does not identify its financing.

Of the 21 coefficients in the government-provided block, only the three 1980 levels and the polyclinic change to 2011 cross a conventional threshold. Without the 1980 level, all 14 changes are statistically insignificant.

Electricity. Grid connection is where the change regressions do move when the 1980 level is left out, and the two margins of it separate cleanly once that level is held fixed. On the extensive margin, without the 1980 level, the share of a city's population living in a village on the state grid falls by 6.4 percentage points per standard deviation of exposure over 1980 to 2003 and by 7.2 percentage points over 1980 to 2011, both statistically significant. That is the less exposed cities catching up. Agriculturally favored places began 10.9 percentage points more connected, a statistically significant difference, so a change regression with no lagged outcome was measuring convergence, and holding the 1980 level fixed, as columns (3) and (4) do, removes the result in both windows, leaving −0.1 and −0.4 percentage points. The 1980 measure is the narrower of the two. The 1980 form records only a village's main source of electricity, so a 1980 village counts as on the grid only where the state utility is its main supplier, while a 2003 or 2011 village counts where any household takes grid electricity.

The intensive margin behaves differently and is the only electricity result that survives the 1980 level. The share of households connected also starts higher in favored places, at 5.8 percentage points per standard deviation with a

Table E17: Public Provision and Agricultural Productivity: City Amenities from the 1980 Baseline

	1980 baseline		Change, 1980 level held fixed	
	Mean in 1980	1980 level on exposure	1980 to 2003	1980 to 2011
	(1)	(2)	(3)	(4)
Panel A. Government-provided: schooling and health, facilities per 1,000 people				
Primary schools	1.159	-0.0150	-0.0156	0.0059
		(0.0787)	(0.0195)	.
		[0.0727]	[0.0298]	[0.0193]
p-value		0.849	0.422	.
Junior-secondary schools	0.144	0.0053	-0.0085	-0.0069
		(0.0110)	(0.0087)	(0.0101)
		[0.0107]	[0.0096]	[0.0088]
p-value		0.628	0.326	0.493
Senior-secondary schools	0.072	0.0149***	0.0008	0.0050
		(0.0042)	(0.0055)	(0.0062)
		[0.0061]	[0.0071]	[0.0064]
p-value		0.000	0.877	0.423
Health centers (puskesmas)	0.081	-0.0001	0.0019	0.0032
		(0.0111)	(0.0016)	(0.0024)
		[0.0106]	[0.0023]	[0.0025]
p-value		0.993	0.219	0.178
Polyclinics	0.073	0.0151***	0.0076	0.0139**
		(0.0051)	(0.0058)	(0.0060)
		[0.0058]	[0.0063]	[0.0065]
p-value		0.003	0.190	0.020
Panel B. Government-provided: road surface, share of population				
Asphalt main road	0.435	-0.0242	-0.0343	0.0141
		(0.0495)	(0.0278)	(0.0289)
		[0.0494]	[0.0283]	[0.0266]
p-value		0.624	0.217	0.625
Passable by four-wheel vehicle	0.797	0.0516*†	0.0010	0.0104
		(0.0300)	(0.0038)	(0.0096)
		[0.0284]	[0.0054]	[0.0109]
p-value		0.085	0.798	0.279
Panel C. Mixed provision: grid electricity				
Villages on the grid, share of population	0.353	0.1091***	-0.0005	-0.0040
		(0.0347)	(0.0128)	(0.0128)
		[0.0377]	[0.0131]	[0.0124]
p-value		0.002	0.968	0.756
Households connected, share of households	0.175	0.0575***	0.0339	0.0449*†
		(0.0181)	(0.0251)	(0.0233)
		[0.0221]	[0.0259]	[0.0249]
p-value		0.002	0.177	0.054
Observations		133	133	133

Notes: Each cell in columns (2) to (4) is one regression of the outcome, from the village census (PODES), on standardized agro-similarity on the cities of the main city sample the census covers in 1980 and in the endline year, in the main specification with the full set of controls and Conley SE within 200km; column (2) leaves out the four 1980 baselines because its outcome is a 1980 value, and columns (3) and (4) add the outcome's own 1980 level. Conley SE (uniform kernel) in parentheses and Bartlett-kernel SE in brackets, with the uniform-kernel p-value beneath, shaded by band (0.10 or more, below 0.10, below 0.05, below 0.01); a dot marks a cell with no uniform-kernel standard error. The stars use the uniform-kernel SE: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. † significant at 10% with the uniform kernel and not significant at 10% with the Bartlett kernel, with province clusters (17 provinces), or with both.

standard error of 1.8, and the plain changes are small and imprecise, 1.1 percentage points (standard error 2.4) by 2003 and 1.9 percentage points (standard error 2.1) by 2011. With the 1980 level held fixed the changes grow to 3.4 percentage points by 2003, statistically insignificant, and 4.5 percentage points by 2011, marginally significant under both kernels and insignificant with province clusters. So the pattern is a modest deepening of connections inside places that were already connected, on a base of about 18% of households connected in 1980.

Electrification reflects both the state's supply and household demand. The grid is a state utility, while the decision to connect a household responds to demand and ability to pay. The connection estimates do not distinguish those explanations.

The allocation rule itself. The sharpest available test examines a central-government program directly. Indonesia's INPRES primary-school construction program ran from 1973 to 1978 and had finished before the 1980 village census was taken (Duflo 2001). That census records INPRES-built schools, INPRES-paid teachers, INPRES-built school buildings and INPRES-built public latrines separately from everything else, so the program is identified by its own label, and the question becomes whether the central government's flagship rural program had already favored agriculturally advantaged places by the start of my sample period.

Table E18: Central Government Programs at the 1980 Baseline

	INPRES primary schooling				INPRES public latrines	Placebo: the schools INPRES did not build
	Share of schools	Schools per 1,000 people	Share of teachers	Share of buildings	Share of latrines	Schools per 1,000 people
	(1)	(2)	(3)	(4)	(5)	(6)
Agro-similarity	-0.0064	-0.0098	-0.0047	-0.0040	0.0206	-0.0023
	(0.0129)	(0.0243)	(0.0115)	(0.0127)	(0.0592)	(0.0658)
p-value	0.623	0.687	0.682	0.752	0.728	0.972
Mean of outcome	0.3355	0.3831	0.3228	0.3384	0.4010	0.7778
Observations	133	133	133	133	120	133

Notes: Each column is one regression of a 1980 level on standardized agro-similarity on the 133 cities of the main city sample that the 1980 village census (PODES) covers, in the main specification with the full set of controls and Conley SE within 200km, without the four 1980 census baselines, since each outcome is itself a 1980 value; the one main-sample city the 1980 village census does not cover is Pemangkat (West Kalimantan); column (5) uses the cities with a public latrine, counted in the Observations row. Standard errors are in parentheses, with the p-value beneath, its shading stepping up at each star threshold. The presidential-instruction (INPRES) primary-school construction program ran from 1973 to 1978, so the 1980 levels show where it had already built. The 1980 village census records INPRES items separately from the rest. Columns (1), (3), (4) and (5) are INPRES shares of the total, column (2) is INPRES schools per 1,000 people, and column (6), the placebo, is the schools the program did not build per 1,000 people. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E18 reports six versions of the test; each estimate is statistically insignificant. The INPRES share of a city's primary schools changes by −0.0064 per standard deviation of exposure (standard error 0.0129) against a mean of 0.336, the INPRES share of primary-school teachers by −0.0047 (standard error 0.0115) against a mean of 0.323, and the INPRES share of school buildings by −0.0040 (standard error 0.0127) against a mean of 0.338, three separate questionnaire items that agree both in their null and in placing about a third of primary schooling under the program. Public latrines give the cleanest form of the contrast, since INPRES and non-INPRES latrines are counted separately within one facility type in one year and one place, and the INPRES share there is 0.0206 (standard error 0.0592) against a mean of 0.401. The placebo is null, with non-INPRES schools per 1,000 people at −0.0023 (standard error 0.0658). One qualification: 13 of the 133 cities have no public latrine in 1980, so the latrine column covers 120 cities.

The road evidence covers road quality on a fixed network in Sumatra and Sulawesi, and the facility evidence covers stocks and connections. These tests do not identify their financing or rule out other public programs. Around the settlements, the state grid extended to more villages on the subdistrict's exposure, and in the settlement villages themselves no provision measure moves on their own agro-similarity (Appendix E.5). Rural public investment remains a possible contributor through food production and trade.

E.5 Tests of the Alternative Channels

Cited in the main text. This section: Section 5.3, p. 41. Table E25: Section 5.3, p. 42. Table E22: Section 5.3, p. 40. Table E21: Section 5.3, p. 40. Table E19: Section 5.3, p. 40.

Table E19: Public Provision in Rural Subdistricts With a Settlement and Roads Around the City

Panel A: Rural subdistricts with a settlement, 2011

	Asphalt main road	Households connected to the grid	Schools per 1,000 people			Health centers per 1,000 people
			Primary	Junior secondary	Senior secondary	
	(1)	(2)	(3)	(4)	(5)	(6)
Agro-similarity	0.0154	0.0115	0.0193	-0.0066	0.0076	-0.0201
	(0.0250)	(0.0217)	(0.0328)	(0.0084)	(0.0075)	(0.0156)
1980 level of the outcome	✓	✓	✓	✓	✓	✓
Subdistrict controls	✓	✓	✓	✓	✓	✓
Outcome mean	0.629	0.651	1.080	0.290	0.129	0.309
Rural subdistricts	644	658	757	780	780	764
with a settlement	168	172	185	191	191	187

Panel B: The city and its catchment

	Log road density on the current map		Road corridors, change 1990 to 2007	
	Catchment	Within 5 km of a settlement village	Roughness	Unpaved share
	(1)	(2)	(3)	(4)
Agro-similarity	-0.0764	-0.1668	0.0576	0.0146
	(0.0894)	(0.1992)	(0.3008)	(0.0202)
1950 road density	✓	✓		
1990 level of the outcome			✓	✓
Main specification controls	✓	✓	✓	✓
Outcome mean	1.664	1.802	-0.991	-0.466
Cities	131	131	132	125

Notes: Panel A: each column regresses a 2011 measure of a rural subdistrict on agro-similarity (per SD over the 787 rural subdistricts of the main sample, 191 of them with a settlement) times an indicator for a settlement village in the subdistrict and times its complement, the indicator, the measure's 1980 level, the settler count, the four 1980 census baselines and the sixteen agro-climatic covariates; the row is the coefficient for subdistricts with a settlement, with Conley SE within 200km in parentheses. The settler count is the number of settlers placed in the subdistrict's own settlement villages where it has one and, where it has none, the settlers of the villages within 300 travel minutes weighted by inverse distance, as for cities. Column 1: the population share living in a village whose main road is asphalt. Column 2: the share of households connected to the state electricity grid among the households of the villages the grid has reached, with the 1980 population and household shares with electricity from the grid and from another supplier held fixed. Columns 3 to 6: facilities per 1,000 people; health centers are community health centers and their branches. Columns 1 and 2 leave out the 71 subdistricts of West Kalimantan, which merged its villages after 1990, and the 51 West Sumatra subdistricts recorded as nagari in 2011, a larger unit; column 2 holds the subdistricts where the grid had reached a village by 2011. Panel B: each column regresses a road measure on standardized agro-similarity on the main city sample (134 cities), in the main specification with the full set of controls and Conley SE within 200km, adding the earlier level its check mark names. Columns 1 and 2: ln(1 + km of road per 100 square km) on the current national and provincial road map, over the rural subdistricts that hold the city's settlement villages within 300 travel minutes (column 1) or the part of them within 5 km of a settlement village (column 2), with the same measure on the 1950 road atlas held fixed. Columns 3 and 4: the change from 1990 to 2007 in roughness (International Roughness Index, meters per km) and in the unpaved share of the road-condition survey posts within 10 km of the straight lines from the city to its settlement villages. A city without a measure leaves its column: 3 cities have no rural subdistrict holding their settlement villages (columns 1 and 2), 2 have no survey post with roughness measured in both years (column 3), and 9 have none with the surface recorded in both years (column 4). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Sources: village censuses (PODES) of 1980 and 2011; population census 1980; the 1950 road atlas, the current national and provincial road network, and the road-condition survey of 1990 and 2007.

This subsection holds the tests behind Section 5.3: public provision by area and by rural subdistrict, the state electricity grid wave by wave, banks, new district seats, the spending of rural households, the road network, and three tests on the cities themselves, the conflict provinces, the exposure of neighboring cities and the city boundary. Every regression is the main specification with the full set of controls and Conley SE within 200km, with each outcome's own 1980 level held fixed where one exists; the Bartlett-kernel standard error appears beside the uniform one, and an estimate statistically significant at the 10% level with the uniform kernel and not with the Bartlett kernel, with province clusters, or with both is marked with a dagger. The INPRES allocation test and the city facilities table are in Appendix E.4, and the city-boundary test in Appendix Table D4.

Public provision in the settlement villages. The electricity, school and health responses to the village's own agro-similarity are imprecise with village geography controls: the population shares with an asphalt main road and in hamlets the state grid has reached, the share of households on the grid, schools per 1,000 people at the three levels and health centers per 1,000 people are statistically insignificant. The share with a main road passable by car is marginally significant under the uniform kernel (Table E21) and becomes imprecise with Bartlett distance weights or province clustering. Without the geography controls, the grid and car-road shares are statistically significant. The villages were settled from 1979 to 1988 and lack comparable pre-settlement provision baselines.

The state electricity grid. The state electricity grid was extended to more of the villages in the more exposed rural subdistricts with a settlement from 1990 on, and households in villages already on the grid did not connect at a higher rate. With the 1980 level measured on the state grid at both ends, the share of people in villages on the grid rises with exposure by 0.0324 (standard error 0.0145) in 1990 and by 0.1048 (0.0221) in 2011, while the household share within those villages changes by 0.0115 (0.0217) in 2011. Electricity from any supplier in 2011 does not move with exposure (-0.0001, standard error 0.0010), so part of the grid's rise replaced other suppliers. A grid line can follow demand that is already there, so the test does not separate the state's supply from household demand (Figure E5).

Public provision by area. At a given 1980 level, the city's roads do not respond to exposure, while the countryside with settlements and the countryside without settlements both gain asphalt roads by 1990 and grid connections by 2011 (the grid gain is marginally significant where the settlements are), and senior secondary schools rise only in the countryside with settlements. The city's asphalt share changes by 0.0271 (standard error 0.0172) to 1990 and 0.0213 (0.0302) to 2011, and its four-wheel access to 1990, 0.0182 (0.0088), is statistically significant with the uniform kernel only. In the countryside with settlements the asphalt share rises by 0.0590 (0.0207) to 1990, the household grid share by 0.0553 (0.0284) to 2011, marginally significant with the Conley kernels and marginally insignificant with province clusters, and senior secondary schools per 1,000 people by 0.0424 (0.0098); in the countryside without settlements the asphalt share rises by 0.0679 (0.0161) and the household grid share by 0.0792 (0.0194), and senior secondary schools change by 0.0030 (0.0037). Health centers per person do not move by 2011 in either countryside, and by 1990 they fall in the countryside with settlements, by -0.0251 per 1,000 people (standard error 0.0127); the city's estimate, 0.0095 (0.0097), is in Table E20 (Table E22).

Public provision by rural subdistrict. With each rural subdistrict as a unit, asphalt roads and household electricity rose with exposure in subdistricts with and without a settlement alike, and the two slopes are statistically indistinguishable: the asphalt share to 1990 rises by 0.0744 with a settlement and 0.0869 without (difference -0.0124, standard error 0.0199), and the household grid share to 2011 by 0.0647 and 0.0520 (difference 0.0127, 0.0172). The senior-school slope is statistically insignificant in both groups, so the senior-school gain of the countryside with settlements shows up in the per-city area only. Four-wheel access to 1990 and the share of people in villages with electricity to 2011 rose more where a settlement sits, the first with the uniform kernel only. In 1980 asphalt roads were already more common in the subdistricts with a settlement, by 0.0473 (0.0185) (Table E23).

Banks. Banks per person and bank presence do not move with exposure in the city at a given 1980 level, and in the countryside with settlements bank presence does not move while banks per person rise, a marginally significant gain. In the city, banks per 1,000 people change by 0.0110 (standard error 0.0092) from 1980 to 2011 and the population share in

Table E20: Alternative Channels in the City and Its Countryside

	The city		The countryside with settlements		The countryside without settlements	
	Estimate	Cities	Estimate	Cities	Estimate	Cities
Panel A: Finance						
Banks	0.0110	133	0.0119*†	131	0.0077***	124
per 1,000 people, 1980 to 2011	(0.0092)	[0.0093]	(0.0072)	[0.0060]	(0.0026)	[0.0034]
Bank presence	0.0251	122	-0.0138	123	0.0185*†	117
population share, 1980 to 2011	(0.0248)	[0.0222]	(0.0137)	[0.0130]	(0.0112)	[0.0118]
Finance and insurance employment	0.0009**	134				
share of employment, 1980 to 2010	(0.0004)	[0.0005]				
Panel B: Public provision						
Asphalt main road	0.0271	131	0.0590***	129	0.0679***	123
population share, 1980 to 1990	(0.0172)	[0.0280]	(0.0207)	[0.0218]	(0.0161)	[0.0175]
Main road passable by car all year	0.0193	122	0.0201*	121	0.0286***	116
population share, 1980 to 2011	(0.0127)	[0.0136]	(0.0106)	[0.0091]	(0.0084)	[0.0083]
Road density around the city	-0.0764	131				
current road map, 1950 density held fixed	(0.0894)	[0.0991]				
Households on the state grid			0.0553*†	131	0.0792***	125
household share, 1980 to 2011			(0.0284)	[0.0262]	(0.0194)	[0.0198]
Population in a village with grid power			0.0719***	123	0.0552***	119
population share, 1980 to 2011			(0.0216)	[0.0215]	(0.0168)	[0.0168]
Senior secondary schools			0.0424***	131	0.0030	125
per 1,000 people, 1980 to 2011			(0.0098)	[0.0120]	(0.0037)	[0.0043]
Health centers	0.0095	128	0.0072	131	-0.0066	125
per 1,000 people, 1980 to 2011	(0.0097)	[0.0106]	(0.0186)	[0.0174]	(0.0155)	[0.0146]
Seat of a district created 1999 to 2010	-0.0064	134				
seat in 1998 held fixed	(0.0518)	[0.0558]				
Log modern-sector employment, 2010	0.1652**†	134				
less education, health, social	(0.0711)	[0.0660]				
and government services						
Panel C: Rural demand						
Log spending per person, 2011			-0.0225	131	-0.0306	128
rural households			(0.0393)	[0.0353]	(0.0273)	[0.0264]
rightmost estimate: whole hinterland						
Non-food share of spending, 2011			-0.0022	131	-0.0060	128
rural households			(0.0078)	[0.0080]	(0.0054)	[0.0047]
rightmost estimate: whole hinterland						

Notes: Each cell regresses the row's outcome, measured over the column's area, on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km plus the outcome's own 1980 value or the earlier value its label names; the modern-sector row, which counts workers with a recorded industry, holds the log of its 1980 count fixed, and the two spending rows do not add a 1980 value, since the household survey waves used here start in 1994. Conley SE within 200km in parentheses (uniform kernel, which the stars use) and Bartlett-kernel SE in brackets; Cities counts the cities in each cell. The countryside with settlements is the rural subdistricts with a settlement village within 300 travel minutes of the city, its own subdistricts left out; the countryside without settlements is the rural subdistricts within the city's 300-minute window (208.7 minutes on the subdistrict travel-time matrix, its calibrated equivalent) with no settlement village, and the whole hinterland is every rural subdistrict within that window. Road density is measured over the subdistricts that hold the city's catchment villages, every subdistrict that belongs to a city left out; electricity and schools are tested in the countryside only. A presence share is the population share in villages with the item (for roads, of villages reached by land); health centers are puskesmas and auxiliary centers (pustu). The presence shares to 2011 leave out the 8 cities of West Kalimantan, whose villages were merged between the 1990 and 1993 village censuses; the car-access and bank-presence shares also leave out the West Sumatra subdistricts recorded as nagari in 2011, a larger unit than the 1980 village, which the grid-village share keeps, measured on their hamlets. Without those subdistricts these estimates of the countryside without settlements change their star level (uniform-kernel SE): senior secondary schools, per 1,000 people, 1980 to 2011: 0.0067* (0.0040) on 123 cities; banks, per 1,000 people, 1980 to 2011: 0.0061** (0.0028) on 122 cities. Sources: village censuses (PODES) 1980 to 2014, population censuses 1980 and 2010, Susenas 2011, the current national and provincial road map and the 1950 atlas, and the World Bank district proliferation crosswalk. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. † significant at 10% with the uniform kernel and not significant at 10% with the Bartlett kernel, with errors clustered by province (a least-squares test not shown), or with both.

villages with a bank by 0.0251 (0.0248); in the countryside with settlements, by 0.0119 (0.0072), marginally significant with the Conley kernels and insignificant with province clusters, and -0.0138 (0.0137). The countryside without settlements gains banks per 1,000 people, 0.0077 (0.0026), and its bank-presence gain is statistically significant with the uniform kernel only. With the log of the 1980 count held fixed instead, the city's gain in banks per 1,000 to 1990 is 0.0106 (0.0046) and statistically significant, and the bank-presence share of the countryside with settlements changes by -0.0151 (0.0085),

Table E21: Public Provision in the Settlement Villages, 2011

	(1) Asphalt road	(2) Car road	(3) Grid reached	(4) Households on grid	(5) Primary	(6) Junior sec.	(7) Senior sec.	(8) Health centers
Agro-similarity of the village	-0.032	0.028*	0.051	0.020	0.009	0.023	-0.003	0.021
	(0.034)	(0.016)	(0.037)	(0.023)	(0.033)	(0.018)	(0.010)	(0.025)
Outcome mean	0.331	0.896	0.705	0.422	0.975	0.314	0.128	0.403
Settlement villages	808	808	813	813	813	813	813	813

Notes: Each column regresses a 2011 measure of a settlement village on the village's own agro-similarity (Bazzi et al. 2016), standardized over the 814 settlement villages, with the village geography controls (ruggedness, distance to the coast, three slope classes, temperature, log altitude); Conley SE within 200km. A village's 2011 records are the 2011 villages that descend from its year-2000 code. Columns 1 and 2: population share with an asphalt main road and with a main road passable by car. Column 3: population share in hamlet units the state grid has reached. Column 4: households on the state grid. Columns 5 to 7: schools per 1,000 people. Column 8: puskesmas and puskesmas pembantu per 1,000 people. The villages were settled from 1979 to 1988, so no 1980 value exists for them. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: village census (PODES) 2011.

a marginally significant decline. With rural subdistricts as units, banks per 1,000 to 1990 rose no more where a settlement sits (difference 0.0017, 0.0020). The village censuses count banks and do not record deposits or lending (Table E24).

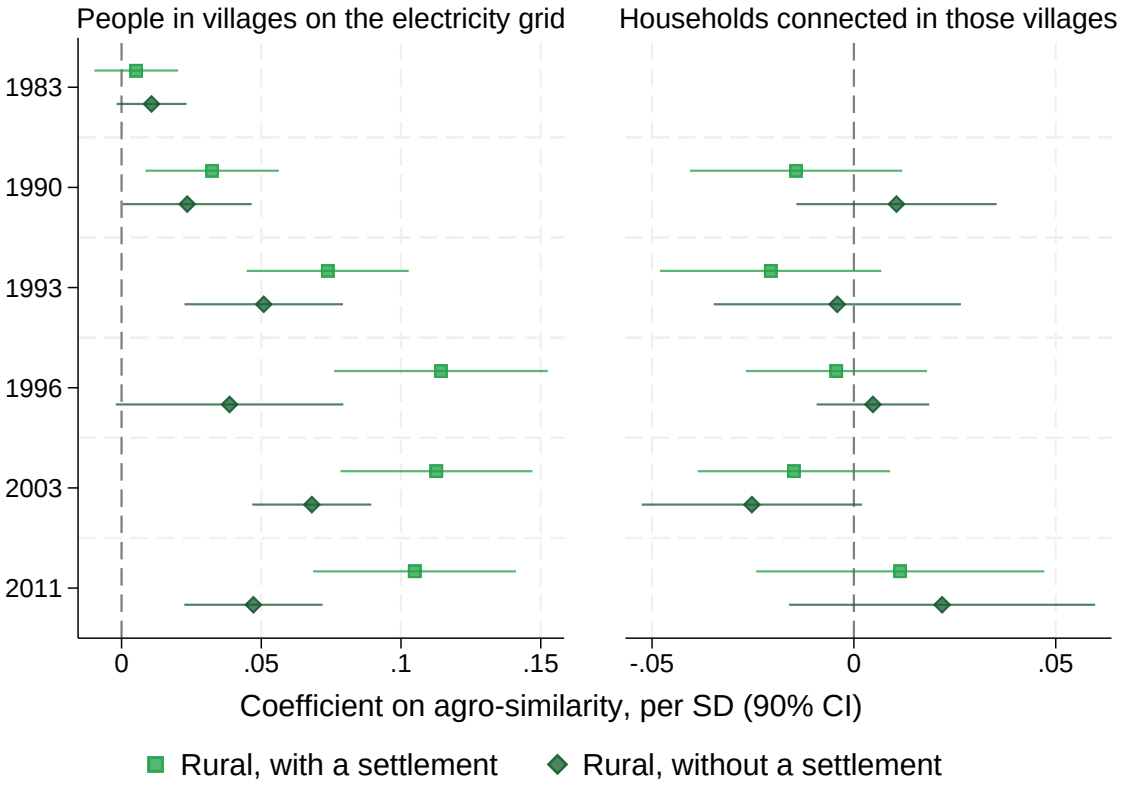
New district seats. Exposure does not predict the seat of a new district. The probability of hosting the seat of a district created 1999 to 2010 changes by -0.006 (standard error 0.052) per standard deviation of exposure, and that of hosting a new kota, the urban type of district, by 0.057 (0.030), statistically significant with the uniform kernel only. Without the 33 cities that host a new seat, the agricultural employment share changes by -0.053 (0.017) and log modern-sector employment by 0.172 (0.064), as on the main city sample. Log modern-sector employment without education, health, social and government services rises by 0.165 (0.071), statistically significant under both Conley kernels and marginally insignificant with province clusters; the industry codes left out hold public and private employers alike, so the cut does not separate public from private employment (Table E25).

Spending of rural households. Rural households around more exposed cities do not spend more per person, and the non-food share of their spending does not move. In 2011, log spending per person changes by -0.0306 (standard error 0.0273) in the whole hinterland and by -0.0225 (0.0393) in the countryside with settlements, and the non-food share by -0.0060 (0.0054) and -0.0022 (0.0078); the 2002 wave and the spending on durables and services are in the table. The household survey (Susenas) starts after 1980, so these regressions do not add a 1980 level (Table E26).

The road network. The road network around more exposed cities did not improve more than its earlier level predicts (Table E27). Road density on the current map, with the 1950 density held fixed, changes by -0.0764 (standard error 0.0894), and the roughness of the surveyed corridors over 1990 to 2007, with its 1990 level held fixed, by 0.0576 (0.3008). Roads were already denser around more exposed cities on the 1950 map, drawn before the program, by 0.245 (0.072) in log density.

The conflict provinces. The effect does not differ measurably in the provinces of the conflicts of around 2000 (Barter & Côté 2015; van Klinken 2007). With exposure interacted with an indicator of Aceh, West Kalimantan, Central Kalimantan and Central Sulawesi on the main city sample, the interaction term is −0.027 (standard error 0.032) on the agricultural employment share and −0.015 (0.084) on log modern-sector employment, and it does not reach the 10% level on any of the seven outcomes under any of the three variances; the slope outside stays within a fifth of the headline, at −0.042 (0.021) and 0.170 (0.081). Leaving the 26 cities of the four provinces out instead gives −0.024 (0.022) on the farm share and 0.080 (0.092) on log modern-sector employment, about half their size on the main city sample, with every standard error larger. The conflict-province cities are about as exposed as the others; the halving comes from estimating the control coefficients on the 108 remaining cities, and the standard deviation of exposure left after the controls falls

Figure E5: Exposure and the State Electricity Grid by Village-Census Wave



Notes: Each marker is the coefficient on agro-similarity (per SD over the 787 in-sample rural subdistricts) for rural subdistricts with and without a settlement, with its 90% interval from Conley SE within 200km. One regression per wave and panel: the outcome on agro-similarity times the indicators of a subdistrict with and without a settlement, the with-settlement indicator, the 1980 population shares in villages whose main source of electricity is the state grid (PLN) and another supplier (right panel: also the 1980 household shares using each), and the controls of the main specification measured at the subdistrict, except that the settler count of a subdistrict with a settlement is the number of settlers placed in the transmigration villages that overlap it. Sample: the 787 in-sample rural subdistricts less the 51 West Sumatra subdistricts recorded on nagari in 2011, 736 in all (191 with a settlement, 545 without), less the 71 of West Kalimantan from 1993 on, when the province merged its villages; the 41 subdistricts in the window that belong to a reclassified city are already outside the 787. Left: the share of the subdistrict's people in villages with a household on PLN. Right: households on PLN over all households of those villages, from 1990. Electricity from any supplier in 2011, with a settlement: -0.0001 (standard error 0.0010). Source: village censuses (PODES) 1980 to 2011 on year-2000 subdistricts; population census 1980.

from 0.46 to 0.30 (Table E28). Where the sources record destinations, part of the displacement left the four provinces, as when Javanese residents fled Aceh for North Sumatra (Barter & Côté 2015), so the drop leaves some of the places that received the displaced in the sample.

The exposure of neighboring cities. The exposure of neighboring cities leaves the shift in sector shares in place, and its estimates on city size are mixed. With the inverse-distance-weighted exposure of the 137 other measured cities added beside a city's own, the farm share and the modern share do not move with it (0.006, standard error 0.075; −0.013, 0.075), and the own-exposure estimates stay at the headline, −0.052 against −0.051 on the farm share. Log population, log employment and log modern-sector employment fall with the exposure of neighbors, by −0.608 (0.281), −0.601 (0.272) and −0.614 (0.307); the weighted measure varies little once own exposure and the controls are known (its standard deviation is 0.29 against 1.00 for own exposure, and the two explain 0.87 of it), and the size estimate becomes

Table E22: Public Provision in the City and Its Countryside

1980 control:	1990 value Level (1)	2011 value Level (2)	1980 value None (3)	1990 value Level (4)	2011 value Level (5)	1980 value None (6)
Panel A: The city						
Asphalt main road (share)	0.0271	0.0213	-0.0526			
	(0.0172)	(0.0302)	(0.0459)			
	[0.0280]	[0.0295]	[0.0519]			
Mean / cities	0.688 / 131	0.817 / 122	0.436 / 122			
Main road passable by car all year (share)	0.0182**†	0.0193	0.0723**†			
	(0.0088)	(0.0127)	(0.0359)			
	[0.0159]	[0.0136]	[0.0333]			
Mean / cities	0.932 / 131	0.971 / 122	0.808 / 122			
Panel B: The countryside						
	With settlements			Without settlements		
Households connected to electricity (share)	0.0108**	0.0553*†	-0.0048	0.0061	0.0792***	0.0020
	(0.0043)	(0.0284)	(0.0050)	(0.0123)	(0.0194)	(0.0042)
	[0.0048]	[0.0262]	[0.0044]	[0.0114]	[0.0198]	[0.0037]
Mean / cities	0.061 / 131	0.512 / 131	0.012 / 131	0.163 / 125	0.629 / 125	0.026 / 125
Population in a village with grid power (share)	0.0460***	0.0719***	-0.0272*	0.0460***	0.0552***	-0.0058
	(0.0137)	(0.0216)	(0.0158)	(0.0165)	(0.0168)	(0.0128)
	[0.0139]	[0.0215]	[0.0130]	[0.0166]	[0.0168]	[0.0125]
Mean / cities	0.192 / 131	0.813 / 123	0.037 / 123	0.432 / 125	0.915 / 119	0.081 / 119
Asphalt main road (share)	0.0590***	-0.0220	-0.0303	0.0679***	-0.0055	0.0095
	(0.0207)	(0.0215)	(0.0258)	(0.0161)	(0.0254)	(0.0302)
	[0.0218]	[0.0191]	[0.0220]	[0.0175]	[0.0227]	[0.0309]
Mean / cities	0.349 / 129	0.513 / 121	0.184 / 121	0.525 / 123	0.696 / 116	0.282 / 116
Main road passable by car all year (share)	0.0310	0.0201*	0.0396	0.0270***	0.0286***	0.0609
	(0.0192)	(0.0106)	(0.0387)	(0.0101)	(0.0084)	(0.0371)
	[0.0198]	[0.0091]	[0.0397]	[0.0107]	[0.0083]	[0.0390]
Mean / cities	0.848 / 129	0.926 / 121	0.591 / 121	0.877 / 123	0.943 / 116	0.714 / 116
Primary schools per 1,000	0.0147	0.0521	0.0470	-0.0131	0.0492**	-0.0665
	(0.0519)	(0.0511)	(0.0976)	(0.0340)	(0.0226)	(0.0712)
	[0.0438]	[0.0465]	[0.0989]	[0.0357]	[0.0206]	[0.0700]
Mean / cities	1.285 / 131	0.902 / 131	1.454 / 131	1.352 / 125	0.962 / 125	1.363 / 125
Junior secondary schools per 1,000	-0.0028	0.0191*	0.0187	-0.0072	-0.0044	0.0176***
	(0.0120)	(0.0104)	(0.0120)	(0.0087)	(0.0112)	(0.0065)
	[0.0104]	[0.0094]	[0.0106]	[0.0077]	[0.0100]	[0.0064]
Mean / cities	0.176 / 131	0.277 / 131	0.103 / 131	0.201 / 125	0.264 / 125	0.114 / 125
Senior secondary schools per 1,000	0.0059	0.0424***	0.0113**	0.0071	0.0030	0.0065
	(0.0064)	(0.0098)	(0.0050)	(0.0045)	(0.0037)	(0.0051)
	[0.0059]	[0.0120]	[0.0042]	[0.0041]	[0.0043]	[0.0048]
Mean / cities	0.044 / 131	0.132 / 131	0.015 / 131	0.075 / 125	0.127 / 125	0.029 / 125
Health centers per 1,000	-0.0251**	0.0072	-0.0039	-0.0140	-0.0066	0.0074
	(0.0127)	(0.0186)	(0.0107)	(0.0148)	(0.0155)	(0.0086)
	[0.0145]	[0.0174]	[0.0104]	[0.0149]	[0.0146]	[0.0075]
Mean / cities	0.290 / 131	0.274 / 131	0.112 / 131	0.210 / 125	0.255 / 125	0.090 / 125

Notes: Each cell is one regression on the main city sample (134 cities), with the outcome measured over the area's subdistricts observed on the item in 1980 and in the end year. Columns 1 and 4 regress the 1990 value and columns 2 and 5 the 2011 value on exposure, each holding the outcome's own 1980 value over the same subdistricts fixed; columns 3 and 6 regress the 1980 value itself, over the subdistricts of columns 2 and 5, on exposure, without the four 1980 census baselines, which are not controls for a 1980 outcome. Every regression is in the main specification with the full set of controls and Conley SE within 200km, adding the 1980 control its column names. Below each estimate: its SE with the uniform kernel in parentheses and with the Bartlett kernel in brackets, then the mean of the outcome and the cities in the column; a city leaves a column when its area has no subdistrict observed on the item in both years. The stars use the uniform-kernel SE: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. † significant at 10% with the uniform kernel and not significant at 10% with the Bartlett kernel, with province clusters, or with both.

statistically insignificant without three pairs of twin cities within 12 km of each other, left out of the sample and still counted as neighbors of the other cities, or with province effects. The plain mean over the other measured cities within 100 km, on the 125 cities with such a neighbor, has the same negative sign on the three size outcomes and is statistically

insignificant on every outcome with the uniform kernel and with province clusters, with log population at −0.133 (0.085) (Table E29).

Table E22: Public Provision in the City and Its Countryside, Continued

1980 control:	1990 value Log (1)	2011 value Log (2)	1990 value Log (4)	2011 value Log (5)
Panel C: The 1980 control as ln(1 + the 1980 numerator), on the cities of Panels A and B				
The city				
Asphalt main road (share)	0.0137	0.0151		
	(0.0176)	(0.0316)		
	[0.0322]	[0.0300]		
Main road passable by car all year (share)	0.0211*†	0.0239**†		
	(0.0119)	(0.0109)		
	[0.0180]	[0.0132]		
The countryside, also holding ln(1 + its 1980 village-census population) fixed				
	With settlements		Without settlements	
Households connected to electricity (share)	0.0107*†	0.0709**	0.0064	0.0524***
	(0.0056)	(0.0310)	(0.0118)	(0.0158)
	[0.0056]	[0.0283]	[0.0112]	[0.0158]
Population in a village with grid power (share)	0.0527***	0.0864***	0.0359**†	0.0431**†
	(0.0146)	(0.0195)	(0.0182)	(0.0186)
	[0.0144]	[0.0210]	[0.0192]	[0.0177]
Asphalt main road (share)	0.0170	-0.0596**	0.0392	-0.0048
	(0.0254)	(0.0257)	(0.0244)	(0.0263)
	[0.0285]	[0.0237]	[0.0252]	[0.0244]
Main road passable by car all year (share)	0.0036	-0.0009	0.0034	0.0359***
	(0.0156)	(0.0092)	(0.0101)	(0.0101)
	[0.0217]	[0.0092]	[0.0114]	[0.0107]
Primary schools per 1,000	-0.0104	0.0281	-0.0296	0.0227
	(0.0567)	(0.0581)	(0.0373)	(0.0231)
	[0.0501]	[0.0514]	[0.0386]	[0.0213]
Junior secondary schools per 1,000	-0.0035	0.0172	-0.0060	0.0012
	(0.0136)	(0.0105)	(0.0082)	(0.0104)
	[0.0116]	[0.0091]	[0.0078]	[0.0098]
Senior secondary schools per 1,000	0.0060	0.0354***	0.0098*†	0.0046
	(0.0071)	(0.0103)	(0.0050)	(0.0036)
	[0.0061]	[0.0118]	[0.0052]	[0.0042]
Health centers per 1,000	-0.0165	-0.0192	-0.0017	0.0045
	(0.0134)	(0.0195)	(0.0152)	(0.0165)
	[0.0146]	[0.0204]	[0.0159]	[0.0162]

Notes, continued: The city is its member subdistricts, each weighted by population times the share of the subdistrict assigned to the city; the countryside with settlements is the rural subdistricts (in no city) with a program village within 300 travel minutes of the city (3 cities have none), the city's own subdistricts left out; the countryside without settlements is the other rural subdistricts within the city's 300-minute window (208.7 minutes on the subdistrict travel-time matrix, its calibrated equivalent), with every subdistrict that holds a program village removed (8 cities have none); both are pooled by population. Road shares are of the population in villages reached by land. Electricity is measured on the state grid (PLN) in every year: in 1980 the household share is the share of households using electricity in villages whose main source is the grid, and a grid village has that main source; in 1990 and 2011 the household share counts households on the grid, and a grid village has at least one of them. Schools are public plus private (1980: the four primary types and the secondary totals). Health centers count health and auxiliary centers, without polyclinics, in 1990 and 2011; the 1980 form has one line for health centers (puskesmas) and no separate line for auxiliary centers (pustu), so it does not show whether the 1980 count includes them. A 1980 count is left missing where the value of any village of the subdistrict is missing (34 outer-island subdistricts for primary schools, 32 for health centers), and a 1990 count where the values of all its villages are missing (4 subdistricts). Rates per 1,000 use each year's village-census population. A presence share rises with the size of its unit. Columns 2, 3, 5 and 6 of the two road shares leave out the West Sumatra subdistricts whose 2011 record includes a nagari, a larger unit than the 1980 village (13 member subdistricts of the cities, 0 in the countryside with settlements and 65 without; 4, 0 and 2 cities with no other subdistrict leave those columns), and those columns of the two road shares and of the grid village in Panel B also leave out the 8 cities of West Kalimantan, whose villages were merged into larger units between the 1990 and 1993 village censuses; the 2011 grid-village share keeps the nagari subdistricts, measured on hamlets of about the size of the 1980 village (1,010 people on average against 948). Without the 62 of those subdistricts that hold a primary-school value in both years, the primary-school estimate of column 5 of Panel B is 0.0359 (0.0240) [0.0223] on 123 cities. Without those subdistricts, 3 other estimates of the countryside without settlements with the 1980 level held fixed change their star level (uniform-kernel SE): senior secondary schools, column 4 of Panel B: 0.0118*** (0.0043) on 123 cities; senior secondary schools, column 5 of Panel B: 0.0067*† (0.0040) on 123 cities; health centers, column 4 of Panel B: -0.0266**† (0.0130) on 123 cities. The 1980 numerator is the 1980 count for a rate and, for a share, the 1980 share times 1980 population (road shares are of land-reached villages, the connection share of households). Source: village census (PODES) 1980, 1990 and 2011. The stars use the uniform-kernel SE: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. † significant at 10% with the uniform kernel and not significant at 10% with the Bartlett kernel, with province clusters, or with both.

Table E23: Public Provision in Rural Subdistricts With and Without a Settlement

	1980 to 1990			1980 to 2011			1980
	With a settlement (1)	Without a settlement (2)	Difference (3)	With a settlement (4)	Without a settlement (5)	Difference (6)	level Difference (7)
Asphalt main road	0.0744***	0.0869***	-0.0124	0.0154	0.0416*	-0.0263	0.0473**
(population share)	(0.0127)	(0.0191)	(0.0199)	(0.0250)	(0.0240)	(0.0237)	(0.0185)
	[0.0203]	[0.0185]	[0.0250]	[0.0246]	[0.0247]	[0.0246]	[0.0197]
Mean		0.4397			0.6291		0.2043
Subdistricts		182 / 563			168 / 476		186 / 578
Four-wheel access all year	0.0805***	0.0370**	0.0436**†	0.0412**	0.0507***	-0.0096	0.0505**
(population share)	(0.0231)	(0.0151)	(0.0217)	(0.0171)	(0.0113)	(0.0131)	(0.0218)
	[0.0279]	[0.0184]	[0.0256]	[0.0171]	[0.0121]	[0.0151]	[0.0249]
Mean		0.8347			0.9179		0.6009
Subdistricts		182 / 563			168 / 476		186 / 578
Households connected	0.0114	0.0170**	-0.0056	0.0647***	0.0520**	0.0127	0.0002
(household share)	(0.0071)	(0.0073)	(0.0065)	(0.0213)	(0.0207)	(0.0172)	(0.0043)
	[0.0073]	[0.0086]	[0.0076]	[0.0200]	[0.0185]	[0.0176]	[0.0045]
Mean		0.1301			0.5832		0.0239
Subdistricts		191 / 586			191 / 589		191 / 589
Village with electricity	0.0381**	0.0282	0.0098	0.1057***	0.0478***	0.0579**	0.0047
(population share)	(0.0185)	(0.0173)	(0.0162)	(0.0214)	(0.0131)	(0.0249)	(0.0103)
	[0.0175]	[0.0188]	[0.0169]	[0.0186]	[0.0119]	[0.0202]	[0.0111]
Mean		0.3439			0.8803		0.0677
Subdistricts		191 / 586			173 / 539		191 / 589
Primary schools	0.0013	-0.0318	0.0331	0.0193	-0.0012	0.0205	0.0642
per 1,000 people	(0.0289)	(0.0336)	(0.0311)	(0.0328)	(0.0234)	(0.0262)	(0.0495)
	[0.0323]	[0.0334]	[0.0321]	[0.0309]	[0.0235]	[0.0262]	[0.0529]
Mean		1.4847			1.0802		1.4662
Subdistricts		185 / 570			185 / 572		185 / 572
Junior secondary schools	0.0025	-0.0048	0.0073	-0.0066	-0.0103	0.0037	-0.0068
per 1,000 people	(0.0101)	(0.0102)	(0.0128)	(0.0084)	(0.0097)	(0.0091)	(0.0092)
	[0.0086]	[0.0084]	[0.0103]	[0.0087]	[0.0088]	[0.0079]	[0.0095]
Mean		0.2053			0.2903		0.1182
Subdistricts		191 / 586			191 / 589		191 / 589
Senior secondary schools	0.0037	0.0005	0.0032	0.0076	0.0051	0.0025	-0.0013
per 1,000 people	(0.0044)	(0.0050)	(0.0034)	(0.0075)	(0.0048)	(0.0063)	(0.0029)
	[0.0045]	[0.0044]	[0.0039]	[0.0070]	[0.0049]	[0.0060]	[0.0025]
Mean		0.0669			0.1293		0.0225
Subdistricts		191 / 586			191 / 589		191 / 589
Puskesmas and pustu	0.0032	-0.0062	0.0094	-0.0201	-0.0100	-0.0101	0.0068
per 1,000 people	(0.0123)	(0.0104)	(0.0086)	(0.0156)	(0.0172)	(0.0108)	(0.0060)
	[0.0111]	[0.0098]	[0.0098]	[0.0146]	[0.0156]	[0.0114]	[0.0063]
Mean		0.2721			0.3092		0.1116
Subdistricts		187 / 574			187 / 577		187 / 577

Notes: Each row is one outcome: columns 1 to 3 come from one regression of the 1990 value and columns 4 to 6 from one regression of the 2011 value, on exposure interacted with the indicators of a subdistrict with and without a settlement, the with-settlement indicator, the outcome's own 1980 value and the controls; columns 3 and 6 subtract the slope without a settlement from the slope with one, and column 7 does the same in one regression of the 1980 value itself, without the four 1980 census baselines, which are not controls for a 1980 outcome. Every regression is in the main specification with the full set of controls and Conley SE within 200km, except that the settler count of a subdistrict with a settlement is the number of settlers placed in the transmigration villages that overlap it. Below each estimate: its SE with the uniform kernel in parentheses and with the Bartlett kernel in brackets, then the mean of the outcome and the subdistricts with and without a settlement in the regression. The stars use the uniform-kernel SE: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. † significant at 10% with the uniform kernel and not significant at 10% with the Bartlett kernel, with least squares clustered by province (20 provinces, t with 19 degrees of freedom; the 1980 to 2011 presence shares, which leave West Kalimantan out, on 19 provinces), or with both.

Table E23: Public Provision in Rural Subdistricts With and Without a Settlement, Continued

Notes, continued: Sample: the 787 in-sample rural subdistricts (191 with a settlement, holding at least one of the 814 settlement villages, and 596 without); the 41 in-sample subdistricts that belong to a reclassified city are left out, and a regression also needs the end-year value and the 1980 value of its outcome. Exposure is agro-similarity within 300 travel minutes, standardized over the 787 (one SD is 0.100). Road shares are of the population in villages reached by land, and missing where every village is reached by water only (2 subdistricts in 2011); the 1993 wave is not used, since its item asks about most village roads. Electricity is measured on the state grid (PLN) in every year: households connected is the share of households using electricity in villages whose main source is the grid in 1980, and PLN households over all households in 1990 and 2011; a village with electricity has the grid as its main source in 1980 and any PLN household in 1990 and 2011, so its change mixes the change of definition with the change in the outcome. The only 1980 control of the two electricity rows is the 1980 state-grid level, while the estimates of the state grid wave by wave from 1983 to 2011 (Figure E5) hold the 1980 levels of the grid and of other suppliers fixed separately, so the two do not form one series. Schools are public plus private in every wave (1980: the four primary types and the secondary totals). Health centers count puskesmas plus pustu, without polyclinics, in 1990 and 2011; the 1980 form has one line for health centers (puskesmas) and no separate line for auxiliary centers (pustu), so it does not show whether the 1980 count includes them. A 1980 count is left missing in a subdistrict that takes any share of a 1980 subdistrict with a missing value for any village (34 outer-island subdistricts for primary schools, 32 for health centers). Shares run from 0 to 1; rates per 1,000 use each wave's own village-census population. The 1980 to 2011 columns of the two road shares leave out the 51 West Sumatra subdistricts whose 2011 record includes a nagari, a larger unit than the village of 1980; the 2011 village-with-electricity share measures them on hamlets (jorong), units of about the size of the 1980 village (a mean of 936 people per hamlet in 2011 and 855 per village in 1980, over the 51 nagari subdistricts), and keeps them; the 1980 to 2011 columns of the three presence shares (asphalt, four-wheel access and village with electricity) also leave out the 71 subdistricts of West Kalimantan (18 with a settlement), whose villages were merged into larger units between the 1990 and 1993 village censuses. Column 7 does not apply either rule. Source: village census (PODES) 1980, 1990 and 2011 on year-2000 subdistricts; population census 1980.

Table E24: Banks in the City and Its Countryside

	Bank presence		Banks per 1,000 people	
	1980 to 1990	1980 to 2011	1980 to 1990	1980 to 2011
Panel A. The city				
Agro-similarity	-0.0194	0.0251	0.0055	0.0110
	(0.0161)	(0.0248)	(0.0039)	(0.0092)
	[0.0168]	[0.0222]	[0.0040]	[0.0093]
Agro-similarity, 1980 log count	-0.0075	0.0337	0.0106**	0.0154
	(0.0194)	(0.0306)	(0.0046)	(0.0098)
	[0.0186]	[0.0283]	[0.0051]	[0.0097]
Mean of outcome	0.0533	0.1047	0.0119	0.0648
Cities	131	122	131	133
Panel B. The countryside with settlements				
Agro-similarity	-0.0085	-0.0138	0.0037	0.0119*†
	(0.0143)	(0.0137)	(0.0025)	(0.0072)
	[0.0115]	[0.0130]	[0.0024]	[0.0060]
Agro-similarity, 1980 log count	-0.0151*†	-0.0102	-0.0007	0.0041
	(0.0085)	(0.0182)	(0.0028)	(0.0062)
	[0.0080]	[0.0169]	[0.0028]	[0.0055]
Mean of outcome	0.0108	0.0527	0.0045	0.0304
Cities	131	123	131	131
Panel C. The countryside without settlements				
Agro-similarity	-0.0050	0.0185*†	0.0016	0.0077***
	(0.0083)	(0.0112)	(0.0024)	(0.0026)
	[0.0079]	[0.0118]	[0.0023]	[0.0034]
Mean of outcome	0.0083	0.0311	0.0005	0.0265
Cities	125	117	124	124
Agro-similarity, 1980 log count	0.0009	0.0114	0.0002	0.0069***
	(0.0082)	(0.0084)	(0.0022)	(0.0023)
	[0.0081]	[0.0101]	[0.0022]	[0.0034]
Cities	124	116	124	124
Panel D. Rural subdistricts with and without a settlement				
Agro-similarity, with a settlement	0.0089	-0.0139	0.0069***	0.0024
	(0.0070)	(0.0125)	(0.0017)	(0.0036)
	[0.0078]	[0.0123]	[0.0022]	[0.0041]
Agro-similarity, without a settlement	-0.0041	-0.0037	0.0052*	0.0050**
	(0.0060)	(0.0084)	(0.0028)	(0.0024)
	[0.0054]	[0.0090]	[0.0025]	[0.0029]
Difference	0.0129	-0.0102	0.0017	-0.0026
	(0.0083)	(0.0141)	(0.0020)	(0.0039)
	[0.0082]	[0.0123]	[0.0020]	[0.0036]
Mean of outcome	-0.0021	0.0202	-0.0016	0.0213
Subdistricts	777	661	768	771
with a settlement	191	173	186	186
Difference, 1980 log count	0.0078	-0.0143	0.0025	-0.0016
	(0.0068)	(0.0128)	(0.0024)	(0.0044)
	[0.0076]	[0.0118]	[0.0026]	[0.0041]
Subdistricts	768	652	768	771
with a settlement	186	168	186	186

Notes: Each column regresses the change in the area's bank measure from 1980 to the year in the column header on standardized agro-similarity, with the outcome's own 1980 level held fixed. Panels A to C use the main city sample (134 cities) and the main specification with the full set of controls and Conley SE within 200km; a row whose label ends in 1980 log count holds fixed instead the log of the area's 1980 banks plus one and, in Panels B and C, the log of its 1980 village-census population plus one. Conley SE within 200km in parentheses (uniform kernel, which the stars use) and Bartlett-kernel SE in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. † significant at 10% with the uniform kernel and not significant at 10% with the Bartlett kernel, with errors clustered by province (a least-squares test not shown), or with both.

Table E24: Banks in the City and Its Countryside, Continued

Notes, continued: Panel A is the city's member subdistricts, weighted by membership share; Panel B the countryside with settlements, the rural subdistricts with a settlement village within 300 travel minutes of the city, its own left out; Panel C the countryside without settlements, the other rural subdistricts in that 300-minute window (208.7 minutes on the subdistrict travel-time matrix, its calibrated equivalent) with no settlement village. Panel D uses the 787 rural subdistricts within 300 travel minutes of a settlement village, measured in the 1980 and 2010 censuses, that do not belong to a reclassified city (191 with a settlement village, 596 without), with agro-similarity, the mean over the settlement villages within 300 minutes, interacted with an indicator for each group, the first indicator, the settler count (for a subdistrict with a settlement, the number of settlers placed in the transmigration villages that overlap it; for one without, the count weighted by inverse travel time within 300 minutes), the four 1980 census baselines and the sixteen agro-climatic means within 300 minutes; Difference is the first slope less the second. A subdistrict with a missing 1980 bank count for any of its villages leaves the per-1,000 columns and the area's 1980 log count. Bank presence is the population share in villages with a bank, and banks per 1,000 is banks over the village-census population; the village censuses count banks and do not record deposits or lending. 1980 counts banks on one line, 2011 commercial and rural banks, and the 1990 line the commercial banks and some of the rural banks. The 1980 to 2011 presence columns leave out West Kalimantan, whose villages were merged between the 1990 and 1993 village censuses, and the West Sumatra subdistricts recorded as nagari in 2011, a larger unit than the 1980 village. Without those subdistricts the other columns change the star level of 3 estimates with the 1980 level held fixed, of the countryside without settlements and the subdistricts without a settlement (uniform-kernel SE): agro-similarity, column 4 of Panel C: 0.0061** (0.0028) on 122 cities; agro-similarity without a settlement, column 3 of Panel D: 0.0037 (0.0025) on 717 subdistricts; agro-similarity without a settlement, column 4 of Panel D: 0.0023 (0.0029) on 720 subdistricts. Sources: village censuses (PODES) of 1980, 1990 and 2011 on year-2000 subdistricts; the 1980 population census; the settlement-village records.

Table E25: New District Seats and Modern-Sector Employment Without Government Services

Panel A. Hosting the seat of a new district, linear probability

	New seat, 1999 to 2010	New seat, 2000 to 2010	New kota seat, 1999 to 2010
Agro-similarity	-0.006	-0.065	0.057*†
	(0.052)	(0.045)	(0.030)
	[0.056]	[0.055]	[0.040]
District seat, end of 1998	-0.281***	-0.151*†	0.079
	(0.088)	(0.091)	(0.079)
	[0.090]	[0.086]	[0.074]
Mean of outcome	0.246	0.179	0.104
Cities	134	134	134

Panel B. Main outcomes without the cities hosting a new seat

	Log population		Agricultural share		Log modern employment	
	Member	Town	Member	Town	Member	Town
Agro-similarity	0.065	0.056	-0.053***	-0.045***	0.172***	0.137***†
	(0.066)	(0.066)	(0.017)	(0.015)	(0.064)	(0.052)
	[0.069]	[0.070]	[0.018]	[0.018]	[0.071]	[0.066]
Mean of outcome	11.856	11.886	0.375	0.364	10.257	10.312
Cities	101	98	101	98	101	98

Panel C. Modern-sector employment without government services, 2010

	All modern	Without codes 14 to 18		Without codes 16 to 18, strict	
	Log	Log	Share	Log	Share
Agro-similarity	0.177***	0.164**†	0.038***	0.165**†	0.039***
	(0.063)	(0.072)	(0.005)	(0.071)	(0.005)
	[0.060]	[0.066]	[0.007]	[0.066]	[0.007]
Own 1980 numerator, log	0.199**	0.055	0.021*†	0.058	0.020*†
	(0.091)	(0.080)	(0.012)	(0.078)	(0.012)
	[0.097]	[0.078]	[0.013]	[0.077]	[0.013]
Mean of outcome	10.232	9.634	0.287	9.675	0.305
Cities	134	134	134	134	134

Notes: Each column is one regression on the main city sample (134 cities), in the main specification with the full set of controls and Conley SE within 200km; Panel A adds the city's district-seat status at the end of 1998, and Panel C adds the outcome's own 1980 numerator as the log of the count plus one. Conley SE within 200km in parentheses (uniform kernel, which the stars use) and Bartlett-kernel SE in brackets. Panel A: linear probability models of hosting the seat of a district created 1999 to 2010 (33 cities), created 2000 to 2010 (24 cities), or of a new kota, the urban type of district, created 1999 to 2010 (14 cities). A city hosts a seat when one of its member subdistricts holds the regent's or mayor's office, located in the village censuses of 2011 and 2014, and a new seat is held in both waves; the status at the end of 1998 is read from the 2011 and 2014 seats of districts established by 1998 and equals 1 for 94 cities. Panel B: log population, the agricultural employment share and log modern-sector employment, 2010, whose estimates on all 134 cities are in Table 3, without the 33 cities hosting a new seat in a member subdistrict (Member, 101 cities), and also without the 3 cities whose town is the capital of such a district although the seat subdistrict is not a member (Town, 98 cities). Panel C: modern-sector employment, 2010, in logs and as a share of employment; column 1 is the headline sector (census industry codes 8 and 11 to 18), columns 2 and 3 keep codes 8 and 11 to 13 (manufacturing, trade, hotels and restaurants, and transport), and columns 4 and 5 leave out codes 16 to 18 (education, health, and social, government and private services) on the strict employment definition, the employed with a recorded industry. Codes 16 to 18 hold public and private employers alike, so columns 4 and 5 do not separate public from private employment. Sources: population censuses of 1980 and 2010; village censuses of 2011 and 2014; World Bank district proliferation crosswalk. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. † significant at 10% with the uniform kernel and not significant at 10% with the Bartlett kernel, with errors clustered by province (a least-squares test not shown), or with both.

Table E26: Spending of Rural Households Around the City

	2002					2011				
	Spending	Non-food	Durables	Any durable	Services	Spending	Non-food	Durables	Any durable	Services
Panel A. The whole hinterland										
Agro-similarity	0.001	0.001	-0.133	-0.017	0.082**	-0.031	-0.006	-0.317**	-0.033**	-0.022
	(0.028)	(0.003)	(0.170)	(0.019)	(0.041)	(0.027)	(0.005)	(0.138)	(0.016)	(0.037)
	[0.025]	[0.004]	[0.162]	[0.018]	[0.038]	[0.026]	[0.005]	[0.136]	[0.015]	[0.038]
Mean of outcome	11.898	0.268	5.226	0.674	8.981	13.057	0.369	4.342	0.475	10.155
Households per city	2248	2248	2248	2248	2248	2814	2112	2112	2112	2112
Cities	123	123	123	123	123	128	128	128	128	128
Panel B. The countryside with settlements										
Agro-similarity	-0.002	0.006	0.479	0.040	0.175	-0.023	-0.002	0.153	0.015	0.039
	(0.052)	(0.009)	(0.368)	(0.035)	(0.116)	(0.039)	(0.008)	(0.314)	(0.030)	(0.060)
	[0.049]	[0.009]	[0.331]	[0.034]	[0.098]	[0.035]	[0.008]	[0.312]	[0.031]	[0.065]
Mean of outcome	11.896	0.263	5.370	0.689	8.952	13.112	0.387	4.979	0.529	10.271
Households per city	541	541	541	541	541	673	508	508	508	508
Cities	118	118	118	118	118	131	131	131	131	131

Notes: Each column regresses one outcome, a mean over the rural households of the area the panel names, on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Panel A: the whole hinterland, every rural subdistrict within the city's 300-minute window (208.7 minutes on the subdistrict travel-time matrix, its calibrated equivalent), the city's own subdistricts left out. Panel B: the countryside with settlements, the rural subdistricts that hold the settlement villages within 300 travel minutes of the city, a subdistrict of an urban center left out. Means are weighted by the household survey weights. Spending is ln(x+1) of monthly spending per person, in rupiah; non-food the non-food share of monthly spending; durables ln(x+1) of monthly spending per person on durable goods (furniture, appliances, electronics, vehicles, jewelry), recalled over 12 months in 2002 and over 3 months in 2011, and any durable the share of households reporting a durable purchase in that period; services ln(x+1) of monthly spending per person on each survey's service items, which differ between 2002 and 2011, so the two service columns are not comparable across years. Households per city is the mean number of surveyed rural households behind a city's means; in 2011 every column but spending uses the households answering the consumption modules, and a city leaves a column when its area has no surveyed rural household. Conley SE within 200km in parentheses (uniform kernel, which the stars use) and Bartlett-kernel SE in brackets. Sources: Susenas 2002 and 2011, the core survey and the consumption modules; population censuses of 1980 and 2010. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E27: Road Density and Road Condition Around the City

	Exposure (1)	Earlier level (2)
Current density, catchment		
1950 density held fixed	-0.0764	0.3263*†
	(0.0894)	(0.1735)
	[0.0991]	[0.1655]
Mean / cities	1.664 / 131	
1992 density held fixed	-0.0449	0.1970
	(0.0933)	(0.1236)
	[0.1004]	[0.1223]
Mean / cities	1.664 / 131	
Current density, within 5 km of a program village		
1950 density held fixed	-0.1668	0.4125***
	(0.1992)	(0.1543)
	[0.1750]	[0.1527]
Mean / cities	1.802 / 131	
1992 density held fixed	-0.0422	0.5451***
	(0.1674)	(0.1469)
	[0.1574]	[0.1459]
Mean / cities	1.802 / 131	
Corridor roughness, change 1990 to 2007, 1990 level held fixed	0.0576	-0.4179***
	(0.3008)	(0.0962)
	[0.3019]	[0.0859]
Mean / cities	-0.991 / 132	
Corridor unpaved share, change 1990 to 2007, 1990 level held fixed	0.0146	-0.9319***
	(0.0202)	(0.0362)
	[0.0238]	[0.0423]
Mean / cities	-0.466 / 125	

Notes: Each row is one regression on the cities of the main city sample (134 cities) that hold the row's measures, in the main specification with the full set of controls and Conley SE within 200km, adding the row's earlier level. The four density rows hold 131 cities (3 cities have no subdistrict hosting their catchment villages once the subdistricts that belong to a city are removed), the roughness row 132 (2 have no corridor survey post with roughness measured in both years) and the unpaved row 125 (9 have no post with the surface recorded in both years). Road density is ln(1 + km of road per 100 square km) on the current national and provincial road map, over the union of the subdistricts hosting the city's catchment villages less every subdistrict that belongs to a city, or over the part of that union within 5 km of a catchment village; the level held fixed is the same density on the 1950 atlas (main and secondary roads) or on the 1992 Digital Chart of the World, and column 2 reports its coefficient. Corridors are the road-condition survey posts within 10 km of the straight line from the city to each catchment village, measured in both 1990 and 2007, with roughness in meters per km (International Roughness Index) and unpaved as a share of posts. Below each estimate: its standard error with the uniform kernel in parentheses and with the Bartlett kernel in brackets, then the mean of the outcome and the cities. The stars use the uniform-kernel standard errors: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. † significant at 10% with the uniform kernel and not significant at 10% with the Bartlett kernel, with province clusters, or with both. Source: road maps and the road-condition survey of the public-provision appendix.

Table E28: Main Results Inside and Outside the Conflict Provinces

	Population	Employment	Unemployment	Agricultural employment		Modern employment	
	Log	Log	Rate	Log	Share	Log	Share
Panel A. Main city sample, agro-similarity interacted with the conflict provinces							
Agro-similarity	0.056	0.045	0.007	-0.122	-0.042**†	0.170**†	0.063***
	(0.084)	(0.079)	(0.007)	(0.119)	(0.021)	(0.081)	(0.020)
	[0.071]	[0.067]	[0.006]	[0.111]	[0.022]	[0.072]	[0.020]
× conflict province	-0.021	-0.066	0.009	-0.125	-0.027	-0.015	0.015
	(0.098)	(0.089)	(0.008)	(0.167)	(0.032)	(0.084)	(0.035)
	[0.091]	[0.087]	[0.008]	[0.145]	[0.033]	[0.096]	[0.032]
Mean of outcome	11.842	10.933	0.122	9.822	0.374	10.232	0.516
Cities	134	134	134	134	134	134	134
Panel B. Without the conflict provinces							
Agro-similarity	0.059	0.040	-0.003	-0.030	-0.024	0.080	0.037
	(0.099)	(0.097)	(0.009)	(0.154)	(0.022)	(0.092)	(0.026)
	[0.102]	[0.098]	[0.009]	[0.150]	[0.028]	[0.099]	[0.027]
Mean of outcome	11.896	10.981	0.122	9.855	0.371	10.279	0.518
Cities	108	108	108	108	108	108	108

Notes: Each column is one regression in the main specification with the full set of controls and Conley SE within 200km. Panel A keeps the main city sample (134 cities in 17 provinces) and adds agro-similarity times an indicator of the four conflict provinces, and the indicator itself: its first row is the slope outside the conflict provinces and its second the difference inside them, with every control coefficient common to both groups. Panel B leaves out the 26 cities of the provinces where communal or separatist violence displaced people around 2000: Aceh, on Sumatra (8 cities; the separatist conflict), West Kalimantan (8) and Central Kalimantan (5) (the Dayak-Madurese violence), and Central Sulawesi (5; the Poso conflict), so 108 cities in 13 provinces remain and every control coefficient is estimated again on them; Maluku and North Maluku, the provinces of the Maluku conflict, hold no city of the main city sample. Panel B's estimates for the agricultural employment share, log modern-sector employment and the modern-sector share are 47% to 56% of their size in Table 3, while in Panel A the slope outside the conflict provinces stays within a fifth of that table's estimate on these three outcomes; without the 26 cities the SD of agro-similarity left after the controls falls from 0.46 to 0.30, and every SE in Panel B is larger than in that table. Agro-similarity is standardized once over every city with a measured exposure, so both panels share one scale. Outcomes are measured in 2010; shares and the unemployment rate are fractions. Conley SE within 200km in parentheses (uniform kernel, which the stars use) and Bartlett-kernel SE in brackets. Source: population censuses of 1980 and 2010. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. † significant at 10% with the uniform kernel and not significant at 10% with the Bartlett kernel, with errors clustered by province (a least-squares test not shown), or with both.

Table E29: Main Results With the Agro-Similarity of Neighboring Cities

	Population	Employment	Unemployment	Agricultural employment		Modern employment	
	Log	Log	Rate	Log	Share	Log	Share
Panel A. Every other measured city, weighted by inverse distance							
Agro-similarity	0.129*†	0.101	0.004	-0.100	-0.052***	0.237***	0.068***
	(0.068)	(0.068)	(0.006)	(0.107)	(0.019)	(0.074)	(0.019)
	[0.067]	[0.065]	[0.006]	[0.105]	[0.020]	[0.072]	[0.018]
Agro-similarity of neighbors	-0.608**	-0.601**	0.035**†	-0.518	0.006	-0.614**†	-0.013
	(0.281)	(0.272)	(0.015)	(0.384)	(0.075)	(0.307)	(0.075)
	[0.289]	[0.282]	[0.023]	[0.362]	[0.073]	[0.320]	[0.072]
Mean of outcome	11.842	10.933	0.122	9.822	0.374	10.232	0.516
Cities	134	134	134	134	134	134	134
Panel B. Measured cities within 100 km, plain mean							
Agro-similarity	0.100	0.058	0.014*†	-0.164	-0.059**	0.210**	0.072***
	(0.085)	(0.085)	(0.008)	(0.121)	(0.024)	(0.101)	(0.021)
	[0.073]	[0.073]	[0.008]	[0.119]	[0.024]	[0.088]	[0.020]
Agro-similarity of neighbors	-0.133	-0.107	-0.006	-0.111	-0.014	-0.063	0.012
	(0.085)	(0.084)	(0.006)	(0.132)	(0.027)	(0.074)	(0.024)
	[0.074]	[0.077]	[0.008]	[0.110]	[0.025]	[0.074]	[0.023]
Mean of outcome	11.853	10.944	0.121	9.848	0.381	10.228	0.510
Cities	125	125	125	125	125	125	125

Notes: Each column is one regression in the main specification with the full set of controls and Conley SE within 200km, with the agro-similarity of neighboring cities added beside the city's own. Panel A: the mean of the standardized agro-similarity of every other city with a measured exposure (137 cities for each city), each weighted by 1 over its great-circle distance in km, defined for all 134 cities of the main city sample. Panel B: the plain mean over the other measured cities within 100 km, which leaves out the 9 cities with no measured city that close, so 125 remain. City centers are the coordinates the Conley SE use. Agro-similarity is standardized once over the 138 cities with a measured exposure, and both neighbor measures are means of it, so each coefficient is the change in the outcome per SD of agro-similarity; on the main city sample the Panel A measure has an SD of 0.29 and the Panel B measure an SD of 0.83. Outcomes are measured in 2010; shares and the unemployment rate are fractions. Conley SE within 200km in parentheses (uniform kernel, which the stars use) and Bartlett-kernel SE in brackets. Source: population censuses of 1980 and 2010. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. † significant at 10% with the uniform kernel and not significant at 10% with the Bartlett kernel, with errors clustered by province (a least-squares test not shown), or with both.

F The Model

This part derives the two-good main specification. The numerical comparisons are separately identified as calculations under partial non-agricultural tradability.

The model in Section 6 has one food good and one non-agricultural good. Both goods are traded. The model retains the common land market, the local land-rent rebate, and location and sector choices. The numerical comparisons in Section F.11 and Section F.12 use a distinct variant with partially tradable non-agricultural output.

F.1 Model Details: Notation, Technology, and Choice Probabilities

Cited in the main text. This section: Section 6.1, p. 44.

This subsection states the parts of the model that Section 6.1 summarizes in words: the notation, the expenditure shares, the production technology with the land market, and the two stages of the worker's choice.

Expenditure shares. With supernumerary income $\tilde{w}_{js} \equiv w_{js} + R_j - p_A^j \bar{a}$ defined as in Section F.2, the implied expenditure shares are

$$s_A^j = \frac{p_A^j c_A}{w_{js} + R_j} = \frac{p_A^j \bar{a} + \psi_A \tilde{w}_{js}}{w_{js} + R_j} \tag{23}$$

$$s_S^j = \frac{\psi_S \tilde{w}_{js}}{w_{js} + R_j} \tag{24}$$

Technology. Total labor $\bar{L}$ is fixed and each worker supplies one unit inelastically. Each location has a fixed land endowment $\bar{T}_j$, split between farming (T_{jA}) and housing ($T_{jH} = \bar{T}_j - T_{jA}$).

- **Agriculture:**

$$Y_{jA} \;=\; (A_j^A \, T_{jA})^{1-\alpha} \, (L_{jA})^{\alpha}, \qquad 0 < \alpha < 1 \tag{25}$$

  Constant returns in (T_{jA}, L_{jA}) jointly, diminishing returns to labor holding land fixed, so land scarcity is the congestion force in farming. A_j^A varies across space and is where the agro-climatic productivity shock from the transmigration program enters the model.

- **Modern / Services:**

$$Y_{jS} \;=\; \bar{A}_j^S \, N_j^{\eta} \, (T_{jH})^{1-\gamma_S} \, (L_{jS})^{\gamma_S}, \qquad 0 < \gamma_S < 1 \tag{26}$$

  where $\bar{A}_j^S$ is the exogenous modern productivity scale and the external term N_j^{η}, entering (26) exactly once, is the agglomeration externality. The agglomeration term enters only non-agricultural production. The main specification sets $\eta = 0$, and Section F.6 separates the price, land, and labor terms in the non-agricultural wage response.

Wages, land, and the rebate. Wages equal the marginal revenue product of labor at the local consumption prices (Section F.3, Section F.5). Labor receives shares α and γ_S of sectoral revenue, with the remainder accruing to land. A common rental market allocates $\bar{T}_j$ across agriculture and non-agricultural uses. The empirical land endowment combines planted crop area and the built-up footprint. The separate numerical comparisons in Section F.11 examine alternative land measures.

Stage 2, sector choice within location j. A worker draws a sector-specific efficiency shock ε_{js} from Fréchet with shape θ_S. With scales equal to the sectoral wages, the sector-choice probability is

$$\pi_{A|j} \;=\; \frac{w_{jA}^{\theta_S}}{w_{jA}^{\theta_S} + w_{jS}^{\theta_S}} \tag{27}$$

Table F1: Key Notation

Symbol	Interpretation
j, k, o	location indices: j the location in question, k a food-supplying origin, o a migration origin; s indexes a sector
J	number of locations in the model economy
$\mathcal{S} = \{A, S\}$	sector set: Agriculture, Modern non-agriculture / Services (manufacturing folded into S)
L_{js}	labor employed in location j sector s
$N_j = \sum_s L_{js}$	total employment in location j (every resident works)
$\bar{L}$	total labor, fixed; each worker supplies one unit
$\bar{T}_j$	total land endowment in location j (crop area + built-up footprint)
T_{jA}, T_{jH}	land allocated to farming and to housing ($T_{jA} + T_{jH} = \bar{T}_j$)
A_j^A, $\bar{A}_j^S$	agricultural productivity parameter; exogenous non-agricultural productivity scale
$\bar{a}$	subsistence food requirement (Stone-Geary), $\bar{a} = 0.108$
ψ_A, ψ_S	expenditure shares on discretionary consumption ($\psi_A + \psi_S = 1$; $\psi_A = 0.550$)
α	labor share in agriculture ($\alpha = 0.5$; land gets $1 - \alpha$)
γ_S	labor share in services ($\gamma_S = 0.7$; housing land gets $1 - \gamma_S$)
η	agglomeration elasticity in services ($\eta = 0$ headline)
τ_{kj}	iceberg cost of shipping food from k to j, $\tau_{kj} = \max(d_{kj}, 1)^\delta$
d_{kj}, d_{oj}	route-aware cost-distance in minutes between the two locations its subscripts name, the same measure in the food-trade and the migration blocks
δ	iceberg cost-distance elasticity ($\delta = 0.5$)
σ	CES / Armington substitution elasticity across food sources ($\sigma = 4$)
R_j	per-person land-rent rebate to residents of j
$\tilde{w}_{js}$	supernumerary income (income net of the subsistence food bill)
Φ_L^j	measured 1980 location amenity (built-up, density, infrastructure)
κ	scale of the standardized 1980 amenity index inside Φ_L^j ($\kappa = 0.5$)
$\varepsilon_{js}(i)$	idiosyncratic sector efficiency shock (scales the worker's effective wage), Fréchet with shape θ_S
$\upsilon_j(i)$	idiosyncratic location-preference shock, Fréchet with shape θ_L
ν	distance elasticity of moving costs in the migration extension
θ_S, θ_L	sector-choice and location-choice elasticities
δ_S	cost-distance elasticity for trade in the whole non-agricultural good; the main specification sets $\delta_S = \delta$
P_j	ideal price index in location j
W_j	sector-inclusive value (option value of living in j)
π_j, $\pi_{s\|j}$	choice probabilities (location; sector conditional on location)

Notes: $\psi_A = 0.550$ is the mean across the model's locations of each location's survey-weighted mean household food expenditure share in Susenas 2011, and $\bar{a} = 0.108$ is BPS's 2000 outer-island food poverty line, scaled to 2011 rupiah, over the median non-agricultural monthly wage across the same locations. The food share is measured on purchased food: the 2011 item file keeps only food the household bought, so food grown at home or received as a gift is left out. On the complete convention, which includes that food, the same share is about 0.58 nationally and 0.61 on the outer islands.

so location agricultural employment is $L_{jA} = \pi_{A|j} N_j$ (and $L_{jS} = (1 - \pi_{A|j}) N_j$). Ranking wages is utility-consistent here because the rebate and the price level are common across sectors within a location, so the wage-maximizing sector is also the utility-maximizing one (Section F.8). The sector-inclusive value $W_j = (w_{jA}^{\theta_S} + w_{jS}^{\theta_S})^{1/\theta_S}$ is the option value of living in j before sector shocks are drawn; its level normalization is stated in the same appendix.

Stage 1, location choice. A worker draws a location taste shock υ_j from Fréchet with shape θ_L. The location stage ranks the Stone-Geary *indirect utility* (Section F.2) evaluated at the location's expected income and scaled by the measured amenity. Writing the location value as $v_j = \Phi_L^j \, \tilde{w}_j / P_j$ with $\tilde{w}_j = W_j + R_j - p_A^j \bar{a}$, the location probability is

$$\pi_j \;=\; \frac{v_j^{\theta_L}}{\sum_{k=1}^{J} v_k^{\theta_L}}, \qquad v_j \;=\; \Phi_L^j \, \frac{\tilde{w}_j}{P_j}, \tag{28}$$

where $\Phi_L^j = \exp(\kappa \, z_j^{\text{amen}})/\text{median}$ is the measured 1980 amenity (standardized built-up, population density, and PODES-1983 infrastructure, with $\kappa = 0.5$), and population and sectoral labor follow as $N_j = \bar{L} \, \pi_j$ and $L_{js} = N_j \, \pi_{s|j}$. The amenity fits where people live in the cross section.

Trade costs. In the limit of large σ the CES aggregate (40) converges to the cheapest-supplier rule $p_A^j = \min_k\{\tau_{kj} \, \text{mc}_k\}$. The cost-distance elasticity δ (how fast τ rises with travel time) is distinct from the trade elasticity $\sigma - 1 = 3$. The model sets $\delta = 0.5$; the measured outer-island price-gap slopes, 0.087 for rice and 0.065 for the food basket, lie below it and are gradients that do not identify the trade cost (Section 4.3). Section F.4 states the posted-cost convention and its simplifications.

F.2 Preferences, Demand, and Indirect Utility

Cited in the main text. This section: Section 6.1, p. 44.

A worker living in location j and working in sector s consumes food c_A and the modern good c_S with generalized Stone-Geary (Kongsamut-Rebelo-Xie) preferences:

$$U(c_A, c_S) \;=\; (c_A - \bar{a})^{\psi_A} \, c_S^{\psi_S}, \qquad \psi_A + \psi_S = 1, \qquad c_A \geq \bar{a}, \tag{29}$$

where $\bar{a} > 0$ is the subsistence food requirement (the single non-homotheticity) and ψ_A, ψ_S are the discretionary-expenditure weights. The first $\bar{a}$ units of food buy survival, not utility, so a poorer household spends a larger share on food and a richer one a larger share on the modern good.

Step 1: the budget. The worker earns the sectoral wage w_{js} and receives the per-person land-rent rebate R_j (derived in Section F.7):

$$p_A^j \, c_A + p_S^j \, c_S \;=\; w_{js} + R_j, \tag{30}$$

with location-specific prices p_A^j (food) and p_S^j (modern good).

Step 2: the Lagrangian. Since ln is monotone, maximizing (29) is the same as maximizing $\psi_A \ln(c_A - \bar{a}) + \psi_S \ln c_S$. With multiplier λ on (30),

$$\mathcal{L} \;=\; \psi_A \ln(c_A - \bar{a}) + \psi_S \ln c_S - \lambda\big(p_A^j c_A + p_S^j c_S - w_{js} - R_j\big).$$

Step 3: first-order conditions.

$$\frac{\partial \mathcal{L}}{\partial c_A} = \frac{\psi_A}{c_A - \bar{a}} - \lambda p_A^j = 0, \qquad \frac{\partial \mathcal{L}}{\partial c_S} = \frac{\psi_S}{c_S} - \lambda p_S^j = 0.$$

Step 4: solve. The two FOCs give $p_A^j(c_A - \bar{a}) = \psi_A/\lambda$ and $p_S^j c_S = \psi_S/\lambda$. Adding them and using the budget yields $1/\lambda = w_{js} + R_j - p_A^j \bar{a}$. Define this object, the income left after the subsistence food bill, as the **supernumerary income**

$$\boxed{\tilde{w}_{js} \;\equiv\; w_{js} + R_j - p_A^j\,\bar{a}} \tag{31}$$

and substitute back to obtain the Marshallian demands

$$c_A^\star = \bar{a} + \frac{\psi_A}{p_A^j}\,\tilde{w}_{js}, \qquad\qquad c_S^\star = \frac{\psi_S}{p_S^j}\,\tilde{w}_{js}. \tag{32}$$

The household first buys the subsistence bundle $\bar{a}$, then splits the remaining income $\tilde{w}_{js}$ in fixed shares $\psi_A : \psi_S$. As income grows, the food expenditure share falls toward ψ_A from above (the Engel curve), which is the demand-side engine of structural transformation.

Step 5: indirect utility and the price index. Substituting (32) back into (29),

$$V_{js} \;=\; \underbrace{\psi_A^{\psi_A}\psi_S^{\psi_S}}_{\text{const.}} \times \frac{\tilde{w}_{js}}{P_j}, \qquad P_j \;=\; (p_A^j)^{\psi_A}\,(p_S^j)^{\psi_S}. \tag{33}$$

The Cobb-Douglas constant multiplies every location's value equally and is dropped. So a location is attractive when supernumerary income is high relative to the price index over the supernumerary basket. Evaluated at the location-level expected income $\tilde{w}_j$ (Section F.8), this indirect utility is the object the location choice ranks: not the nominal wage, but $\tilde{w}_j/P_j$.

F.3 Agriculture: Wage and the Return to Land

Farming at j combines effective land $A_j^A T_{jA}$ (quality-adjusted hectares allocated to farming) with labor:

$$Y_{jA} \;=\; (A_j^A\,T_{jA})^{1-\alpha}\,L_{jA}^{\alpha}, \qquad 0<\alpha<1. \tag{34}$$

The function has constant returns in (T_{jA}, L_{jA}) jointly and diminishing returns to labor holding land fixed, so land scarcity is the congestion force in farming.

The wage (labor FOC). A competitive farmer takes p_A^j as given and hires labor until the value of the marginal product equals the wage:

$$w_{jA} \;=\; p_A^j\,\frac{\partial Y_{jA}}{\partial L_{jA}} \;=\; p_A^j\,\alpha\,(A_j^A T_{jA})^{1-\alpha}\,L_{jA}^{\alpha-1}. \tag{35}$$

The land payment. Land collects the value of its marginal product times land used. With Cobb-Douglas, factor payments are revenue shares:

$$v_{jA} \;\equiv\; T_{jA}\,p_A^j\,\frac{\partial Y_{jA}}{\partial T_{jA}} \;=\; (1-\alpha)\,p_A^j\,Y_{jA}. \tag{36}$$

Product exhaustion. Because (34) has constant returns, the two payments exactly exhaust revenue (Euler's theorem):

$$w_{jA}L_{jA} + v_{jA} \;=\; p_A^j\,Y_{jA}. \tag{37}$$

The land-market accounting in Section F.7 therefore closes each location's budget.

F.4 Trade in Food: Cost-Distance Enters Here

Cited in the main text. This section: Section 6.1, p. 45.

The iceberg cost. Shipping one unit of food from k to j melts part of it away, at an iceberg factor that is a power law in the route-aware cost-distance d_{kj} (minutes):

$$\tau_{kj} \;=\; \max(d_{kj}, 1)^{\,\delta}, \qquad \delta > 0. \tag{38}$$

The floor at one minute makes the own-location cost exactly $\tau_{jj} = 1$ (the diagonal of d is zero); the smallest off-diagonal distance in the sample is about five minutes, so the floor binds only own-location shipping. This is the within-location $k = j$, $\tau_{jj} \geq 1$ normalization that closes the system of food prices: every location can always source food from itself at unit trade cost, so p_A^j is well defined at each j even when the location imports nothing.

The farm-gate supply price. Each origin k posts a farm-gate marginal cost obtained by inverting the labor marginal product in (34):

$$\mathrm{mc}_k \;=\; \frac{1}{\partial Y_{kA}/\partial L_{kA}} \;=\; \frac{L_{kA}^{1-\alpha}}{\alpha\,(A_k^A T_{kA})^{1-\alpha}}. \tag{39}$$

Supply-price convention. The food supply-price index in Equation 39 is inverse marginal labor productivity. The non-agricultural good uses the same convention. The indices omit the origin's wage. With unit own-location trade costs and complete self-sourcing, the price index equals the own supply-price index and the corresponding sectoral wage equals one. Away from that limit, an origin's wage affects its factor payments without entering this supply-price index. The equations therefore describe price transmission through productivity and input allocation; they do not impose goods-market clearing.

The food price at each location (CES over origins). The food price at j aggregates the costs of food from all origins, trade costs included, with elasticity σ:

$$p_A^j \;=\; \Big[\sum_k \big(\tau_{kj}\,\mathrm{mc}_k\big)^{1-\sigma}\Big]^{1/(1-\sigma)}, \qquad \sigma > 1, \tag{40}$$

with import shares $\lambda_{kj} \propto (\tau_{kj}\,\mathrm{mc}_k)^{1-\sigma}$. Two limits: as $\sigma \to \infty$ the CES aggregate converges to the cheapest-supplier rule $p_A^j = \min_k\{\tau_{kj}\,\mathrm{mc}_k\}$; and with a single origin the food price at j is exactly $\tau \cdot \mathrm{mc}$. Equation (40) is the smooth "buy from the cheapest place, shipping included" rule in differentiable form. This is the channel that transmits the agro-similarity shock: a productivity gain at transmigration-village hosts lowers their mc_k, lowering p_A^j at every location within shipping reach. The non-agricultural good uses the corresponding trade-price equation in Section F.9.

The food trade cost. The baseline assumes $\delta = 0.5$ and applies the same distance elasticity to the non-agricultural good. The measured price-gap gradients in Section 4.3 lie below this value and do not directly identify trade costs. Figure 19 displays separate numerical comparisons under partial non-agricultural tradability; its fitted coefficients are specific to those calculations.

Implied trade-cost levels and the openness of the solved equilibrium. With own-location trade costs fixed at one and distance measured in minutes, off-diagonal costs are determined by the distance elasticity and the time unit. Changing the time unit while holding the own-location cost fixed changes the level of trade costs relative to self-supply. The numerical comparisons therefore require an explicit distance convention. Their implied sourcing shares do not establish sourcing shares in the restored two-good specification.

F.5 Services: Housing Land, Agglomeration, Wage

Services at j combine labor with housing land T_{jH} (the land the city occupies) and an agglomeration external effect, with $\bar{A}_j^S$ the exogenous modern productivity scale and N_j^η the external term, entering once:

$$Y_{jS} \;=\; \bar{A}_j^S \, N_j^\eta \, T_{jH}^{1-\gamma_S} \, L_{jS}^{\gamma_S}, \qquad 0 < \gamma_S < 1, \quad \eta \geq 0. \tag{41}$$

The N_j^η term is external: an individual firm takes city size as given, so its private labor FOC contains γ_S only, and the land bid is the residual revenue share:

$$w_{jS} \;=\; p_S^j \, \gamma_S \, \bar{A}_j^S \, N_j^\eta \, T_{jH}^{1-\gamma_S} \, L_{jS}^{\gamma_S - 1}, \qquad v_{jS} \;=\; (1-\gamma_S) \, p_S^j \, Y_{jS}, \tag{42}$$

and the private payments again exhaust revenue, $w_{jS}L_{jS} + v_{jS} = p_S^j Y_{jS}$. The non-agricultural good is tradable, and p_S^j is determined by the trade-price equation in Section F.9.

F.6 The Flat Modern Wage With and Without Agglomeration

Differentiating the non-agricultural wage condition gives

$$\frac{\partial \ln w_{jS}}{\partial \ln N_j} = \eta + (\gamma_S - 1)\frac{\partial \ln L_{jS}}{\partial \ln N_j} + (1-\gamma_S)\frac{\partial \ln T_{jH}}{\partial \ln N_j} + \frac{\partial \ln p_S^j}{\partial \ln N_j}. \tag{43}$$

The terms describe agglomeration, labor allocation, land allocation, and the traded-good price response. At $\eta = 0$, price and land adjustments can offset diminishing returns to labor. Their signs and magnitudes follow from the joint allocation; local demand does not impose a positive price response when the whole non-agricultural good is traded.

F.7 The Land Market: von Thünen Split, Rebate, Built-up Endowment

Cited in the main text. This section: Section 6.1, p. 44.

One land market. The location's endowment $\bar{T}_j$ is allocated between farming (T_{jA}) and housing ($T_{jH} = \bar{T}_j - T_{jA}$) by a single rental market. Each use bids its average land value per hectare: farming bids v_{jA}/T_{jA} from (36), housing bids v_{jS}/T_{jH} from (42).

Rents equalize. If farming paid more per hectare, land would flow to farming, and vice versa. Setting the two bids equal, $v_{jA}/T_{jA} = v_{jS}/(\bar{T}_j - T_{jA})$, and solving,

$$T_{jA} \;=\; \bar{T}_j \, \frac{v_{jA}}{v_{jA} + v_{jS}}, \qquad T_{jH} \;=\; \bar{T}_j - T_{jA}. \tag{44}$$

Land shares follow value shares: the sector generating the larger land value occupies the larger share of the location. This single equation is the von Thünen mechanism, and it is why a booming city's agriculture can shrink through land, not just through labor.

The rebate. Land income accrues to local residents: each location's residents receive equal shares of their own location's land income:

$$R_j \;=\; \frac{v_{jA} + v_{jS}}{N_j}. \tag{45}$$

This closes the budget (all local value added becomes local income). Substituting $L_{jA} = \pi_{A|j} N_j$ into (36) shows the agricultural part of the rebate, v_{jA}/N_j, scales as $N_j^{\alpha-1}$ at given prices, land, and sector shares, falling in city size, so the rebate is also a dispersion force.

The total land endowment. The endowment is measured as

$$\bar{T}_j \;=\; T_j^{\text{crop}} + T_j^{\text{built}}, \tag{46}$$

the PODES planted crop area plus the GHS built-up footprint (the housing land the satellite observes). Two rules keep the economics straight: agricultural productivity is recovered on the crop base only, $A_j^A = (L_{jA,0}/T_j^{\text{crop}})^{\alpha}$, winsorized at the 1st and 99th percentiles across the J locations and normalized by its median (built-up is housing, not farmland), and the split (44) operates on the full $\bar{T}_j$. This built-up endowment removes the crop-area floor artifact at cities without diluting the agricultural base; a pure-crop-area endowment is the land robustness.

F.8 Worker Choice: the Two Fréchet Stages

Cited in the main text. This section: Section 6.1, p. 45.

A Fréchet primer. A random variable X with scale u is Fréchet with shape θ if $\Pr(X \le x) = \exp\!\big(-(x/u)^{-\theta}\big)$. Two properties make it the workhorse of discrete choice: it is max-stable, and the probability that draw s is the largest is $u_s^{\theta}/\sum_{s'} u_{s'}^{\theta}$. A larger θ means less idiosyncratic dispersion, so choices chase the scales u_s more aggressively.

Stage 2: the sector choice

After settling at j, a worker draws sector tastes and picks the sector with the higher taste-scaled wage. With scales $u_s = w_{js}$, direct integration of the Fréchet maximum gives

$$\pi_{A|j} \;=\; \frac{w_{jA}^{\theta_S}}{w_{jA}^{\theta_S} + w_{jS}^{\theta_S}}, \qquad L_{jA} = \pi_{A|j} N_j, \quad L_{jS} = (1-\pi_{A|j}) N_j. \tag{47}$$

The sector stage ranks utility, not income. The choice (47) uses wages, which is utility-consistent under a productivity-shock interpretation of the Fréchet draw: if the idiosyncratic term is labor efficiency, a worker's income is $\varepsilon_s w_{js} + R_j$ and utility is $(\varepsilon_s w_{js} + R_j - p_A^j \bar{a})/P_j$; the additive rebate and the price level are common across sectors *within* a location, so the utility-maximizing sector is the wage-maximizing sector. The wage-ranked sector stage is therefore utility maximization with efficiency shocks, not income maximization.

The expected income of a worker at j, before sector tastes are drawn, is the Fréchet expected maximum

$$\mathbb{E}[\max_s w_{js}\varepsilon_s] \;=\; \Gamma\!\Big(1 - \tfrac{1}{\theta_S}\Big) \underbrace{\big(w_{jA}^{\theta_S} + w_{jS}^{\theta_S}\big)^{1/\theta_S}}_{\equiv\, W_j}. \tag{48}$$

Remark (the Gamma factor). The expected maximum in Equation 48 is finite for $\theta_S > 1$. The allocation rules use the index $W_j = (w_{jA}^{\theta_S} + w_{jS}^{\theta_S})^{1/\theta_S}$, defined for any positive θ_S, without treating it as an observed average wage. Its use in expenditure and location value is the income convention stated in Section F.9. The parameter estimates in the numerical comparison below do not fix an estimate for the restored main specification.

Stage 1: the location choice

The location stage ranks the indirect utility (33) evaluated at the location's expected income and scaled by the measured amenity: the location value is

$$v_j \;=\; \Phi_L^j \,\frac{\tilde{w}_j}{P_j}, \qquad \tilde{w}_j = W_j + R_j - p_A^j \bar{a}, \qquad \Phi_L^j = \frac{\exp(\kappa\, z_j^{\text{amen}})}{\text{median}}, \tag{49}$$

where z^{amen} is the standardized 1980 amenity index (GHS built-up, population density, PODES-1983 infrastructure) and κ scales its dispersion. The location choice therefore ranks Stone-Geary indirect utility $\tilde{w}_j/P_j$, not the nominal wage.

A worker draws a location taste shock $\upsilon_j \sim \text{Gumbel}(0, 1/\theta_L)$ (equivalently a Fréchet location value), so realized location utility is $\ln \widetilde{U}_j = \ln \upsilon_j + \upsilon_j$. The probability of choosing j is $\pi_j = \Pr[\ln \widetilde{U}_j \geq \ln \widetilde{U}_k \ \forall k]$. Conditioning on $\upsilon_j = u$ and integrating,

$$\pi_j = \int_{-\infty}^{\infty} \prod_{k \neq j} F_\upsilon\big(u + \ln \upsilon_j - \ln \upsilon_k\big)\, f_\upsilon(u)\, du, \tag{50}$$

with $F_\upsilon(x) = \exp[-e^{-\theta_L x}]$ and $f_\upsilon(x) = \theta_L e^{-\theta_L x} \exp[-e^{-\theta_L x}]$. Defining $S_j \equiv \sum_{k \neq j} (v_k/v_j)^{\theta_L}$, the integrand collapses to $\exp[-e^{-\theta_L u}(1 + S_j)] \cdot \theta_L e^{-\theta_L u}$. The change of variables $z = e^{-\theta_L u}$ (so $dz = -\theta_L e^{-\theta_L u}\, du$) turns (50) into an elementary integral,

$$\pi_j = \int_0^{\infty} e^{-z(1+S_j)}\, dz = \frac{1}{1+S_j} = \frac{v_j^{\theta_L}}{\sum_k v_k^{\theta_L}},$$

so the location probability is

$$\pi_j \;=\; \frac{v_j^{\theta_L}}{\sum_{k=1}^{J} v_k^{\theta_L}}, \qquad N_j = \bar{L}\, \pi_j. \tag{51}$$

The costly-migration generalization. The migration-cost extension replaces (51) with an origin-destination allocation over the 1980 origin population, $\pi_{oj} \propto \left(v_j\, d_{oj}^{-\nu}\right)^{\theta_L}$, with the composite $\theta_L \nu = 1.865$ pinned by the IFLS migration gravity. At $\nu = 0$ it collapses exactly to (51). It is a robustness extension and stays out of the headline.

F.9 Market Clearing, the Value Balance, and Equilibrium

Cited in the main text. This section: Section 6.1, p. 44.

Trade in the non-agricultural good. The supply-price index of origin k is the inverse marginal product of labor:

$$q_k^S = \frac{L_{kS}^{1-\gamma_S}}{\gamma_S \bar{A}_k^S N_k^{\eta} T_{kH}^{1-\gamma_S}}. \tag{52}$$

The price of the non-agricultural good at destination j aggregates all origins,

$$p_S^j = \left[\sum_k (\tau_{kj}^S q_k^S)^{1-\sigma}\right]^{1/(1-\sigma)}, \qquad \tau_{kj}^S = \max(d_{kj}, 1)^{\delta_S}. \tag{53}$$

Own-location trade costs equal one. The main specification sets $\delta_S = \delta$. Both price levels are determined by quantity-based supply indices. A separate reference-wage normalization would add a restriction to this system and is not imposed.

Income and expenditure. The location value and expenditure rules use $W_j + R_j$, with $W_j = (w_{jA}^{\theta_S} + w_{jS}^{\theta_S})^{1/\theta_S}$. Production pays average wages $\bar{w}_j = \pi_{A|j} w_{jA} + (1 - \pi_{A|j}) w_{jS}$. With local land rents, factor payments exhaust the value of production at the model's prices:

$$N_j \bar{w}_j + v_{jA} + v_{jS} = p_A^j Y_{jA} + p_S^j Y_{jS}. \tag{54}$$

Consequently, aggregate spending $N_j(W_j + R_j)$ differs from aggregate factor income by $N_j(W_j - \bar{w}_j)$. Trading both goods does not remove this income convention. Food purchases can be financed by non-agricultural sales, but the price

and allocation equations do not impose a complete balance between goods expenditure and production. Goods-market clearing is not an additional equation for either good.

Equilibrium conditions. Given productivities, amenities, land endowments, distances, and the model parameters, wages satisfy Equation 35 and Equation 42; food and non-agricultural prices satisfy Equation 40 and Equation 53; land allocation and rebates satisfy Equation 44 and Equation 45; sector shares satisfy Equation 47, with $L_{jA} + L_{jS} = N_j$; and location shares satisfy Equation 51, with $\sum_j N_j = \bar{L}$. These conditions apply to the two-good main specification.

The direct comparative static. At fixed employment, population, and land inputs in every location, a change in village agricultural productivity leaves all non-agricultural supply-price indices fixed. The direct change in the urban wage ratio then equals the change in the food price. The food sourcing derivative and the sector-choice derivative in the main text imply

$$\frac{\partial \pi_{A|j}}{\partial \ln A_v^A} = -\theta_S \pi_{A|j}(1 - \pi_{A|j})(1 - \alpha) s_{vj}.$$

This expression holds all location inputs fixed. In the full adjustment both traded-good price indices, sectoral wages, land allocation, and population can change, so the total response uses the derivative of the relative wage in the main text.

F.10 From Equilibrium Conditions to Estimating Equations

The equilibrium conditions map directly to the main empirical specifications.[30]

Sectoral employment share. From the sector-choice condition (47) and the equilibrium wages, the agricultural employment share falls with a lower local food price, motivating

$$\underbrace{\text{AgShare}_j}_{\approx L_{jA}/N_j} = \beta_0 + \beta_1 \underbrace{\text{AgSim}_j}_{\propto A_j^{A,\text{nearby}}} + \beta_2 \,\text{SettlerCount}_j + \beta_3 \, X_j + \varepsilon_j \tag{55}$$

The agricultural-share response is negative when the productivity change lowers the city's agricultural wage relative to its non-agricultural wage. The direct food-price effect gives this sign with all location inputs and the non-agricultural supply indices fixed. The full allocation also changes both traded-good price indices and factor inputs.

Population. The location-choice condition relates population to relative location values. Rural productivity can change a city's income and consumption prices while also changing the attractiveness of other locations. The population response therefore reflects these joint relative changes; a small location-choice elasticity limits adjustment for a given change in relative location values.

Non-agricultural price. The price of the non-agricultural composite follows Equation 53. Supply-price changes across all origins determine its response. The traded-composite price is distinct from local housing rent, so that equation does not impose a sign on the empirical rent response.

Non-agricultural wage. The wage condition in Equation 42 combines the non-agricultural price, employment, land, and any agglomeration effect. Its total response to rural agricultural productivity is determined jointly with those quantities. The price contribution is given by the traded-good equation, so an increase in local demand does not establish a positive non-agricultural price response.

[30] The calibration targets are the standardized (z-on-z) agro-similarity-response slopes reported in Table B4; the model's predicted responses are computed in the same z-on-z units, so the model's slopes are compared with the data's in sign and relative size.

F.11 Numerical Comparison Under Partial Non-Agricultural Tradability

Cited in the main text. This section: Section 6.2, p. 47. Figure F1: Section 6.2, p. 49.

The numerical variant used in the retained figures divides non-agricultural output into a local activity and a traded activity. The quantities and parameter values reported here refer to that variant and its comparison with a food-only-trade economy. They are not estimates of the two-good main specification derived above.

City characteristics in the comparison. Agricultural productivity is constructed from agricultural employment per unit of crop land, local-modern productivity from the modern employment share and land scale, and location amenities from geographic and infrastructure measures. The comparison's traded-modern productivity uses observed census employment valued at the reference allocation's model wages, with total sales scaled to expenditure on the traded component.

Parameter selection in the comparison. The channel calculations raise agricultural productivity by a common amount at the rural settlement hosts and measure each urban response as the simple slope of the standardized change on standardized agro-similarity across the model's 118 urban locations. The elasticities minimize the precision-weighted distance between three such slopes, for population, the agricultural employment share and modern employment, and the same slopes in the data. The squares of Figure 19 are separate re-estimates, one for each food trade cost and land-income convention, each fitted only to the population and agricultural-share slopes, so modern employment is untargeted in that figure.

Target interpretation. Population and agricultural-share gradients are informative about location and sectoral adjustment jointly. The target set and outcome standardization must be held explicit for each numerical comparison. A count of parameters and targets does not establish separate identification of the elasticity levels.

Main estimate. On the built-up $\bar{T}_j$ endowment, under the local land-rent rebate, with the food decay at 0.5 and the traded-modern decay at 1.0, the estimates are $\theta = (\theta_L, \theta_S) = (0.25, 0.82)$ in the services basket and (0.25, 0.84) with prepared foods in the traded basket. At these values the model's standardized responses in the services basket are $+0.10$ for log population and -0.17 for the agricultural employment share, against $+0.06$ (s.e. 0.07) and -0.28 (s.e. 0.10) on the main city sample (Table B4; standard errors in the units of Figure 19), and $+0.13$ for log modern employment against $+0.16$ (s.e. 0.06), each inside the data's 90% interval (Figure 19). The nest holds ($\theta_S > \theta_L$): workers respond more strongly to within-location sectoral wage gaps than to across-location real-income gaps, the ordering under which structural transformation is strong and population reallocation modest. The restricted economy with food as its only traded good cannot reproduce the agricultural-share response through trade, because a city there has no claim on outside output with which to finance a net food purchase (Section F.12); its agricultural-share response falls short of the data at every trade cost in Figure 19, which is the case for the traded-modern activity. Section F.12 sets out that activity.

Robustness on the land specification. A pure-crop-area endowment reverses the ordering $\theta_S > \theta_L$ because of the floor imposed on the crop-area endowment, so the built-up endowment is the baseline specification.

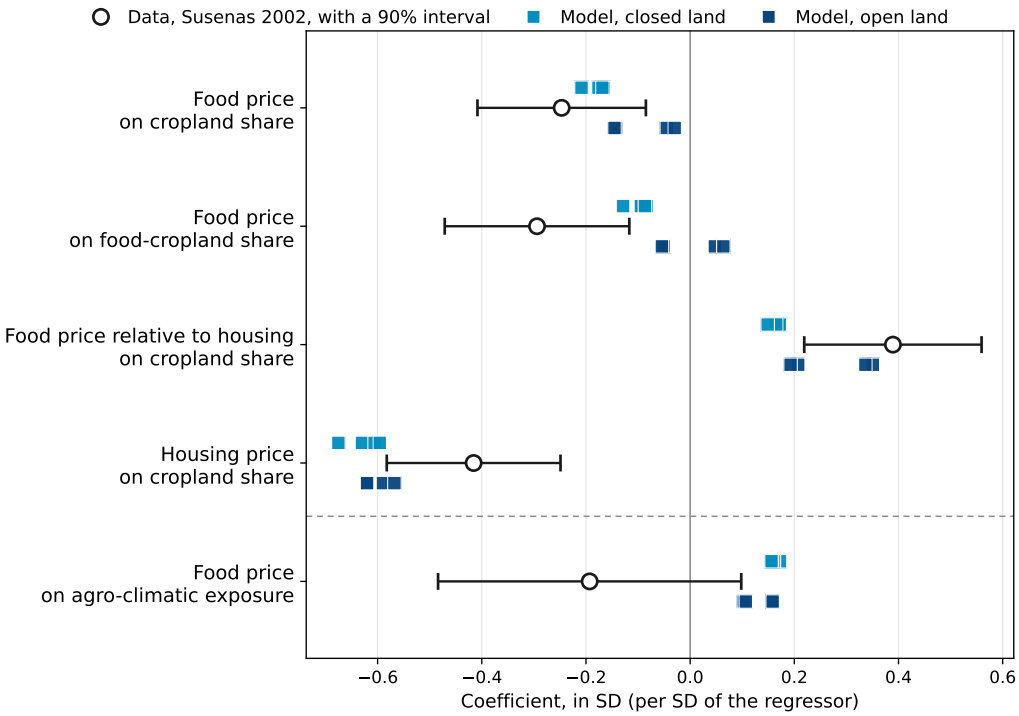
Fit on the untargeted moments. The food price is untargeted in both sets of estimates, and modern employment is untargeted in the re-estimates of Figure 19; the figure below is the untargeted price test that Section 6.2 discusses.

F.12 Channel Comparisons Under Partial Non-Agricultural Tradability

Cited in the main text. This section: Section 6.2, p. 48; Section 6.3, p. 50.

Scope of the comparison. The calculations in this subsection use the partially tradable variant, with local and traded non-agricultural activities. They retain the sectoral earnings index in expenditure and the fixed-price restrictions specified in the figure notes. These calculations explain the numerical figures; they do not introduce a third good into the

Figure F1: The Food Price Falls in the Agricultural Density of a City's Surroundings, in the Data and in the Model



Notes: The untargeted price test implied by the von Thünen logic: a city near abundant cropland faces a cheaper food price. Each row is one cross-sectional regression on the same 102 urban centers with a 2002 Susenas price, with the price and the agricultural density standardized and no controls; controls are left out because the agro-climatic covariates are themselves determinants of how much land is cropped. The hollow circle is the data estimate with a 90% interval on Conley standard errors at a 200-kilometer cutoff; each filled square is the model re-estimated at one of the four assumed food trade costs, with the model side handed the same measured density so the two sides run the identical regression; the light squares rebate land income locally, as the economy of the text does, and the dark squares pool it nationally. Neither side is a targeted moment: the estimator never saw this regression. On the headline row, the food price on the cropland share, the local-rebate model sits inside the data's 90% interval at all four food trade costs. Because the model's land base is all crops, the gradient it responds to is the all-crops one.

main specification. In the main model, trade in the whole non-agricultural good provides the source of export earnings for food purchases.

The comparison variant's traded-modern activity. With food as the only traded good, a local land-rent rebate, and non-traded modern output, a city holds no claim on outside output with which to finance a persistent net food purchase, so no city can move out of agriculture through trade. In that restricted economy the city's declining agricultural share is driven by migration, and shutting migration down leaves the trade-only response wrong-signed. The mechanism this paper studies, a city importing food and reallocating labor to the modern sector, requires a way for the city to run a net food deficit, and trade in modern output provides it, because urban services are reached by travel: the modern composite is split into a catchment-traded component (census trade, transport, and skilled private services, plus manufacturing) and a purely local component, and the traded part takes an Armington structure at a steep distance decay. The block has two parameters. The traded expenditure share is $\psi_M = 0.082$, the Susenas-2011 traded-services share of household spending (0.200 with prepared foods in the traded basket, where the discretionary food weight is $\psi_A = 0.432$ and the subsistence requirement $\bar{a} = 0.096$), so the expenditure side is calibrated services-only while the employment side of the traded bundle also includes manufacturing. The distance decay is measured directly from IFLS household service-flow gravity at $\delta_S = 1.091$, with a standard error of 0.033 (Online Supplement); the model solves at $\delta_S = 1.0$, the highest decay below the measured estimate at which the traded-modern block solves. A city can then

export modern output to pay for net food imports. Each urban region with its rural hinterland is the closed economy of Matsuyama (1992), and within it the villages specialize in food while the city, connected through cost-distance to the villages that feed it, imports food and moves its own workers into the modern sector.

For the agricultural employment share all three counterfactuals are informative, because a city can change what it does whether or not anyone moves. For city size the response with migration shut is zero by construction, since shutting migration means fixing populations, so the informative comparison there runs between the full response and the response with trade shut.

The decomposition at the estimated θ. In the services basket, the trade-only response of the agricultural employment share is 96% of the full response and the migration-only response 8%, with an interaction of −4%; with prepared foods in the traded basket the trade-only response is 100% of the full response, the migration-only response 67% and the interaction −67% (the services basket is Figure 20). Even with migration shut down, the response to the agricultural productivity gain runs almost entirely through trade. For city size the pattern reverses. Its trade-only response is zero by construction, so the informative report is the other two counterfactuals: the migration-only counterfactual reproduces 67% of the full population response in the services basket, with the interaction 33%, and 112% with prepared foods traded, with the interaction −12%. The two baskets bracket how the channels share the remainder, while the trade result does not depend on the basket. Consistent with the friction estimated in Section F.13, holding population fixed leaves the structural transformation intact, while in the restricted economy the same exercise leaves the trade-only response wrong-signed, which is the case for the traded-modern activity.

F.13 Migration Friction: The Gravity Estimate and the Costly-Migration Variant

Cited in the main text. This section: Section 6.1, p. 45.

The frictionless location choice (28) is generalized to an origin-destination allocation over the 1980 origin population, $\pi_{oj} \propto \left(v_j\, d_{oj}^{-\nu}\right)^{\theta_L}$, where the composite $\theta_L\,\nu$ is pinned by the subdistrict-to-subdistrict migration gravity estimated in the IFLS:

$$\theta_L\,\nu = 1.865$$

At the estimated $\theta_L = 0.25$ the structural friction elasticity is $\nu \approx 7.5$, against Bryan and Morten's effective $\tilde{\nu} \approx 3.18$; the choice-probability distance term $d_{oj}^{-1.865}$ does not depend on θ_L, so only the reported ν scales with it.

The costly-migration model, in which both the baseline and the shock bear the moving cost from 1980 origins, is a robustness check. With the moving cost on, the rural productivity boom holds people near the transmigration-village hosts and city growth does not absorb them (a stay-home effect), so the population response to the shock turns negative, while the modern-employment response and the trade-only response of the agricultural share keep their signs. The migration cost is a friction that dampens over-mobility. A dynamic migration model, motivated by the gap between the per-decade gravity estimate and the 30-year stay share, is left for future work.